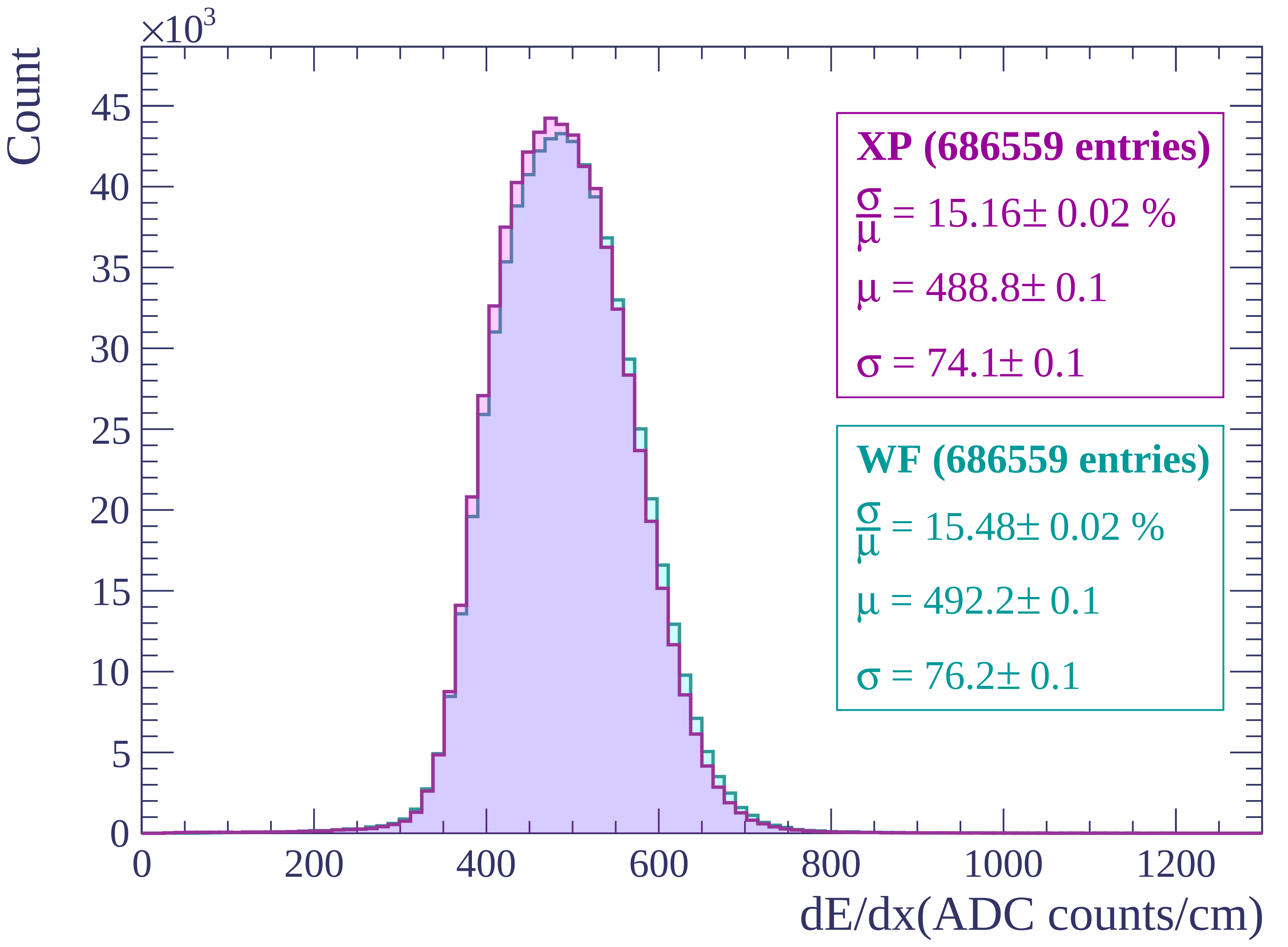
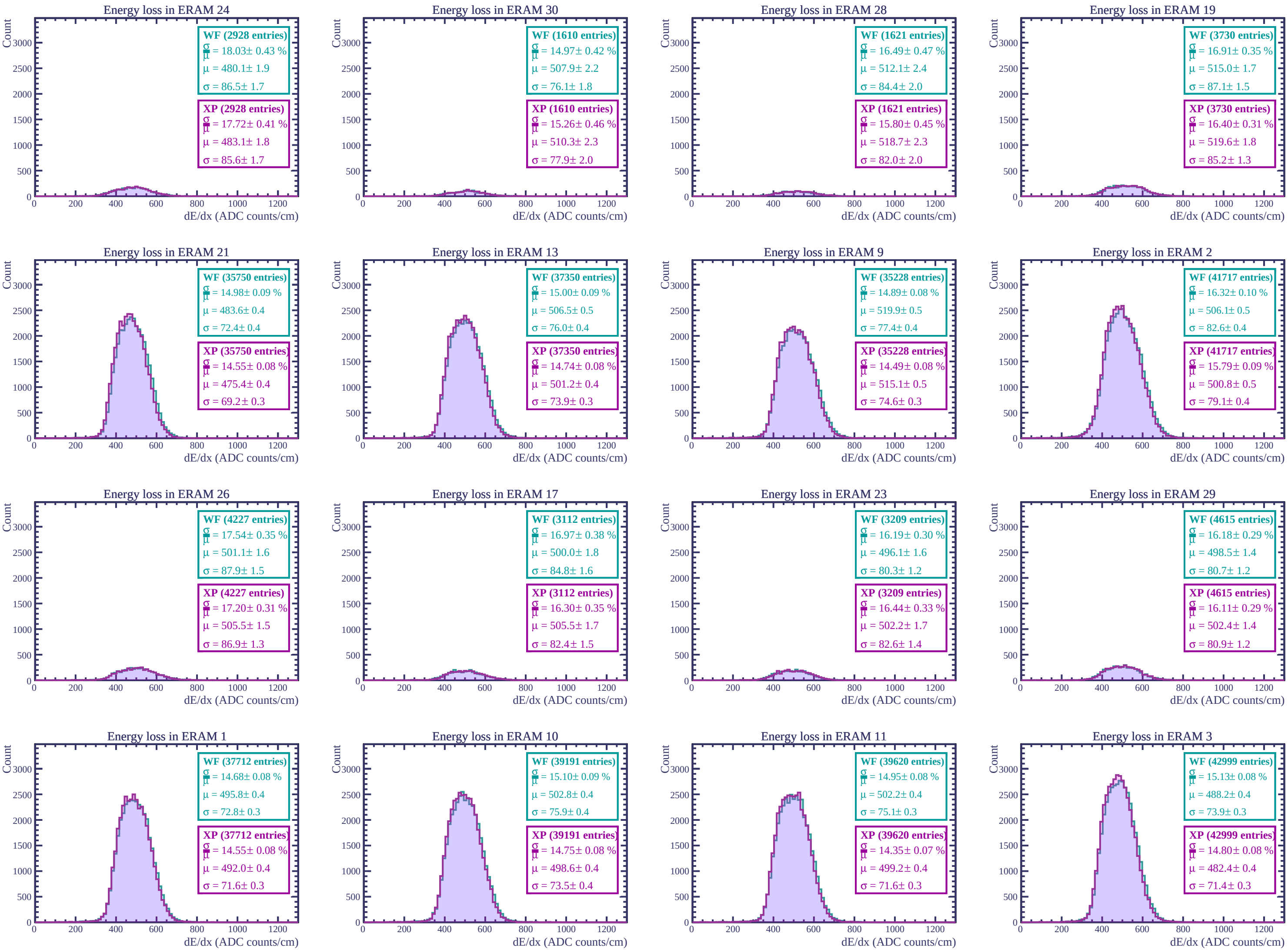
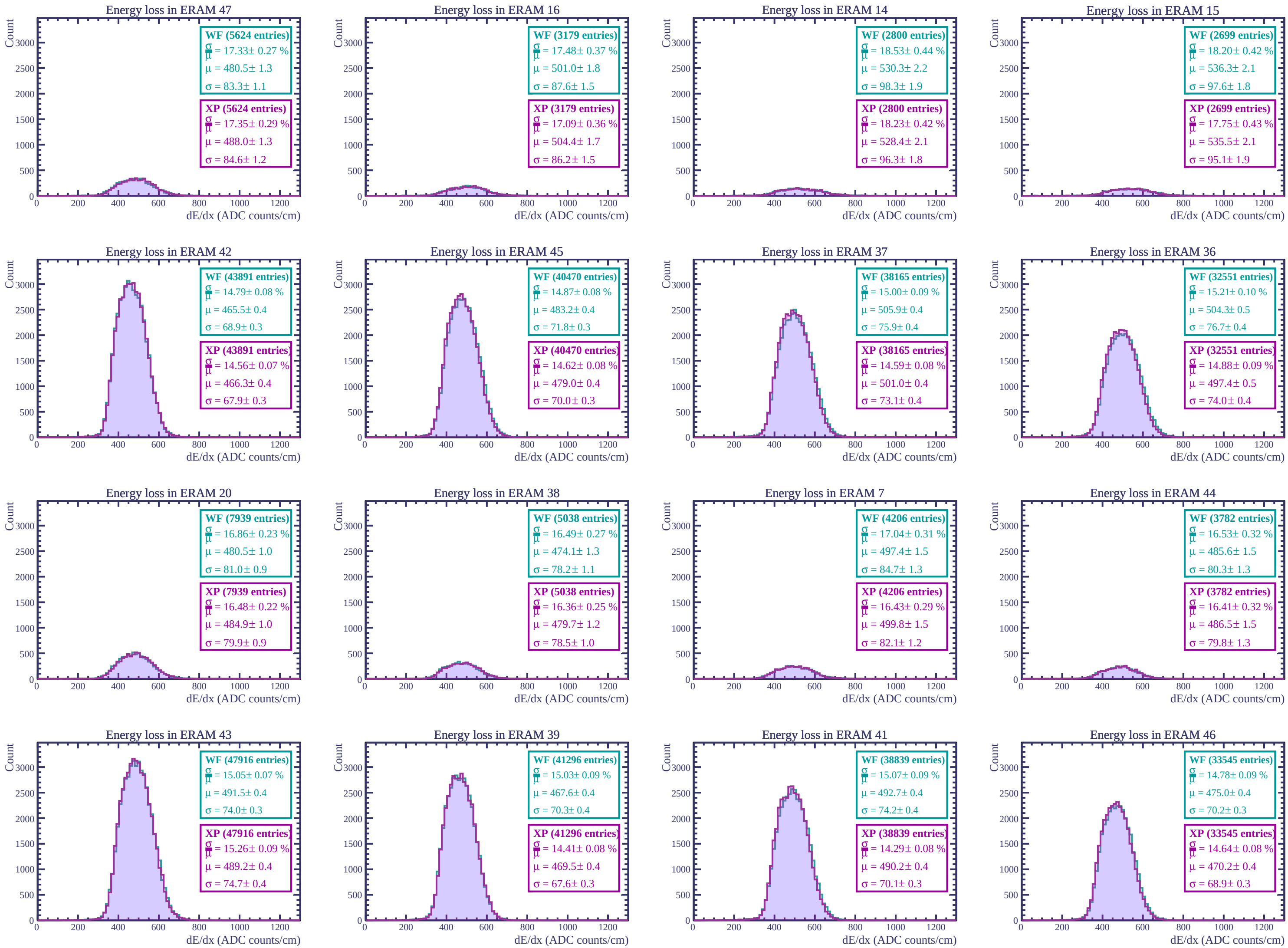
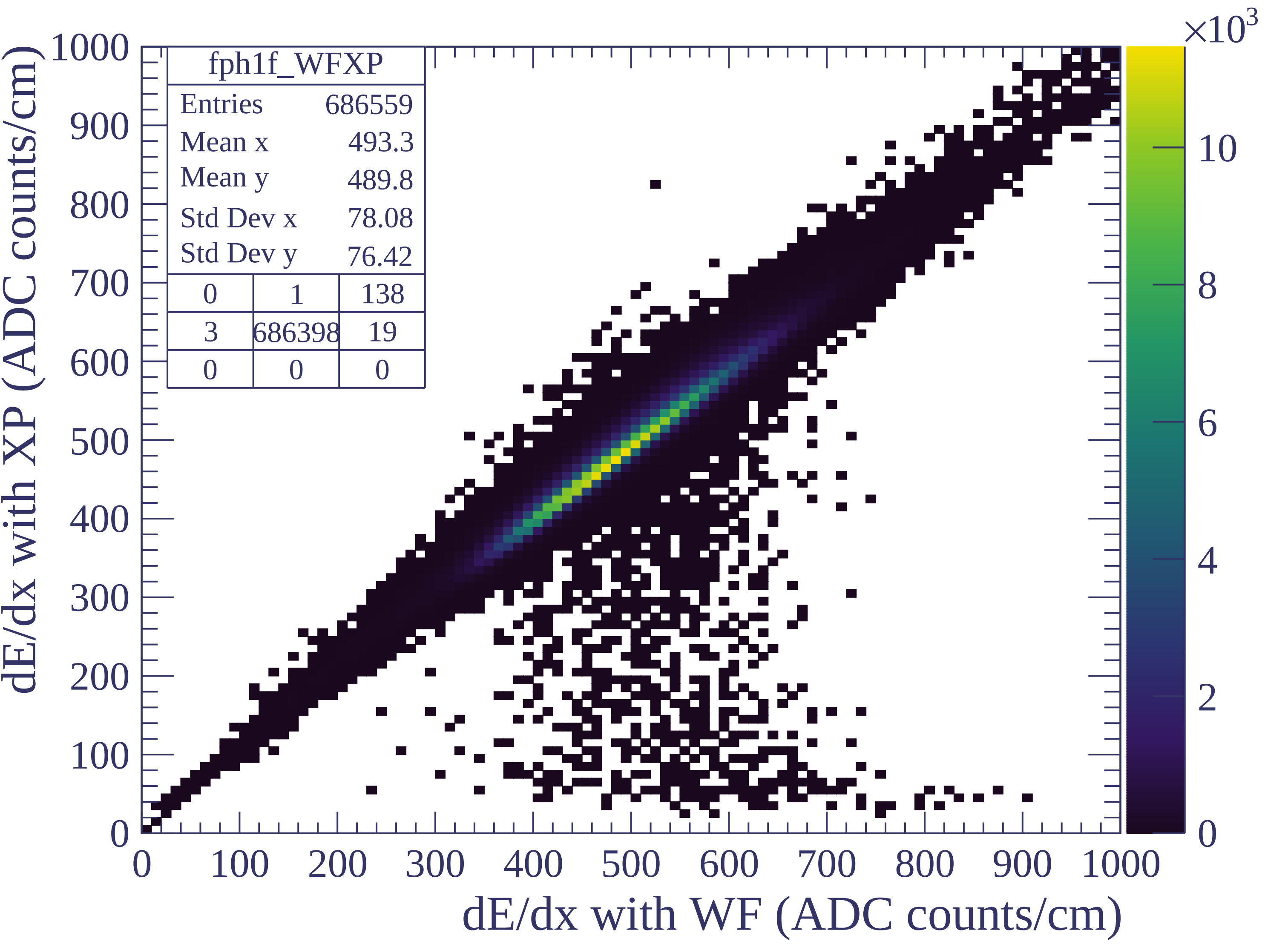
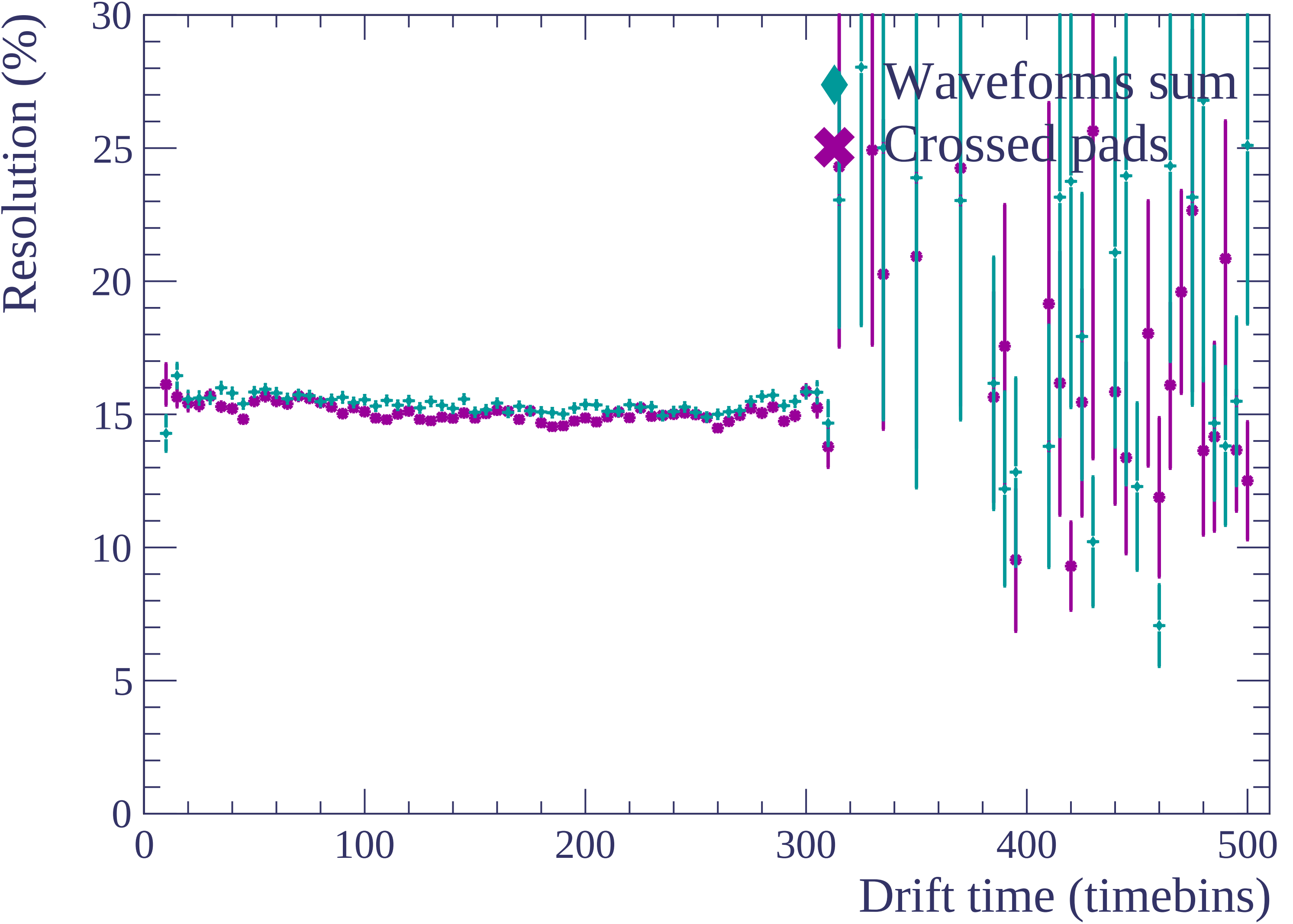

×10^3
Count
45
40
35
30
25
20
15
10
5
0
0
200
400
600
800
1000
1200
dE/dx(ADC counts/cm)
XP (686559 entries)
σ/μ = 15.16± 0.02 %
μ = 488.8± 0.1
σ = 74.1± 0.1
WF (686559 entries)
σ/μ = 15.48± 0.02 %
μ = 492.2± 0.1
σ = 76.2± 0.1

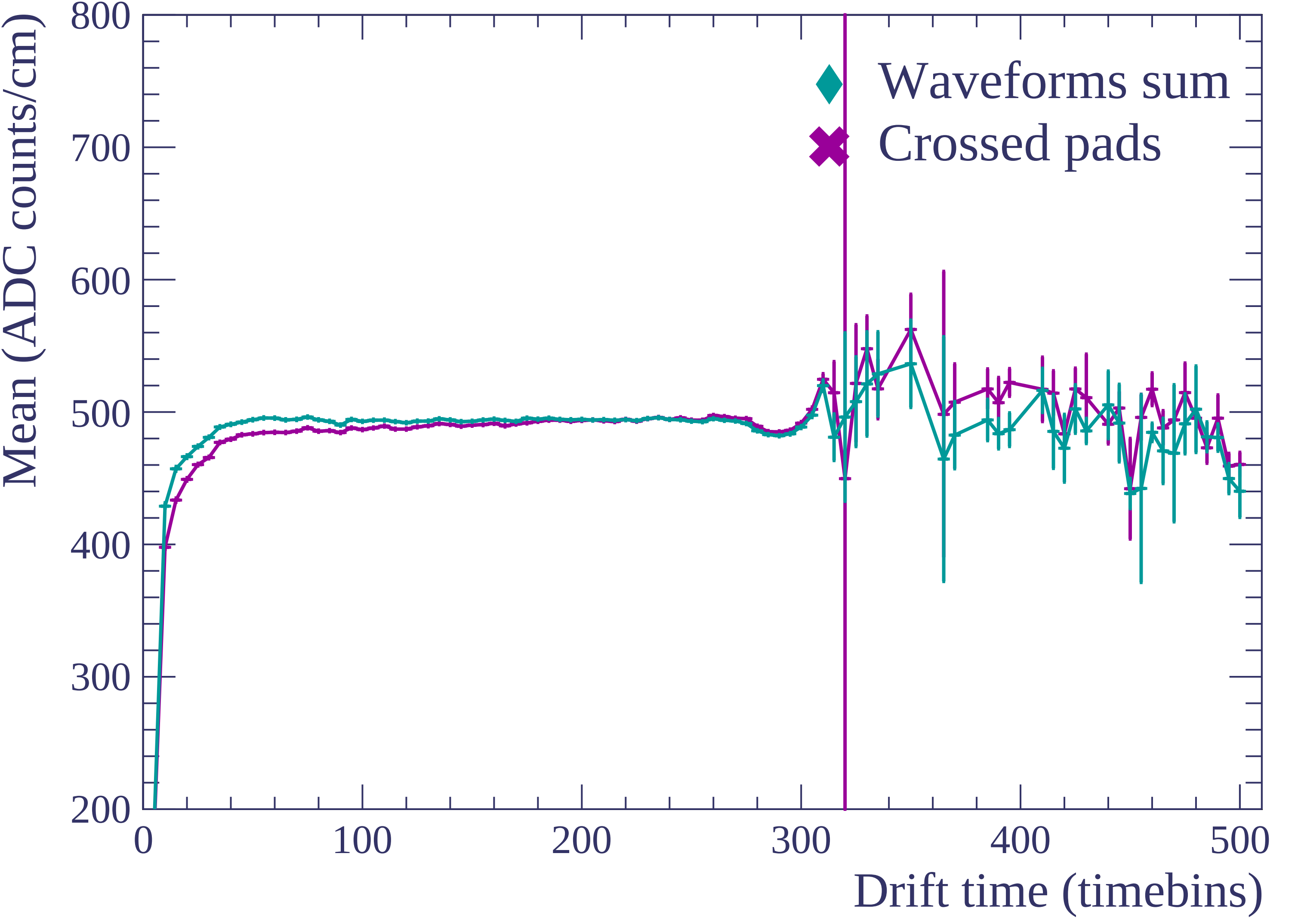

Energy loss in ERAM 24
WF (2928 entries)
σ/μ = 18.03± 0.43 %
μ = 480.1± 1.9
σ = 86.5± 1.7
XP (2928 entries)
σ/μ = 17.72± 0.41 %
μ = 483.1± 1.8
σ = 85.6± 1.7
Energy loss in ERAM 30
WF (1610 entries)
σ/μ = 14.97± 0.42 %
μ = 507.9± 2.2
σ = 76.1± 1.8
XP (1610 entries)
σ/μ = 15.26± 0.46 %
μ = 510.3± 2.3
σ = 77.9± 2.0
Energy loss in ERAM 28
WF (1621 entries)
σ/μ = 16.49± 0.47 %
μ = 512.1± 2.4
σ = 84.4± 2.0
XP (1621 entries)
σ/μ = 15.80± 0.45 %
μ = 518.7± 2.3
σ = 82.0± 2.0
Energy loss in ERAM 19
WF (3730 entries)
σ/μ = 16.91± 0.35 %
μ = 515.0± 1.7
σ = 87.1± 1.5
XP (3730 entries)
σ/μ = 16.40± 0.31 %
μ = 519.6± 1.8
σ = 85.2± 1.3
Energy loss in ERAM 21
WF (35750 entries)
σ/μ = 14.98± 0.09 %
μ = 483.6± 0.4
σ = 72.4± 0.4
XP (35750 entries)
σ/μ = 14.55± 0.08 %
μ = 475.4± 0.4
σ = 69.2± 0.3
Energy loss in ERAM 13
WF (37350 entries)
σ/μ = 15.00± 0.09 %
μ = 506.5± 0.5
σ = 76.0± 0.4
XP (37350 entries)
σ/μ = 14.74± 0.08 %
μ = 501.2± 0.4
σ = 73.9± 0.3
Energy loss in ERAM 9
WF (35228 entries)
σ/μ = 14.89± 0.08 %
μ = 519.9± 0.5
σ = 77.4± 0.4
XP (35228 entries)
σ/μ = 14.49± 0.08 %
μ = 515.1± 0.5
σ = 74.6± 0.3
Energy loss in ERAM 2
WF (41717 entries)
σ/μ = 16.32± 0.10 %
μ = 506.1± 0.5
σ = 82.6± 0.4
XP (41717 entries)
σ/μ = 15.79± 0.09 %
μ = 500.8± 0.5
σ = 79.1± 0.4
Energy loss in ERAM 26
WF (4227 entries)
σ/μ = 17.54± 0.35 %
μ = 501.1± 1.6
σ = 87.9± 1.5
XP (4227 entries)
σ/μ = 17.20± 0.31 %
μ = 505.5± 1.5
σ = 86.9± 1.3
Energy loss in ERAM 17
WF (3112 entries)
σ/μ = 16.97± 0.38 %
μ = 500.0± 1.8
σ = 84.8± 1.6
XP (3112 entries)
σ/μ = 16.30± 0.35 %
μ = 505.5± 1.7
σ = 82.4± 1.5
Energy loss in ERAM 23
WF (3209 entries)
σ/μ = 16.19± 0.30 %
μ = 496.1± 1.6
σ = 80.3± 1.2
XP (3209 entries)
σ/μ = 16.44± 0.33 %
μ = 502.2± 1.7
σ = 82.6± 1.4
Energy loss in ERAM 29
WF (4615 entries)
σ/μ = 16.18± 0.29 %
μ = 498.5± 1.4
σ = 80.7± 1.2
XP (4615 entries)
σ/μ = 16.11± 0.29 %
μ = 502.4± 1.4
σ = 80.9± 1.2
Energy loss in ERAM 1
WF (37712 entries)
σ/μ = 14.68± 0.08 %
μ = 495.8± 0.4
σ = 72.8± 0.3
XP (37712 entries)
σ/μ = 14.55± 0.08 %
μ = 492.0± 0.4
σ = 71.6± 0.3
Energy loss in ERAM 10
WF (39191 entries)
σ/μ = 15.10± 0.09 %
μ = 502.8± 0.4
σ = 75.9± 0.4
XP (39191 entries)
σ/μ = 14.75± 0.08 %
μ = 498.6± 0.4
σ = 73.5± 0.4
Energy loss in ERAM 11
WF (39620 entries)
σ/μ = 14.95± 0.08 %
μ = 502.2± 0.4
σ = 75.1± 0.3
XP (39620 entries)
σ/μ = 14.35± 0.07 %
μ = 499.2± 0.4
σ = 71.6± 0.3
Energy loss in ERAM 3
WF (42999 entries)
σ/μ = 15.13± 0.08 %
μ = 488.2± 0.4
σ = 73.9± 0.3
XP (42999 entries)
σ/μ = 14.80± 0.08 %
μ = 482.4± 0.4
σ = 71.4± 0.3
Count
dE/dx (ADC counts/cm)

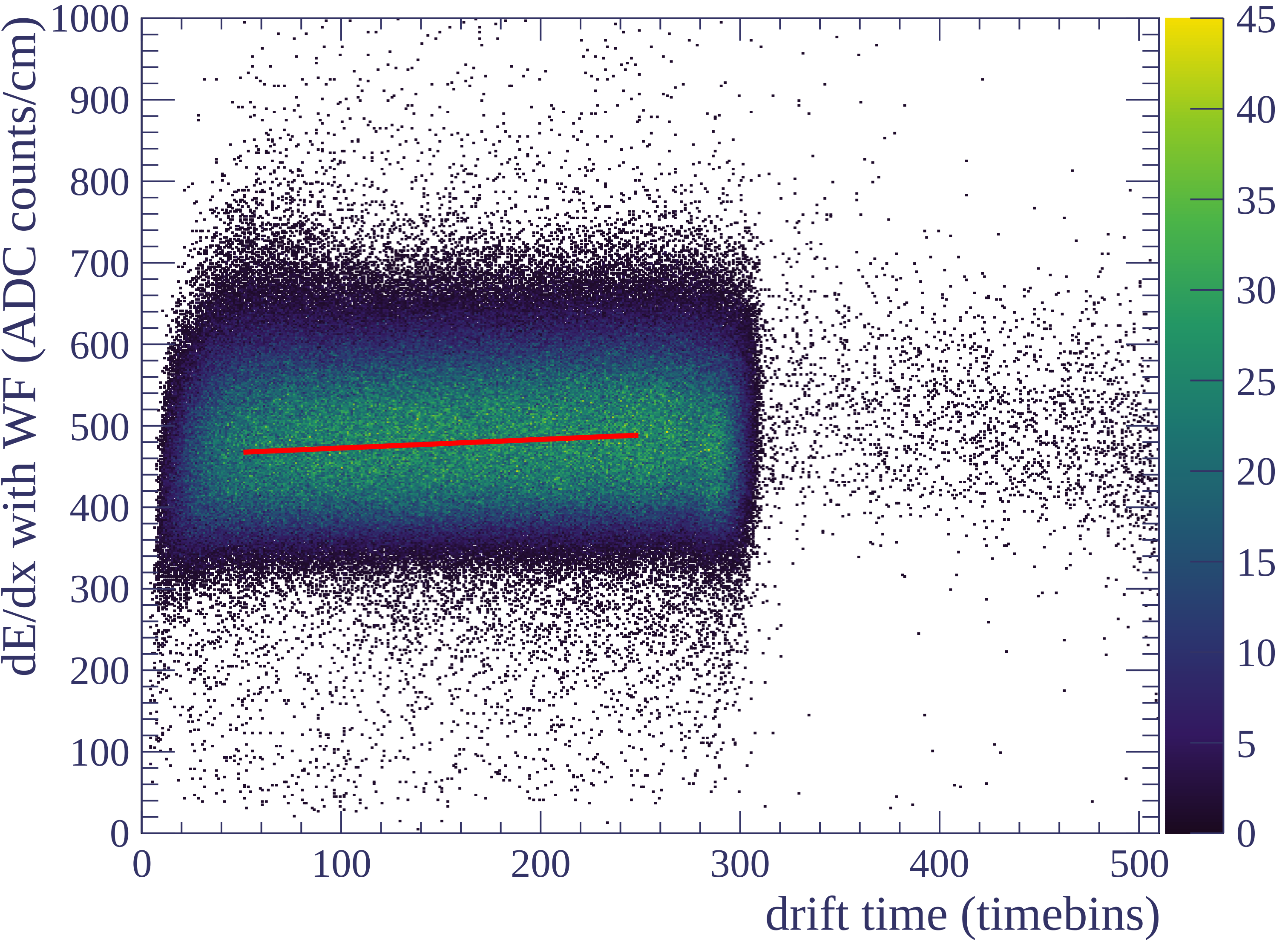

Energy loss in ERAM 47
WF (5624 entries)
σ/μ = 17.33± 0.27 %
μ = 480.5± 1.3
σ = 83.3± 1.1
XP (5624 entries)
σ/μ = 17.35± 0.29 %
μ = 488.0± 1.3
σ = 84.6± 1.2
Energy loss in ERAM 16
WF (3179 entries)
σ/μ = 17.48± 0.37 %
μ = 501.0± 1.8
σ = 87.6± 1.5
XP (3179 entries)
σ/μ = 17.09± 0.36 %
μ = 504.4± 1.7
σ = 86.2± 1.5
Energy loss in ERAM 14
WF (2800 entries)
σ/μ = 18.53± 0.44 %
μ = 530.3± 2.2
σ = 98.3± 1.9
XP (2800 entries)
σ/μ = 18.23± 0.42 %
μ = 528.4± 2.1
σ = 96.3± 1.8
Energy loss in ERAM 15
WF (2699 entries)
σ/μ = 18.20± 0.42 %
μ = 536.3± 2.1
σ = 97.6± 1.8
XP (2699 entries)
σ/μ = 17.75± 0.43 %
μ = 535.5± 2.1
σ = 95.1± 1.9
Energy loss in ERAM 42
WF (43891 entries)
σ/μ = 14.79± 0.08 %
μ = 465.5± 0.4
σ = 68.9± 0.3
XP (43891 entries)
σ/μ = 14.56± 0.07 %
μ = 466.3± 0.4
σ = 67.9± 0.3
Energy loss in ERAM 45
WF (40470 entries)
σ/μ = 14.87± 0.08 %
μ = 483.2± 0.4
σ = 71.8± 0.3
XP (40470 entries)
σ/μ = 14.62± 0.08 %
μ = 479.0± 0.4
σ = 70.0± 0.3
Energy loss in ERAM 37
WF (38165 entries)
σ/μ = 15.00± 0.09 %
μ = 505.9± 0.4
σ = 75.9± 0.4
XP (38165 entries)
σ/μ = 14.59± 0.08 %
μ = 501.0± 0.4
σ = 73.1± 0.4
Energy loss in ERAM 36
WF (32551 entries)
σ/μ = 15.21± 0.10 %
μ = 504.3± 0.5
σ = 76.7± 0.4
XP (32551 entries)
σ/μ = 14.88± 0.09 %
μ = 497.4± 0.5
σ = 74.0± 0.4
Energy loss in ERAM 20
WF (7939 entries)
σ/μ = 16.86± 0.23 %
μ = 480.5± 1.0
σ = 81.0± 0.9
XP (7939 entries)
σ/μ = 16.48± 0.22 %
μ = 484.9± 1.0
σ = 79.9± 0.9
Energy loss in ERAM 38
WF (5038 entries)
σ/μ = 16.49± 0.27 %
μ = 474.1± 1.3
σ = 78.2± 1.1
XP (5038 entries)
σ/μ = 16.36± 0.25 %
μ = 479.7± 1.2
σ = 78.5± 1.0
Energy loss in ERAM 7
WF (4206 entries)
σ/μ = 17.04± 0.31 %
μ = 497.4± 1.5
σ = 84.7± 1.3
XP (4206 entries)
σ/μ = 16.43± 0.29 %
μ = 499.8± 1.5
σ = 82.1± 1.2
Energy loss in ERAM 44
WF (3782 entries)
σ/μ = 16.53± 0.32 %
μ = 485.6± 1.5
σ = 80.3± 1.3
XP (3782 entries)
σ/μ = 16.41± 0.32 %
μ = 486.5± 1.5
σ = 79.8± 1.3
Energy loss in ERAM 43
WF (47916 entries)
σ/μ = 15.05± 0.07 %
μ = 491.5± 0.4
σ = 74.0± 0.3
XP (47916 entries)
σ/μ = 15.26± 0.09 %
μ = 489.2± 0.4
σ = 74.7± 0.4
Energy loss in ERAM 39
WF (41296 entries)
σ/μ = 15.03± 0.09 %
μ = 467.6± 0.4
σ = 70.3± 0.4
XP (41296 entries)
σ/μ = 14.41± 0.08 %
μ = 469.5± 0.4
σ = 67.6± 0.3
Energy loss in ERAM 41
WF (38839 entries)
σ/μ = 15.07± 0.09 %
μ = 492.7± 0.4
σ = 74.2± 0.4
XP (38839 entries)
σ/μ = 14.29± 0.08 %
μ = 490.2± 0.4
σ = 70.1± 0.3
Energy loss in ERAM 46
WF (33545 entries)
σ/μ = 14.78± 0.09 %
μ = 475.0± 0.4
σ = 70.2± 0.3
XP (33545 entries)
σ/μ = 14.64± 0.08 %
μ = 470.2± 0.4
σ = 68.9± 0.3
Count
dE/dx (ADC counts/cm)

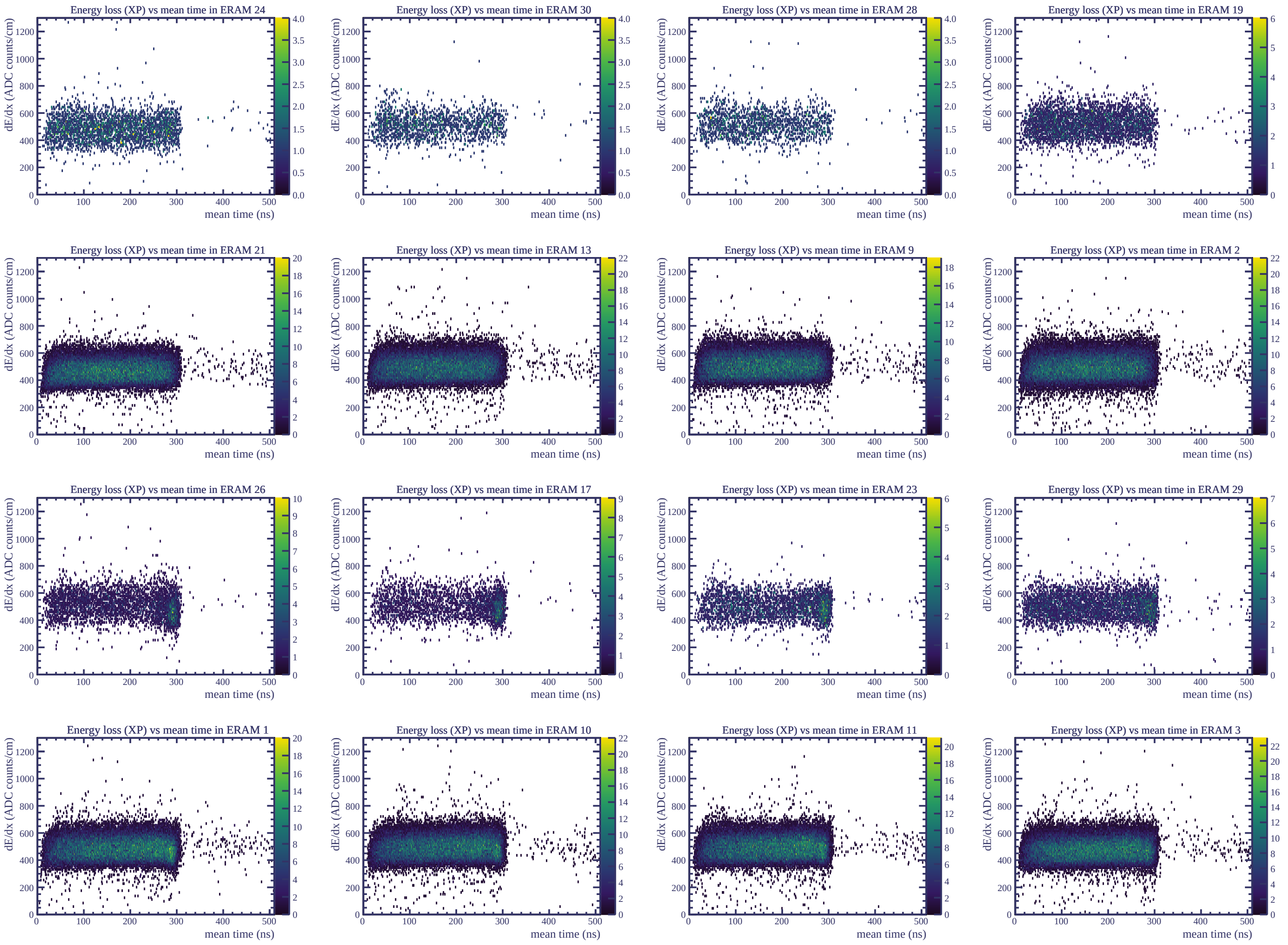

fph1f_WFXP
Entries 686559
Mean x 493.3
Mean y 489.8
Std Dev x 78.08
Std Dev y 76.42
0 1 138
3 686398 19
0 0 0
dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
×10^3
0
100
200
300
400
500
600
700
800
900
1000
2
4
6
8
10



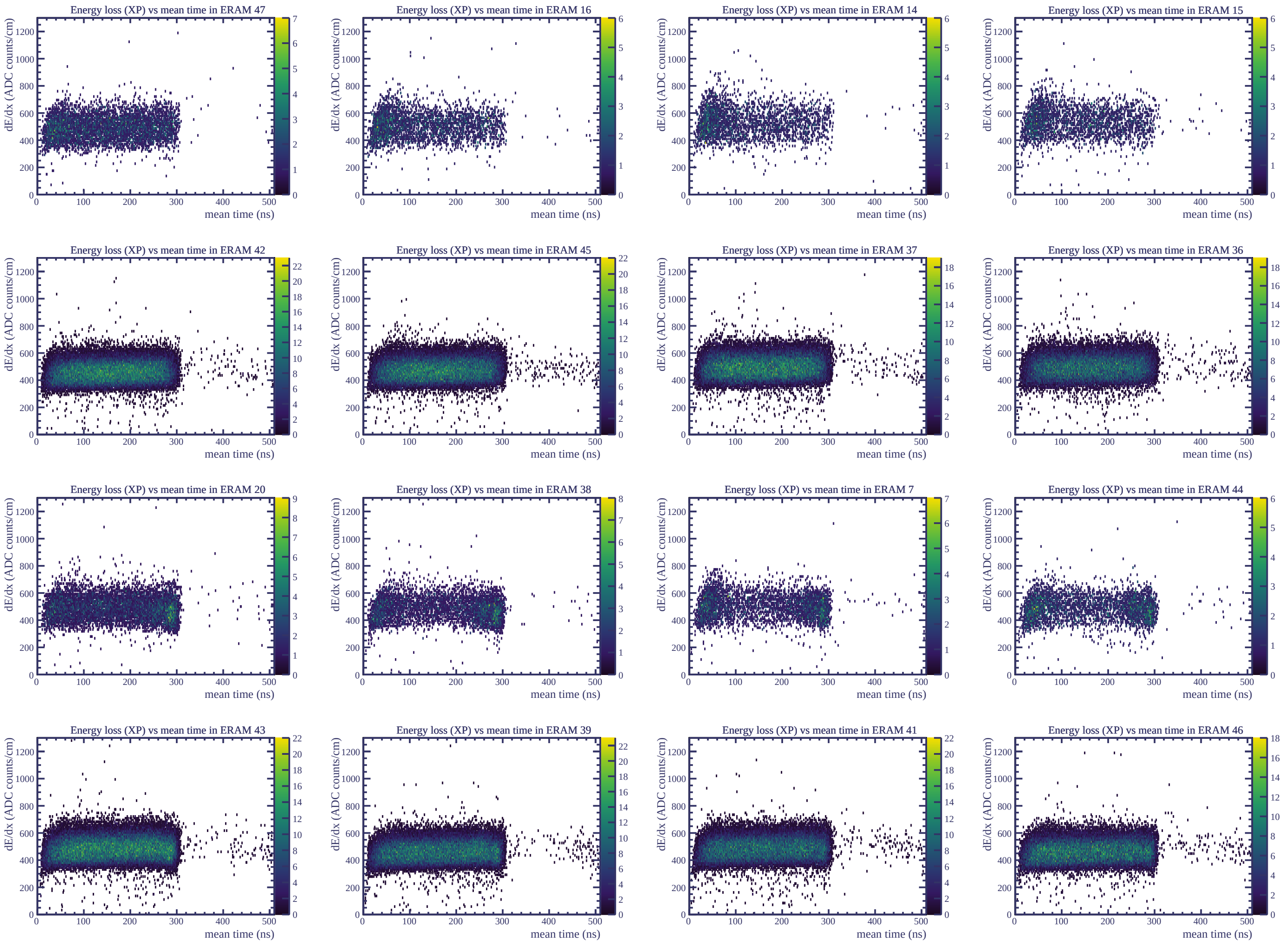
Resolution (%)
Drift time (timebins)
Waveforms sum
Crossed pads
0
5
10
15
20
25
30
0
100
200
300
400
500

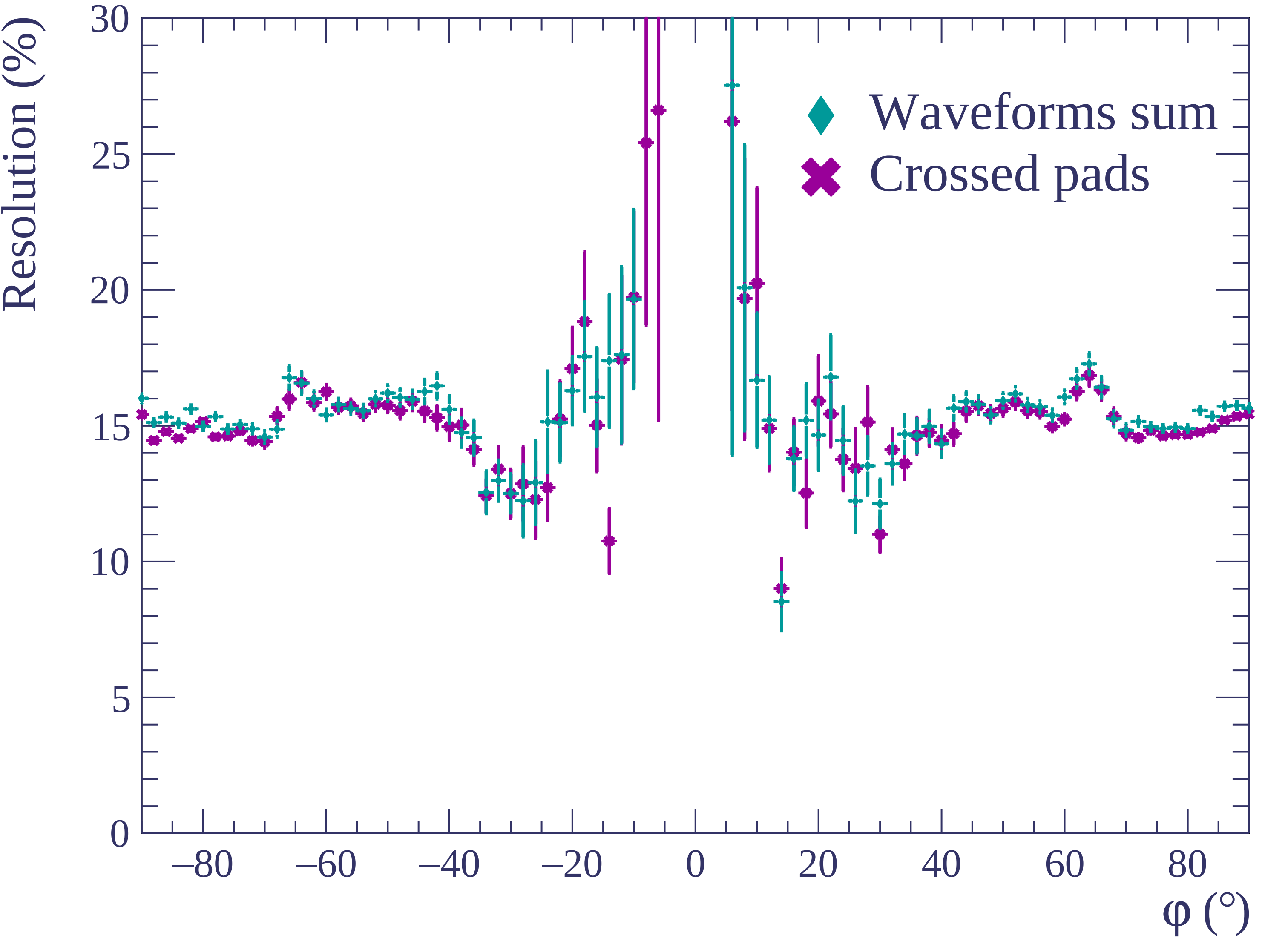

Waveforms sum
Crossed pads
Mean (ADC counts/cm)
Drift time (timebins)
800
700
600
500
400
300
200
0
100
200
300
400
500


dE/dx with WF (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
1000
drift time (timebins)
0
100
200
300
400
500
0
5
10
15
20
25
30
35
40
45

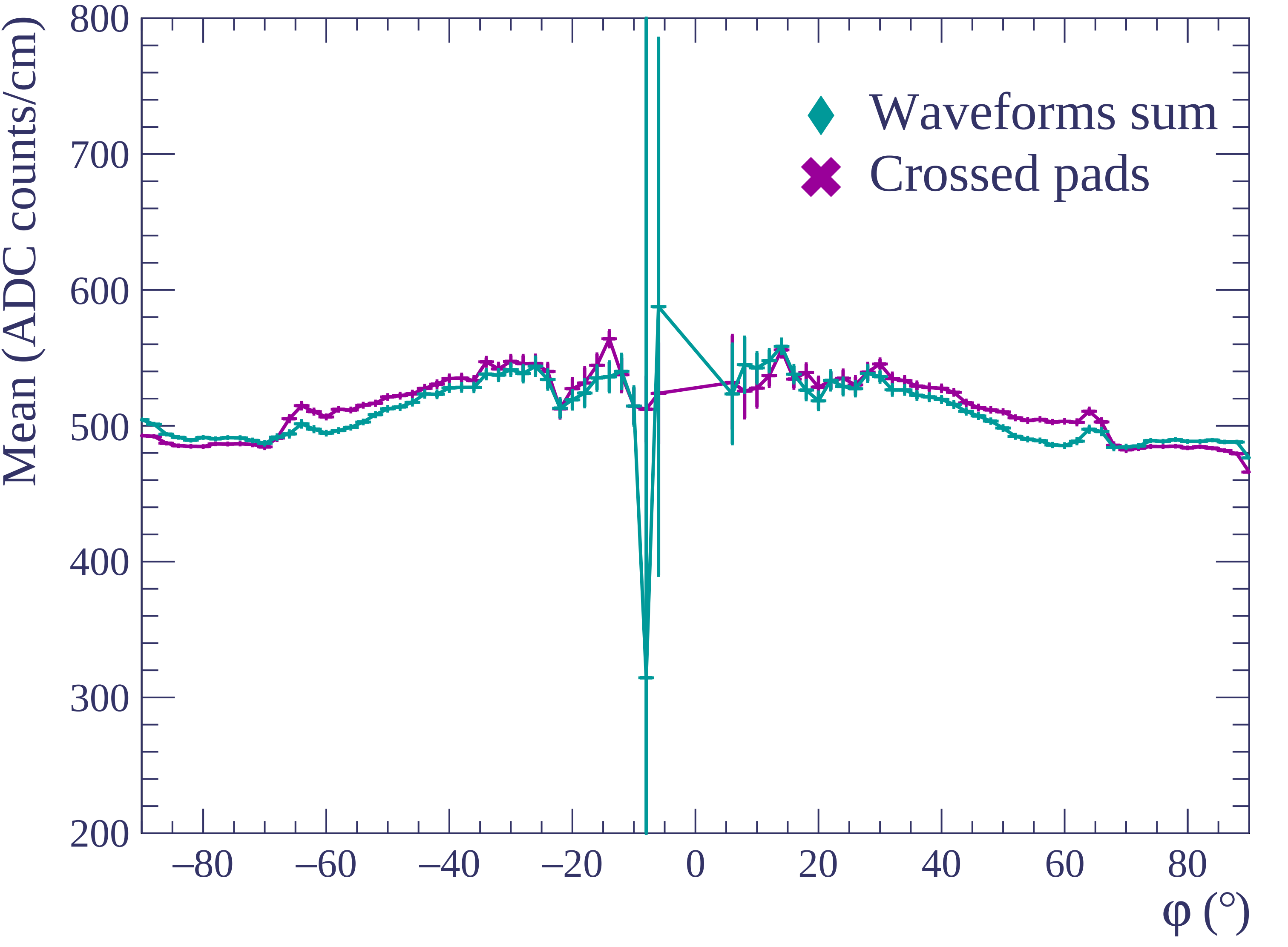


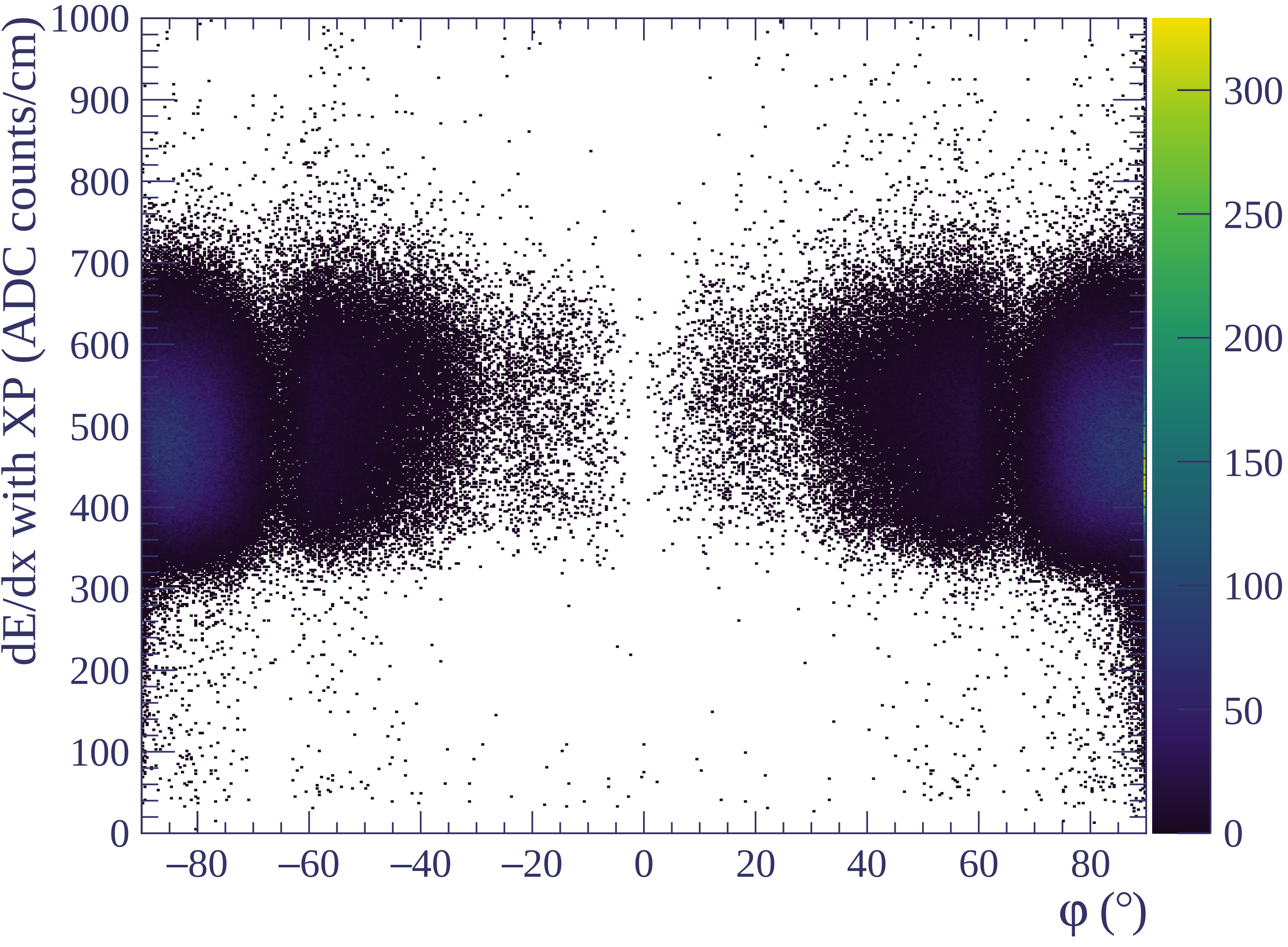
Energy loss (XP) vs mean time in ERAM 24
Energy loss (XP) vs mean time in ERAM 30
Energy loss (XP) vs mean time in ERAM 28
Energy loss (XP) vs mean time in ERAM 19
Energy loss (XP) vs mean time in ERAM 21
Energy loss (XP) vs mean time in ERAM 13
Energy loss (XP) vs mean time in ERAM 9
Energy loss (XP) vs mean time in ERAM 2
Energy loss (XP) vs mean time in ERAM 26
Energy loss (XP) vs mean time in ERAM 17
Energy loss (XP) vs mean time in ERAM 23
Energy loss (XP) vs mean time in ERAM 29
Energy loss (XP) vs mean time in ERAM 1
Energy loss (XP) vs mean time in ERAM 10
Energy loss (XP) vs mean time in ERAM 11
Energy loss (XP) vs mean time in ERAM 3
dE/dx (ADC counts/cm)
mean time (ns)

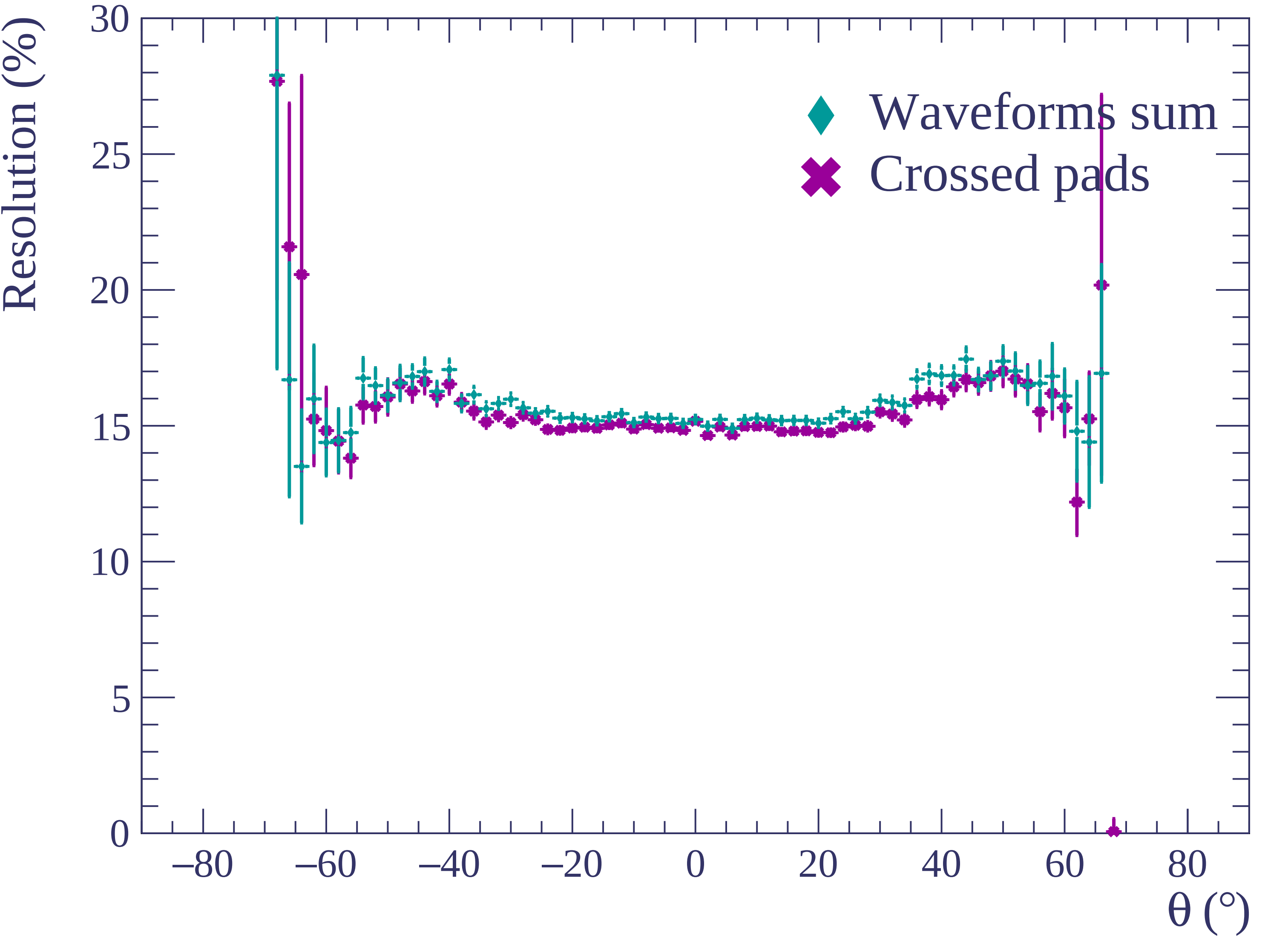

Energy loss (XP) vs mean time in ERAM 47
Energy loss (XP) vs mean time in ERAM 16
Energy loss (XP) vs mean time in ERAM 14
Energy loss (XP) vs mean time in ERAM 15
Energy loss (XP) vs mean time in ERAM 42
Energy loss (XP) vs mean time in ERAM 45
Energy loss (XP) vs mean time in ERAM 37
Energy loss (XP) vs mean time in ERAM 36
Energy loss (XP) vs mean time in ERAM 20
Energy loss (XP) vs mean time in ERAM 38
Energy loss (XP) vs mean time in ERAM 7
Energy loss (XP) vs mean time in ERAM 44
Energy loss (XP) vs mean time in ERAM 43
Energy loss (XP) vs mean time in ERAM 39
Energy loss (XP) vs mean time in ERAM 41
Energy loss (XP) vs mean time in ERAM 46
dE/dx (ADC counts/cm)
mean time (ns)

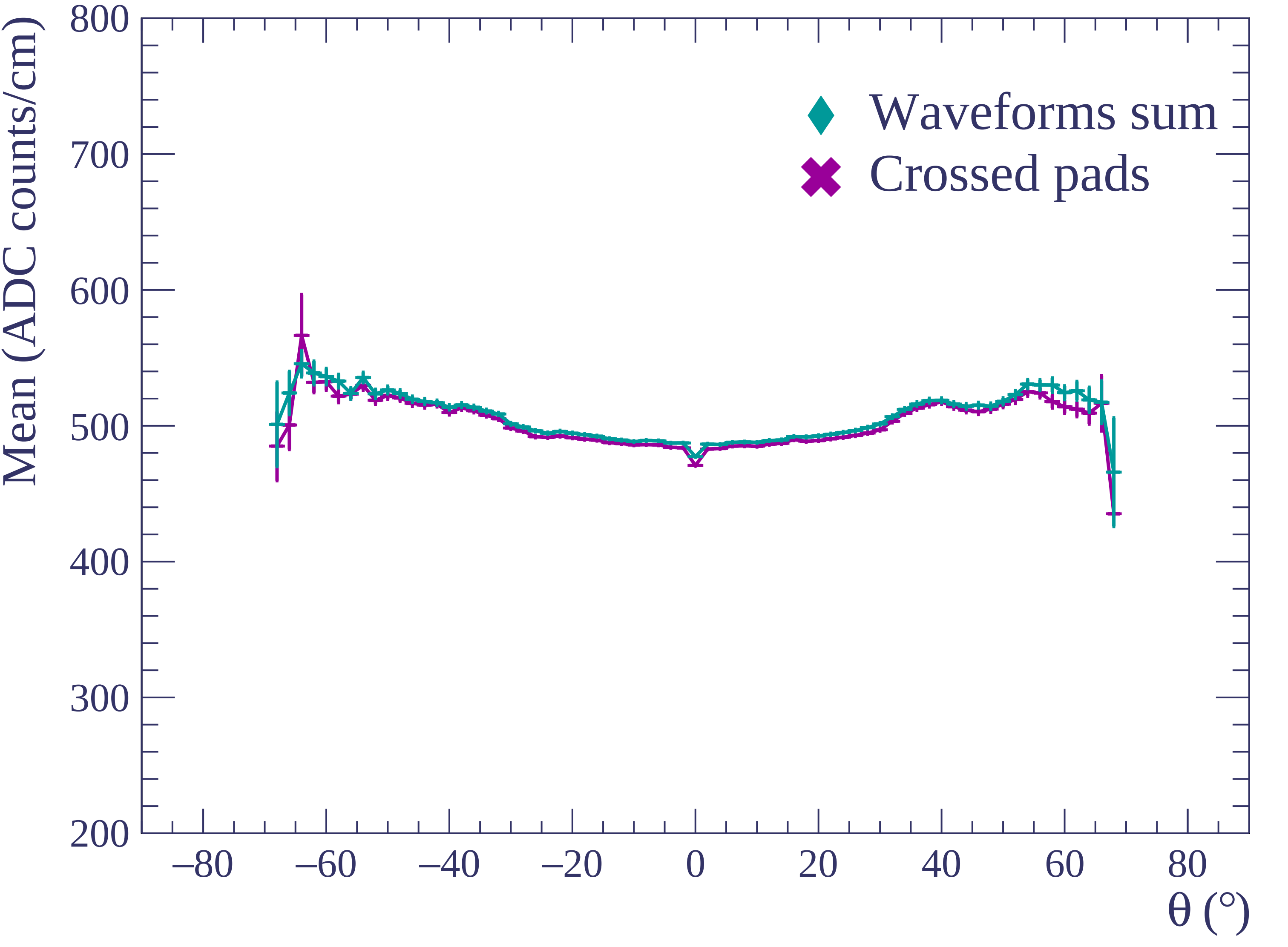


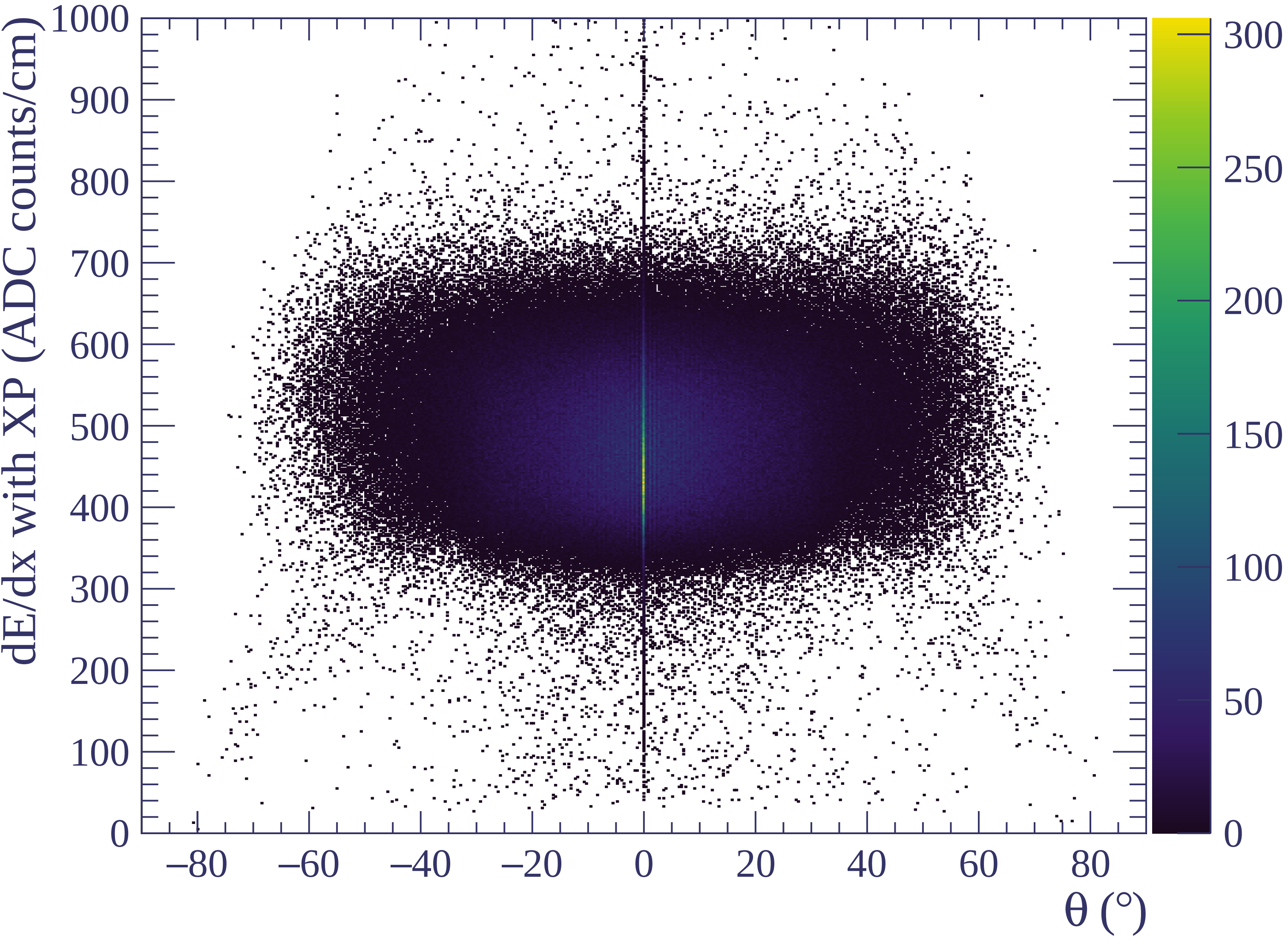
Resolution (%)
φ (°)
Waveforms sum
Crossed pads
0
5
10
15
20
25
30
−80
−60
−40
−20
0
20
40
60
80


Mean (ADC counts/cm)
800
700
600
500
400
300
200
−80
−60
−40
−20
0
20
40
60
80
φ (°)
Waveforms sum
Crossed pads

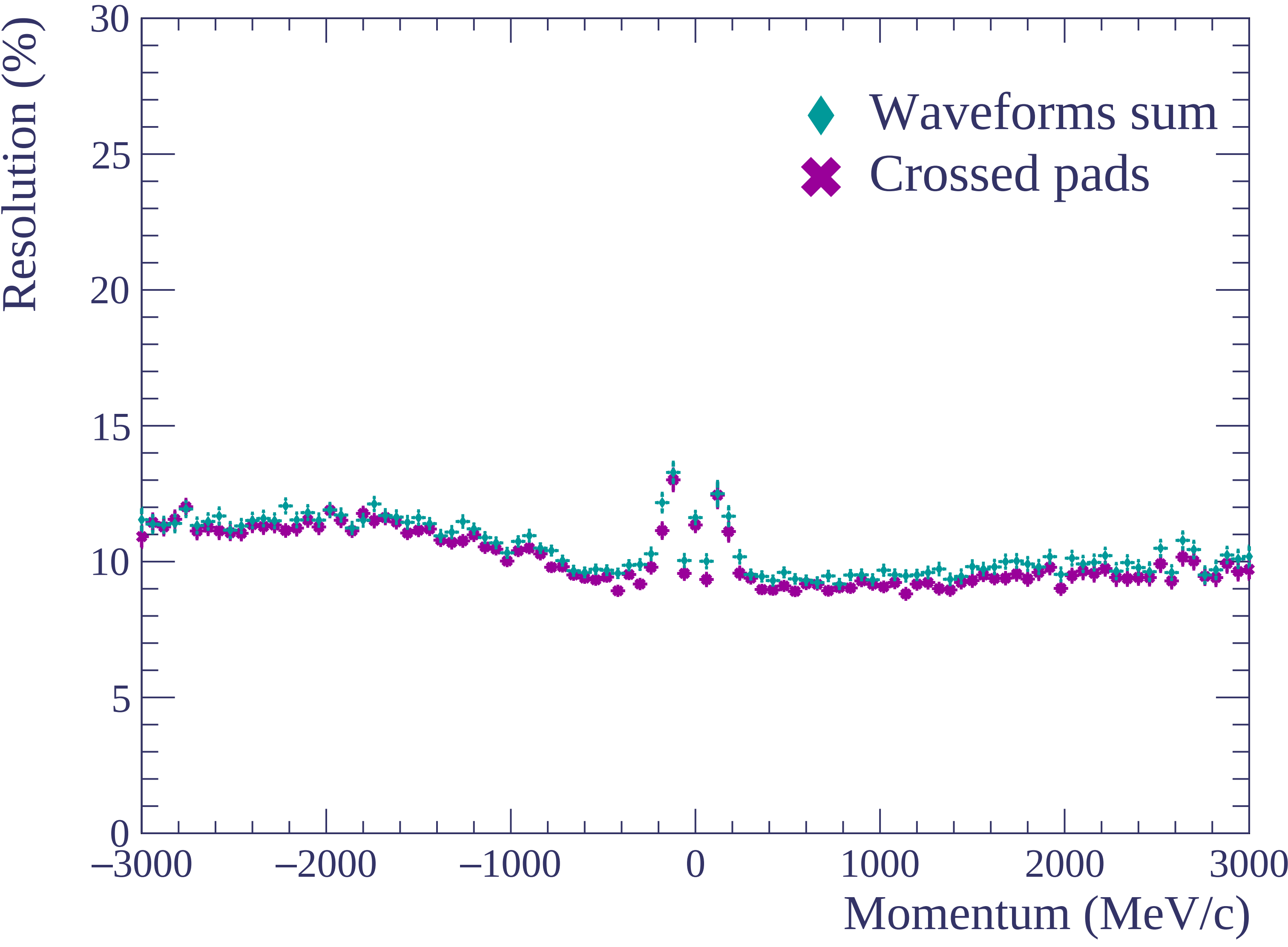

dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
φ (°)
300
250
200
150
100
50
0


Resolution (%)
30
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
θ (°)
Waveforms sum
Crossed pads

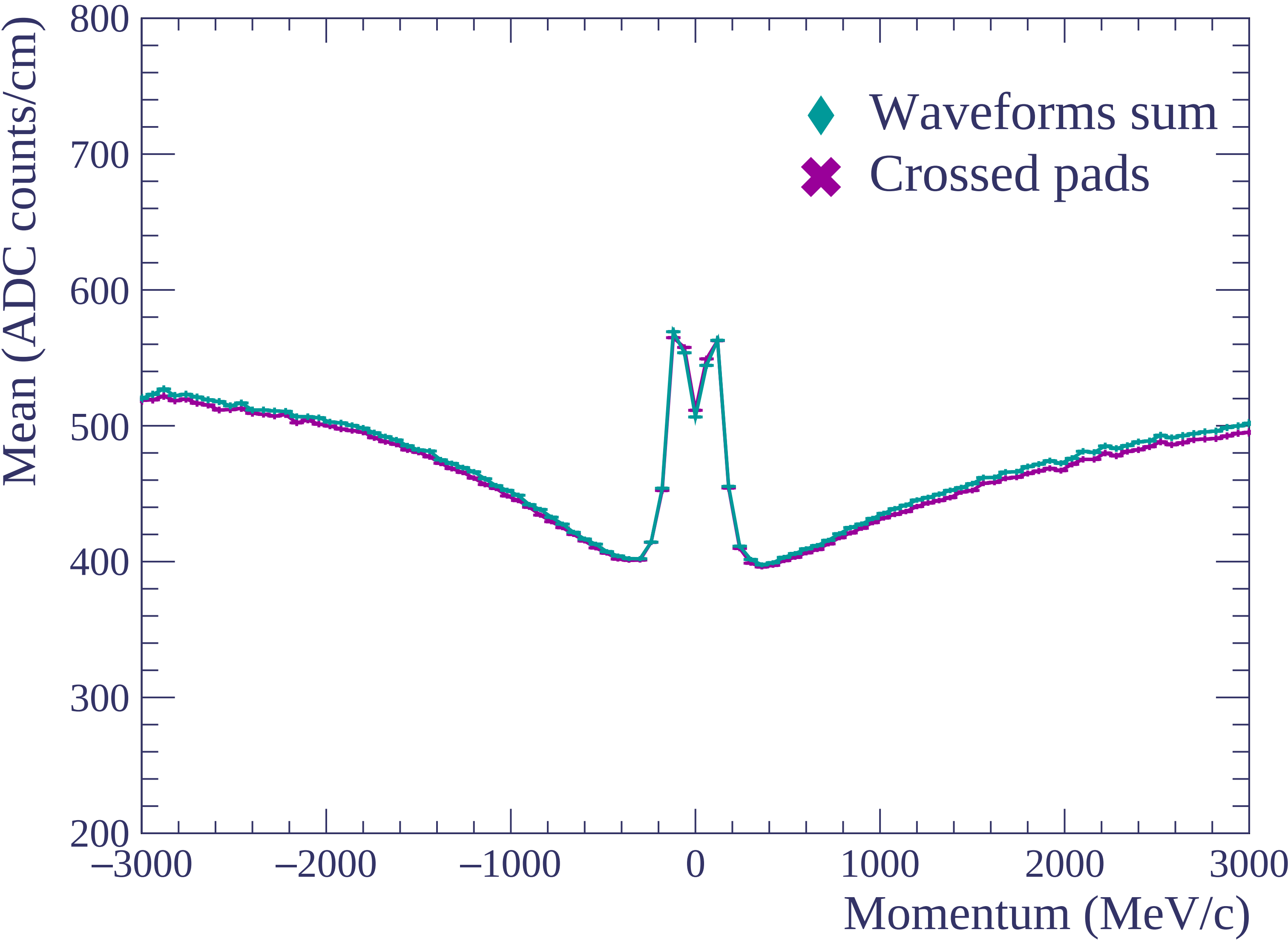

Waveforms sum
Crossed pads
Mean (ADC counts/cm)
θ (°)
800
700
600
500
400
300
200
−80
−60
−40
−20
0
20
40
60
80



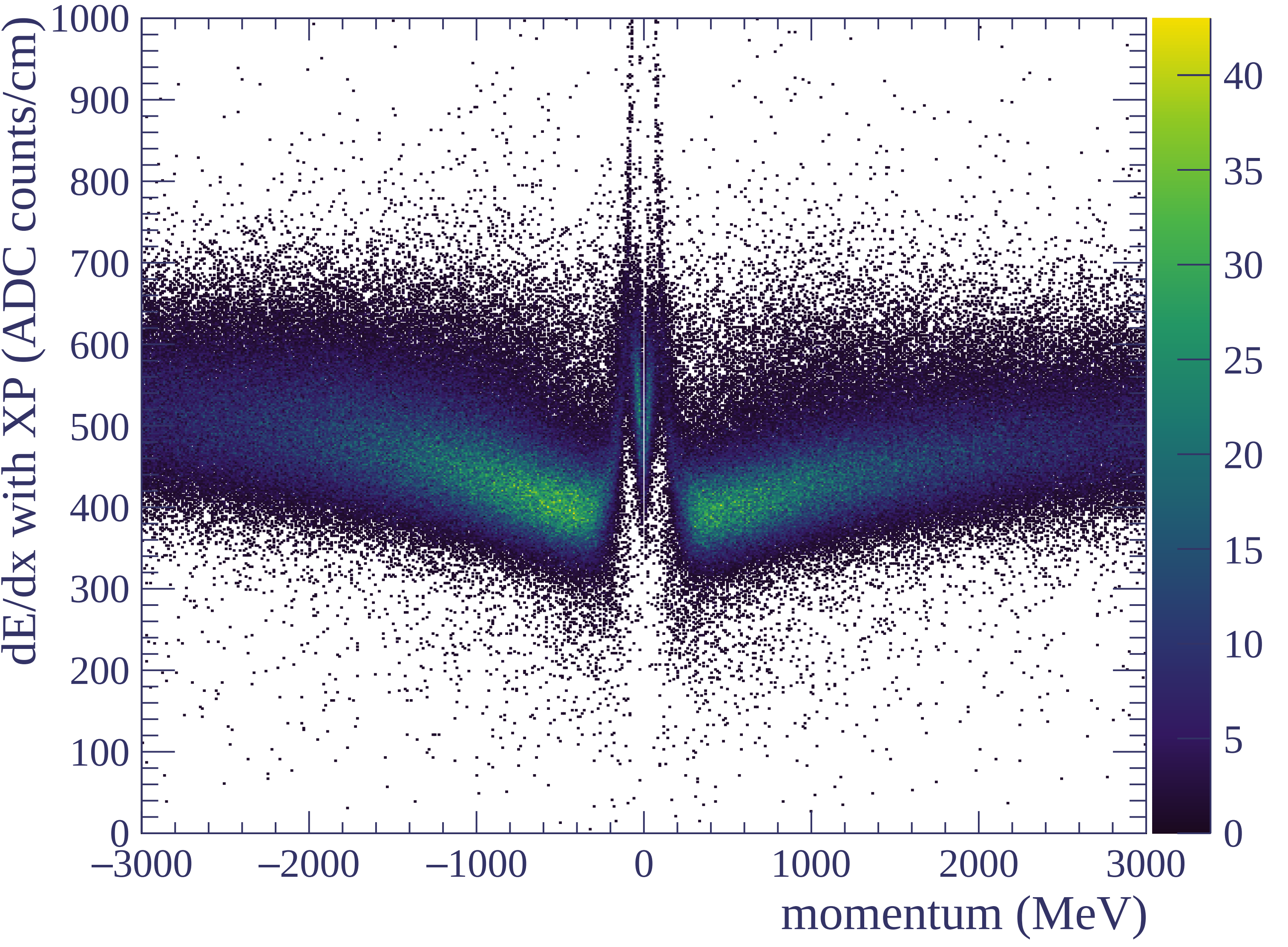
dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
θ (°)
300
250
200
150
100
50
0

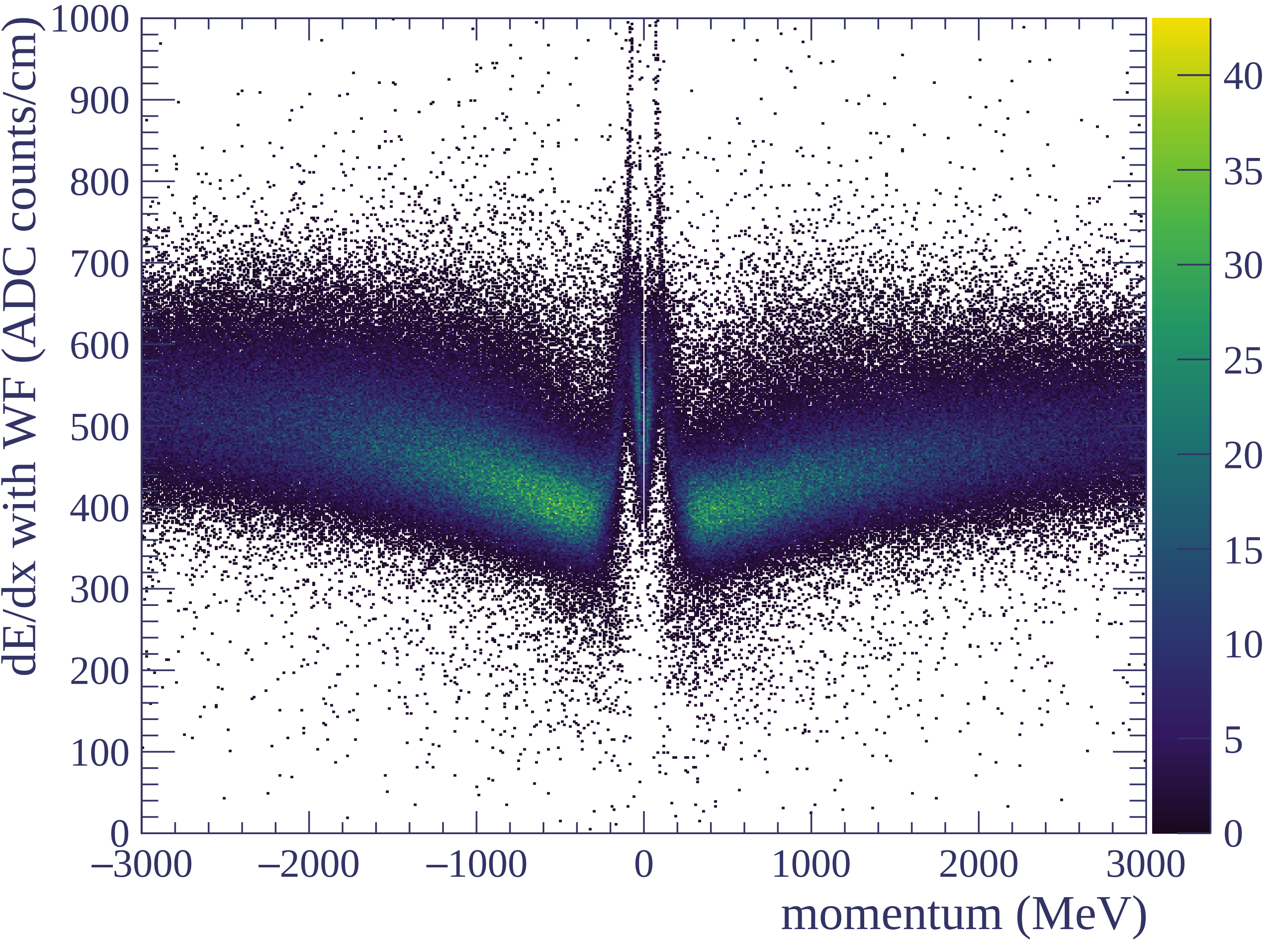


Resolution (%)
Momentum (MeV/c)
Waveforms sum
Crossed pads
0
5
10
15
20
25
30
−3000
−2000
−1000
0
1000
2000
3000

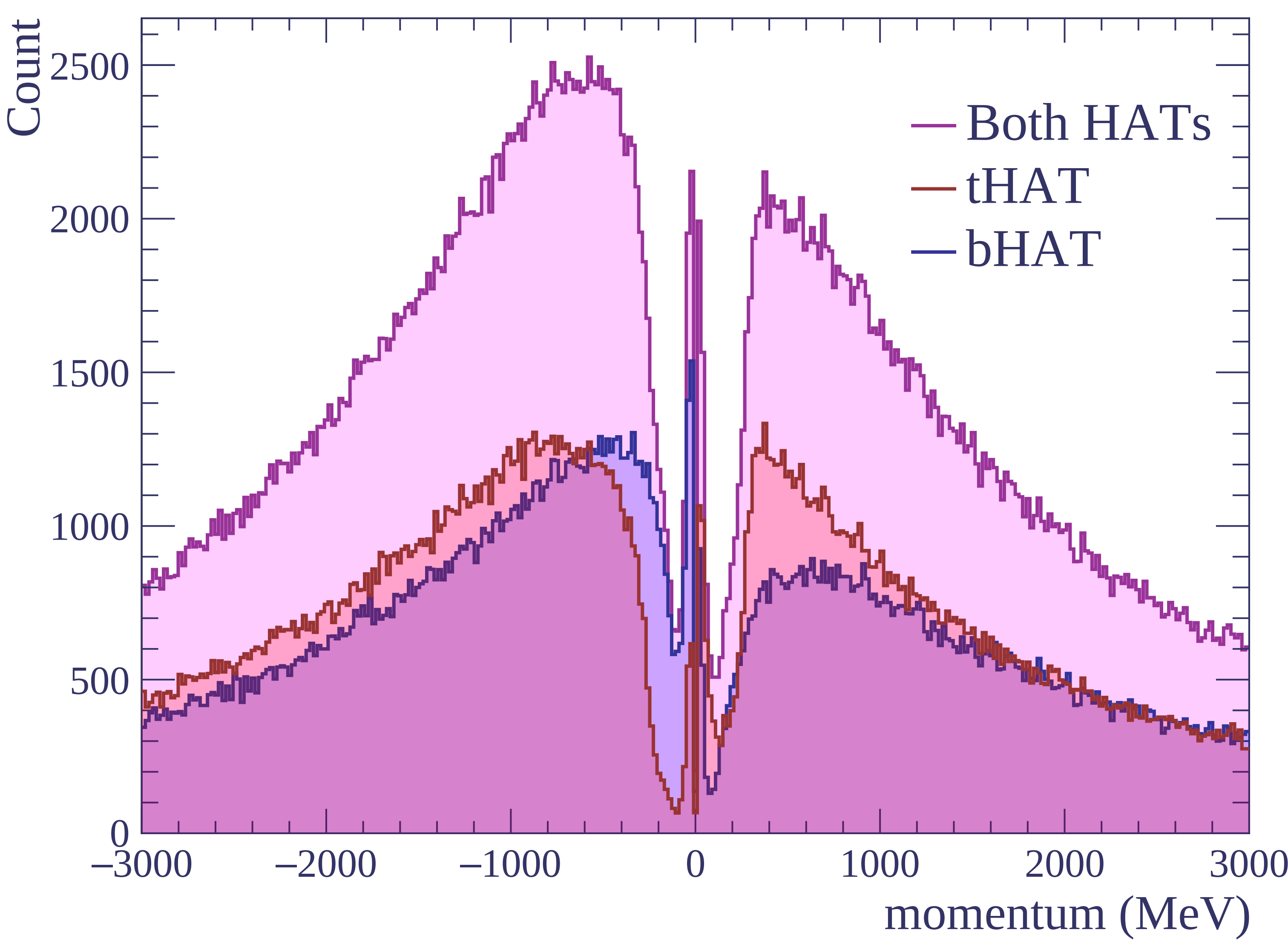


Mean (ADC counts/cm)
800
700
600
500
400
300
200
−3000
−2000
−1000
0
1000
2000
3000
Momentum (MeV/c)
Waveforms sum
Crossed pads

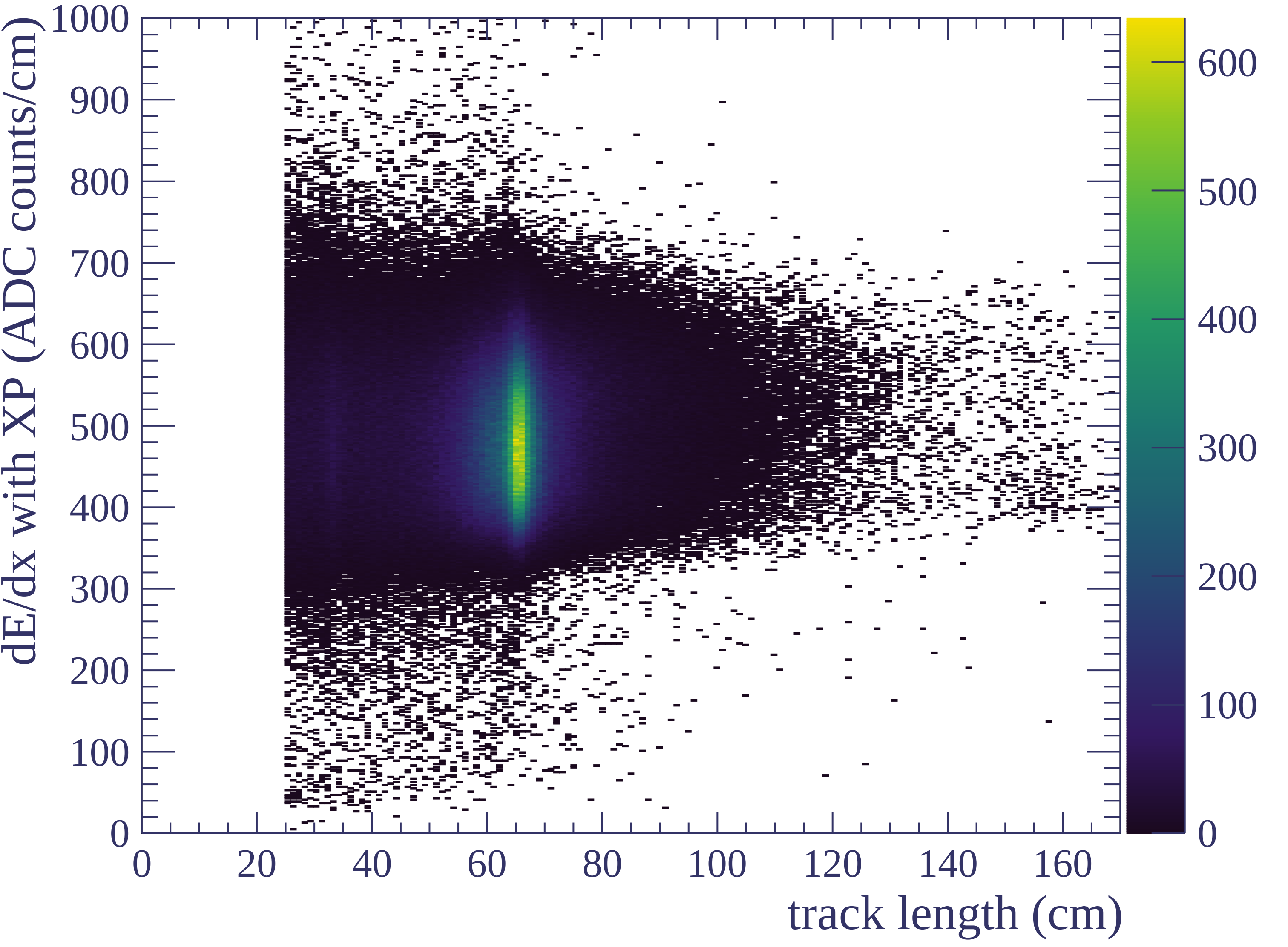

dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
40
35
30
25
20
15
10
5
0



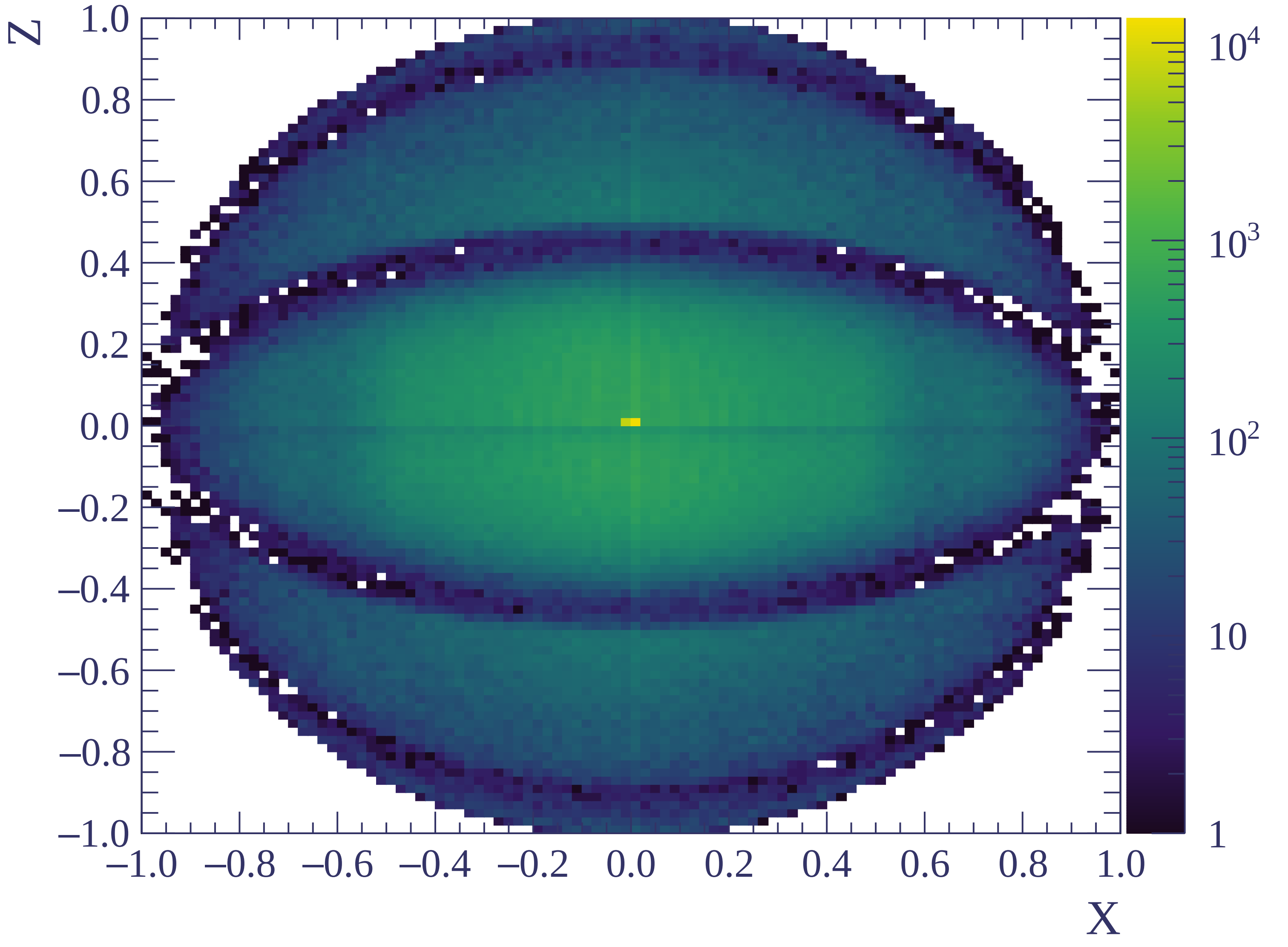
dE/dx with WF (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
40
35
30
25
20
15
10
5
0

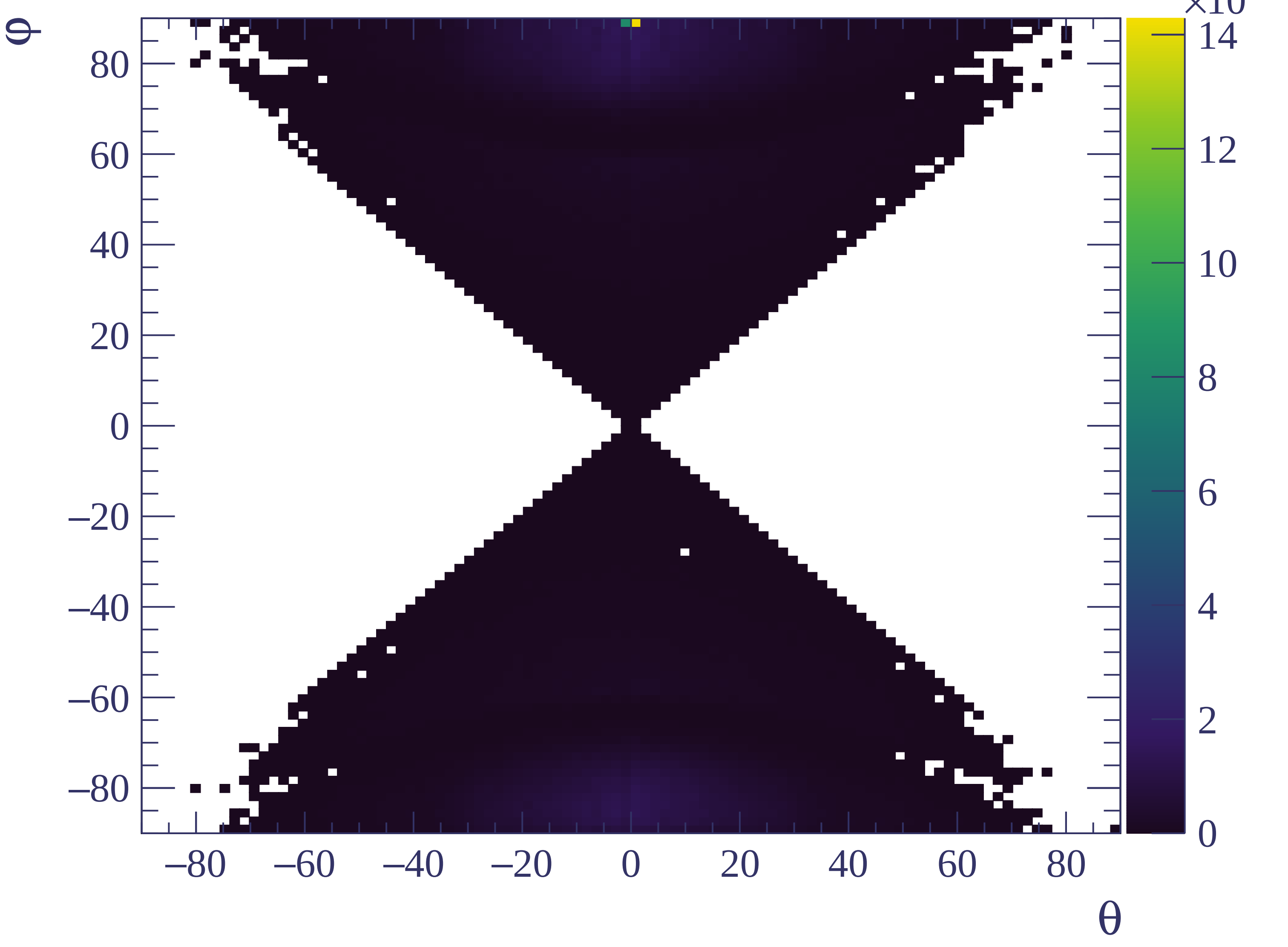


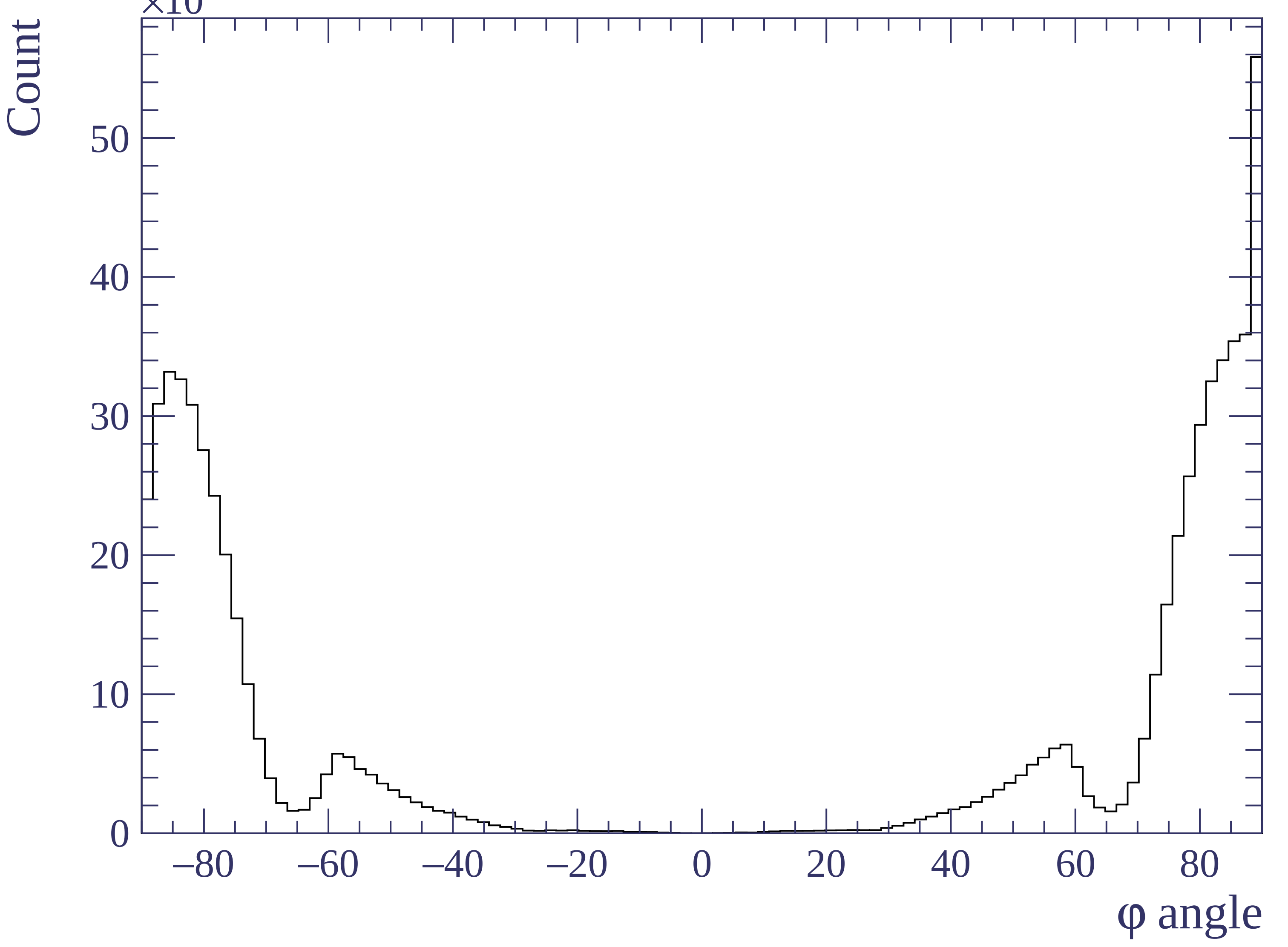
Count
momentum (MeV)
Both HATs
tHAT
bHAT
0
500
1000
1500
2000
2500
−3000
−2000
−1000
0
1000
2000
3000



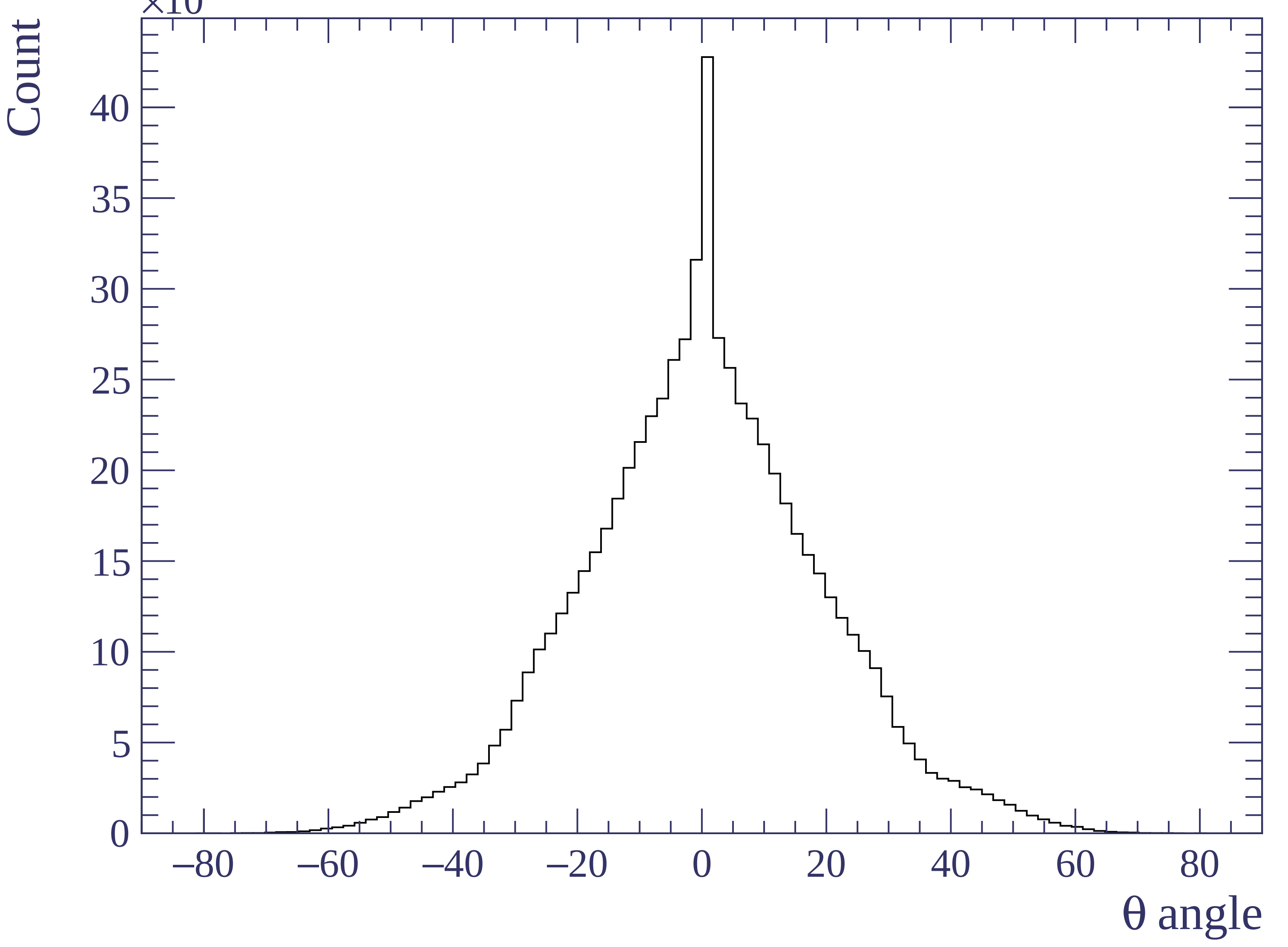
dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
0
20
40
60
80
100
120
140
160
track length (cm)
600
500
400
300
200
100
0


Z
X
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
10^4
10^3
10^2
10
1

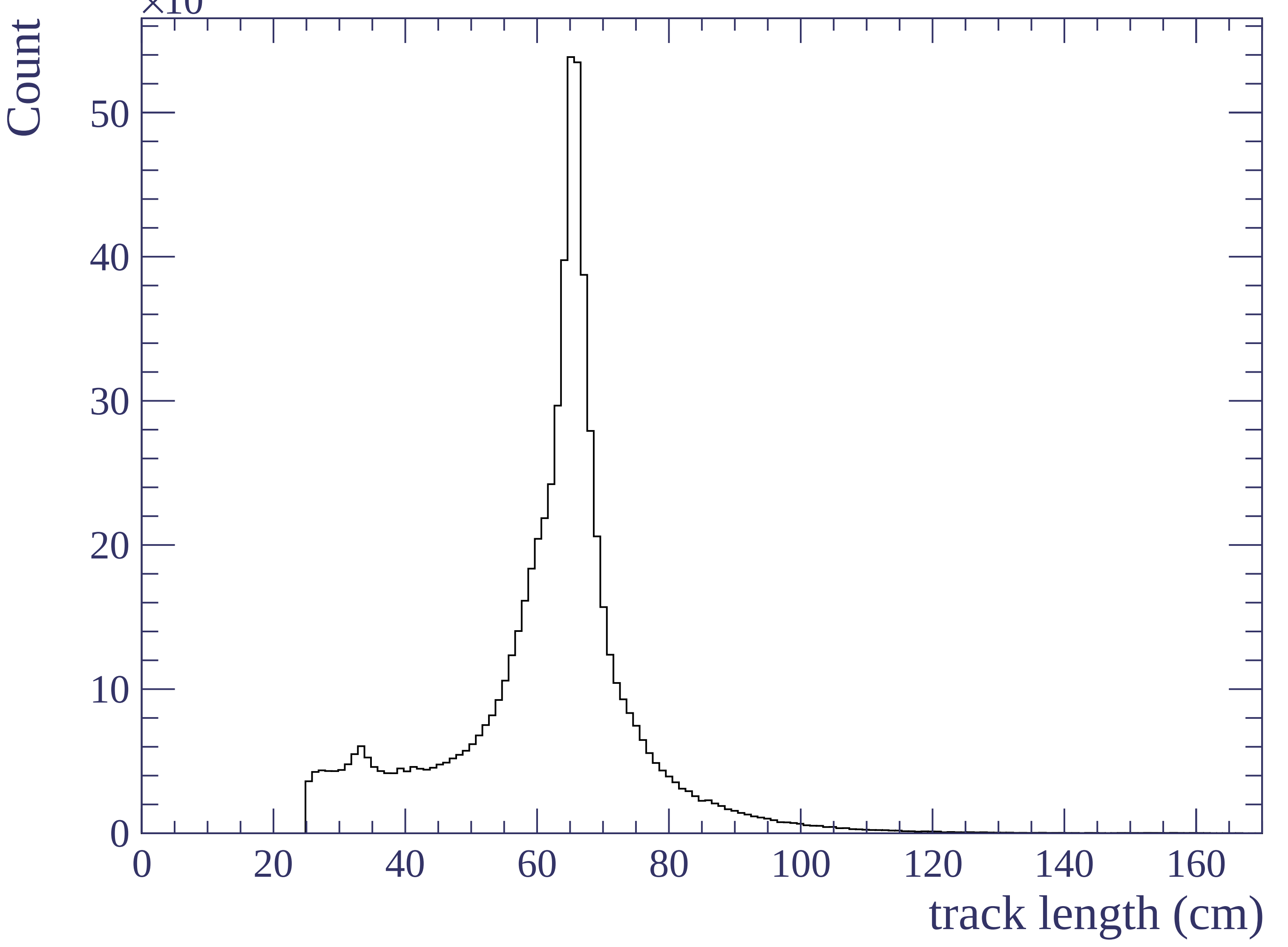

φ
80
60
40
20
0
−20
−40
−60
−80
−80
−60
−40
−20
0
20
40
60
80
θ
×10
14
12
10
8
6
4
2
0

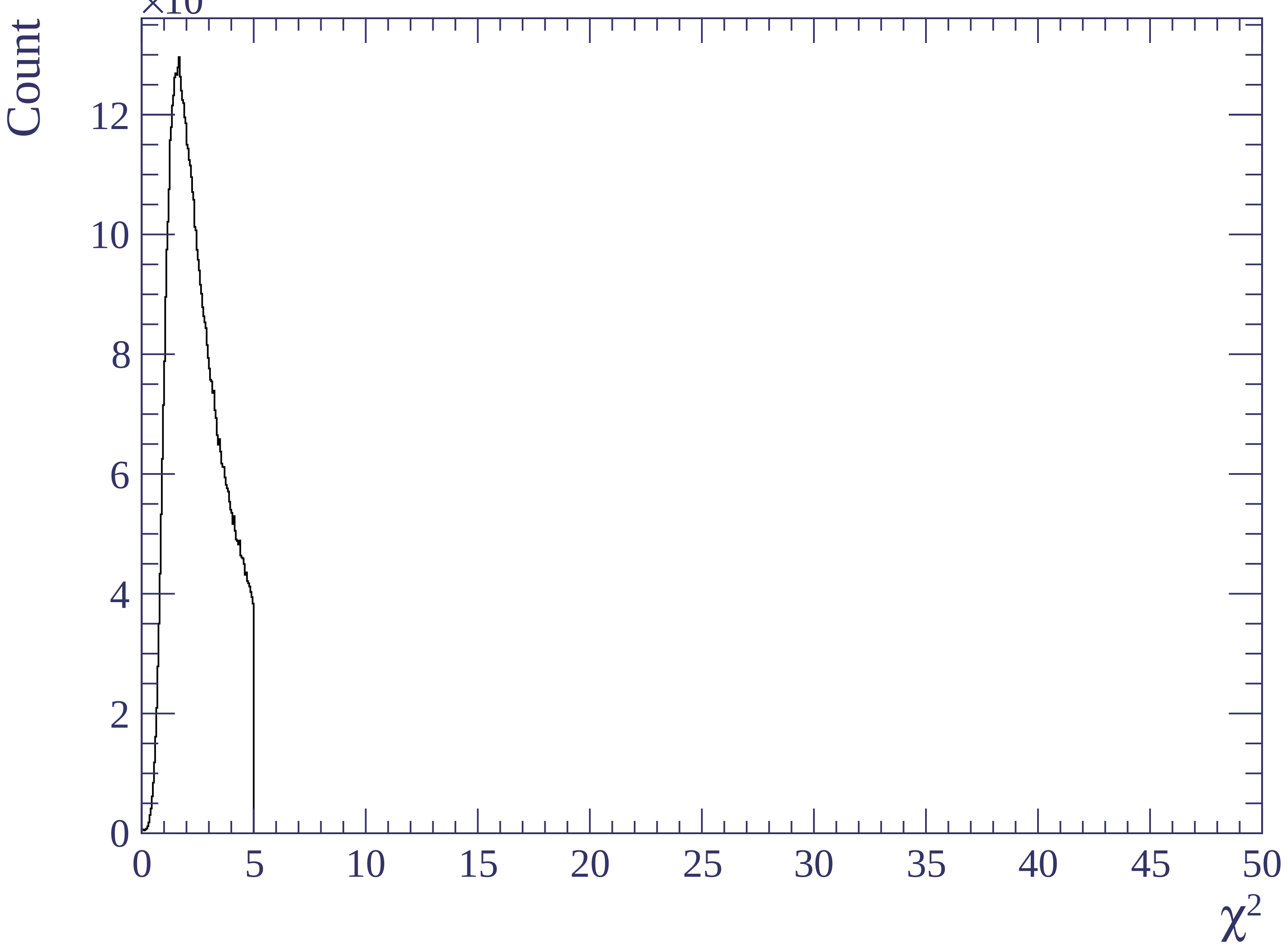


×10
Count
50
40
30
20
10
0
−80
−60
−40
−20
0
20
40
60
80
φ angle

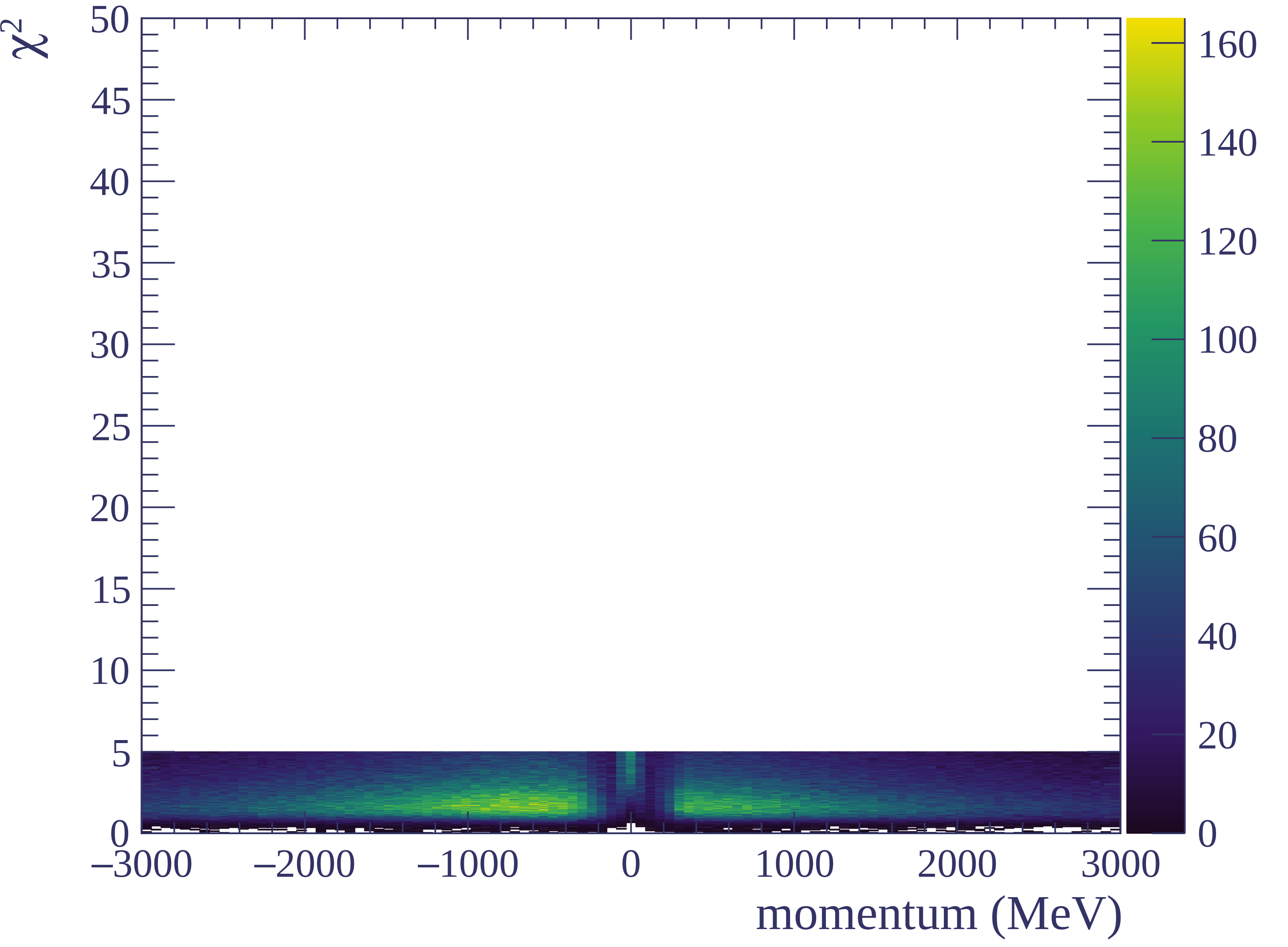


×10
Count
40
35
30
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
θ angle

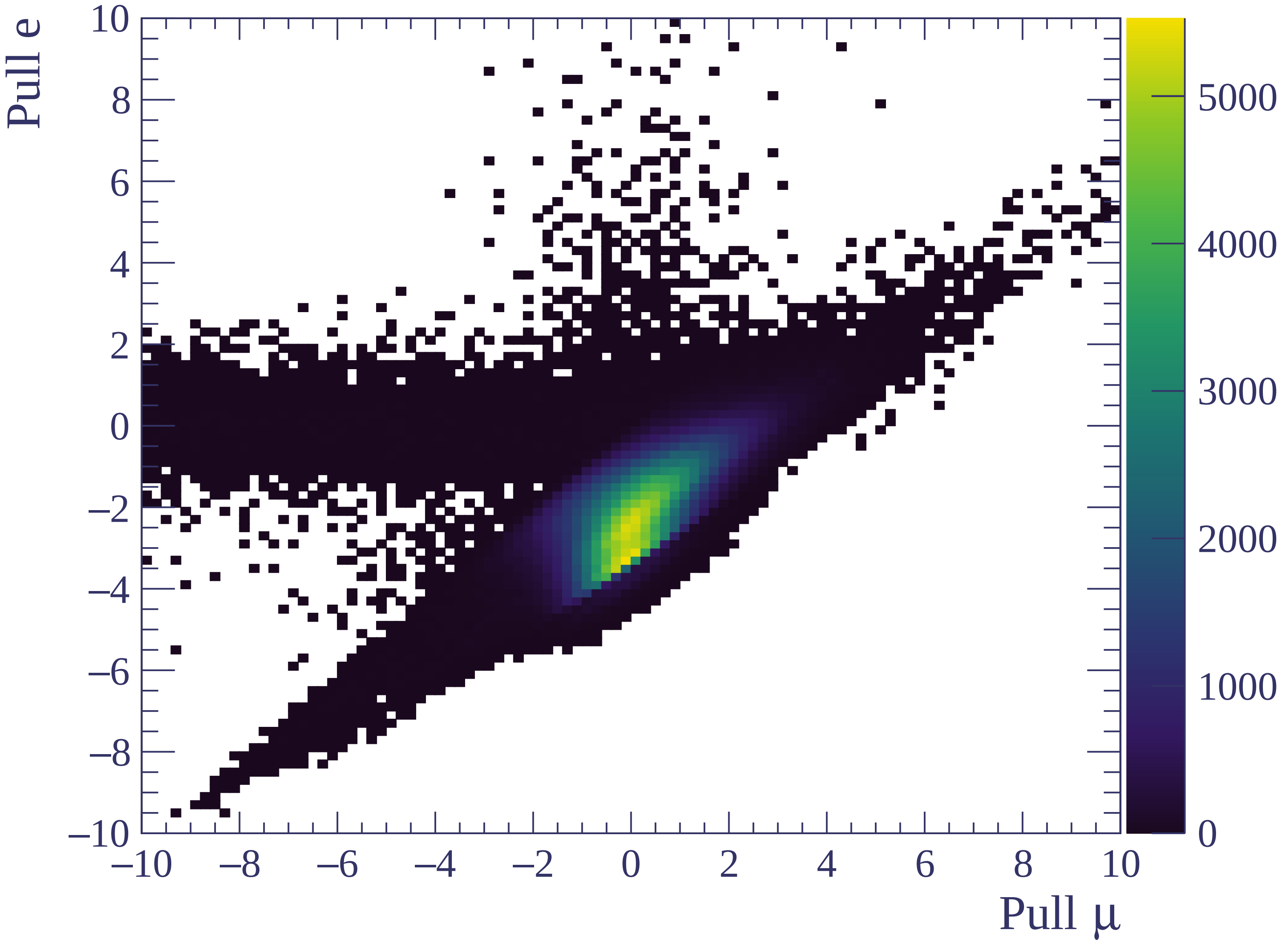

×10
Count
0
10
20
30
40
50
0
20
40
60
80
100
120
140
160
track length (cm)

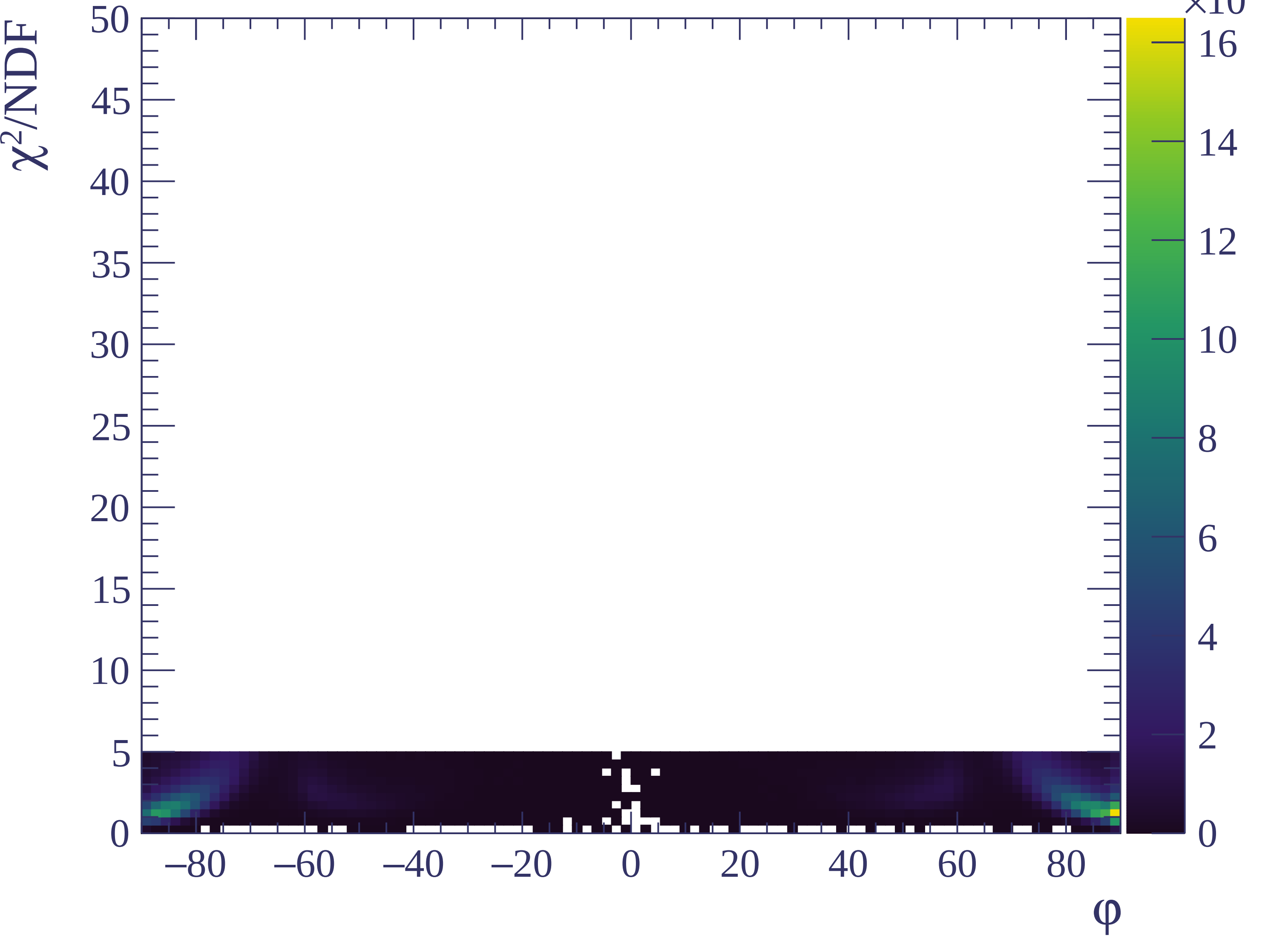


Count
×10
12
10
8
6
4
2
0
0
5
10
15
20
25
30
35
40
45
50
χ²

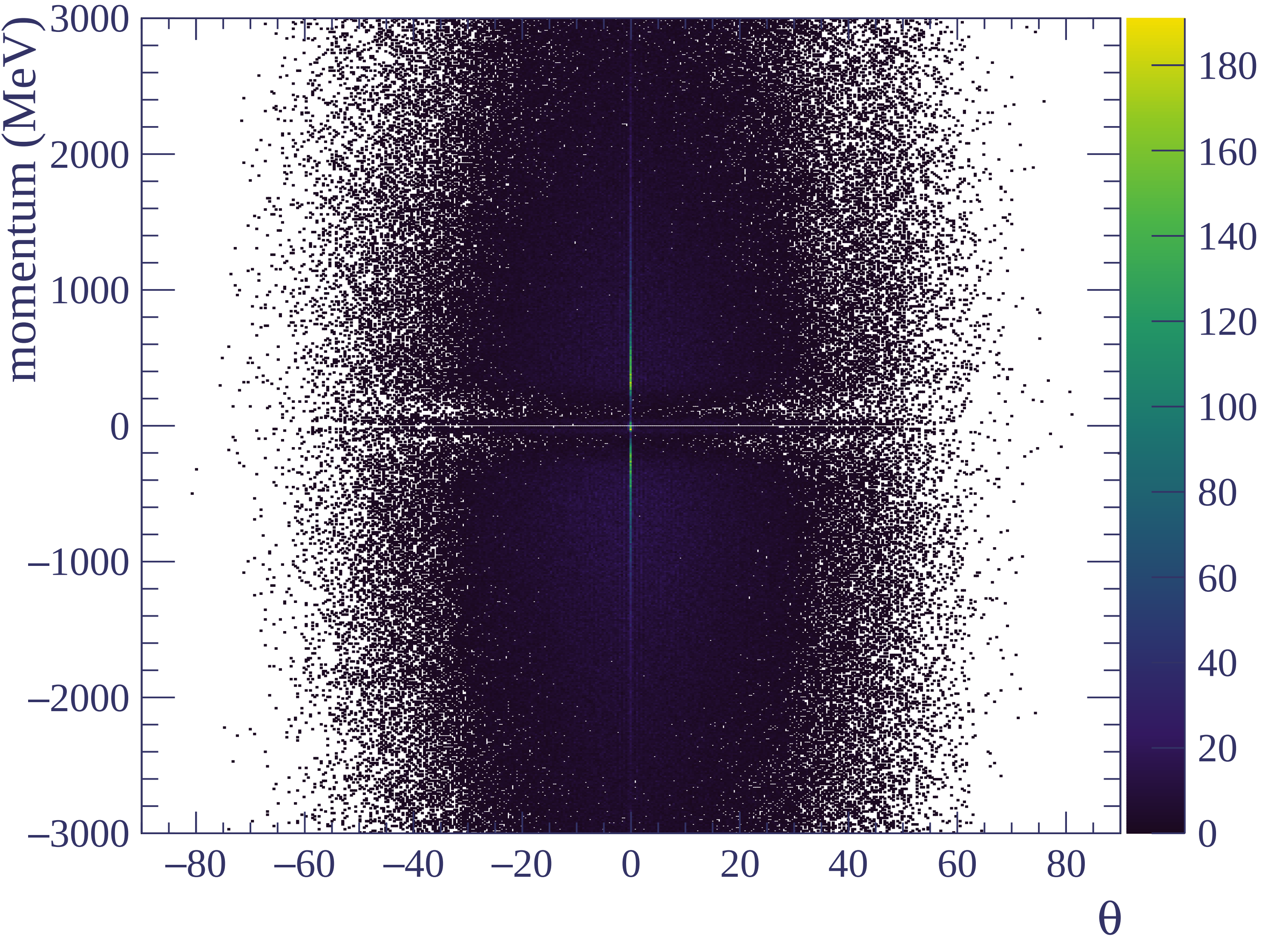


χ^2
50
45
40
35
30
25
20
15
10
5
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
160
140
120
100
80
60
40
20
0

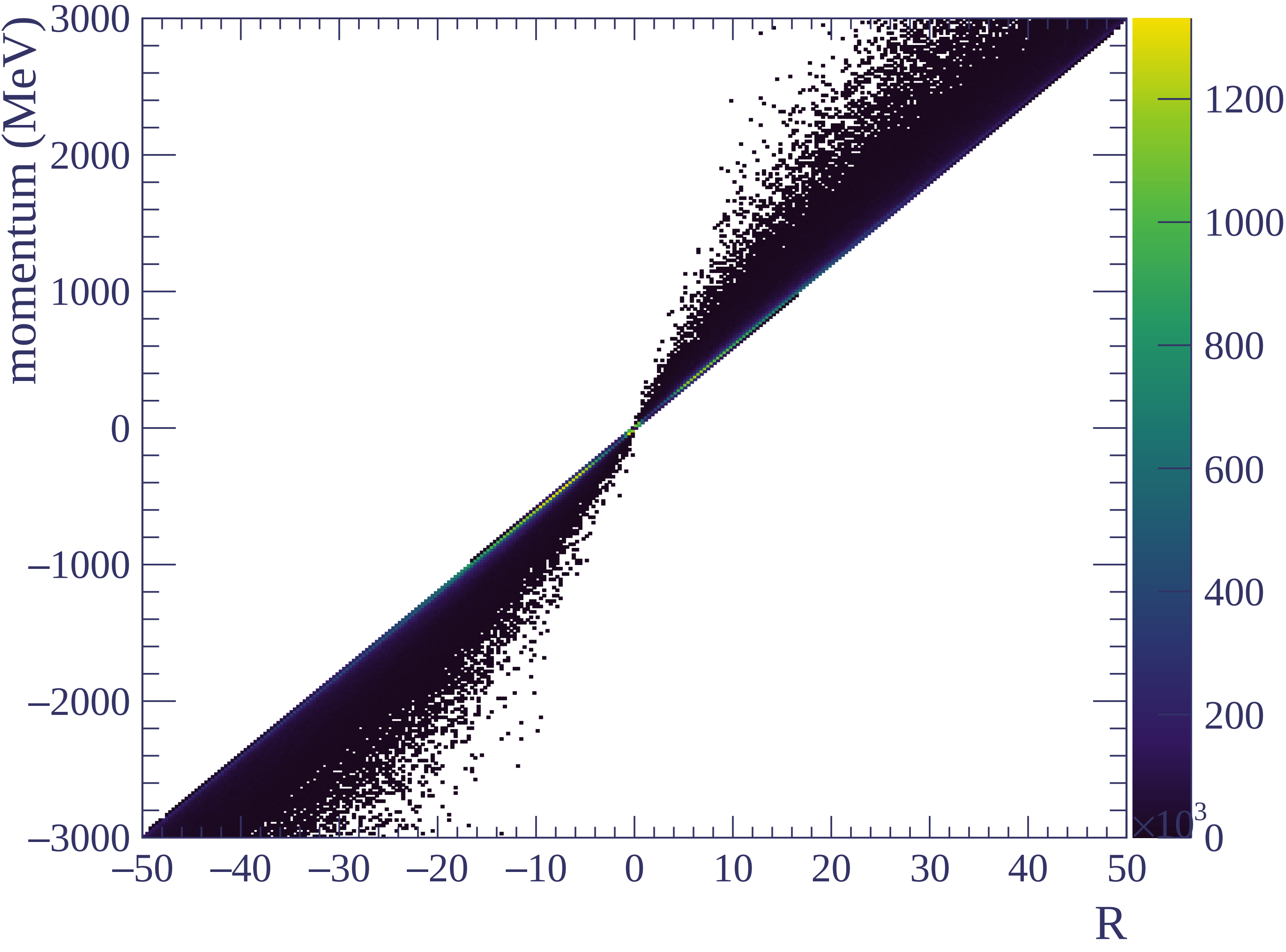

Pull e
Pull μ
10
8
6
4
2
0
−2
−4
−6
−8
−10
5000
4000
3000
2000
1000
0



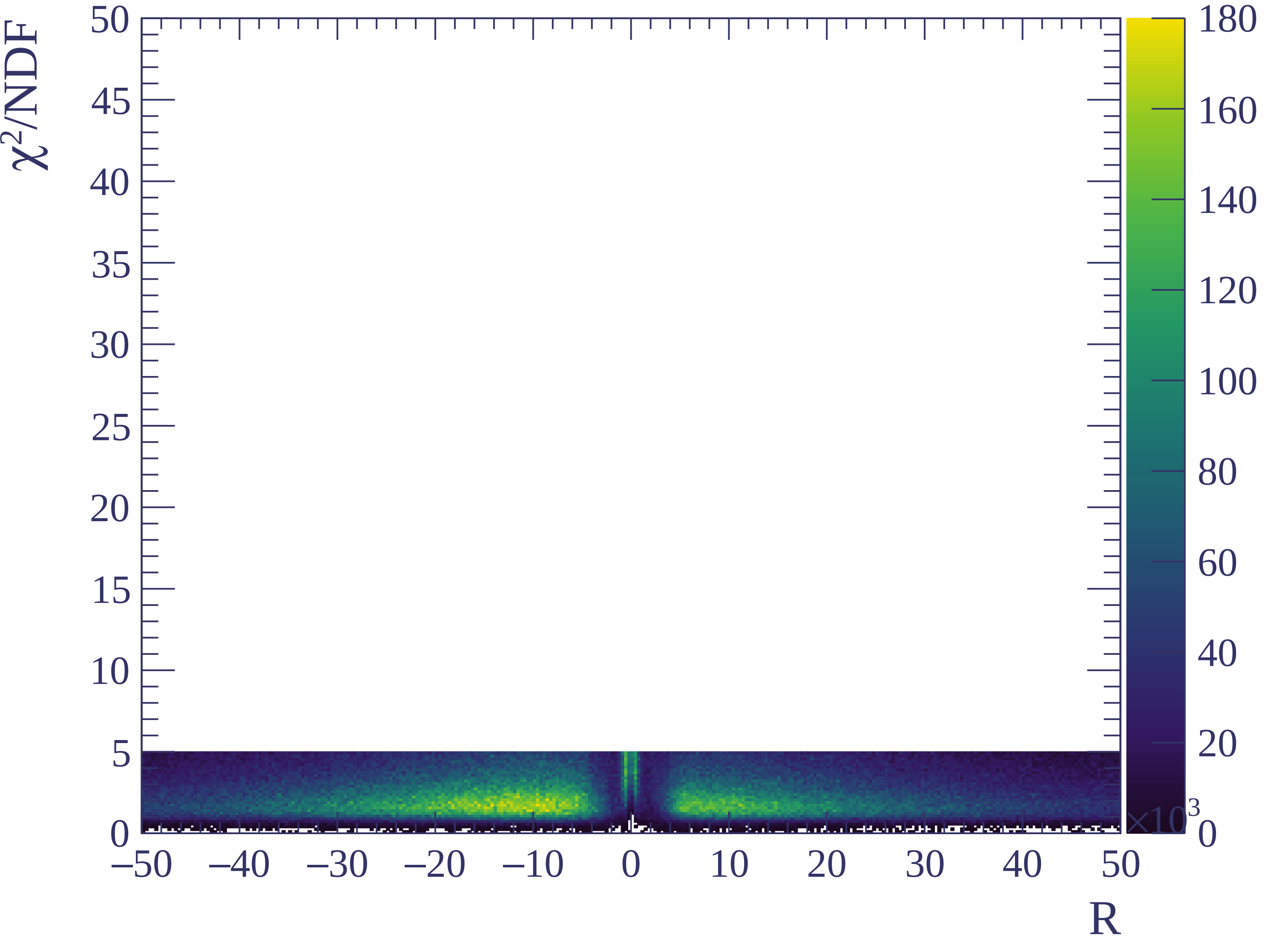
χ^2/NDF
φ
×10



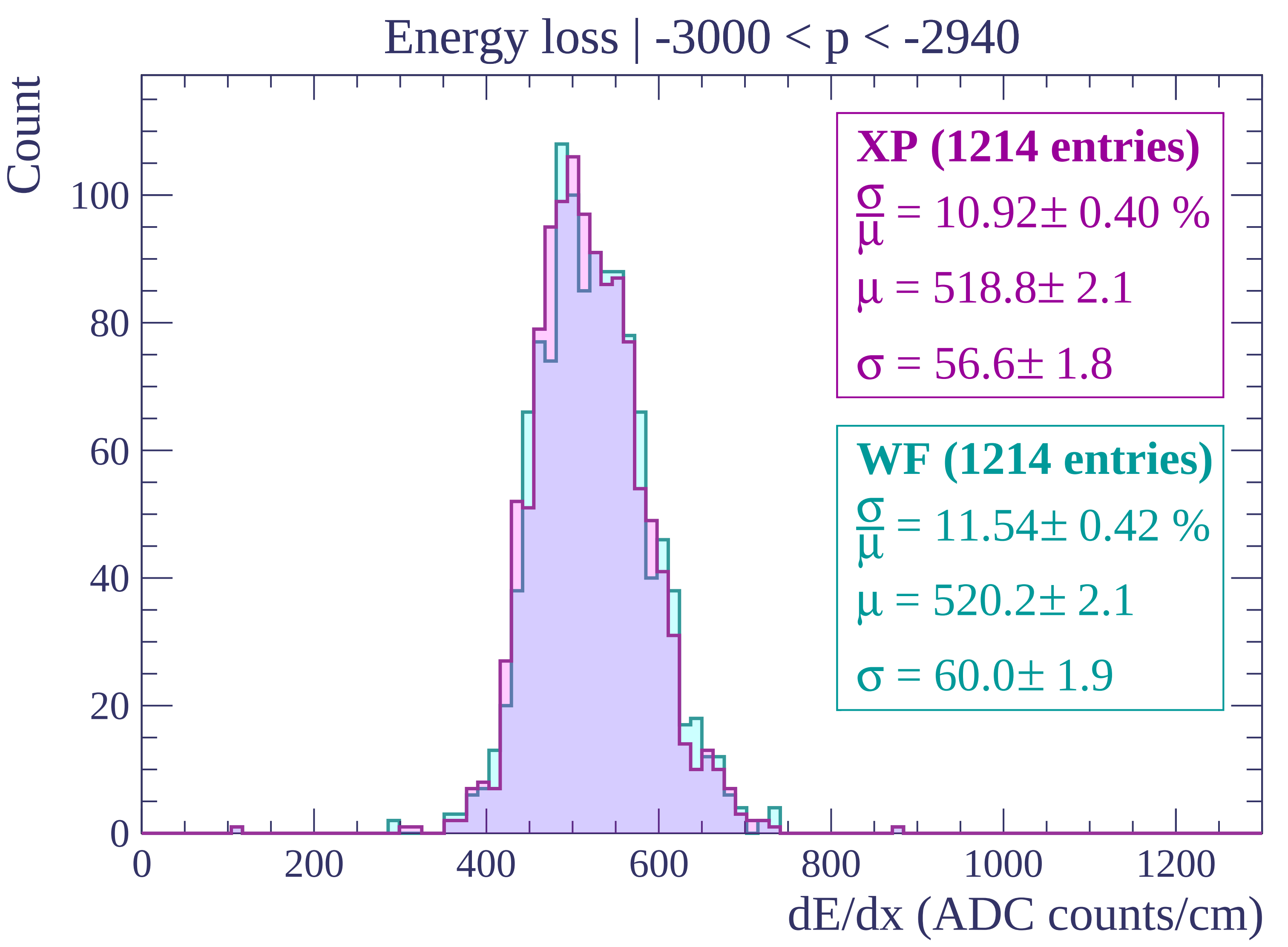
momentum (MeV)
3000
2000
1000
0
−1000
−2000
−3000
−80
−60
−40
−20
0
20
40
60
80
θ
180
160
140
120
100
80
60
40
20
0

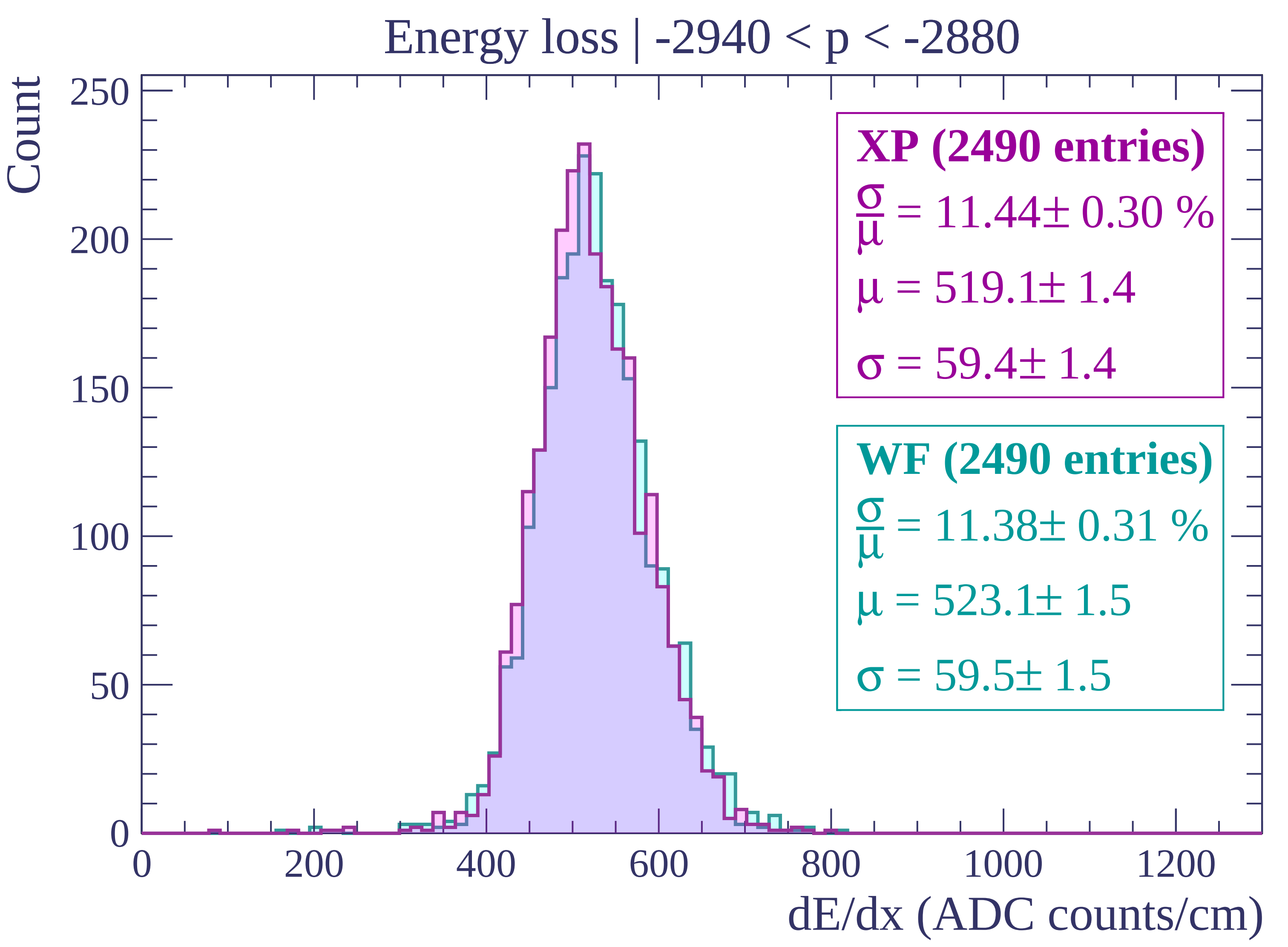


momentum (MeV)
3000
2000
1000
0
−1000
−2000
−3000
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
1200
1000
800
600
400
200
0
×10³

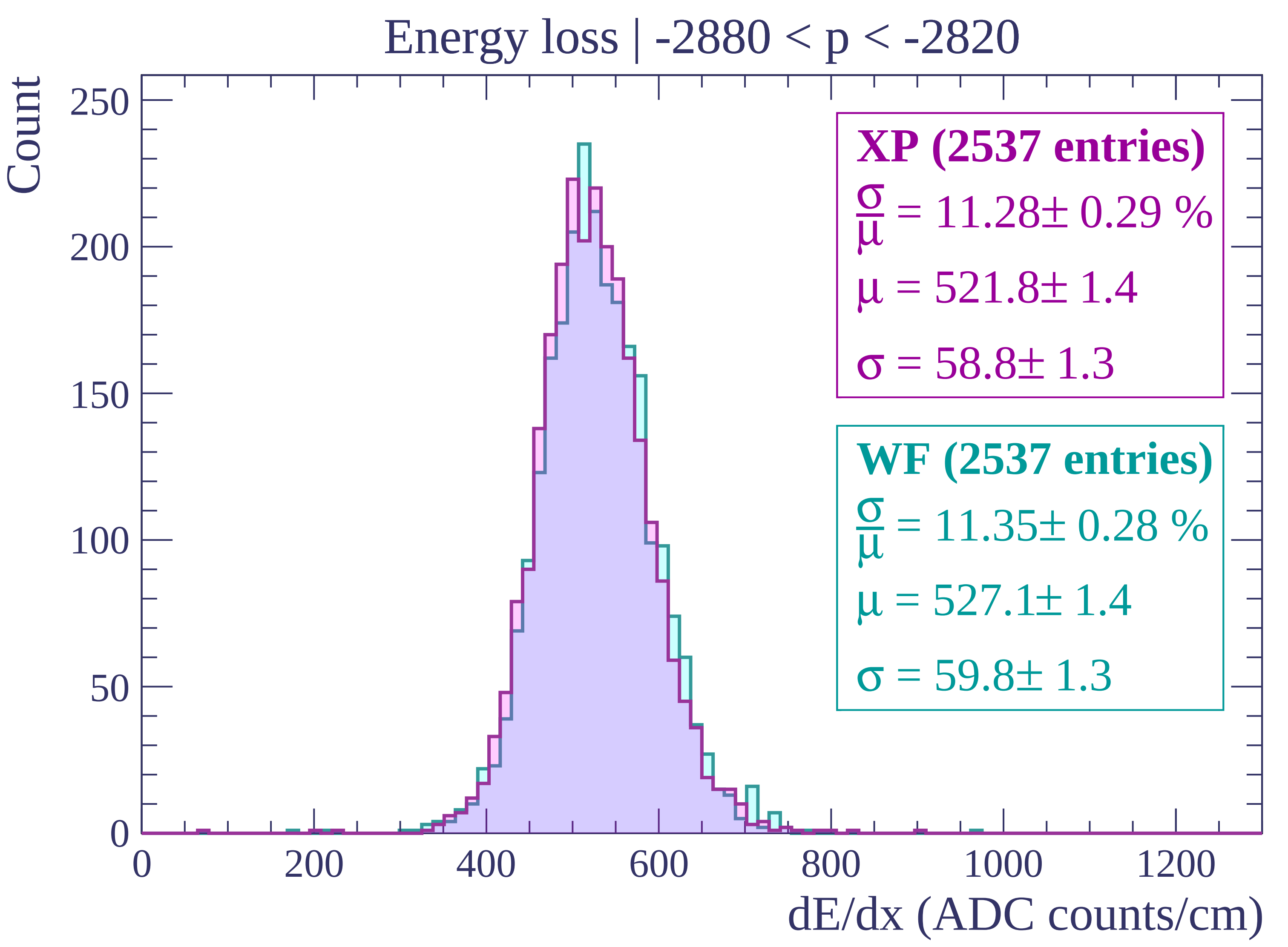


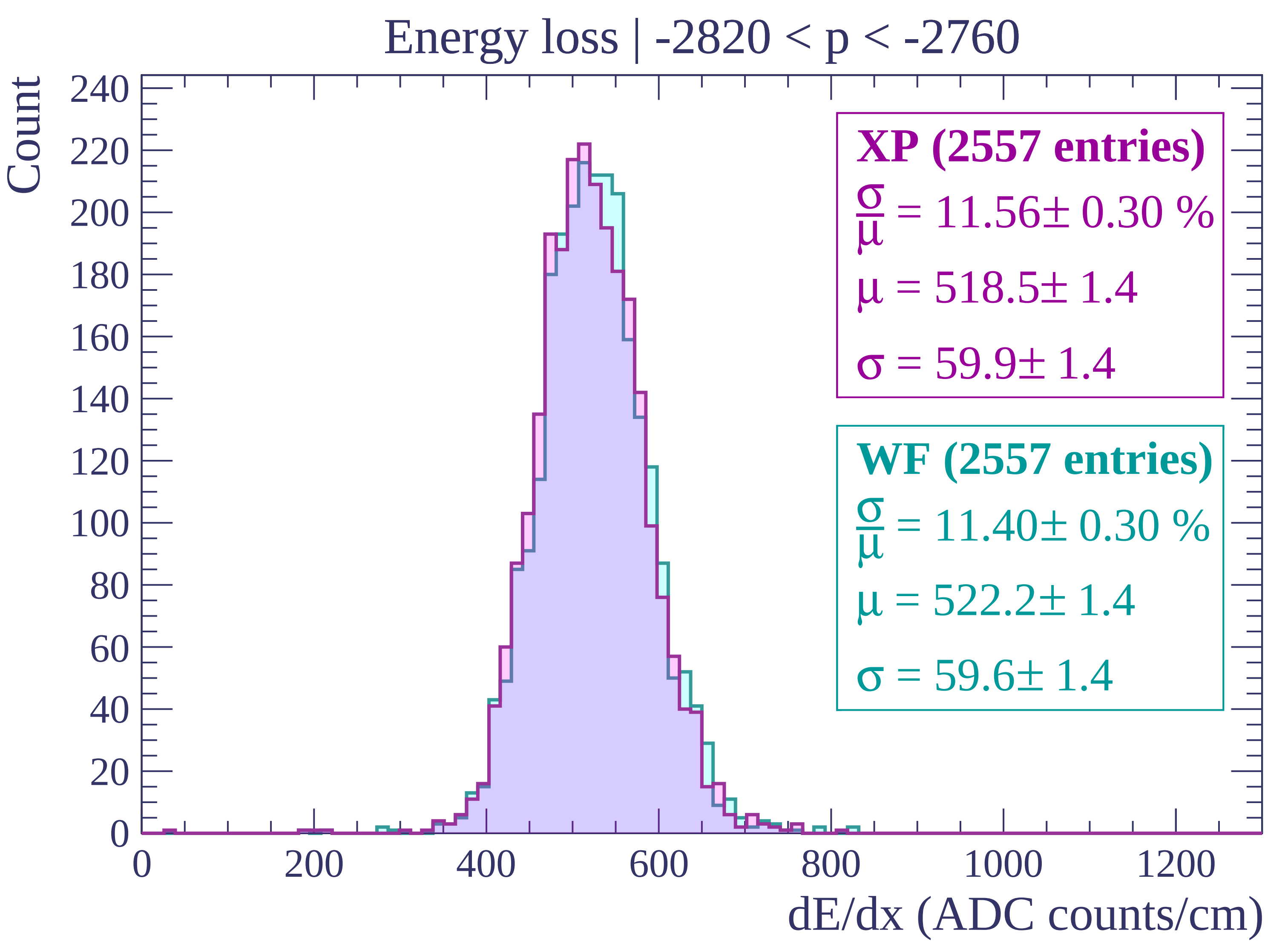
χ^2/NDF
R
$\times10^3$


Energy loss | -3000 < p < -2940
Count
dE/dx (ADC counts/cm)
XP (1214 entries)
$\frac{\sigma}{\mu}$ = 10.92± 0.40 %
μ = 518.8± 2.1
σ = 56.6± 1.8
WF (1214 entries)
$\frac{\sigma}{\mu}$ = 11.54± 0.42 %
μ = 520.2± 2.1
σ = 60.0± 1.9
0
20
40
60
80
100
0
200
400
600
800
1000
1200

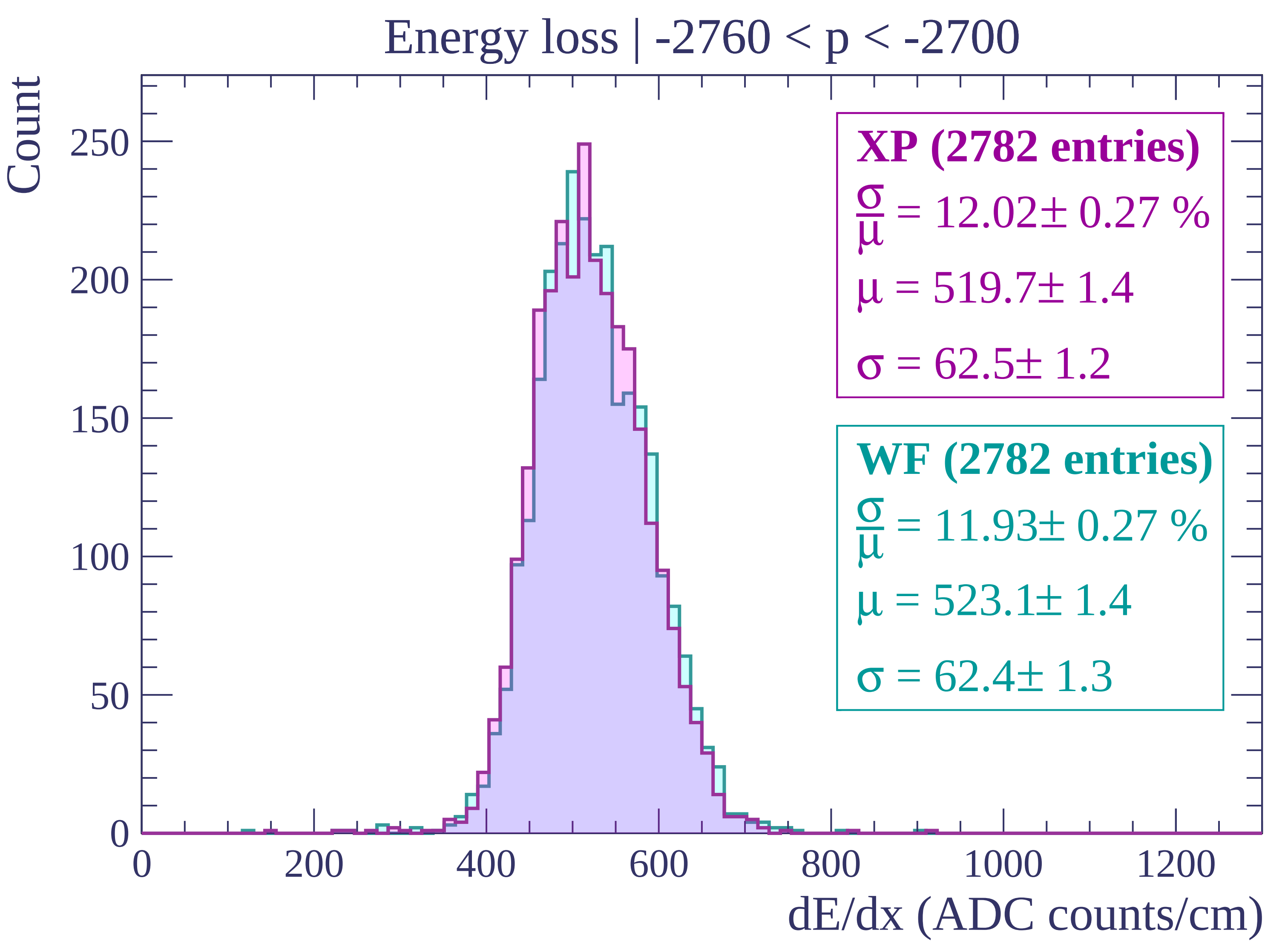


Energy loss | -2940 < p < -2880
Count
dE/dx (ADC counts/cm)
XP (2490 entries)
σ/μ = 11.44± 0.30 %
μ = 519.1± 1.4
σ = 59.4± 1.4
WF (2490 entries)
σ/μ = 11.38± 0.31 %
μ = 523.1± 1.5
σ = 59.5± 1.5

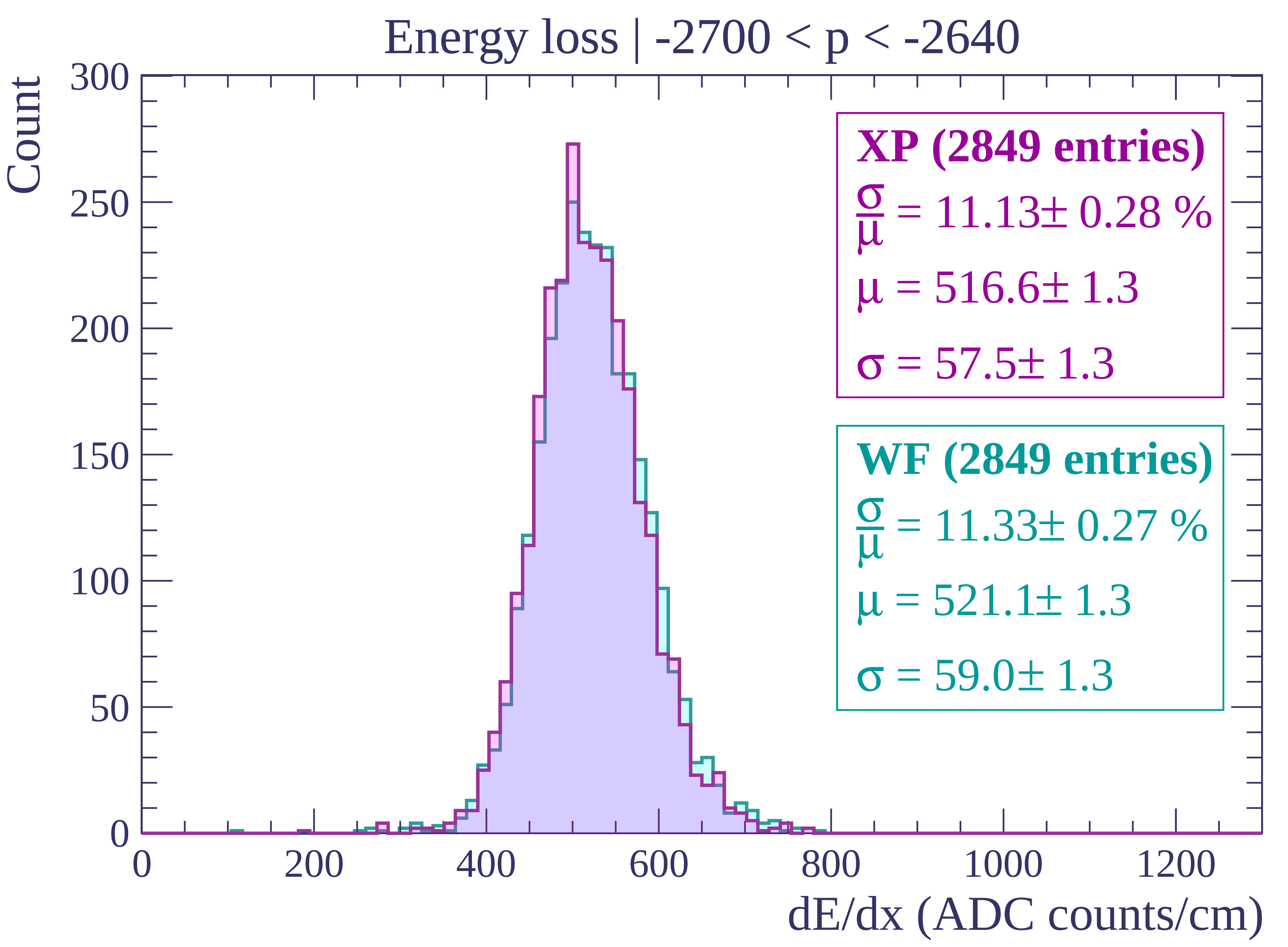

Energy loss | -2880 < p < -2820
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2537 entries)
$\frac{\sigma}{\mu}$ = 11.28± 0.29 %
μ = 521.8± 1.4
σ = 58.8± 1.3
WF (2537 entries)
$\frac{\sigma}{\mu}$ = 11.35± 0.28 %
μ = 527.1± 1.4
σ = 59.8± 1.3



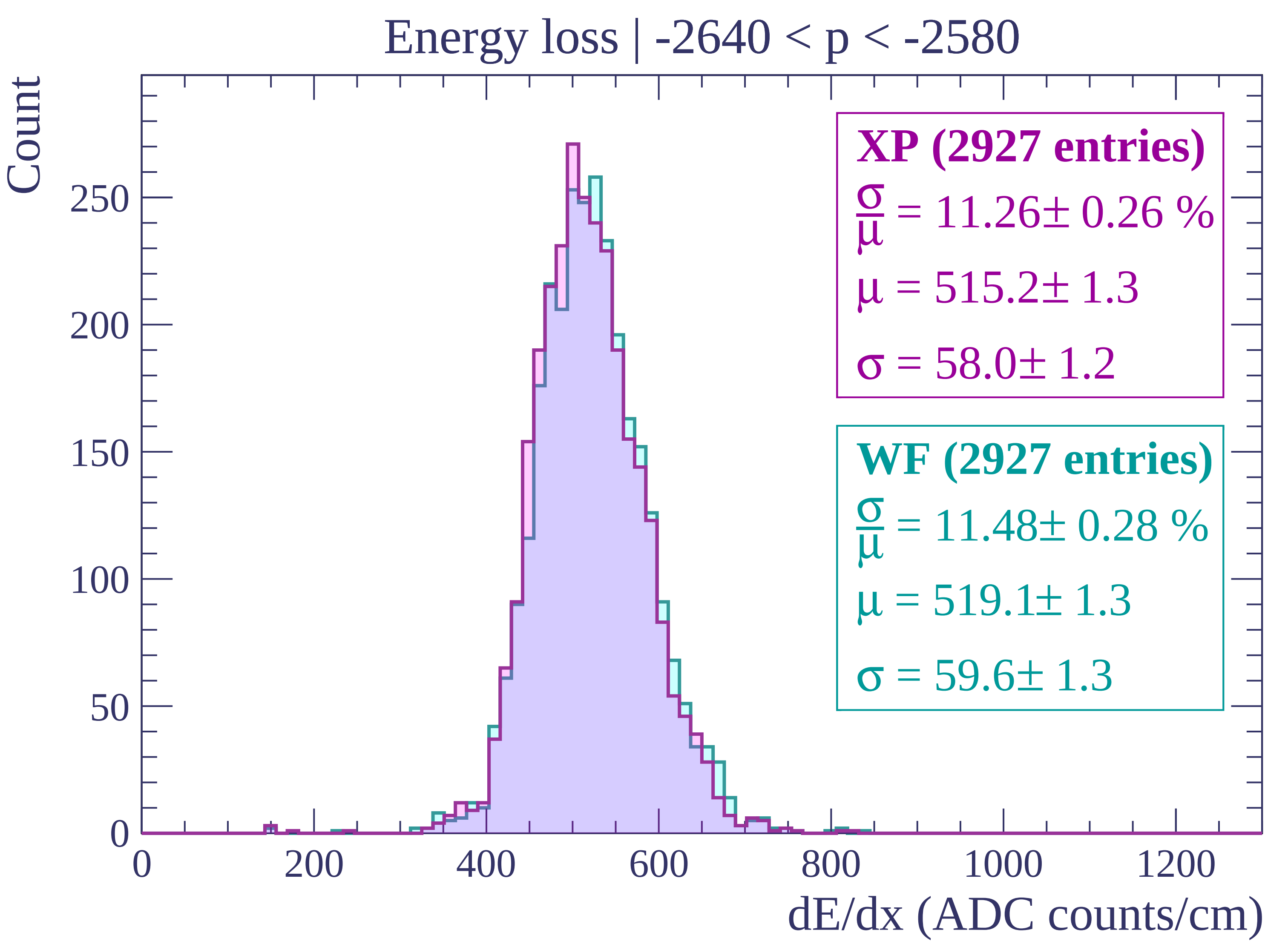
Energy loss | -2820 < p < -2760
Count
240
220
200
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2557 entries)
$\frac{\sigma}{\mu}$ = 11.56± 0.30 %
μ = 518.5± 1.4
σ = 59.9± 1.4
WF (2557 entries)
$\frac{\sigma}{\mu}$ = 11.40± 0.30 %
μ = 522.2± 1.4
σ = 59.6± 1.4



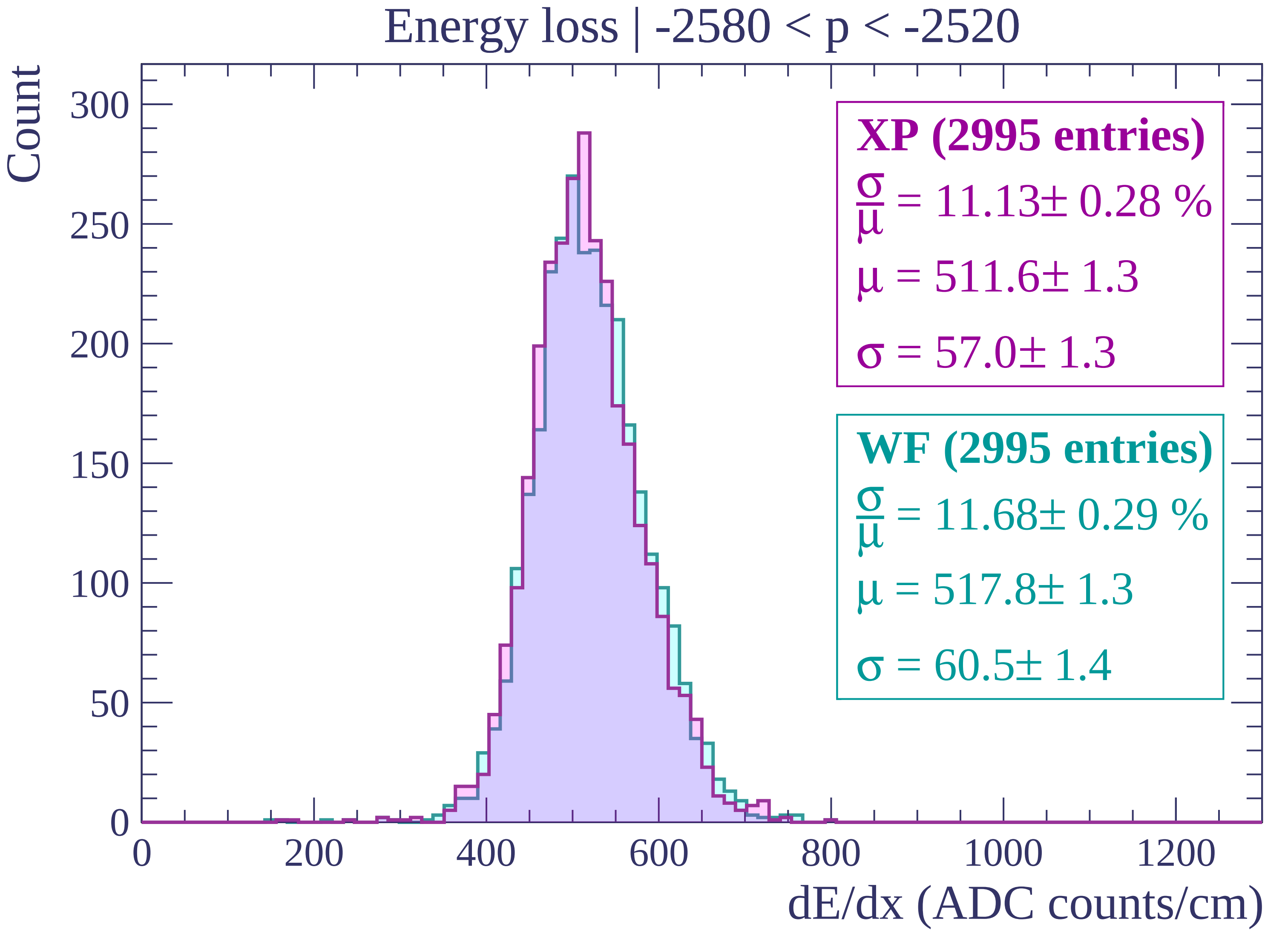
Energy loss | -2760 < p < -2700
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
XP (2782 entries)
σ/μ = 12.02± 0.27 %
μ = 519.7± 1.4
σ = 62.5± 1.2
WF (2782 entries)
σ/μ = 11.93± 0.27 %
μ = 523.1± 1.4
σ = 62.4± 1.3



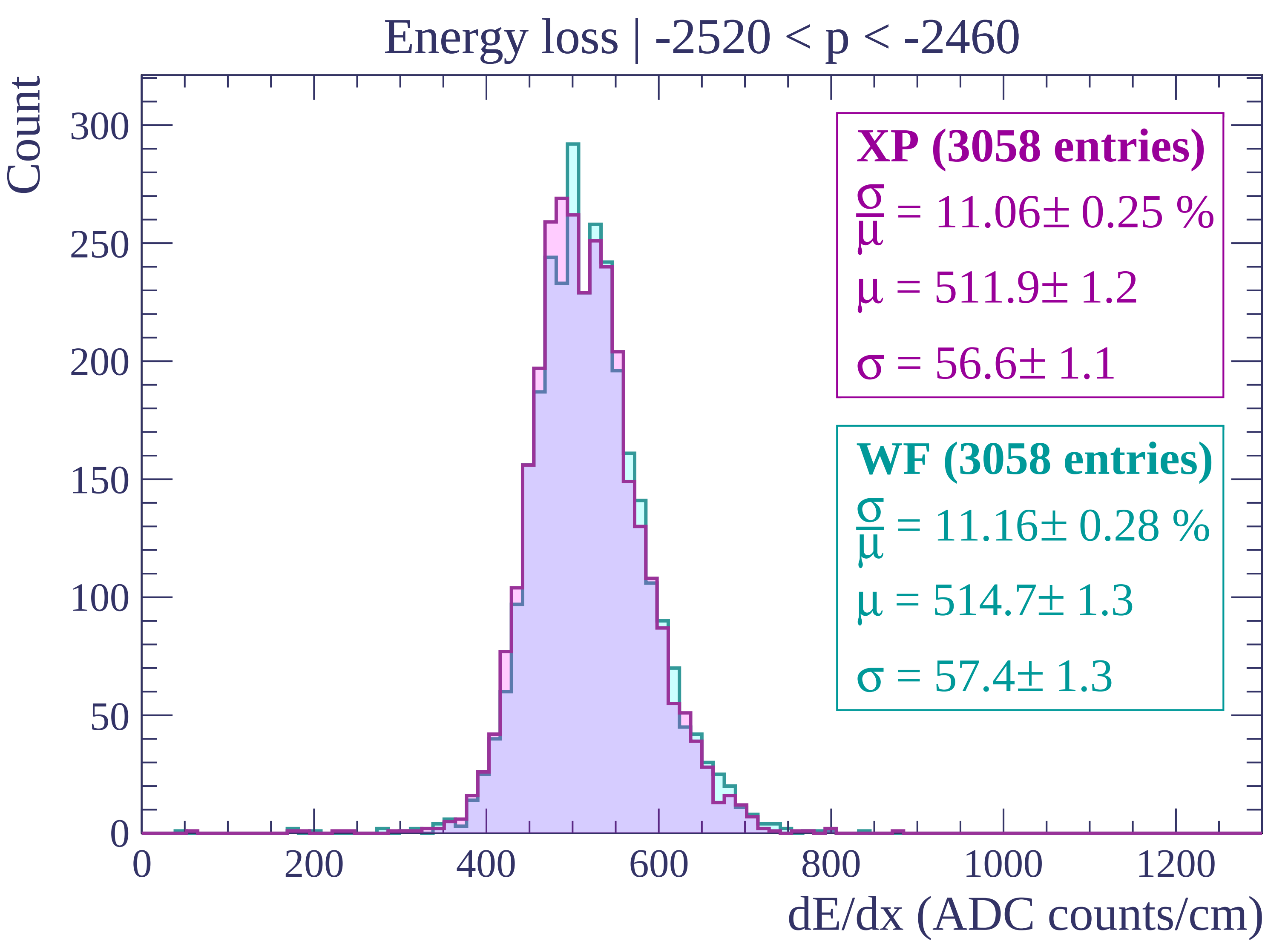
Energy loss | -2700 < p < -2640
Count
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2849 entries)
σ/μ = 11.13± 0.28 %
μ = 516.6± 1.3
σ = 57.5± 1.3
WF (2849 entries)
σ/μ = 11.33± 0.27 %
μ = 521.1± 1.3
σ = 59.0± 1.3

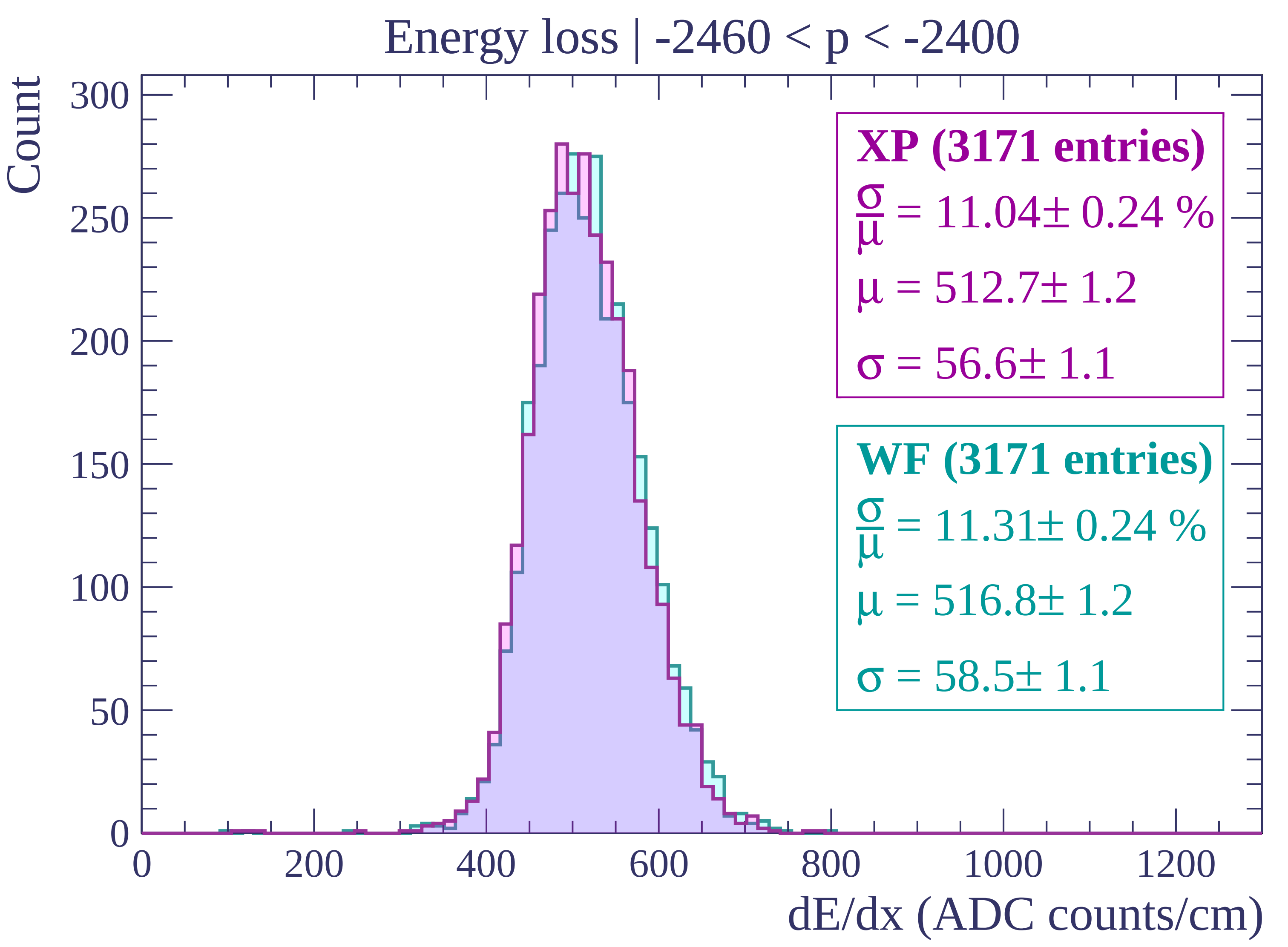


Energy loss | -2640 < p < -2580
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP (2927 entries)
$\frac{\sigma}{\mu}$ = 11.26± 0.26 %
μ = 515.2± 1.3
σ = 58.0± 1.2
WF (2927 entries)
$\frac{\sigma}{\mu}$ = 11.48± 0.28 %
μ = 519.1± 1.3
σ = 59.6± 1.3

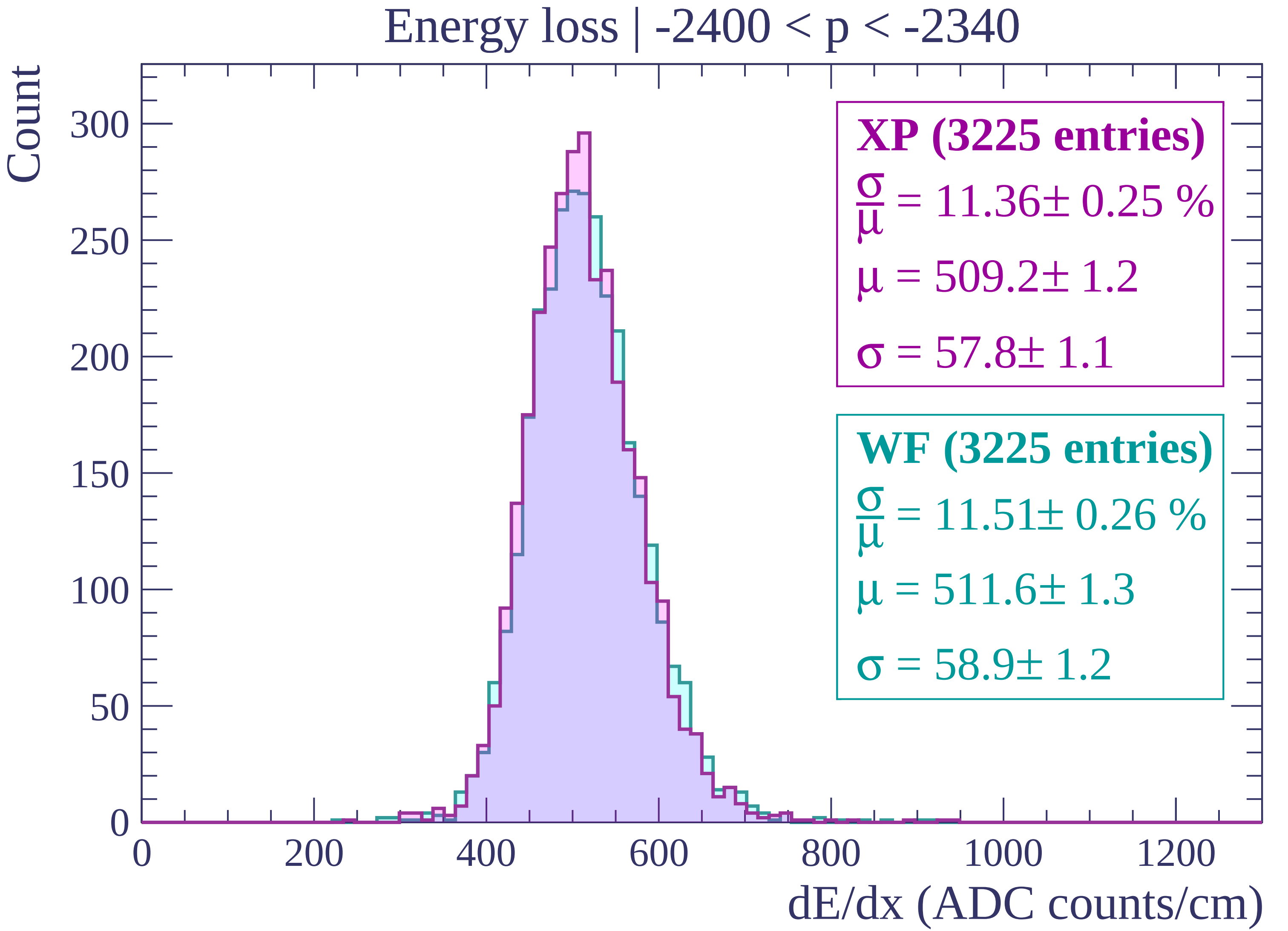


Energy loss | -2580 < p < -2520
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (2995 entries)
$\frac{\sigma}{\mu}$ = 11.13± 0.28 %
μ = 511.6± 1.3
σ = 57.0± 1.3
WF (2995 entries)
$\frac{\sigma}{\mu}$ = 11.68± 0.29 %
μ = 517.8± 1.3
σ = 60.5± 1.4

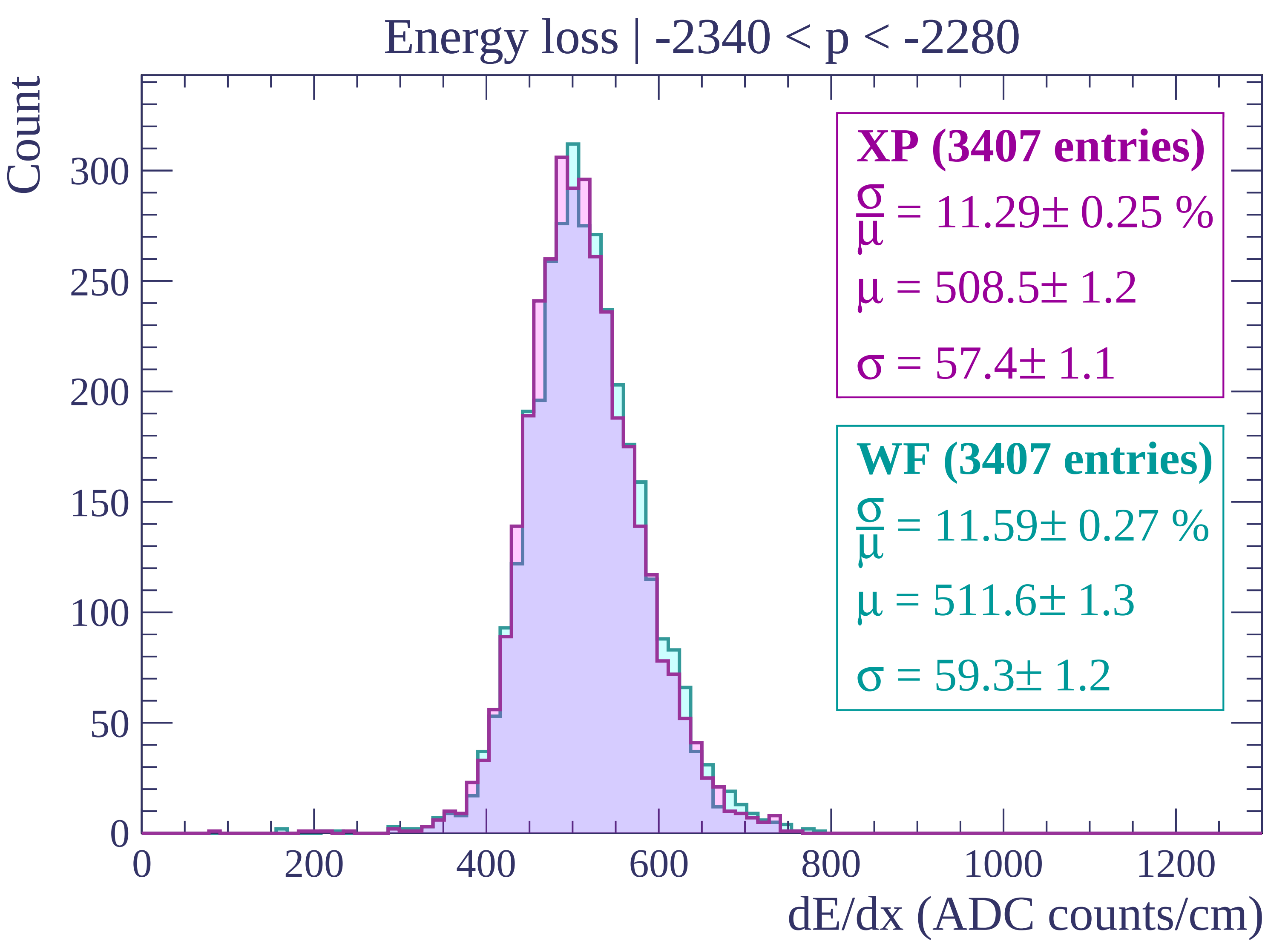

Energy loss | -2520 < p < -2460
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (3058 entries)
σ/μ = 11.06± 0.25 %
μ = 511.9± 1.2
σ = 56.6± 1.1
WF (3058 entries)
σ/μ = 11.16± 0.28 %
μ = 514.7± 1.3
σ = 57.4± 1.3

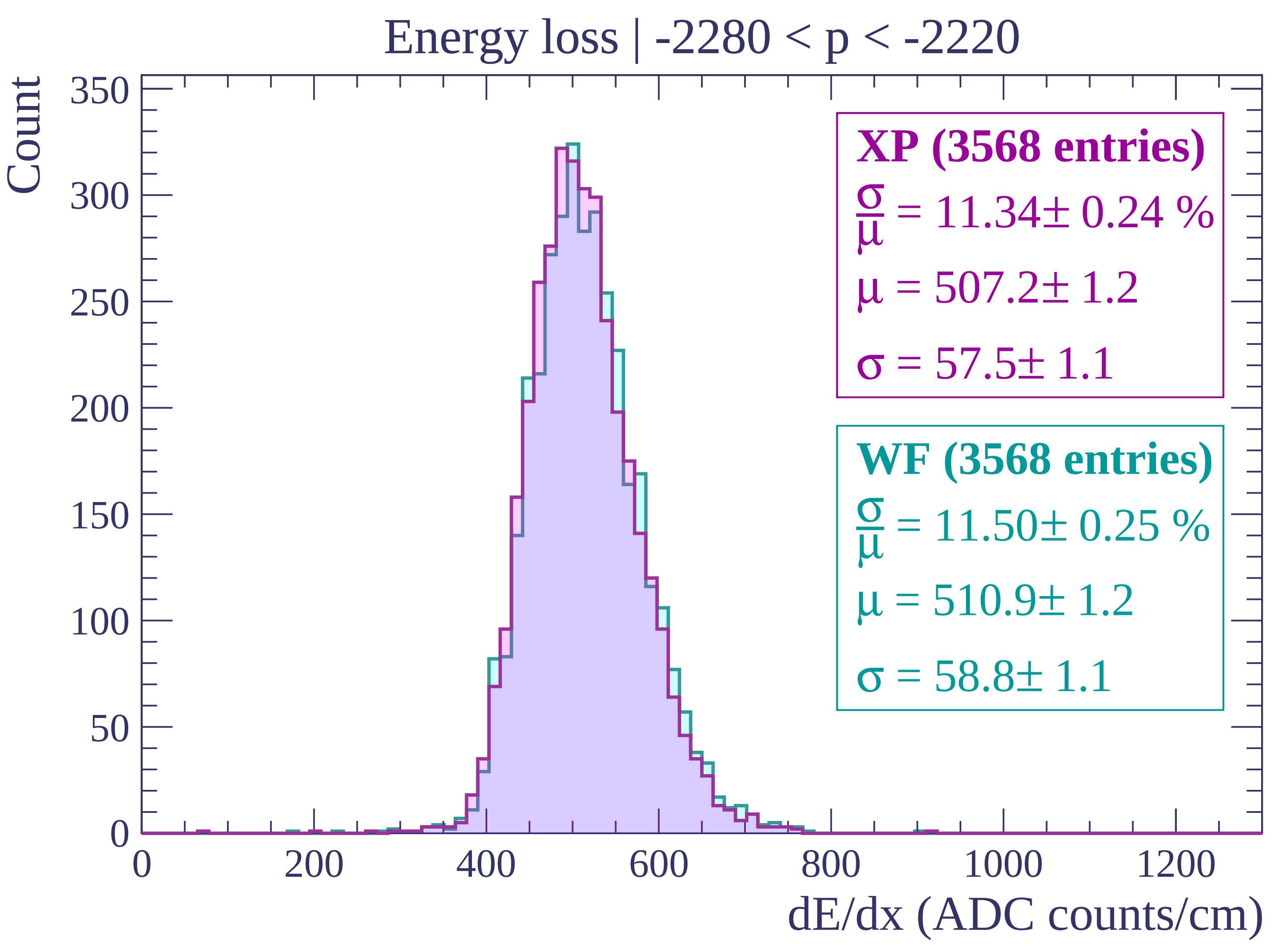


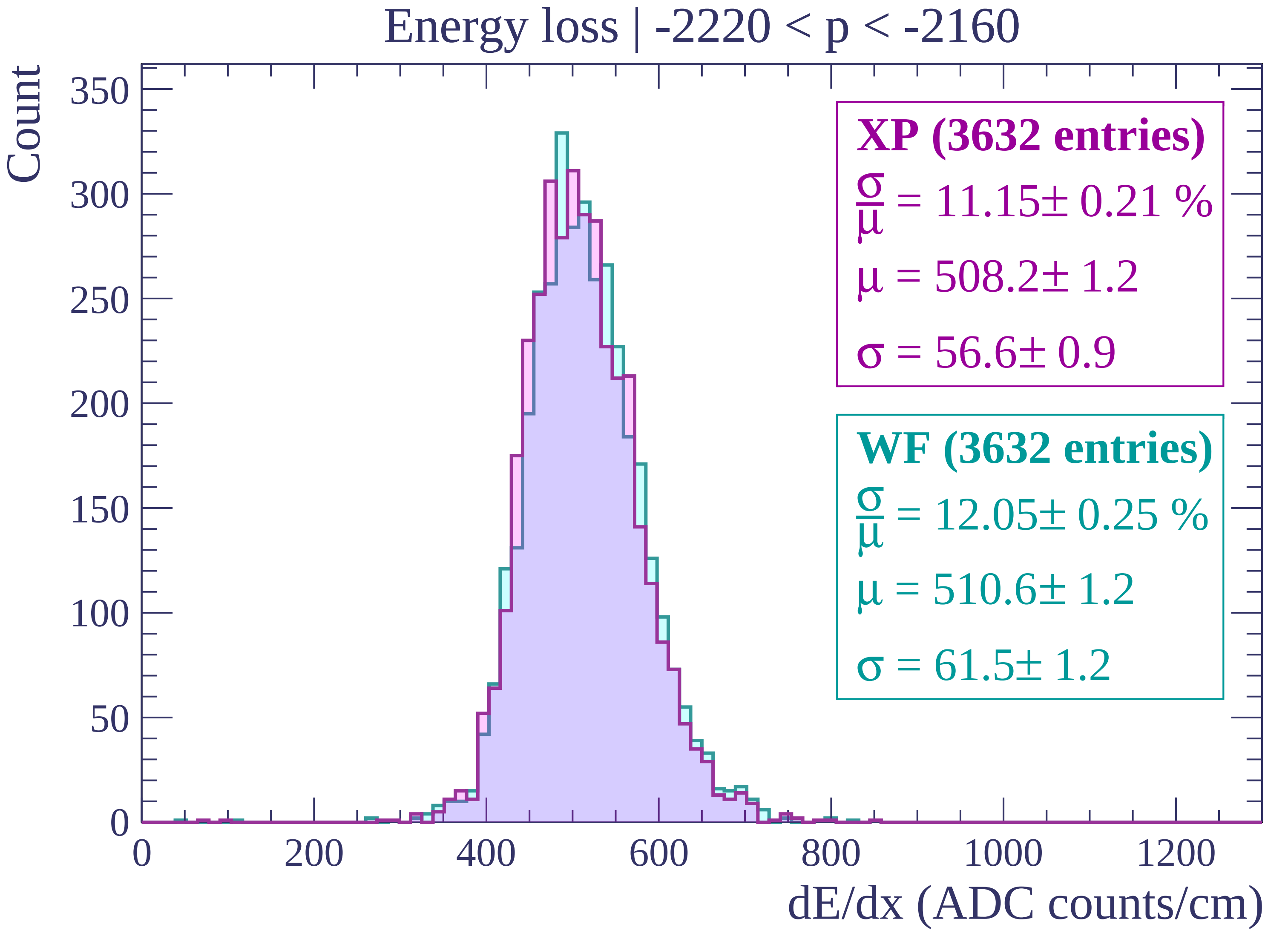
Energy loss | -2460 < p < -2400
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (3171 entries)
$\frac{\sigma}{\mu}$ = 11.04± 0.24 %
μ = 512.7± 1.2
σ = 56.6± 1.1
WF (3171 entries)
$\frac{\sigma}{\mu}$ = 11.31± 0.24 %
μ = 516.8± 1.2
σ = 58.5± 1.1



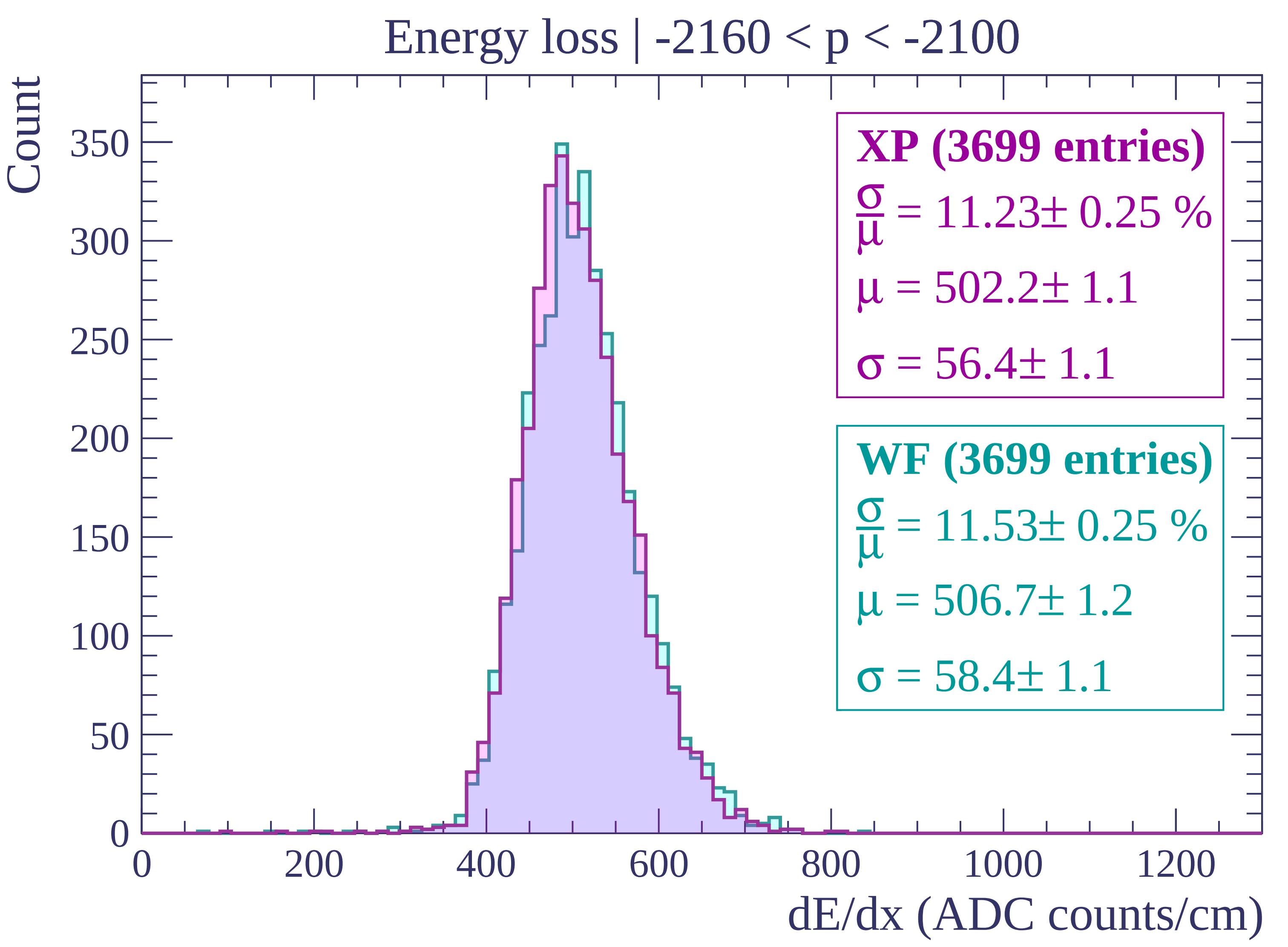
Energy loss | -2400 < p < -2340
Count
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (3225 entries)
σ/μ = 11.36± 0.25 %
μ = 509.2± 1.2
σ = 57.8± 1.1
WF (3225 entries)
σ/μ = 11.51± 0.26 %
μ = 511.6± 1.3
σ = 58.9± 1.2

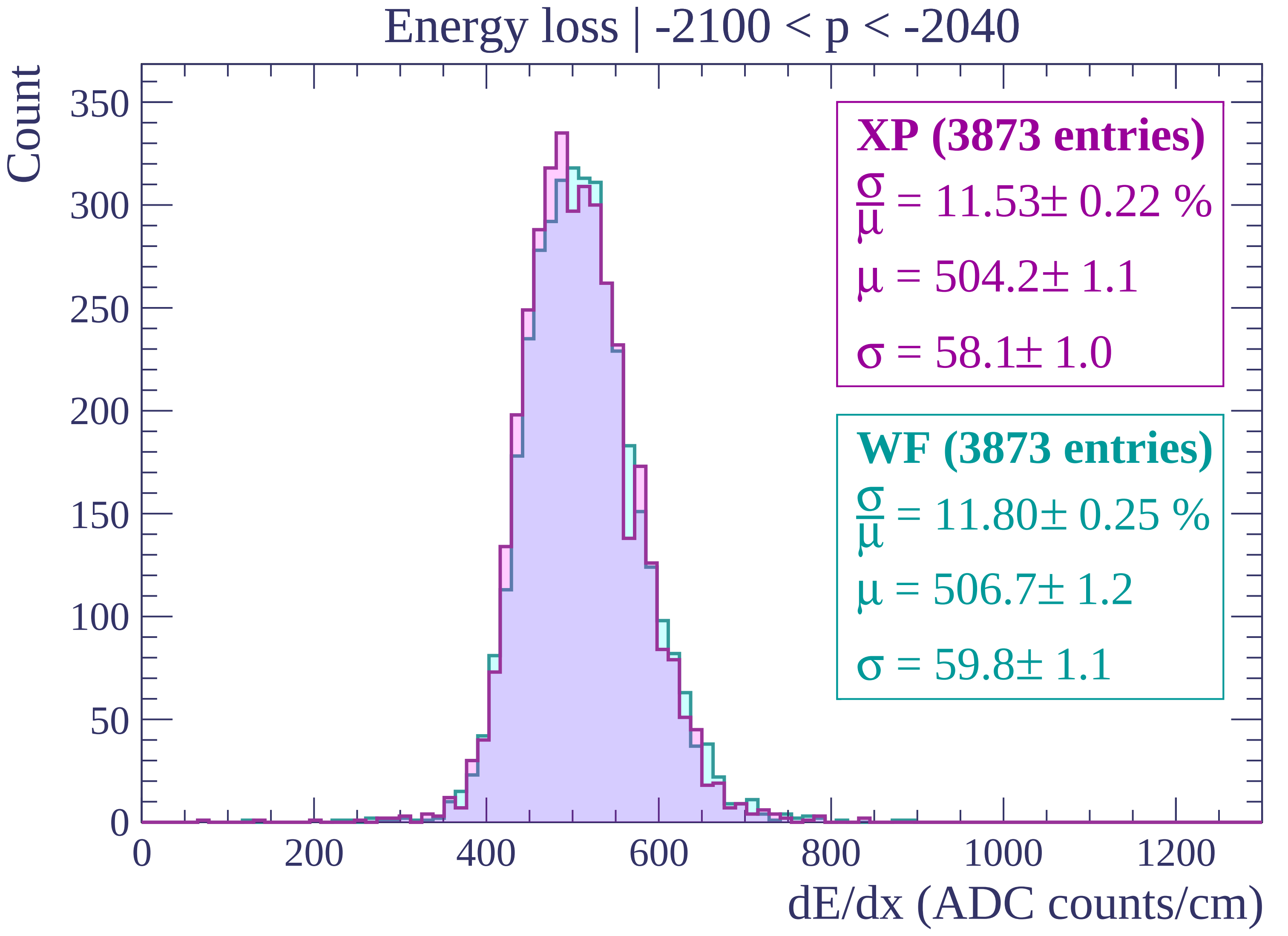

Energy loss | -2340 < p < -2280
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (3407 entries)
$\frac{\sigma}{\mu}$ = 11.29± 0.25 %
μ = 508.5± 1.2
σ = 57.4± 1.1
WF (3407 entries)
$\frac{\sigma}{\mu}$ = 11.59± 0.27 %
μ = 511.6± 1.3
σ = 59.3± 1.2


Energy loss | -2280 < p < -2220
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (3568 entries)
σ/μ = 11.34± 0.24 %
μ = 507.2± 1.2
σ = 57.5± 1.1
WF (3568 entries)
σ/μ = 11.50± 0.25 %
μ = 510.9± 1.2
σ = 58.8± 1.1

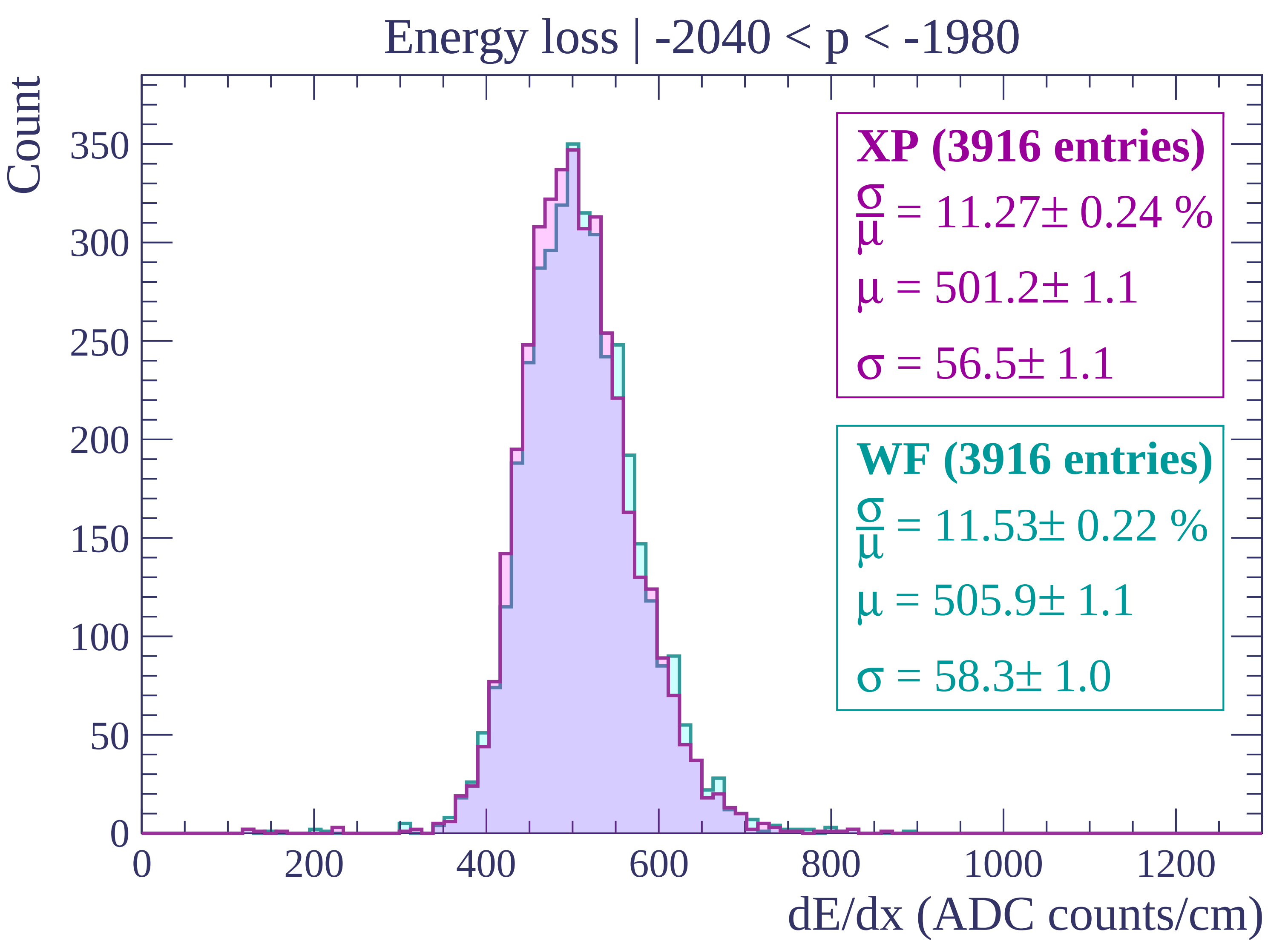


Energy loss | -2220 < p < -2160
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
350
XP (3632 entries)
$\frac{\sigma}{\mu}$ = 11.15± 0.21 %
μ = 508.2± 1.2
σ = 56.6± 0.9
WF (3632 entries)
$\frac{\sigma}{\mu}$ = 12.05± 0.25 %
μ = 510.6± 1.2
σ = 61.5± 1.2

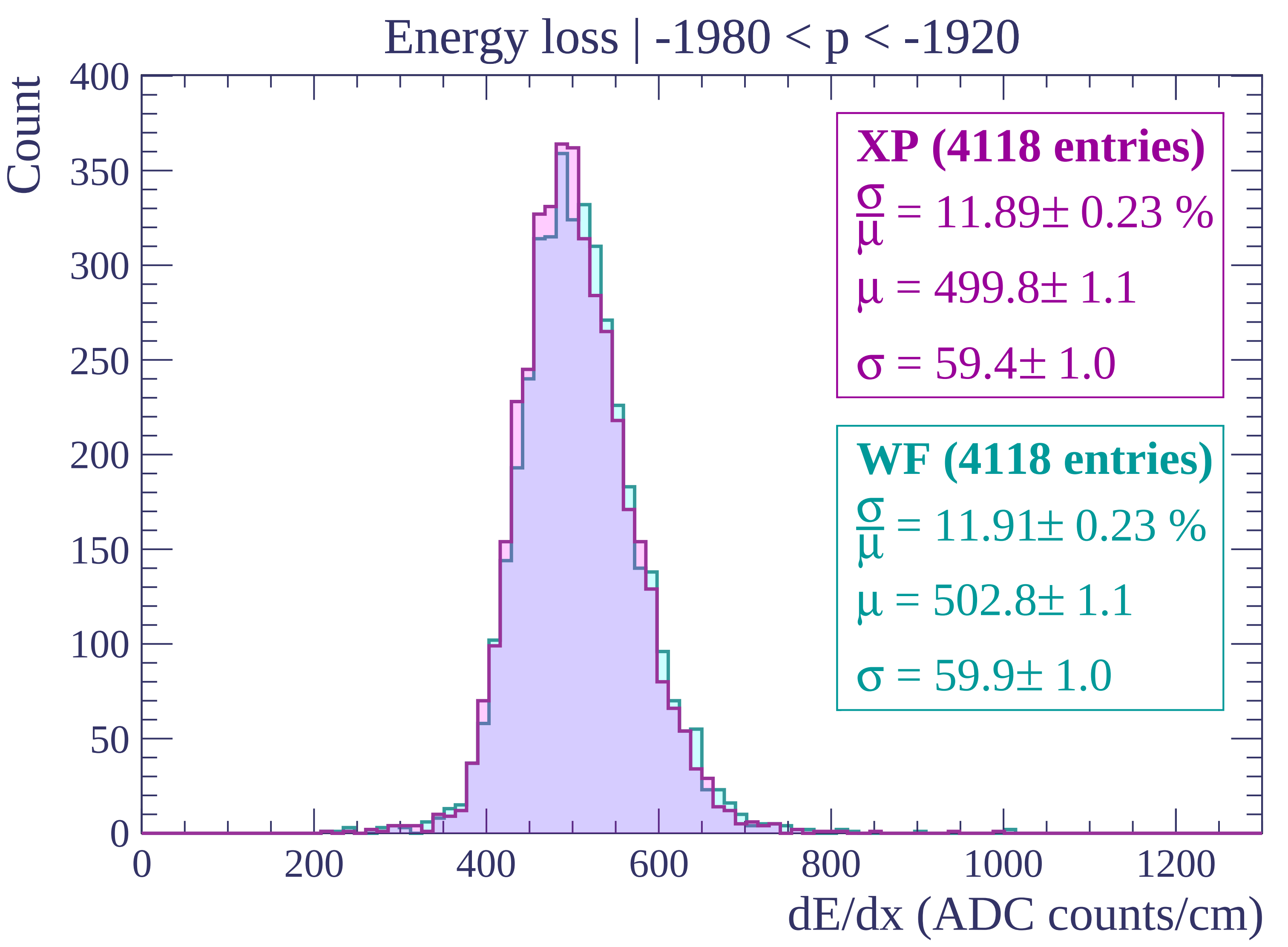

Energy loss | -2160 < p < -2100
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (3699 entries)
σ/μ = 11.23± 0.25 %
μ = 502.2± 1.1
σ = 56.4± 1.1
WF (3699 entries)
σ/μ = 11.53± 0.25 %
μ = 506.7± 1.2
σ = 58.4± 1.1



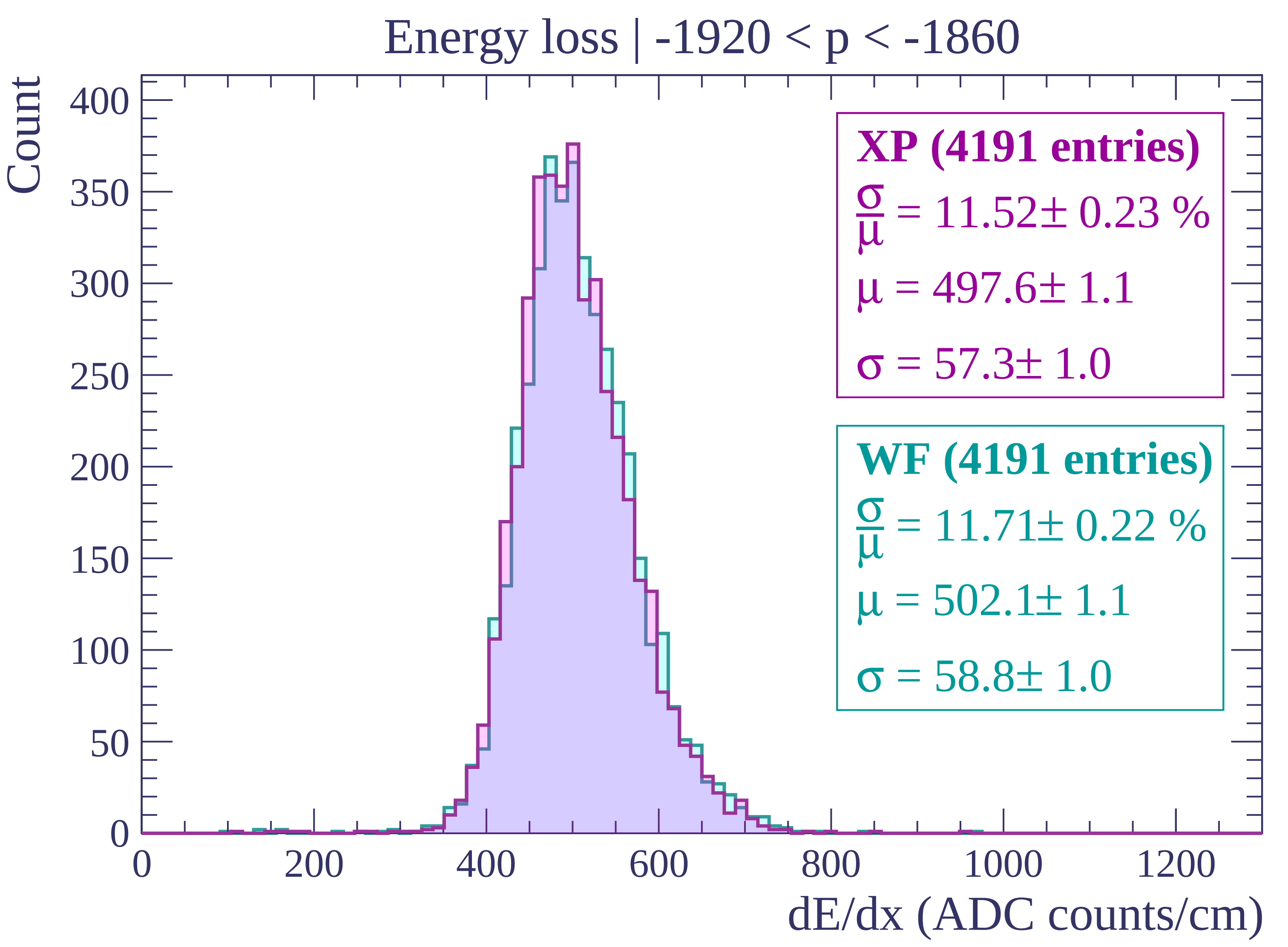
Energy loss | -2100 < p < -2040
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (3873 entries)
σ/μ = 11.53± 0.22 %
μ = 504.2± 1.1
σ = 58.1± 1.0
WF (3873 entries)
σ/μ = 11.80± 0.25 %
μ = 506.7± 1.2
σ = 59.8± 1.1



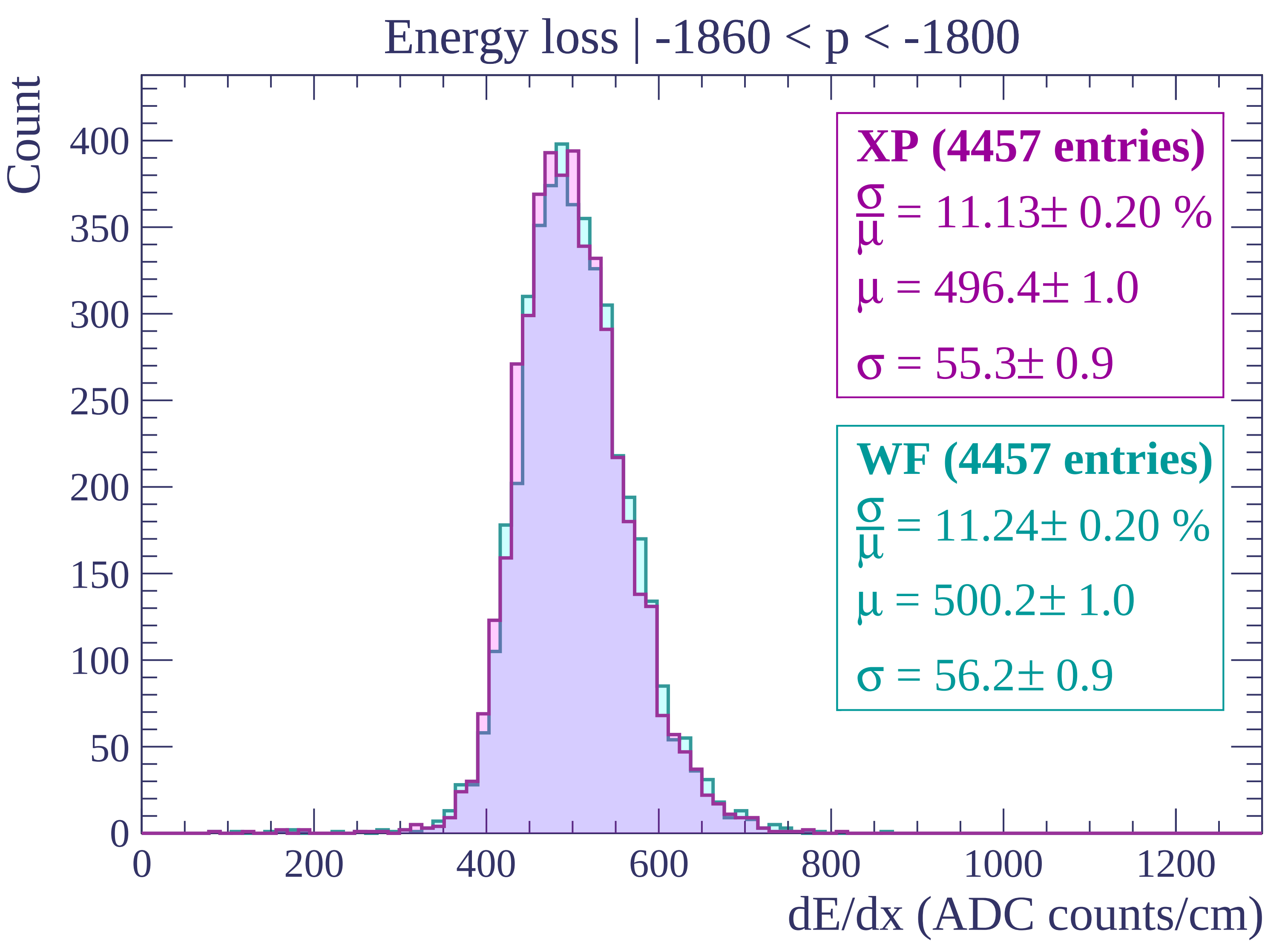
Energy loss | -2040 < p < -1980
Count
dE/dx (ADC counts/cm)
XP (3916 entries)
σ/μ = 11.27± 0.24 %
μ = 501.2± 1.1
σ = 56.5± 1.1
WF (3916 entries)
σ/μ = 11.53± 0.22 %
μ = 505.9± 1.1
σ = 58.3± 1.0
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200



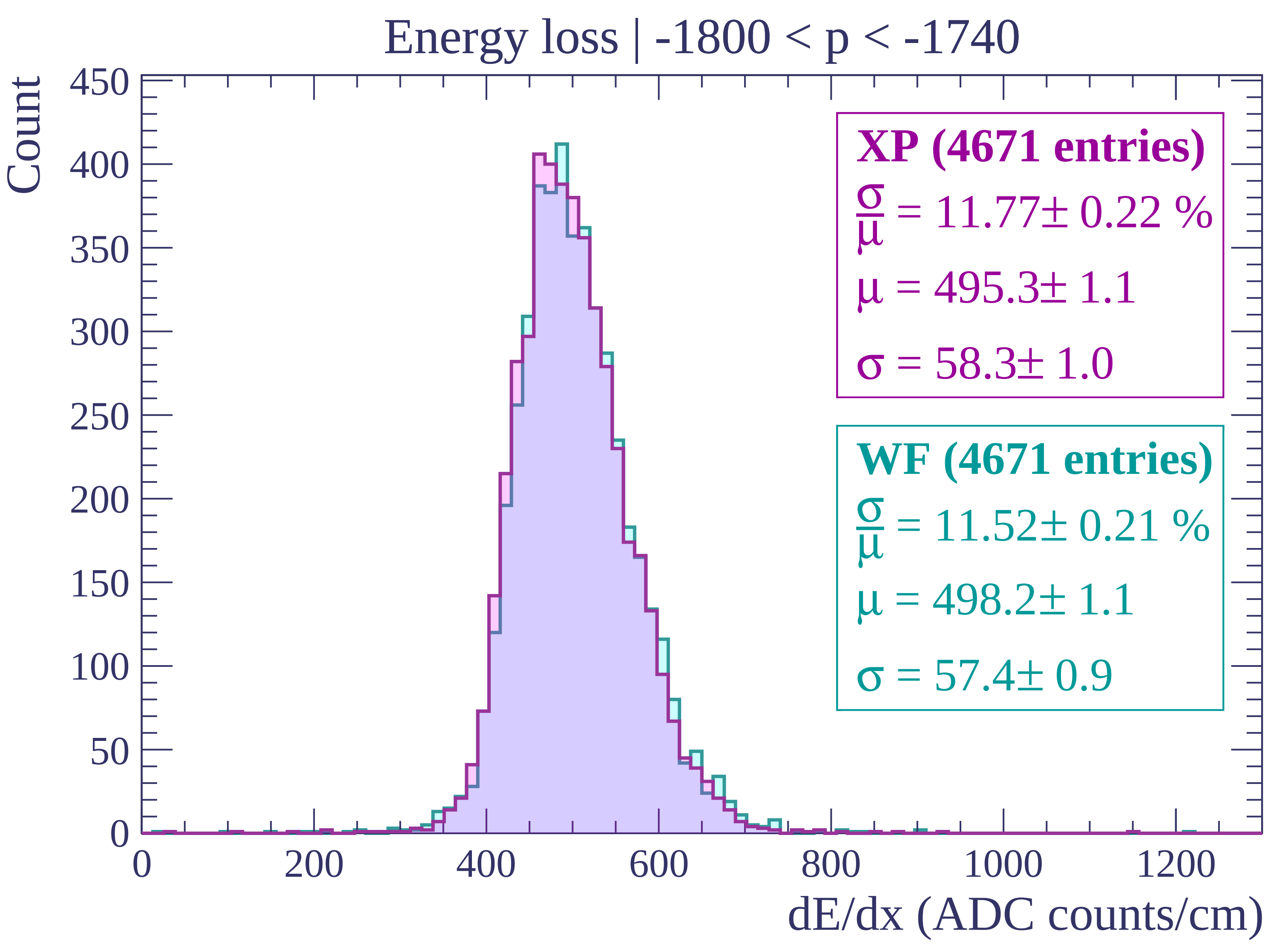
Energy loss | -1980 < p < -1920
Count
dE/dx (ADC counts/cm)
XP (4118 entries)
σ/μ = 11.89± 0.23 %
μ = 499.8± 1.1
σ = 59.4± 1.0
WF (4118 entries)
σ/μ = 11.91± 0.23 %
μ = 502.8± 1.1
σ = 59.9± 1.0
0
200
400
600
800
1000
1200
50
100
150
250
300
350
400



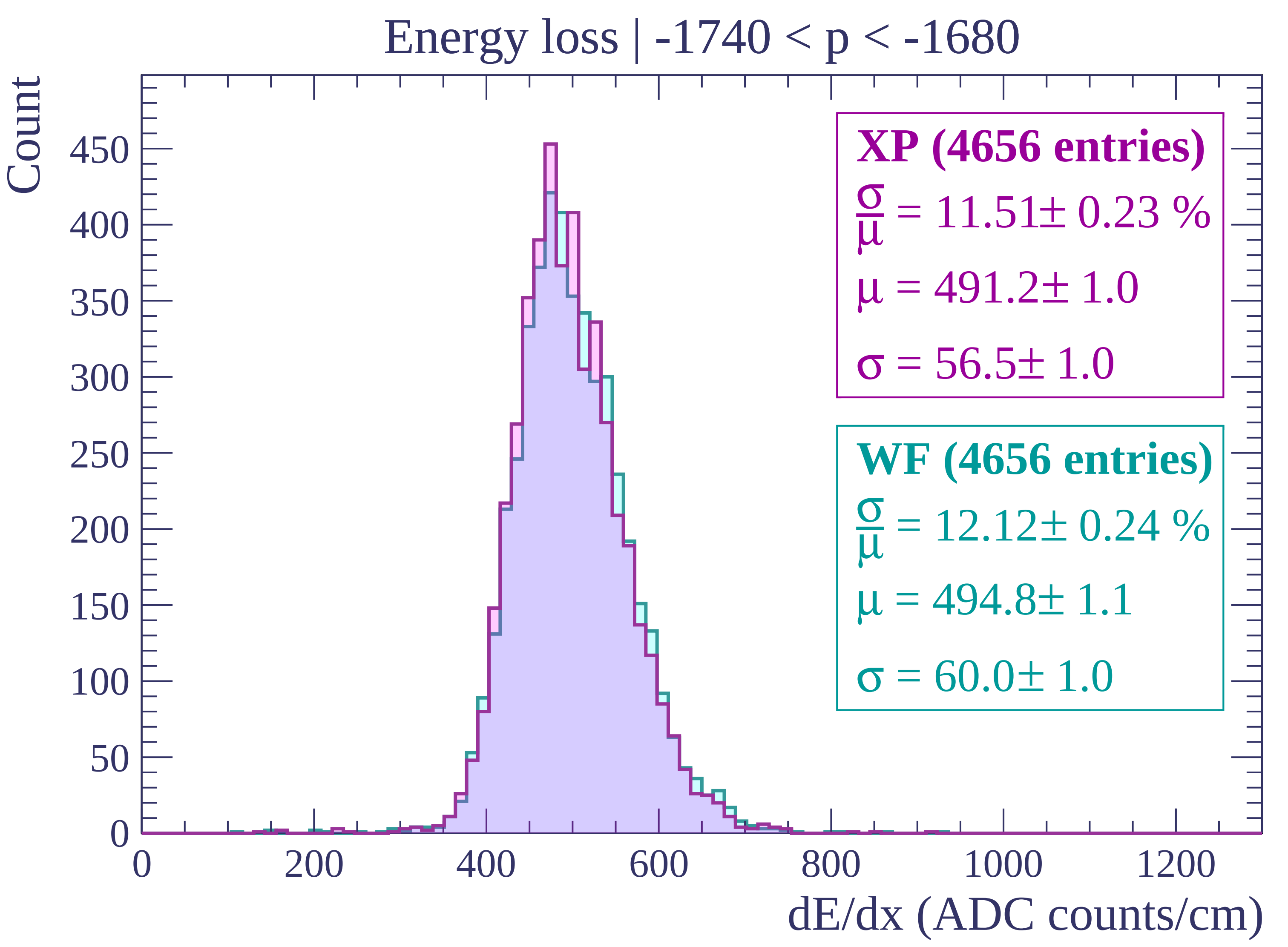
Energy loss | -1920 < p < -1860
Count
dE/dx (ADC counts/cm)
XP (4191 entries)
$\frac{\sigma}{\mu}$ = 11.52± 0.23 %
μ = 497.6± 1.1
σ = 57.3± 1.0
WF (4191 entries)
$\frac{\sigma}{\mu}$ = 11.71± 0.22 %
μ = 502.1± 1.1
σ = 58.8± 1.0
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200



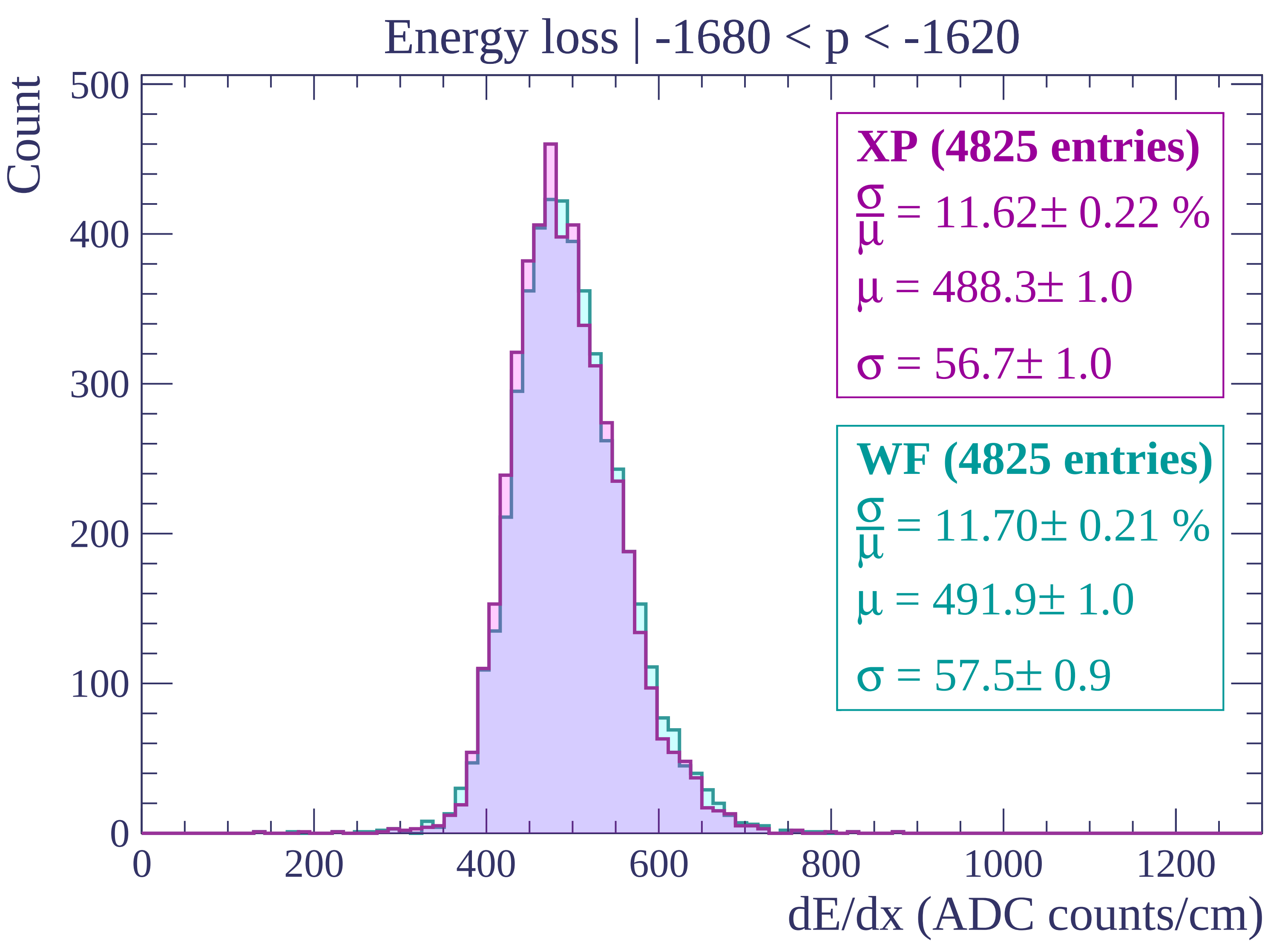
Energy loss | -1860 < p < -1800
Count
dE/dx (ADC counts/cm)
XP (4457 entries)
σ/μ = 11.13± 0.20 %
μ = 496.4± 1.0
σ = 55.3± 0.9
WF (4457 entries)
σ/μ = 11.24± 0.20 %
μ = 500.2± 1.0
σ = 56.2± 0.9



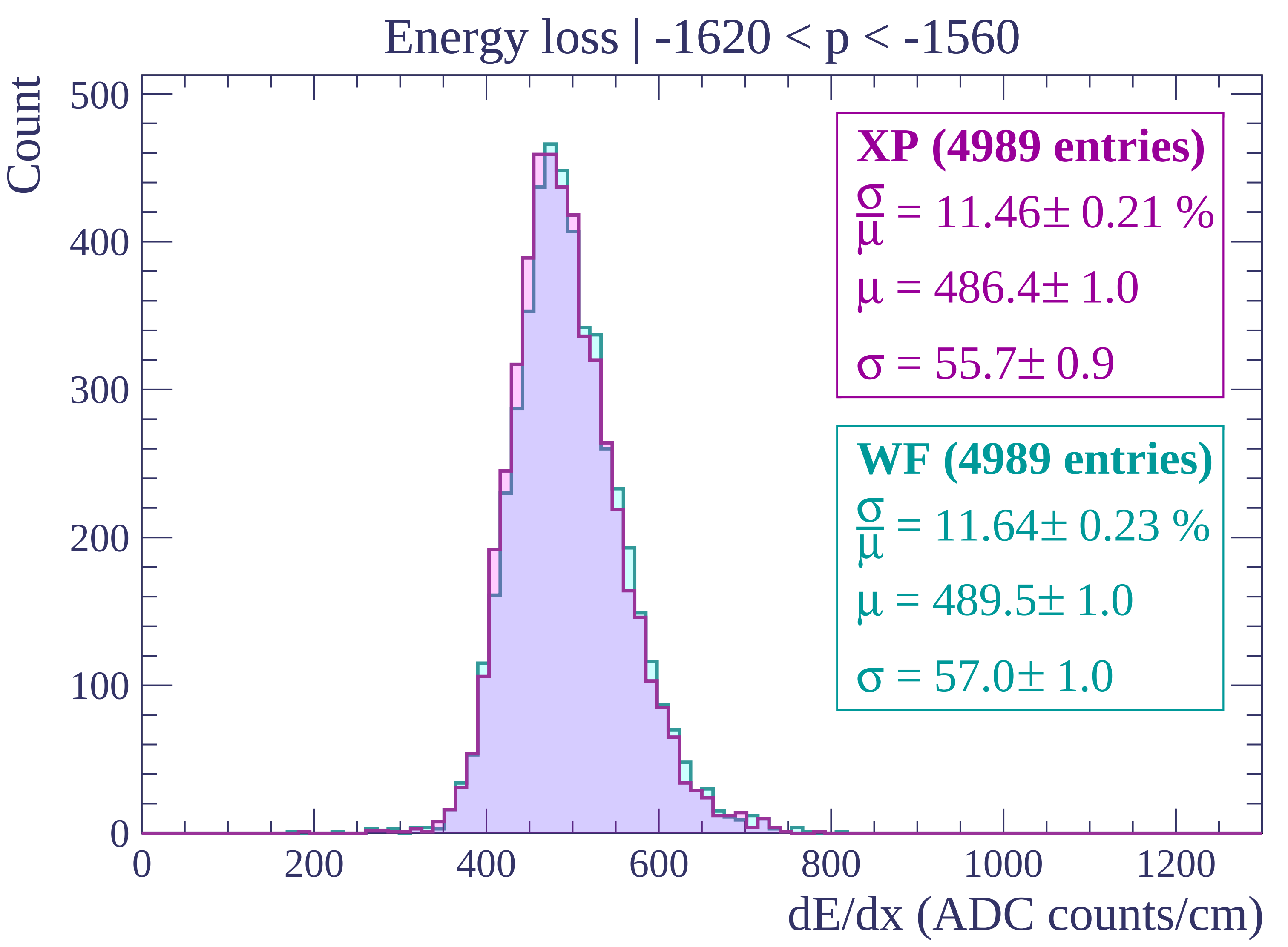
Energy loss | -1800 < p < -1740
Count
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (4671 entries)
$\frac{\sigma}{\mu}$ = 11.77± 0.22 %
μ = 495.3± 1.1
σ = 58.3± 1.0
WF (4671 entries)
$\frac{\sigma}{\mu}$ = 11.52± 0.21 %
μ = 498.2± 1.1
σ = 57.4± 0.9



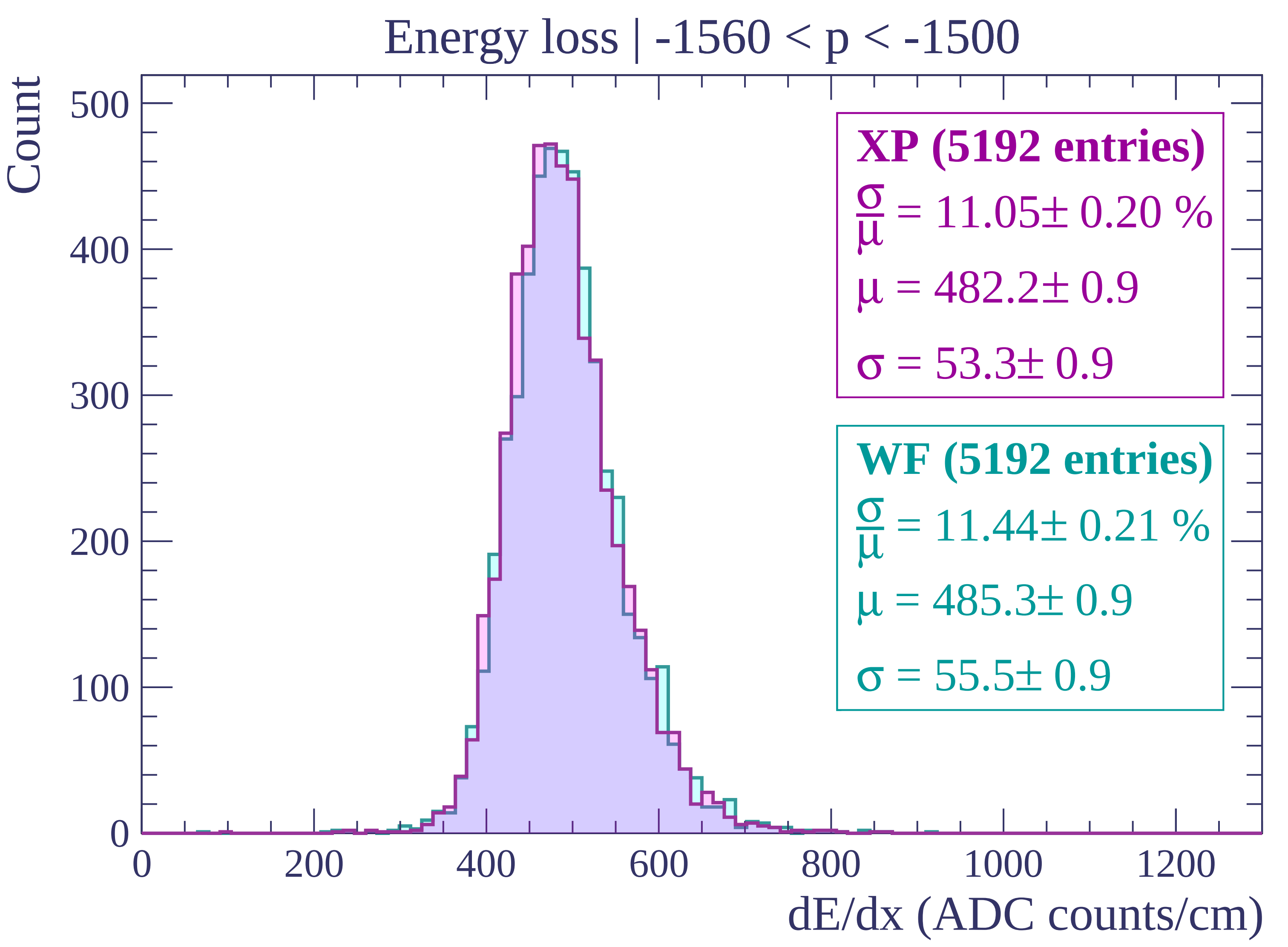
Energy loss | -1740 < p < -1680
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP (4656 entries)
σ/μ = 11.51± 0.23 %
μ = 491.2± 1.0
σ = 56.5± 1.0
WF (4656 entries)
σ/μ = 12.12± 0.24 %
μ = 494.8± 1.1
σ = 60.0± 1.0



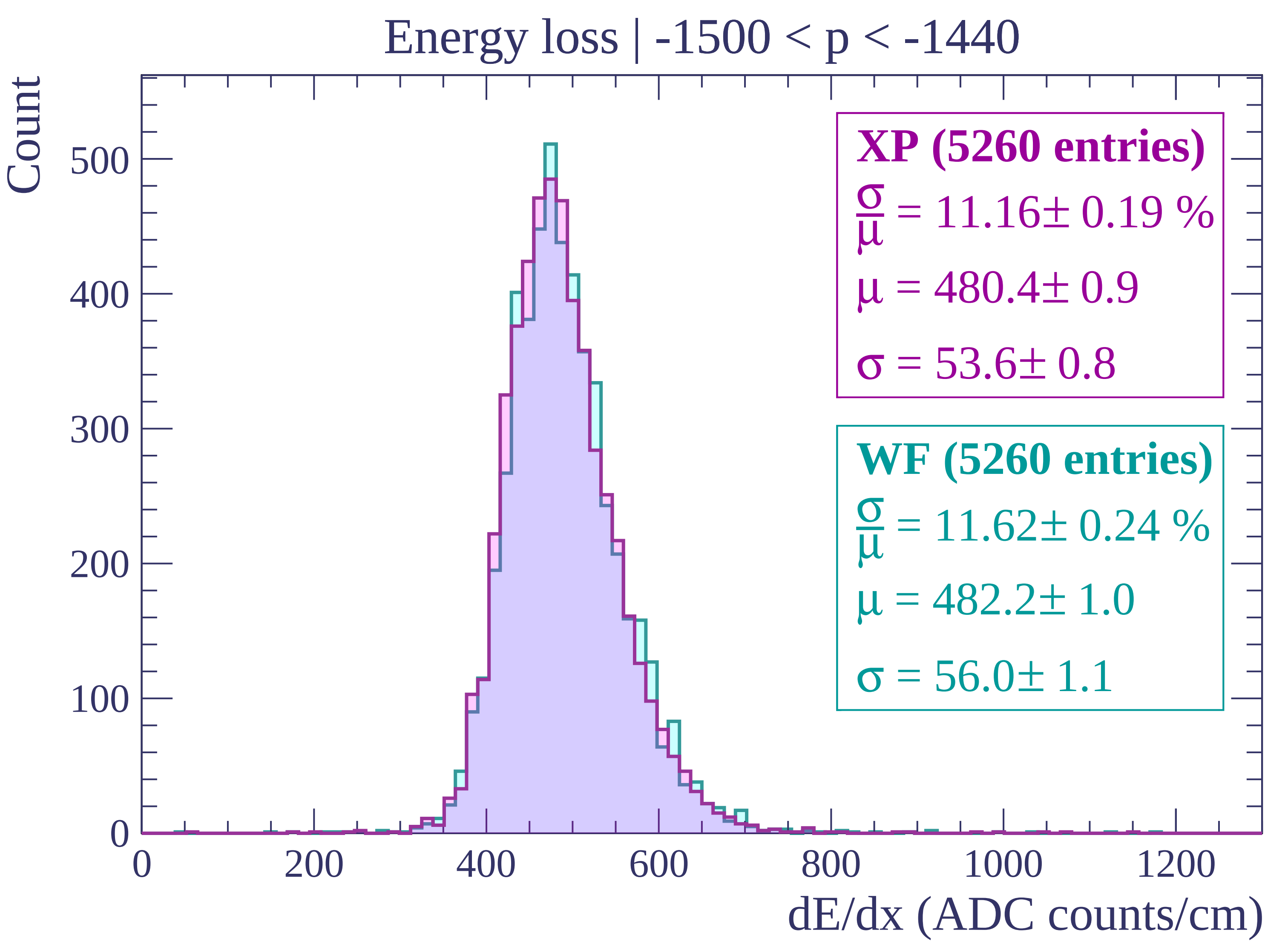
Energy loss | -1680 < p < -1620
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (4825 entries)
σ/μ = 11.62± 0.22 %
μ = 488.3± 1.0
σ = 56.7± 1.0
WF (4825 entries)
σ/μ = 11.70± 0.21 %
μ = 491.9± 1.0
σ = 57.5± 0.9

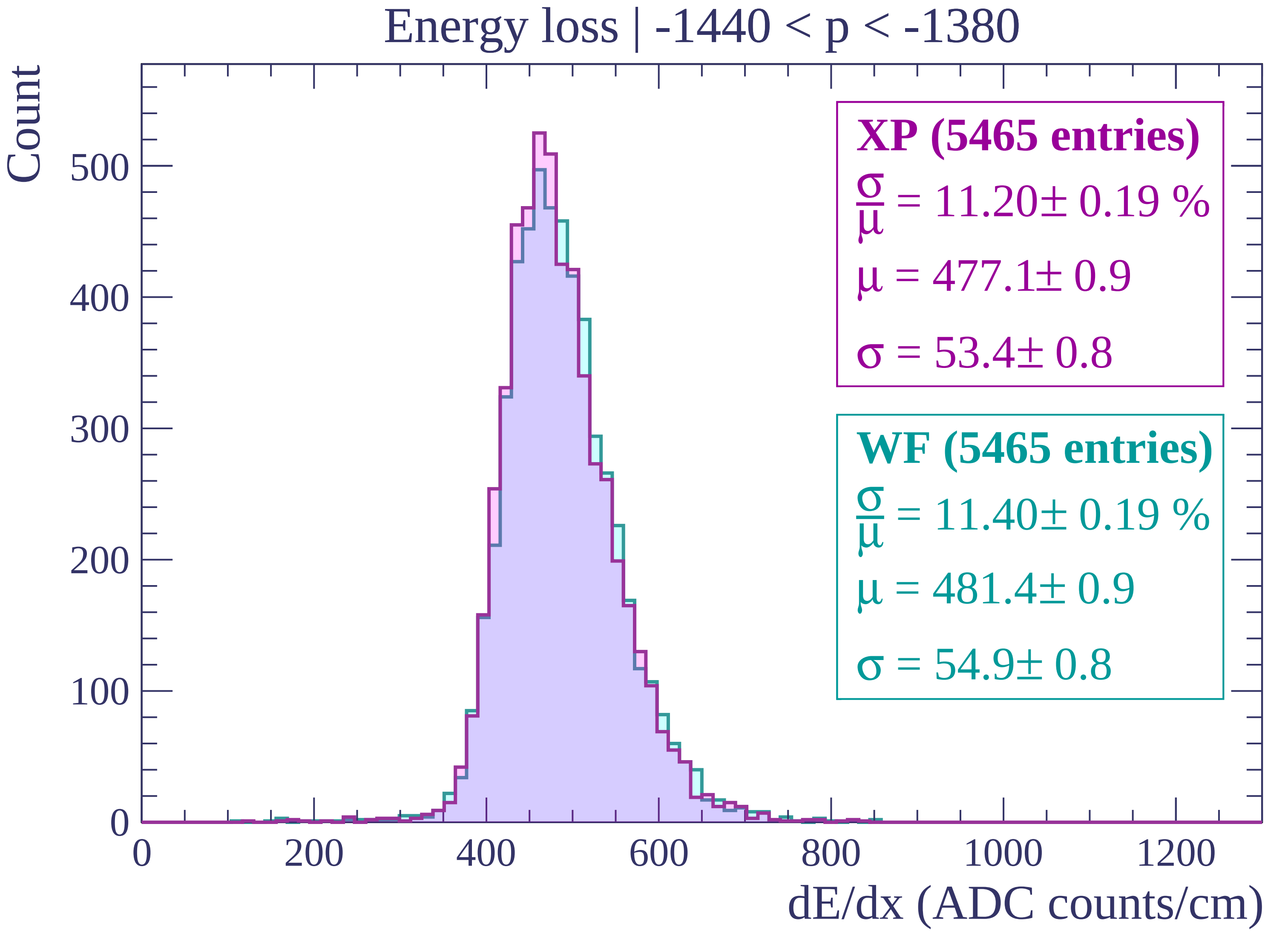


Energy loss | -1620 < p < -1560
Count
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (4989 entries)
$\frac{\sigma}{\mu}$ = 11.46± 0.21 %
μ = 486.4± 1.0
σ = 55.7± 0.9
WF (4989 entries)
$\frac{\sigma}{\mu}$ = 11.64± 0.23 %
μ = 489.5± 1.0
σ = 57.0± 1.0

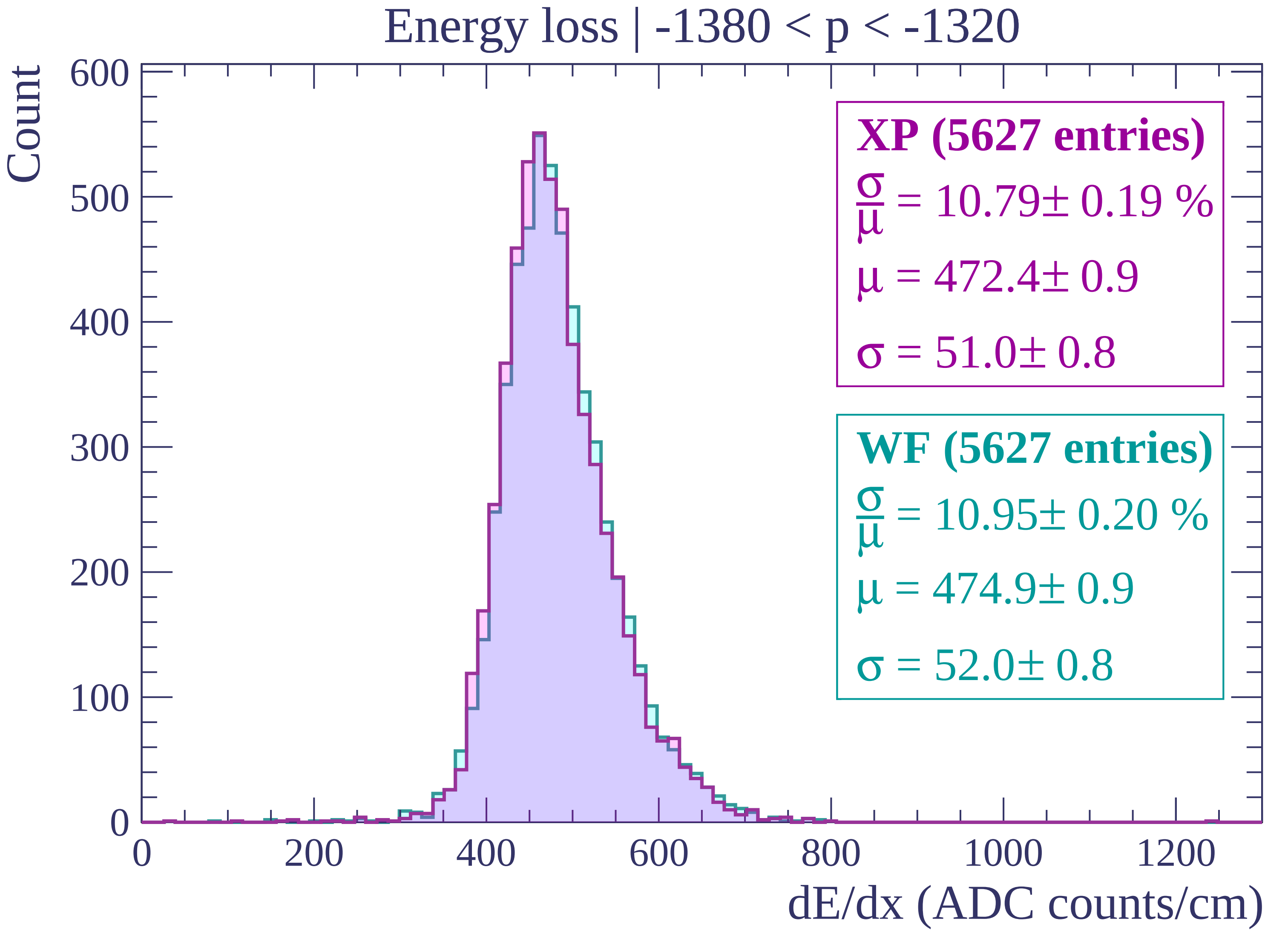


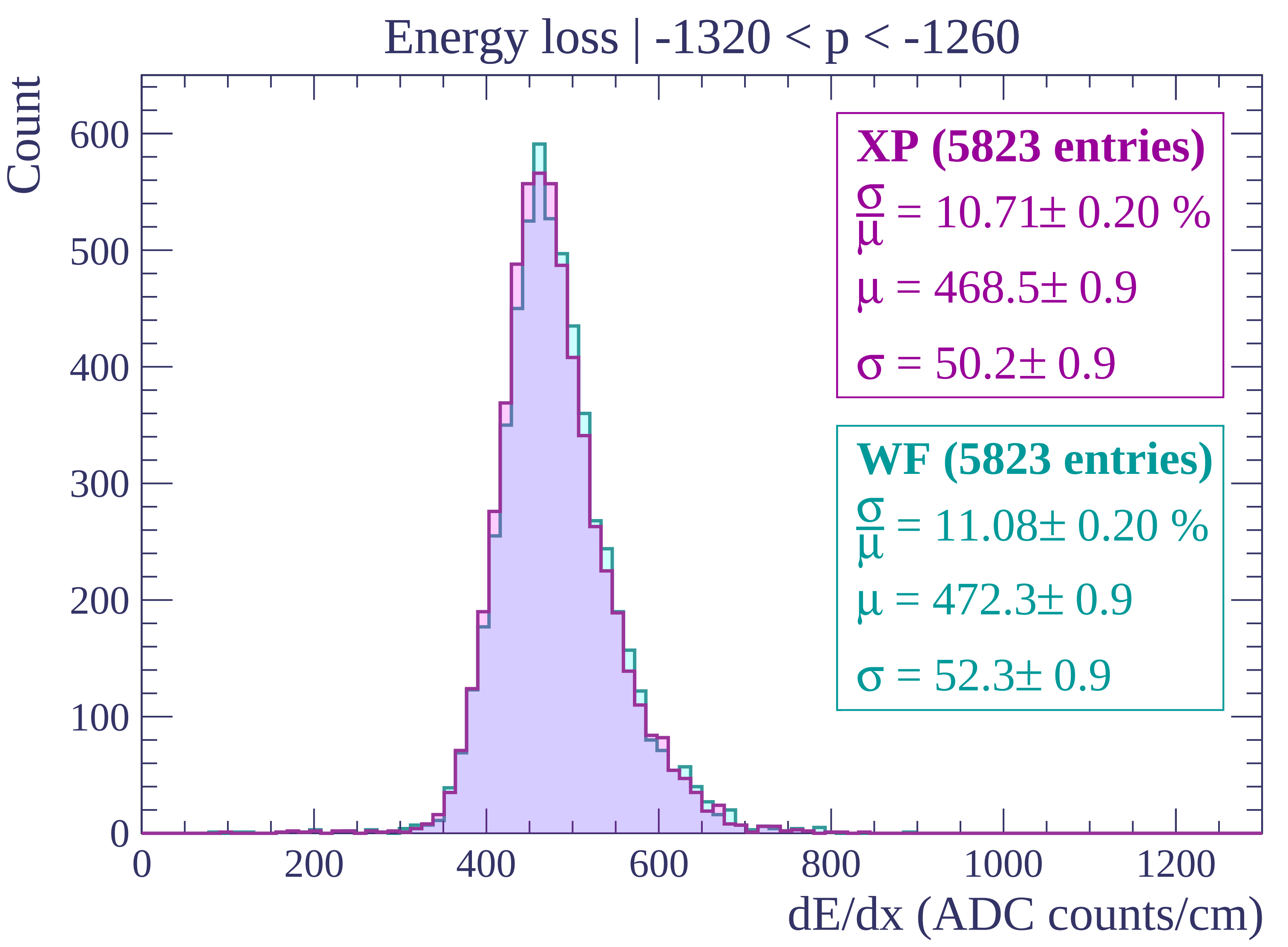
Energy loss | -1560 < p < -1500
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (5192 entries)
σ/μ = 11.05± 0.20 %
μ = 482.2± 0.9
σ = 53.3± 0.9
WF (5192 entries)
σ/μ = 11.44± 0.21 %
μ = 485.3± 0.9
σ = 55.5± 0.9

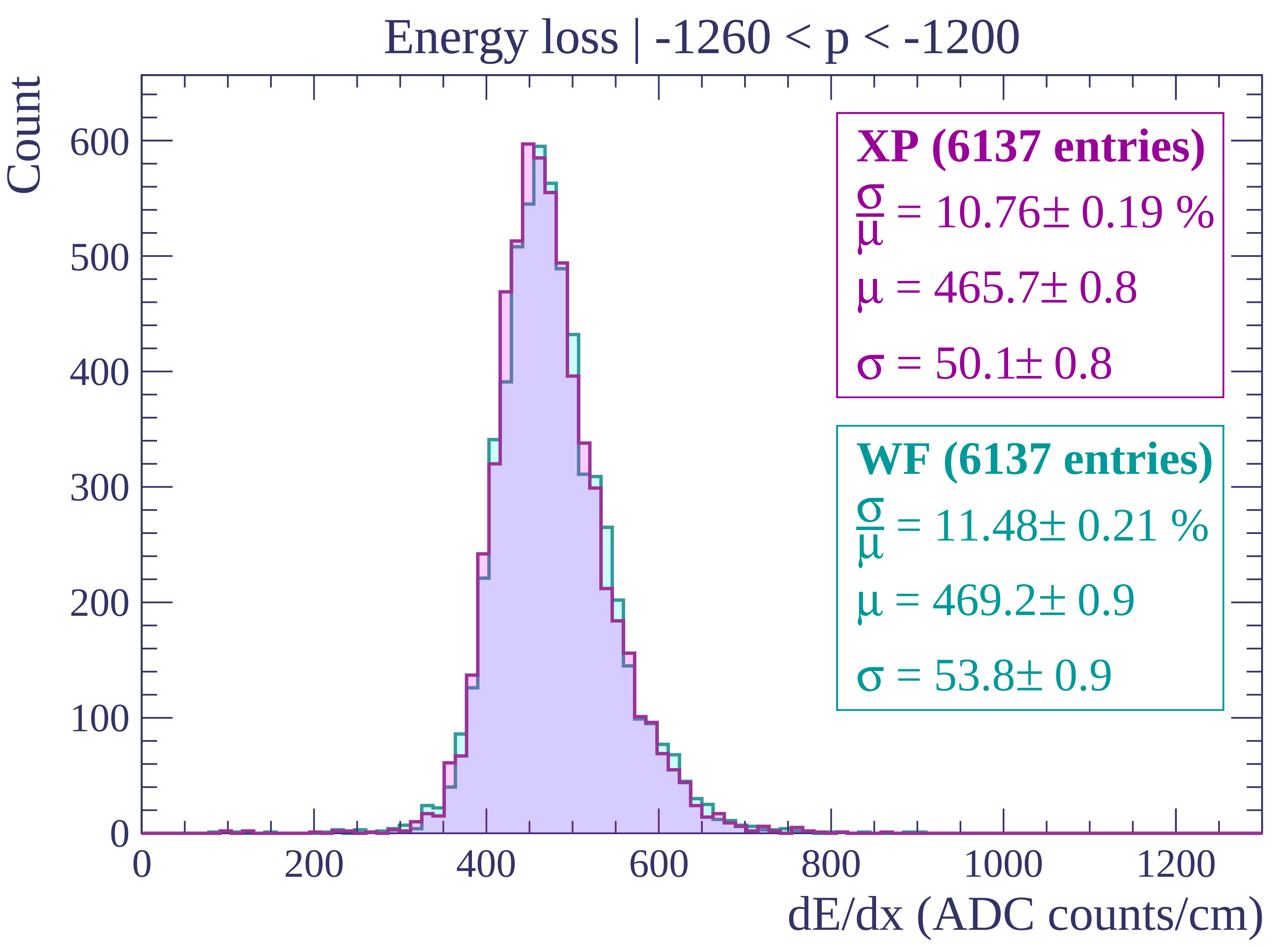

Energy loss | -1500 < p < -1440
Count
dE/dx (ADC counts/cm)
XP (5260 entries)
σ/μ = 11.16± 0.19 %
μ = 480.4± 0.9
σ = 53.6± 0.8
WF (5260 entries)
σ/μ = 11.62± 0.24 %
μ = 482.2± 1.0
σ = 56.0± 1.1
0
100
200
300
400
500
0
200
400
600
800
1000
1200



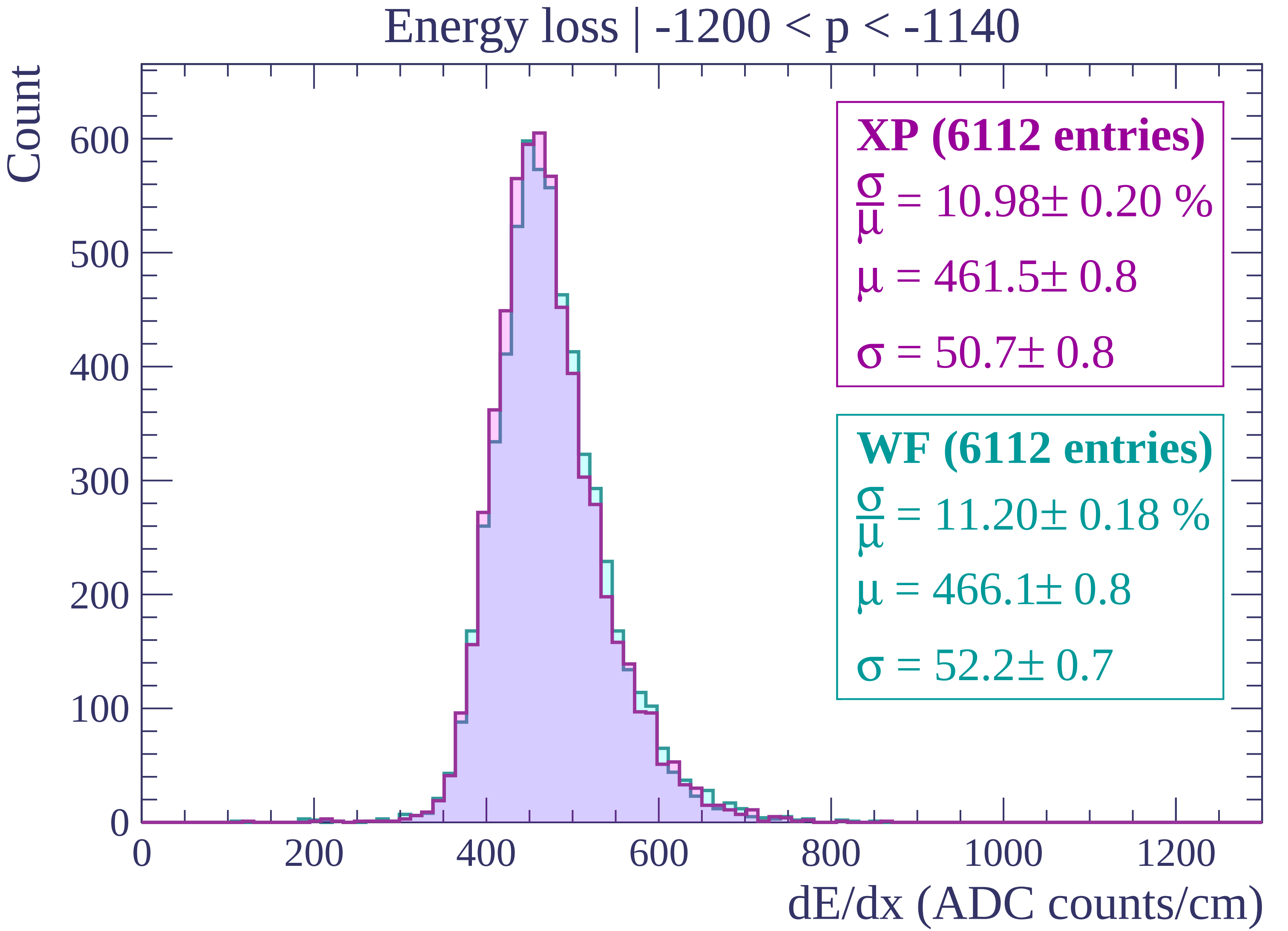
Energy loss | -1440 < p < -1380
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (5465 entries)
σ/μ = 11.20± 0.19 %
μ = 477.1± 0.9
σ = 53.4± 0.8
WF (5465 entries)
σ/μ = 11.40± 0.19 %
μ = 481.4± 0.9
σ = 54.9± 0.8

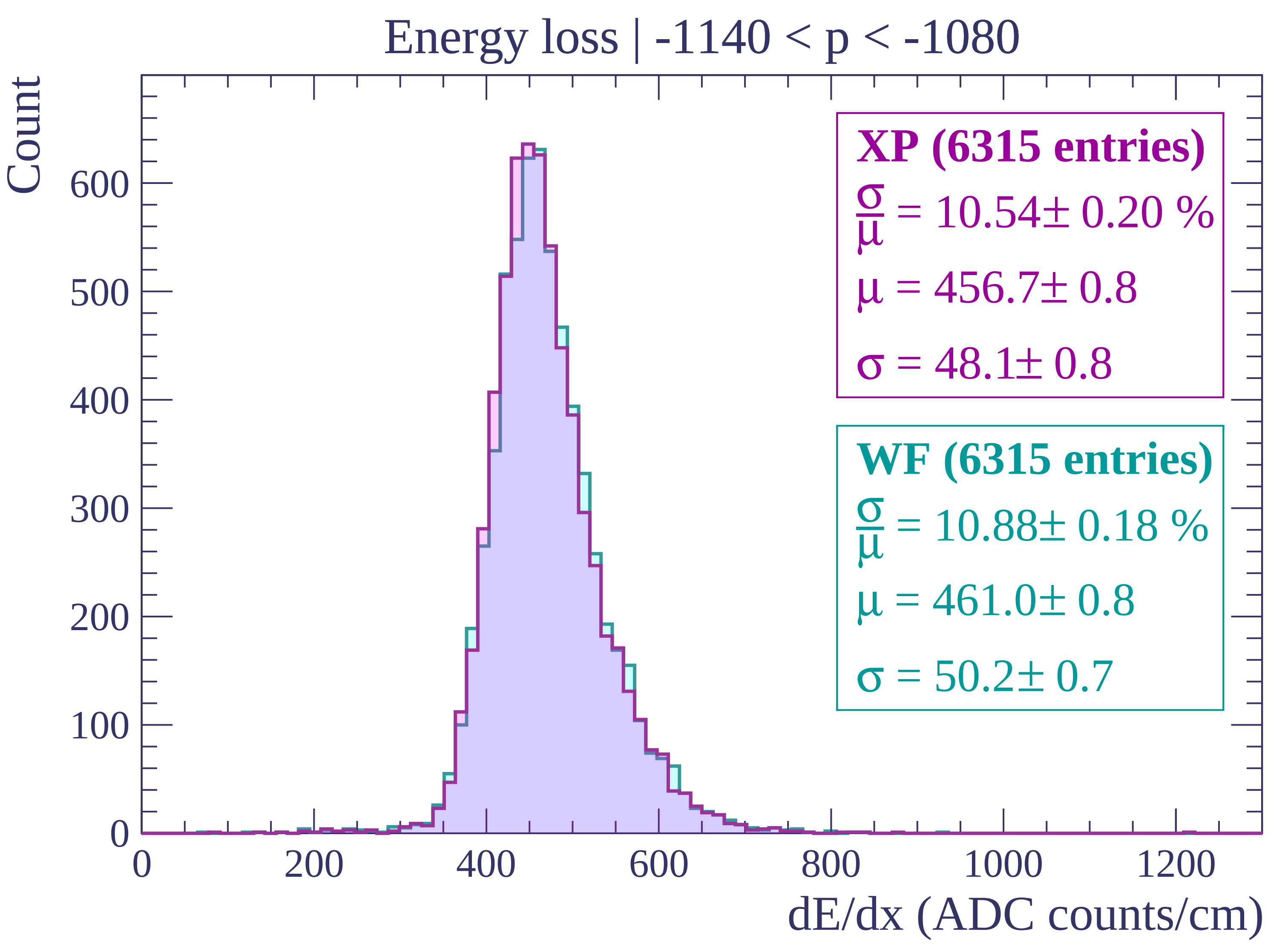

Energy loss | -1380 < p < -1320
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP (5627 entries)
$\frac{\sigma}{\mu}$ = 10.79± 0.19 %
μ = 472.4± 0.9
σ = 51.0± 0.8
WF (5627 entries)
$\frac{\sigma}{\mu}$ = 10.95± 0.20 %
μ = 474.9± 0.9
σ = 52.0± 0.8


Energy loss | -1320 < p < -1260
Count
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (5823 entries)
σ/μ = 10.71± 0.20 %
μ = 468.5± 0.9
σ = 50.2± 0.9
WF (5823 entries)
σ/μ = 11.08± 0.20 %
μ = 472.3± 0.9
σ = 52.3± 0.9

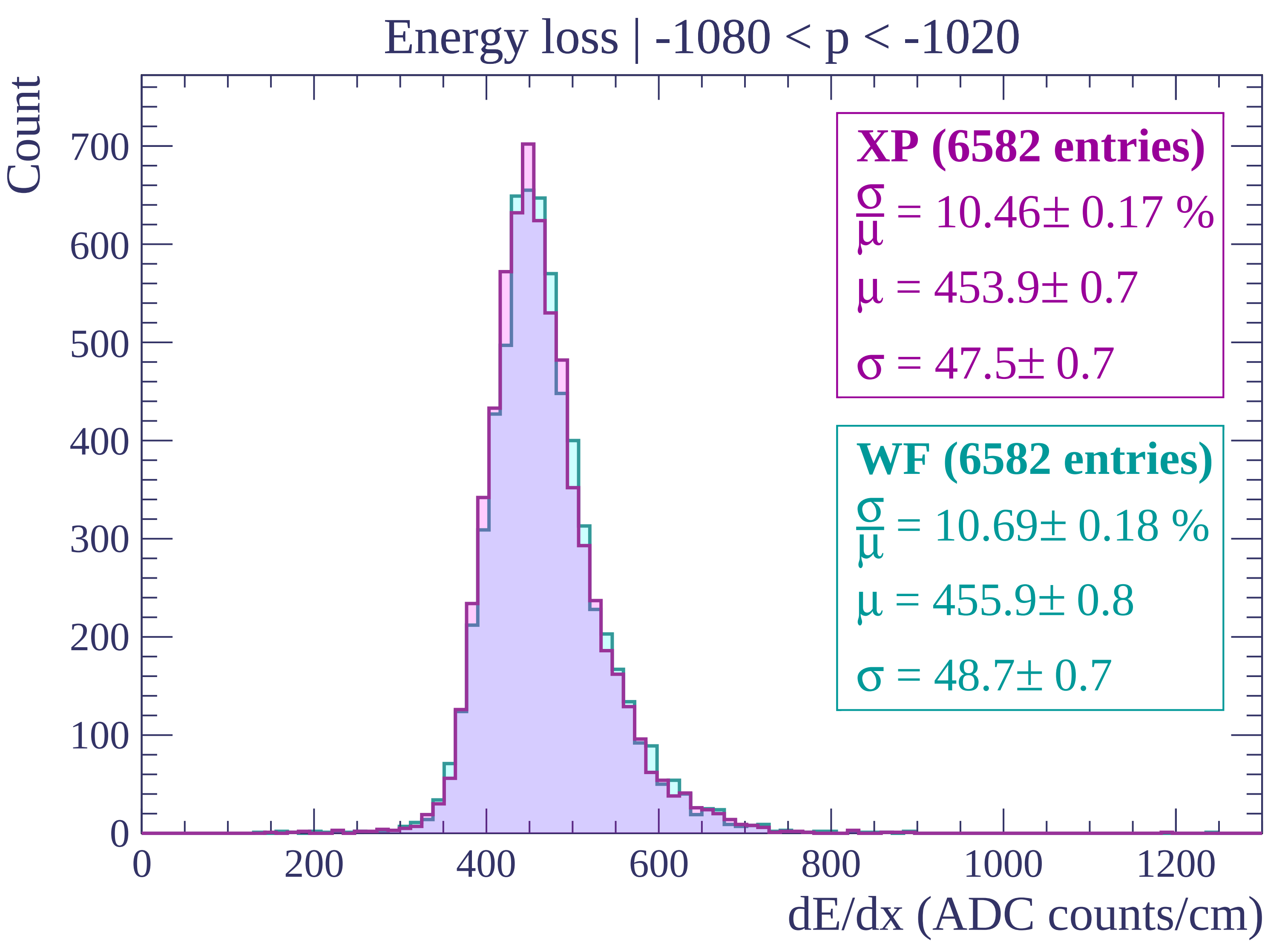


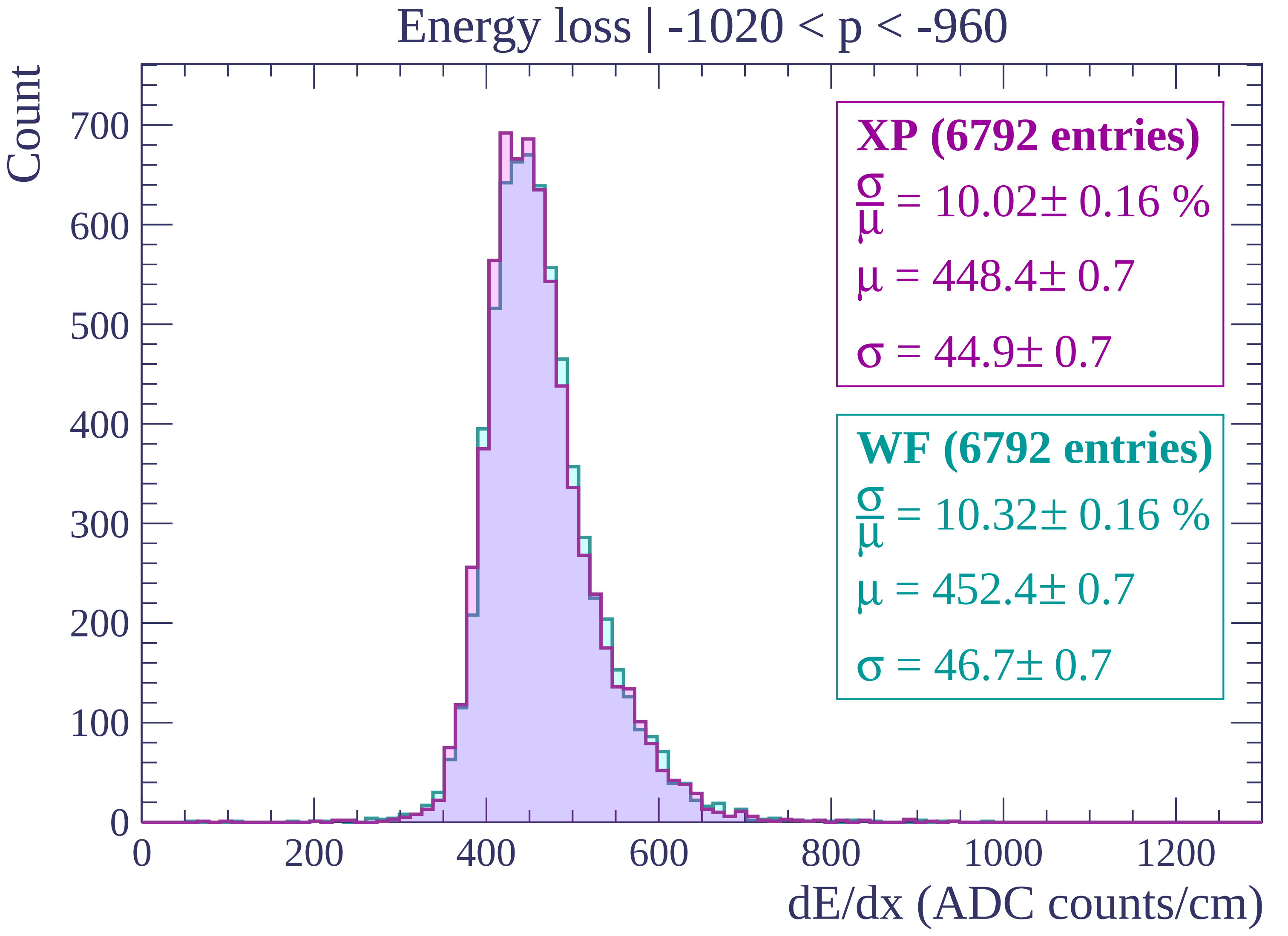
Energy loss | -1260 < p < -1200
Count
dE/dx (ADC counts/cm)
XP (6137 entries)
$\frac{\sigma}{\mu}$ = 10.76± 0.19 %
μ = 465.7± 0.8
σ = 50.1± 0.8
WF (6137 entries)
$\frac{\sigma}{\mu}$ = 11.48± 0.21 %
μ = 469.2± 0.9
σ = 53.8± 0.9
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200


Energy loss | -1200 < p < -1140
Count
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6112 entries)
σ/μ = 10.98± 0.20 %
μ = 461.5± 0.8
σ = 50.7± 0.8
WF (6112 entries)
σ/μ = 11.20± 0.18 %
μ = 466.1± 0.8
σ = 52.2± 0.7

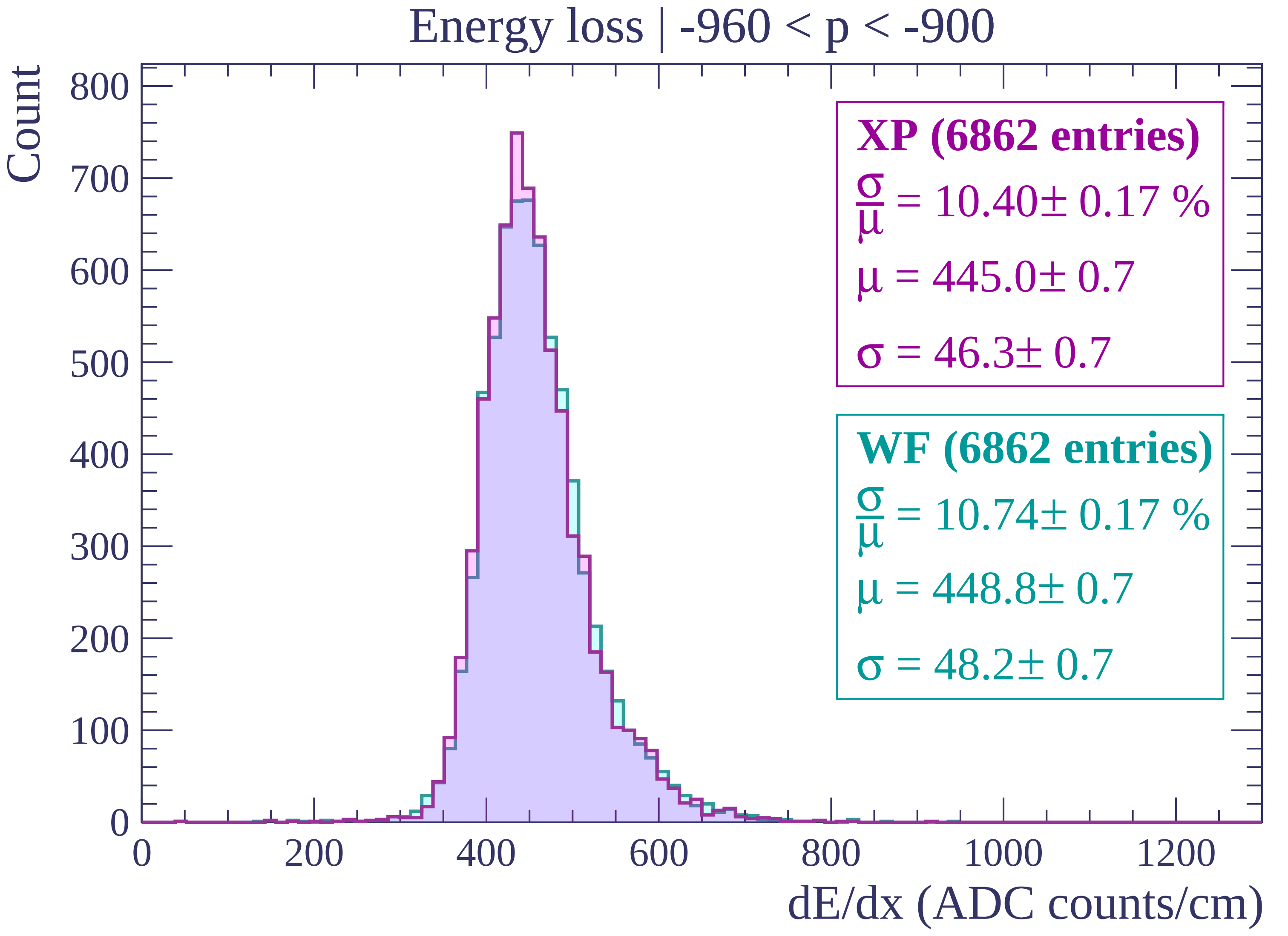


Energy loss | -1140 < p < -1080
Count
dE/dx (ADC counts/cm)
XP (6315 entries)
σ/μ = 10.54± 0.20 %
μ = 456.7± 0.8
σ = 48.1± 0.8
WF (6315 entries)
σ/μ = 10.88± 0.18 %
μ = 461.0± 0.8
σ = 50.2± 0.7
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200

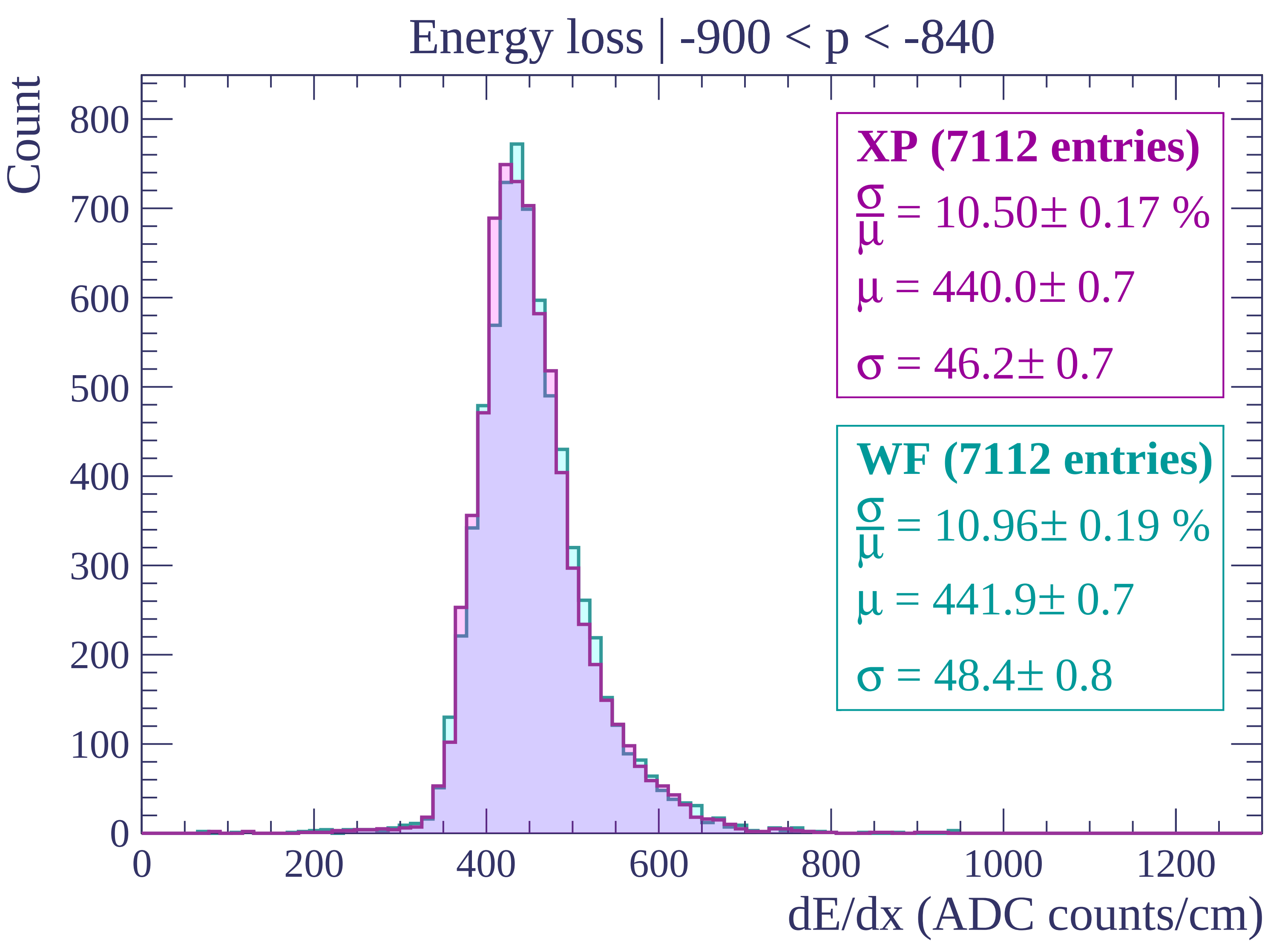

Energy loss | -1080 < p < -1020
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP (6582 entries)
$\frac{\sigma}{\mu}$ = 10.46± 0.17 %
μ = 453.9± 0.7
σ = 47.5± 0.7
WF (6582 entries)
$\frac{\sigma}{\mu}$ = 10.69± 0.18 %
μ = 455.9± 0.8
σ = 48.7± 0.7



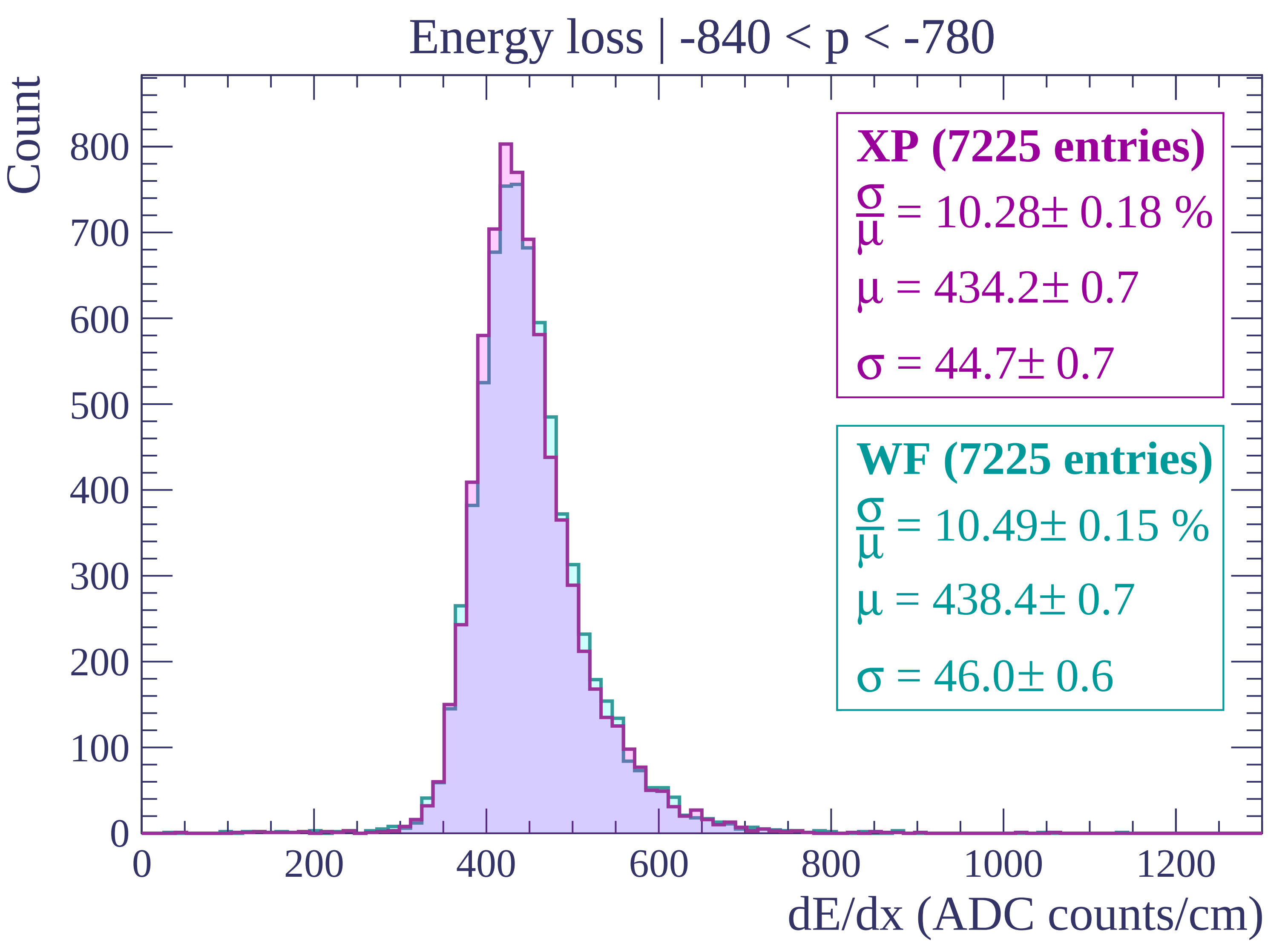
Energy loss | -1020 < p < -960
Count
dE/dx (ADC counts/cm)
XP (6792 entries)
$\frac{\sigma}{\mu}$ = 10.02± 0.16 %
μ = 448.4± 0.7
σ = 44.9± 0.7
WF (6792 entries)
$\frac{\sigma}{\mu}$ = 10.32± 0.16 %
μ = 452.4± 0.7
σ = 46.7± 0.7
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200

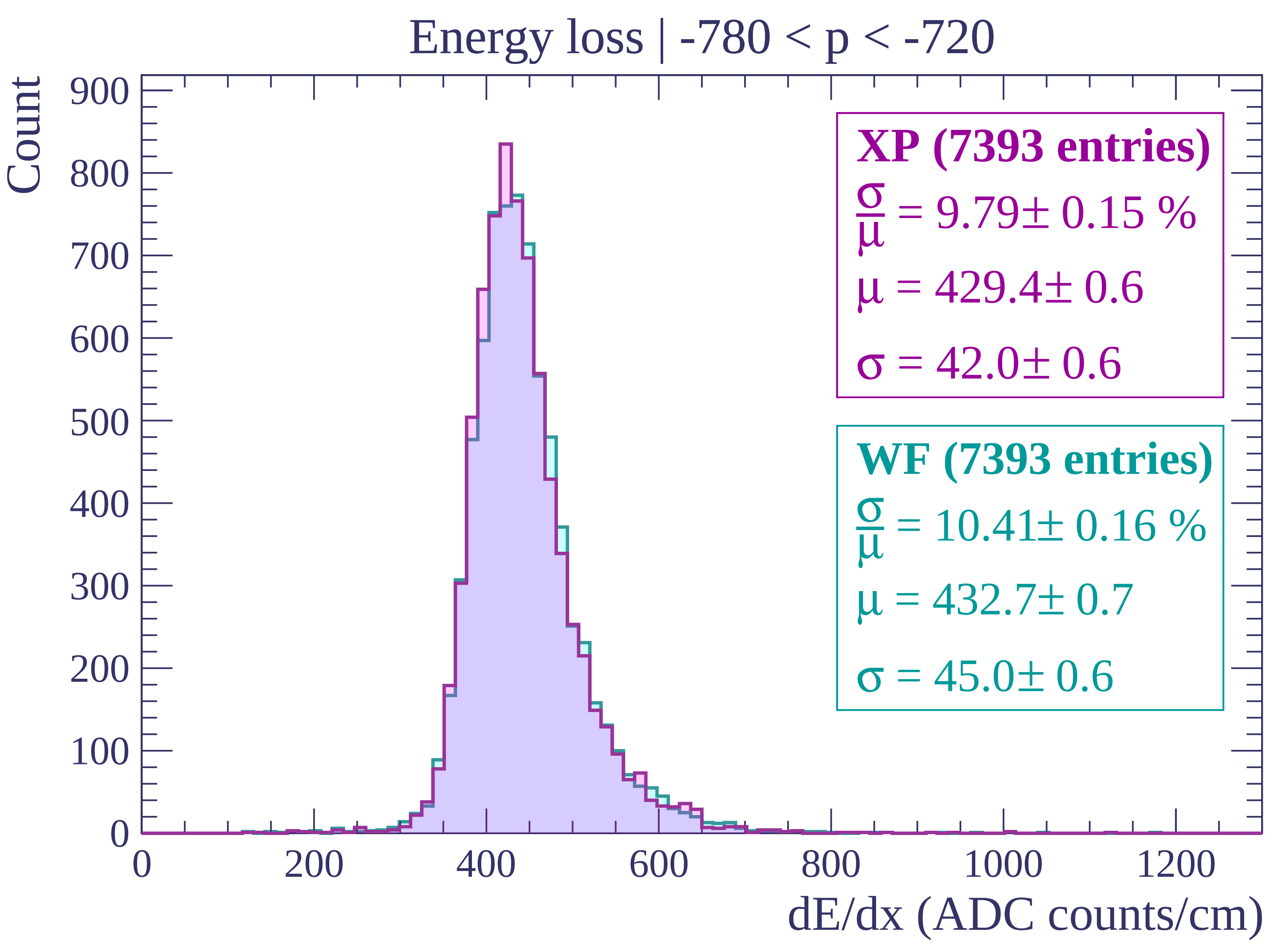


Energy loss | -960 < p < -900
Count
dE/dx (ADC counts/cm)
XP (6862 entries)
$\frac{\sigma}{\mu}$ = 10.40± 0.17 %
μ = 445.0± 0.7
σ = 46.3± 0.7
WF (6862 entries)
$\frac{\sigma}{\mu}$ = 10.74± 0.17 %
μ = 448.8± 0.7
σ = 48.2± 0.7

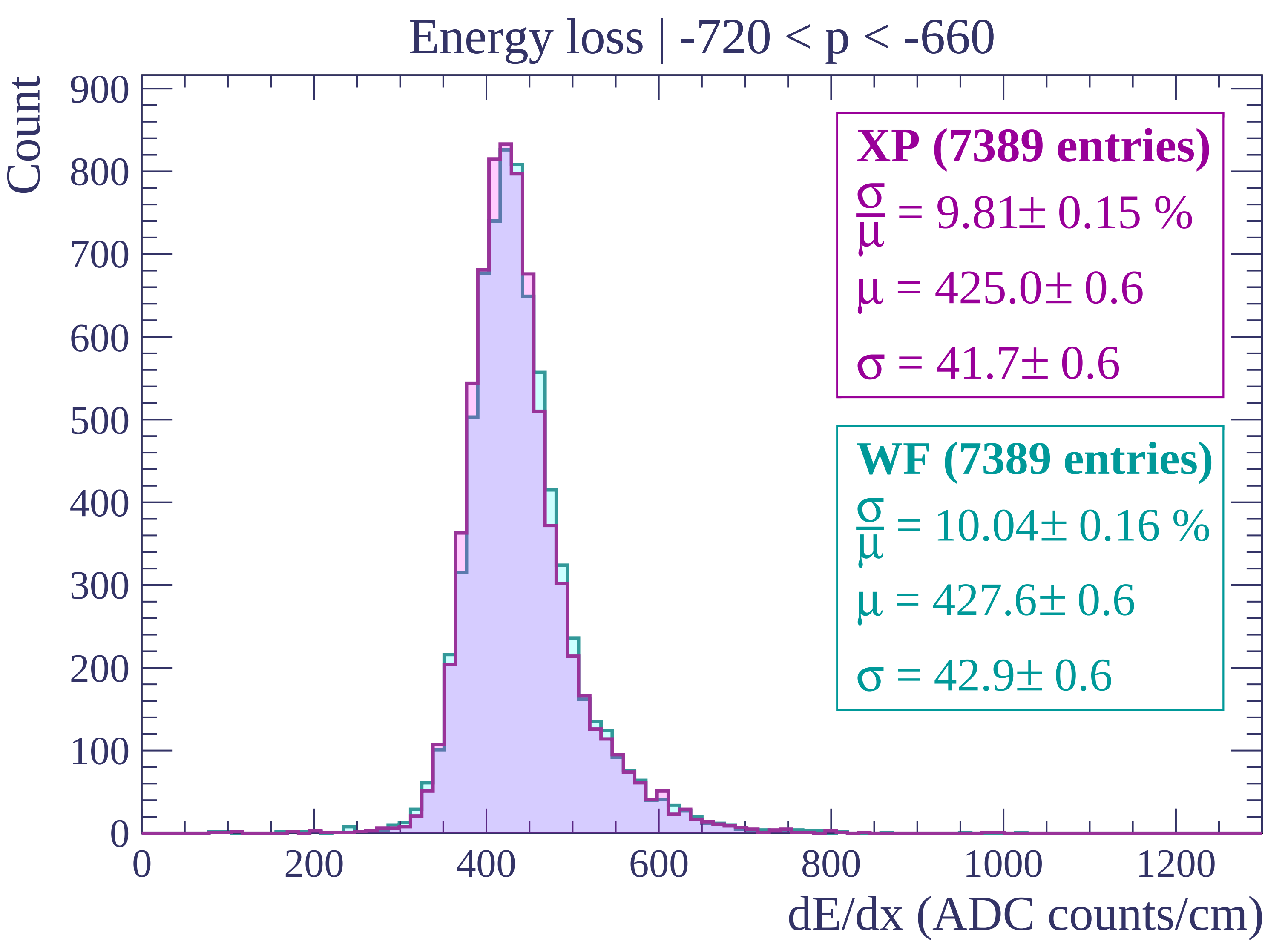


Energy loss | -900 < p < -840
Count
dE/dx (ADC counts/cm)
XP (7112 entries)
σ/μ = 10.50± 0.17 %
μ = 440.0± 0.7
σ = 46.2± 0.7
WF (7112 entries)
σ/μ = 10.96± 0.19 %
μ = 441.9± 0.7
σ = 48.4± 0.8

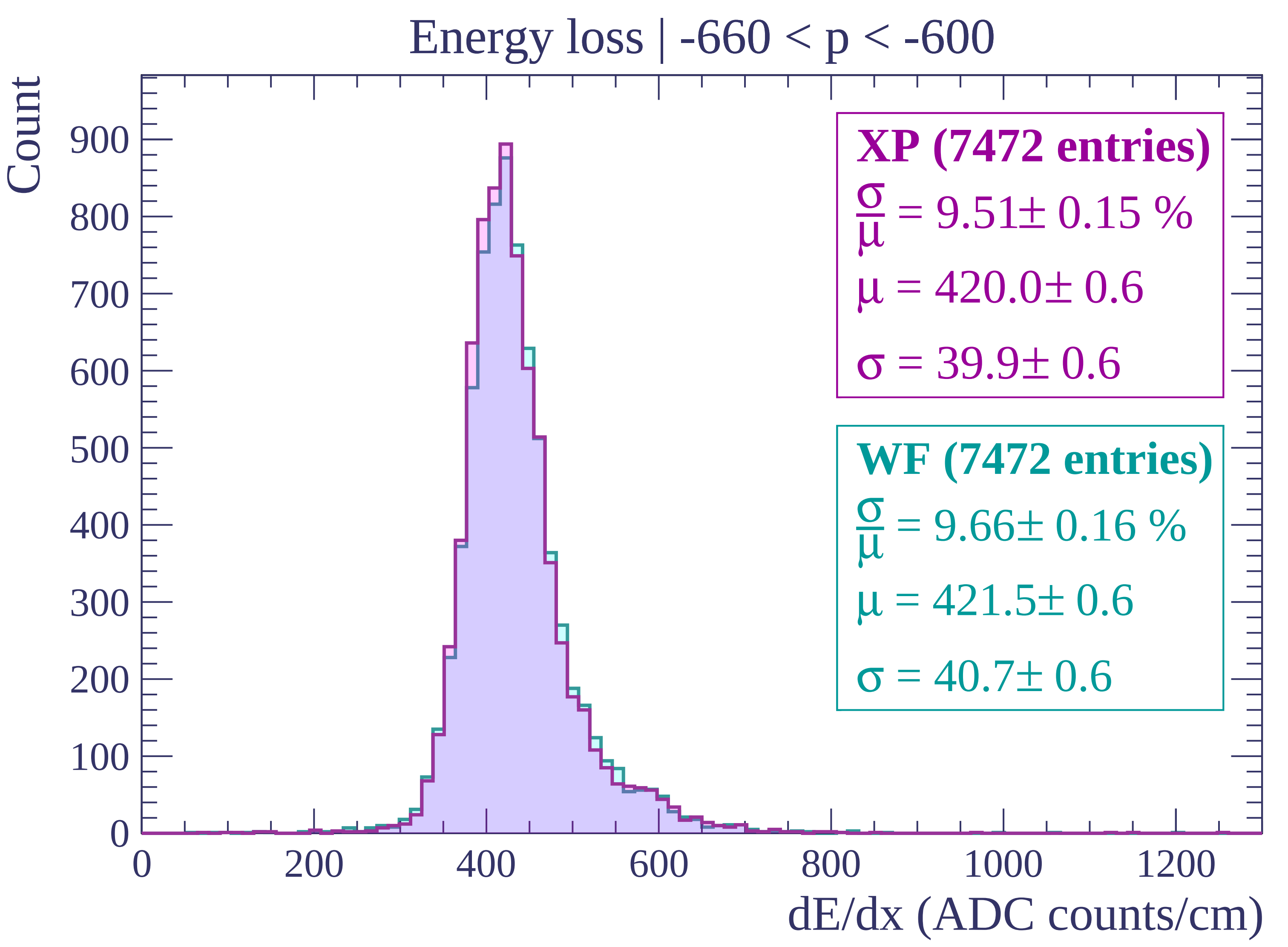

Energy loss | -840 < p < -780
Count
dE/dx (ADC counts/cm)
XP (7225 entries)
σ/μ = 10.28± 0.18 %
μ = 434.2± 0.7
σ = 44.7± 0.7
WF (7225 entries)
σ/μ = 10.49± 0.15 %
μ = 438.4± 0.7
σ = 46.0± 0.6
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200



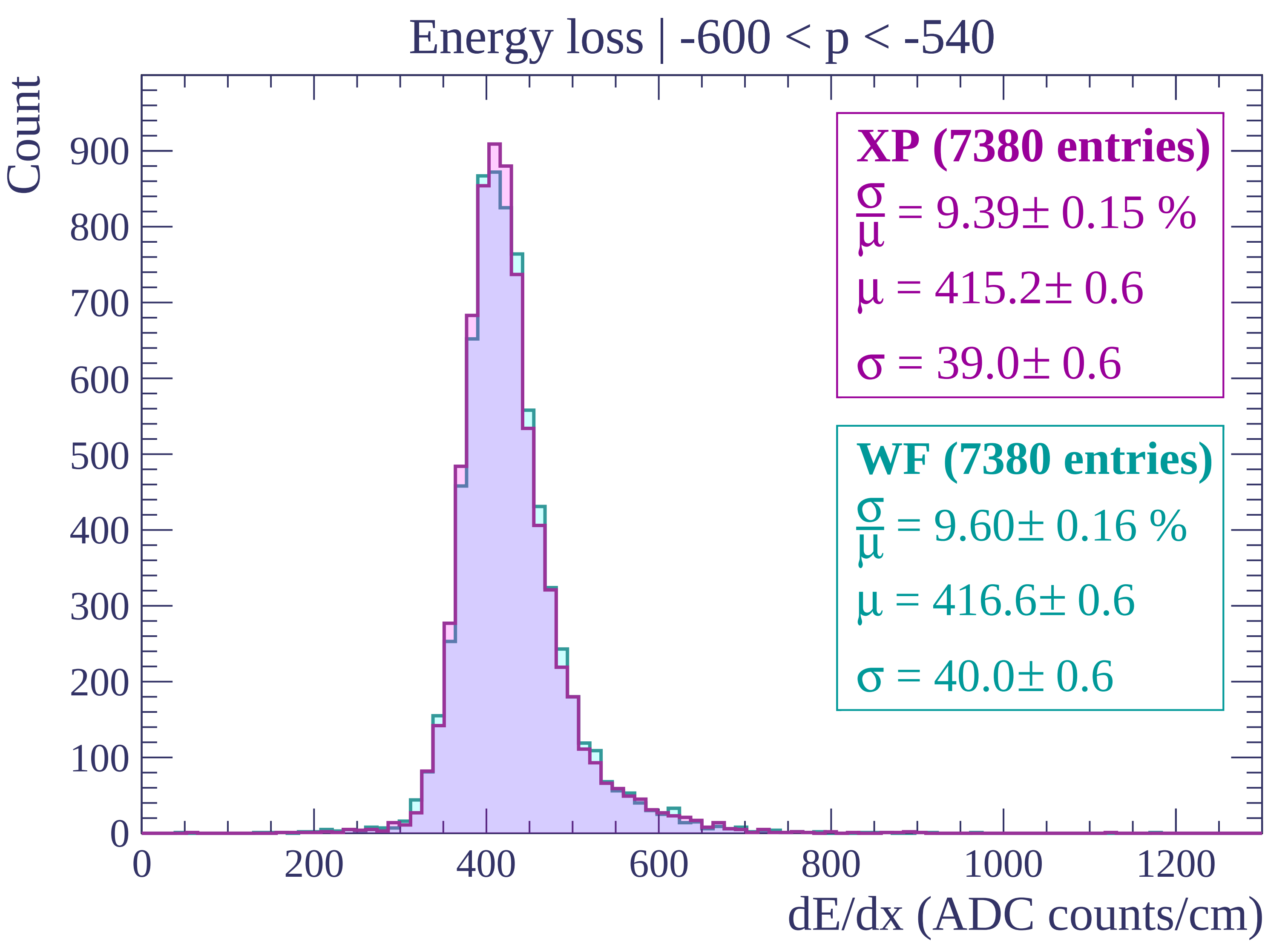
Energy loss | -780 < p < -720
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP (7393 entries)
$\frac{\sigma}{\mu}$ = 9.79± 0.15 %
μ = 429.4± 0.6
σ = 42.0± 0.6
WF (7393 entries)
$\frac{\sigma}{\mu}$ = 10.41± 0.16 %
μ = 432.7± 0.7
σ = 45.0± 0.6



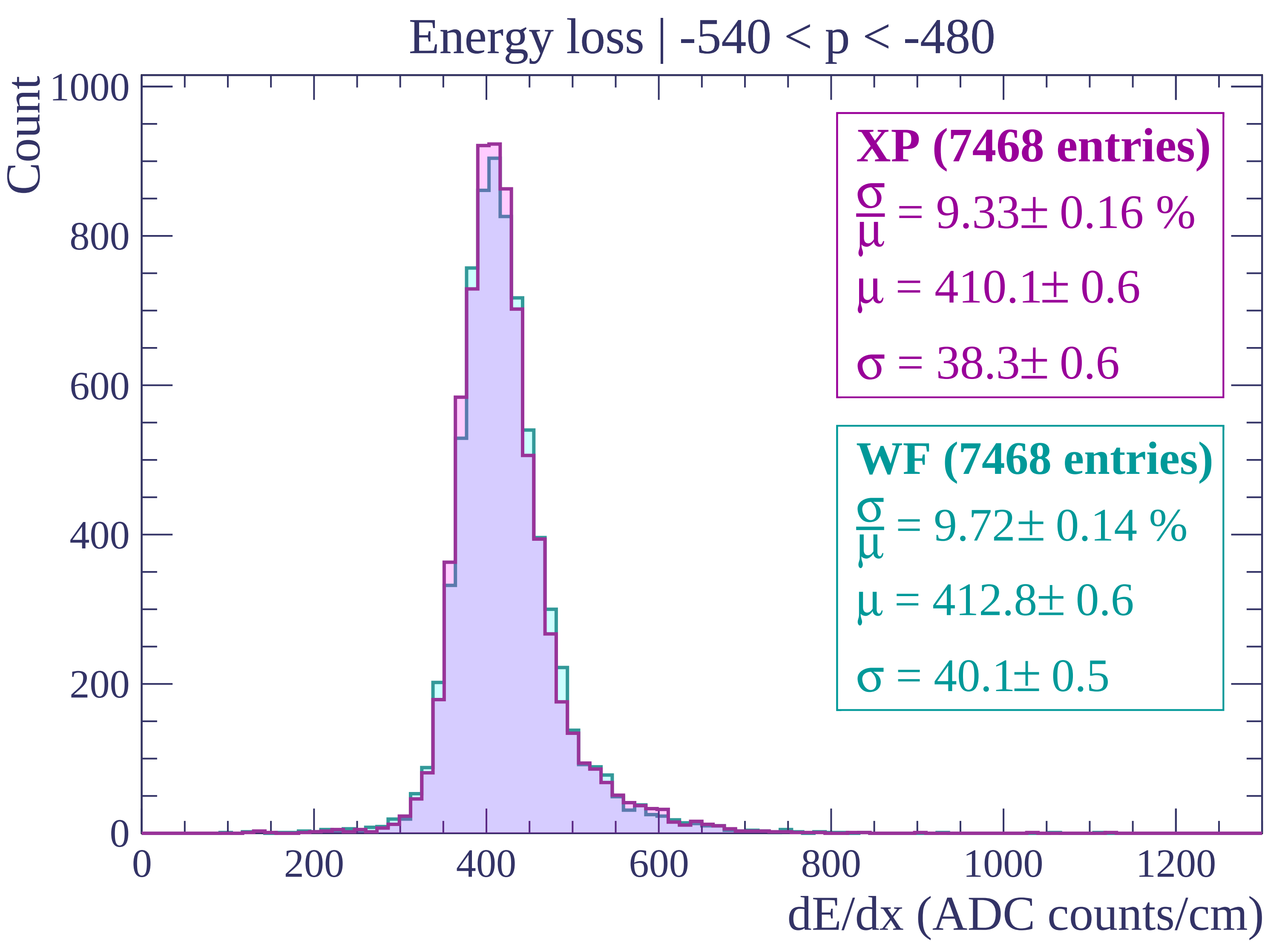
Energy loss | -720 < p < -660
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP (7389 entries)
σ/μ = 9.81± 0.15 %
μ = 425.0± 0.6
σ = 41.7± 0.6
WF (7389 entries)
σ/μ = 10.04± 0.16 %
μ = 427.6± 0.6
σ = 42.9± 0.6

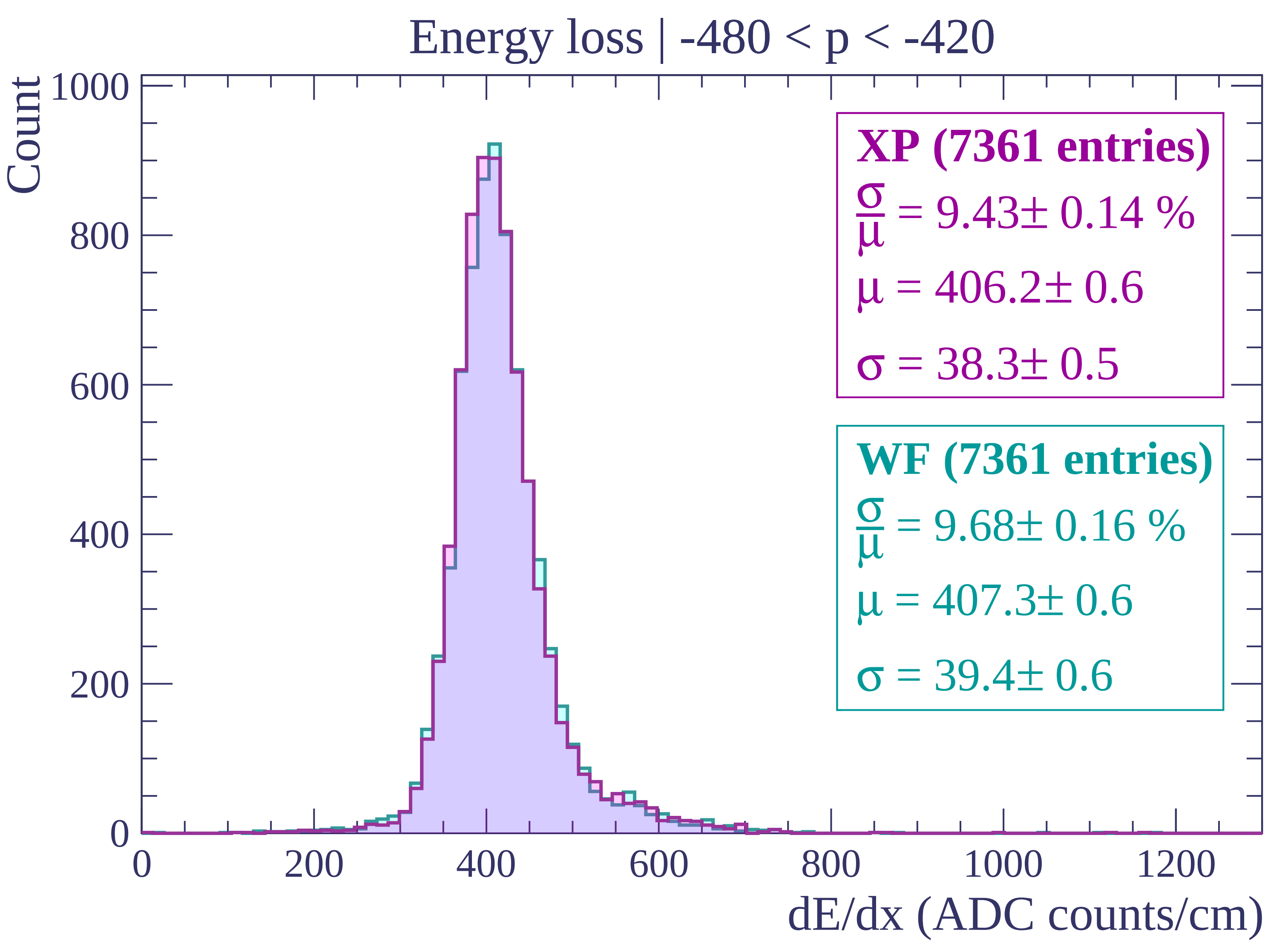


Energy loss | -660 < p < -600
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP (7472 entries)
σ/μ = 9.51± 0.15 %
μ = 420.0± 0.6
σ = 39.9± 0.6
WF (7472 entries)
σ/μ = 9.66± 0.16 %
μ = 421.5± 0.6
σ = 40.7± 0.6

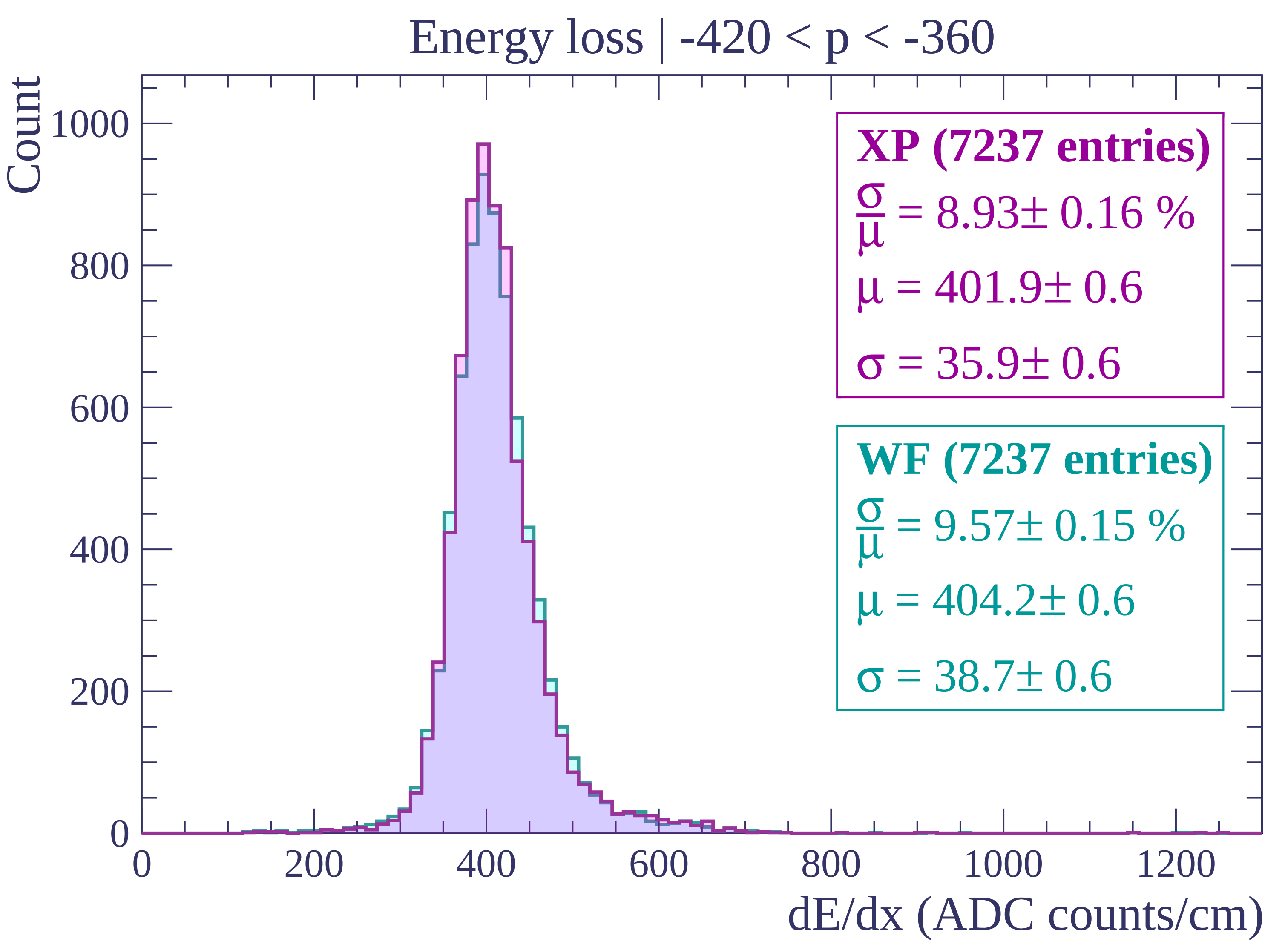


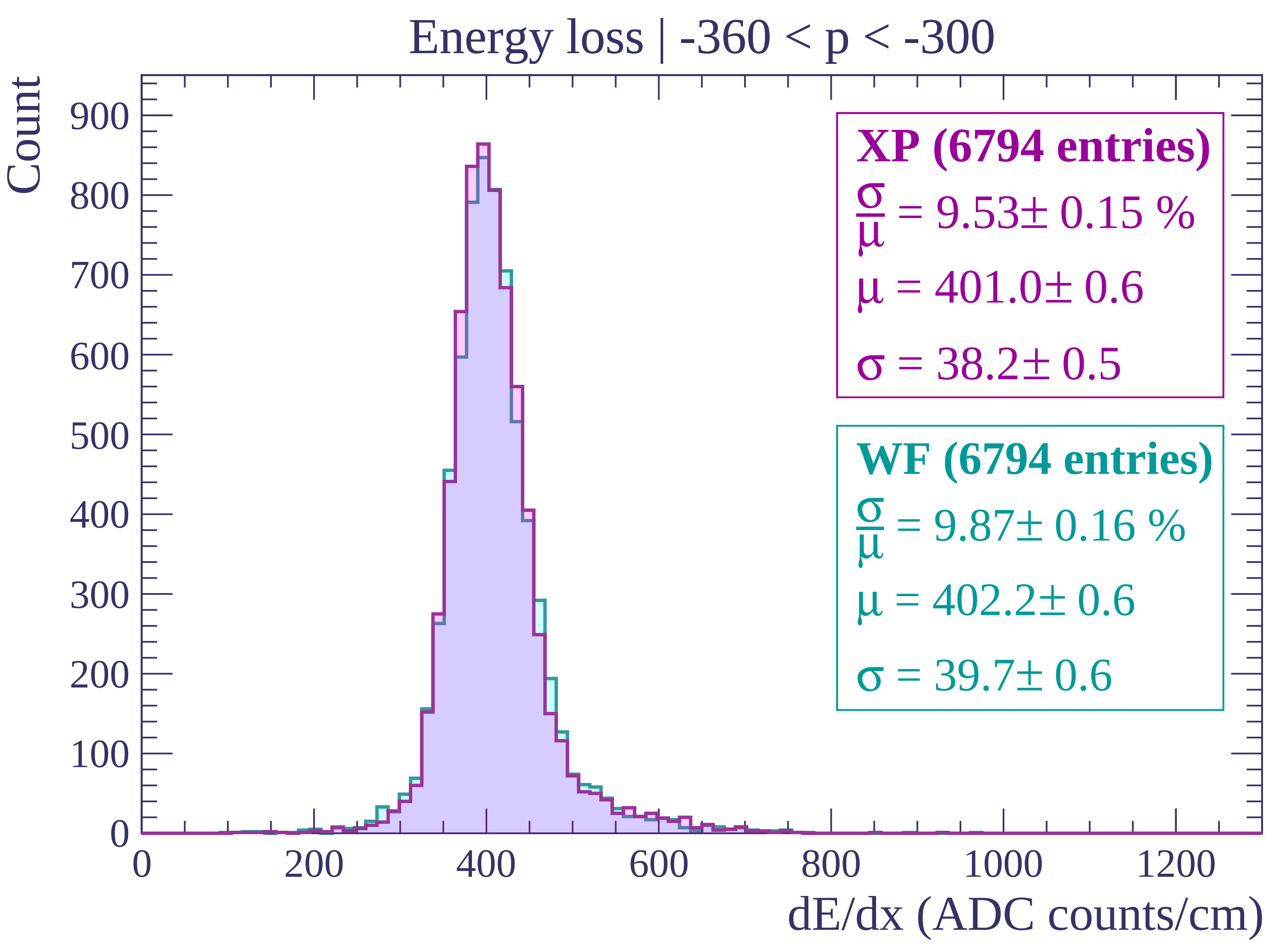
Energy loss | -600 < p < -540
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP (7380 entries)
σ/μ = 9.39± 0.15 %
μ = 415.2± 0.6
σ = 39.0± 0.6
WF (7380 entries)
σ/μ = 9.60± 0.16 %
μ = 416.6± 0.6
σ = 40.0± 0.6

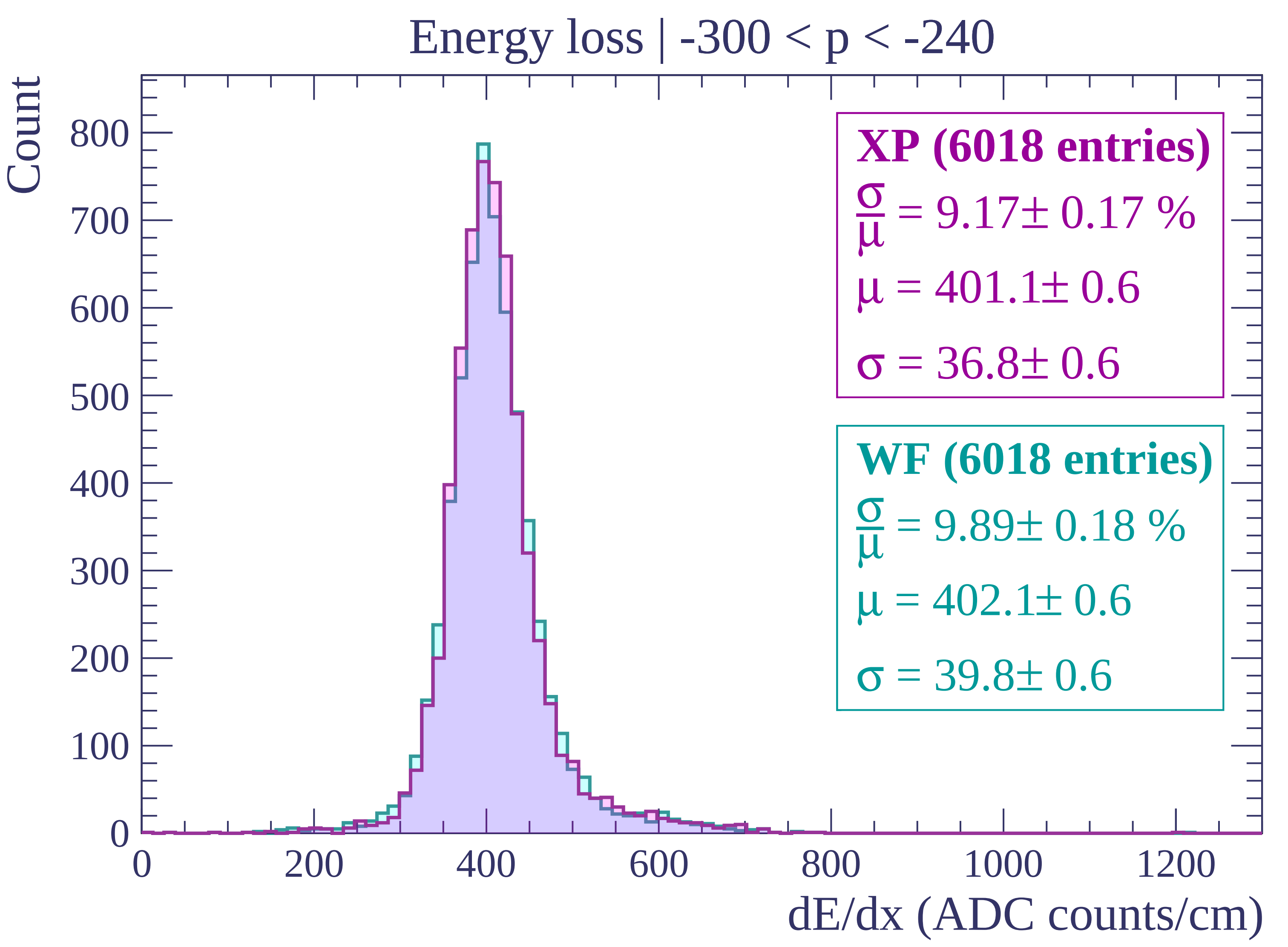

Energy loss | -540 < p < -480
Count
1000
800
600
400
200
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (7468 entries)
σ/μ = 9.33± 0.16 %
μ = 410.1± 0.6
σ = 38.3± 0.6
WF (7468 entries)
σ/μ = 9.72± 0.14 %
μ = 412.8± 0.6
σ = 40.1± 0.5



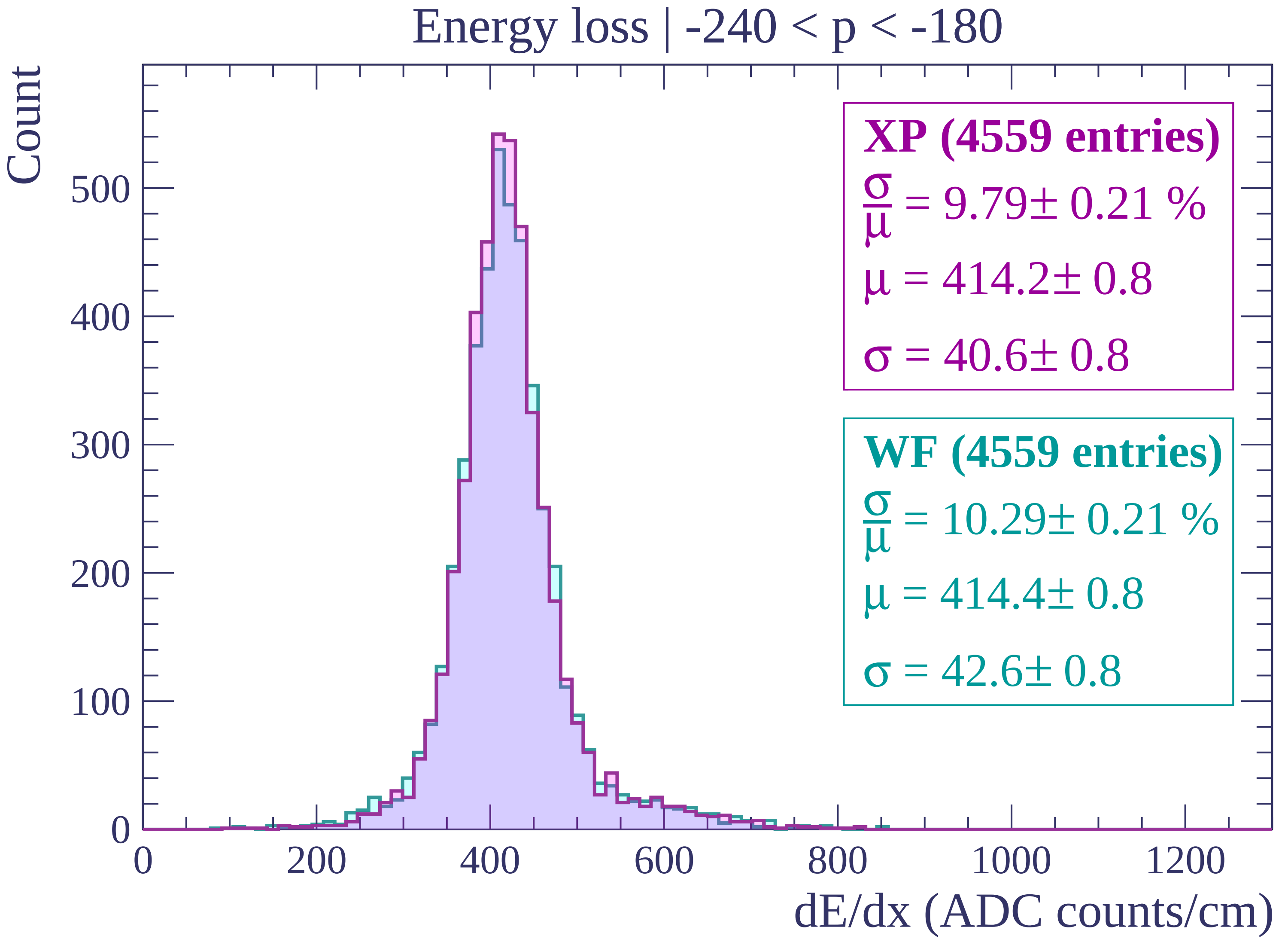
Energy loss | -480 < p < -420
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (7361 entries)
$\frac{\sigma}{\mu}$ = 9.43± 0.14 %
μ = 406.2± 0.6
σ = 38.3± 0.5
WF (7361 entries)
$\frac{\sigma}{\mu}$ = 9.68± 0.16 %
μ = 407.3± 0.6
σ = 39.4± 0.6

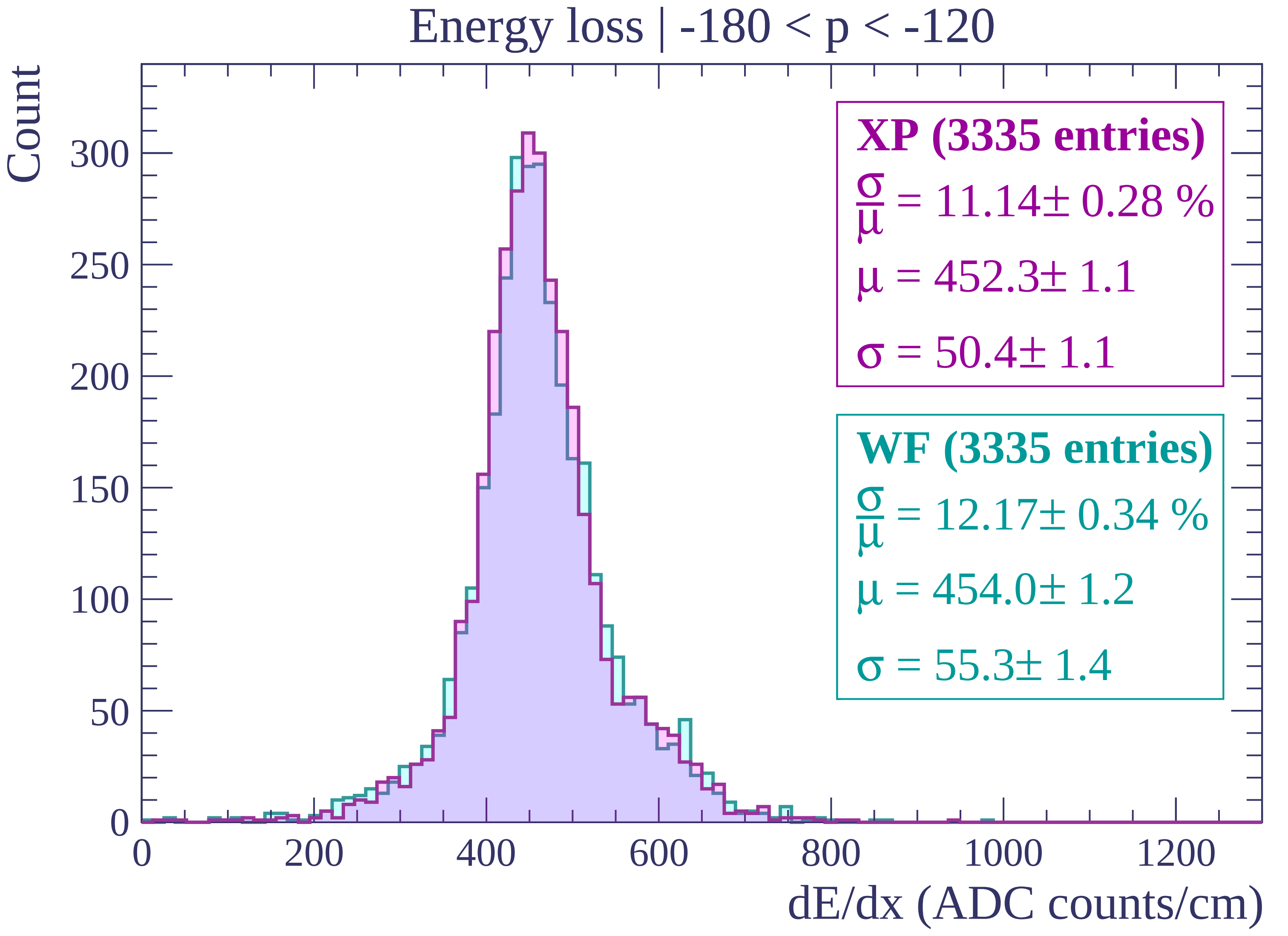

Energy loss | -420 < p < -360
Count
dE/dx (ADC counts/cm)
XP (7237 entries)
σ/μ = 8.93± 0.16 %
μ = 401.9± 0.6
σ = 35.9± 0.6
WF (7237 entries)
σ/μ = 9.57± 0.15 %
μ = 404.2± 0.6
σ = 38.7± 0.6

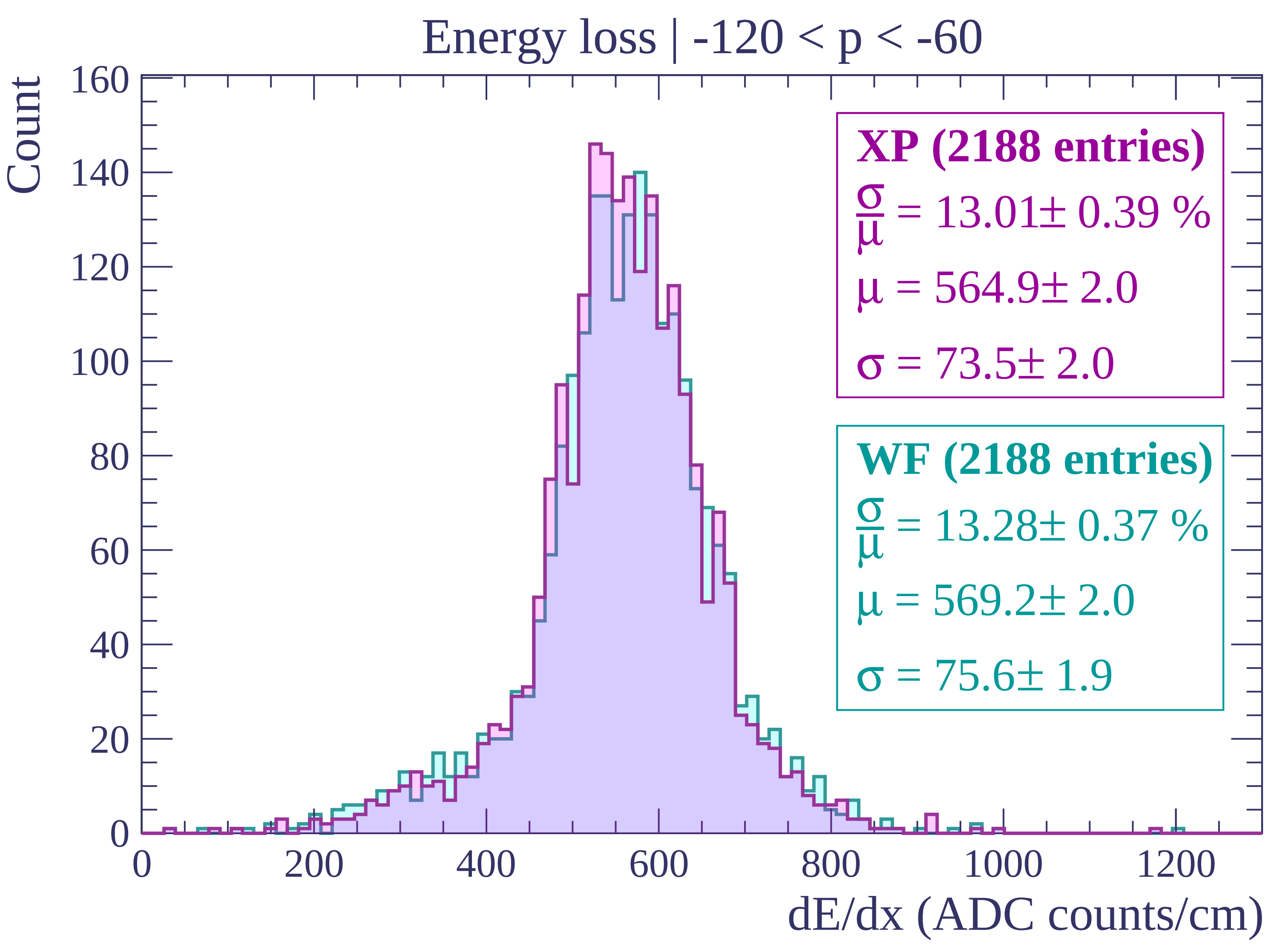

Energy loss | -360 < p < -300
Count
dE/dx (ADC counts/cm)
XP (6794 entries)
σ/μ = 9.53± 0.15 %
μ = 401.0± 0.6
σ = 38.2± 0.5
WF (6794 entries)
σ/μ = 9.87± 0.16 %
μ = 402.2± 0.6
σ = 39.7± 0.6
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200



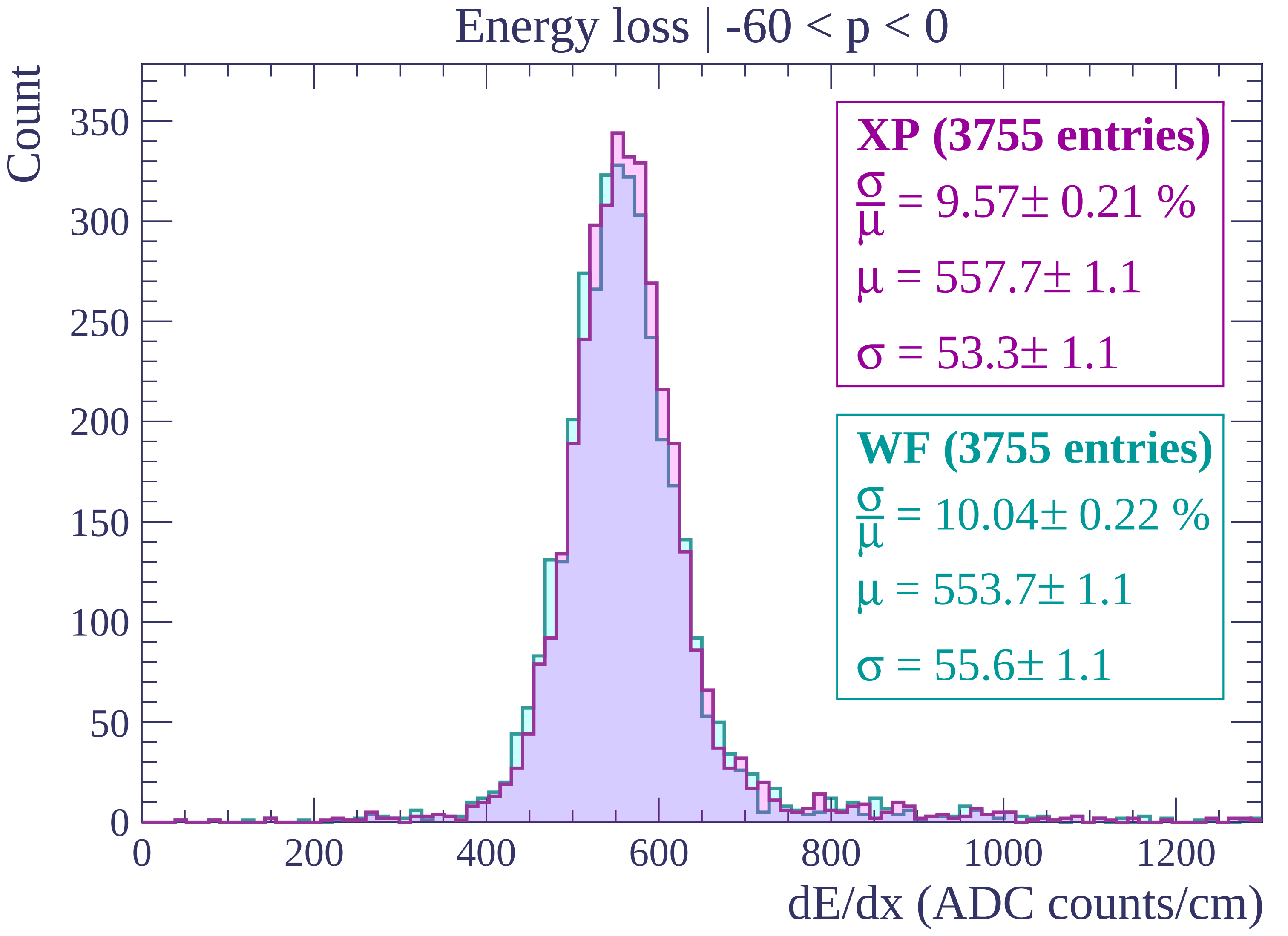
Energy loss | -300 < p < -240
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP (6018 entries)
σ/μ = 9.17± 0.17 %
μ = 401.1± 0.6
σ = 36.8± 0.6
WF (6018 entries)
σ/μ = 9.89± 0.18 %
μ = 402.1± 0.6
σ = 39.8± 0.6


Energy loss | -240 < p < -180
Count
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (4559 entries)
σ/μ = 9.79± 0.21 %
μ = 414.2± 0.8
σ = 40.6± 0.8
WF (4559 entries)
σ/μ = 10.29± 0.21 %
μ = 414.4± 0.8
σ = 42.6± 0.8

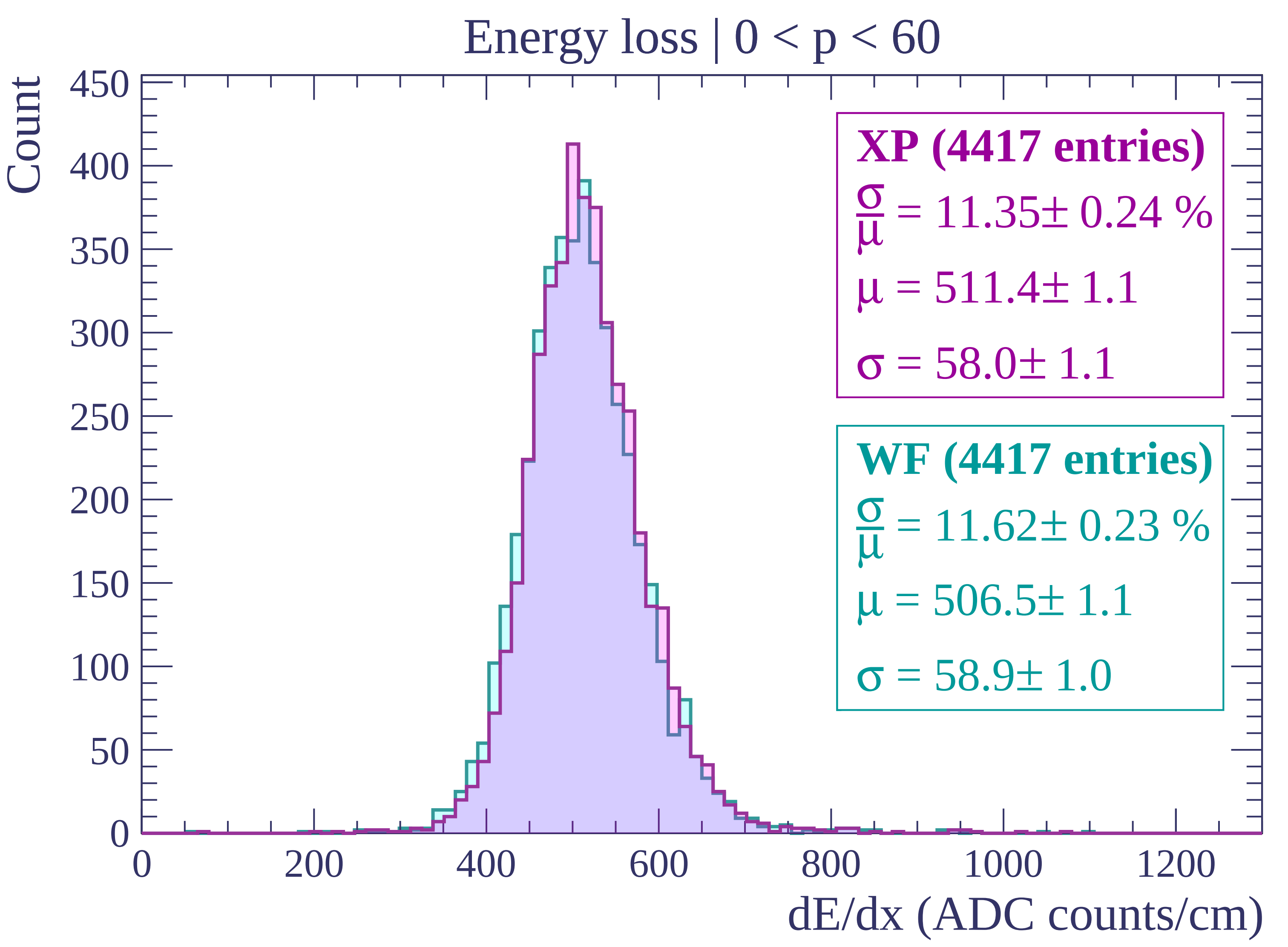

Energy loss | -180 < p < -120
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (3335 entries)
σ/μ = 11.14± 0.28 %
μ = 452.3± 1.1
σ = 50.4± 1.1
WF (3335 entries)
σ/μ = 12.17± 0.34 %
μ = 454.0± 1.2
σ = 55.3± 1.4

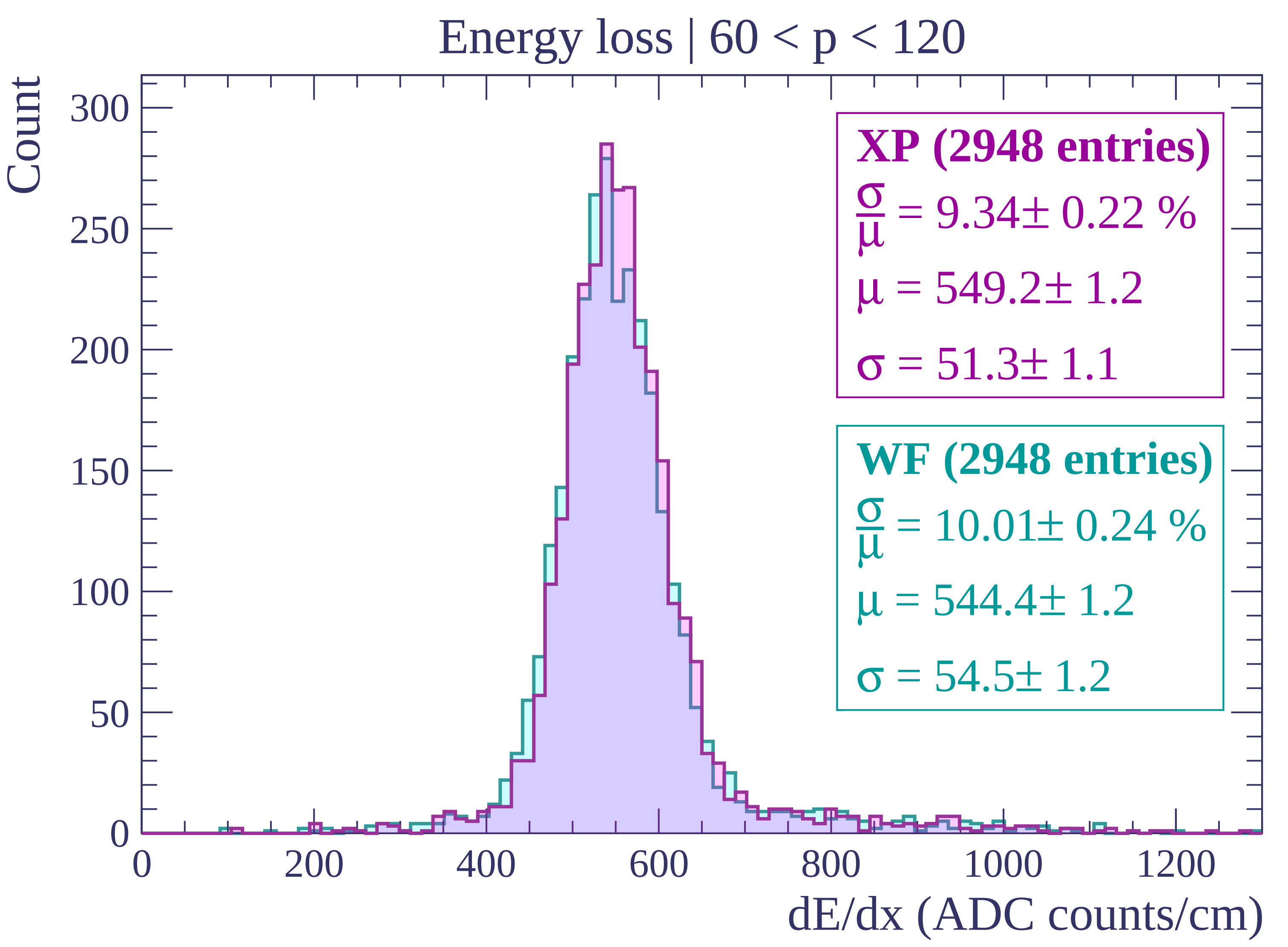


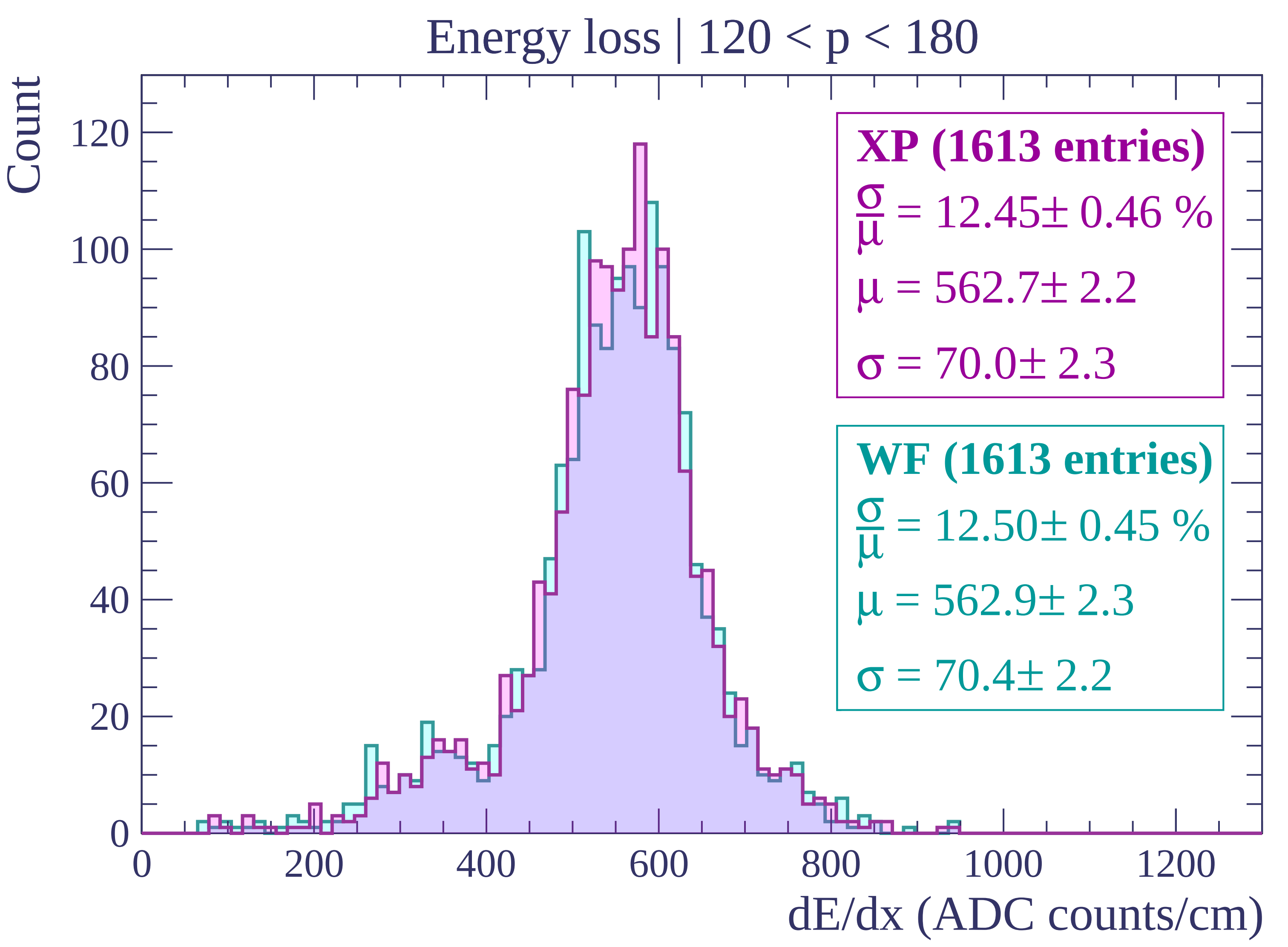
Energy loss | -120 < p < -60
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP (2188 entries)
$\frac{\sigma}{\mu}$ = 13.01± 0.39 %
μ = 564.9± 2.0
σ = 73.5± 2.0
WF (2188 entries)
$\frac{\sigma}{\mu}$ = 13.28± 0.37 %
μ = 569.2± 2.0
σ = 75.6± 1.9

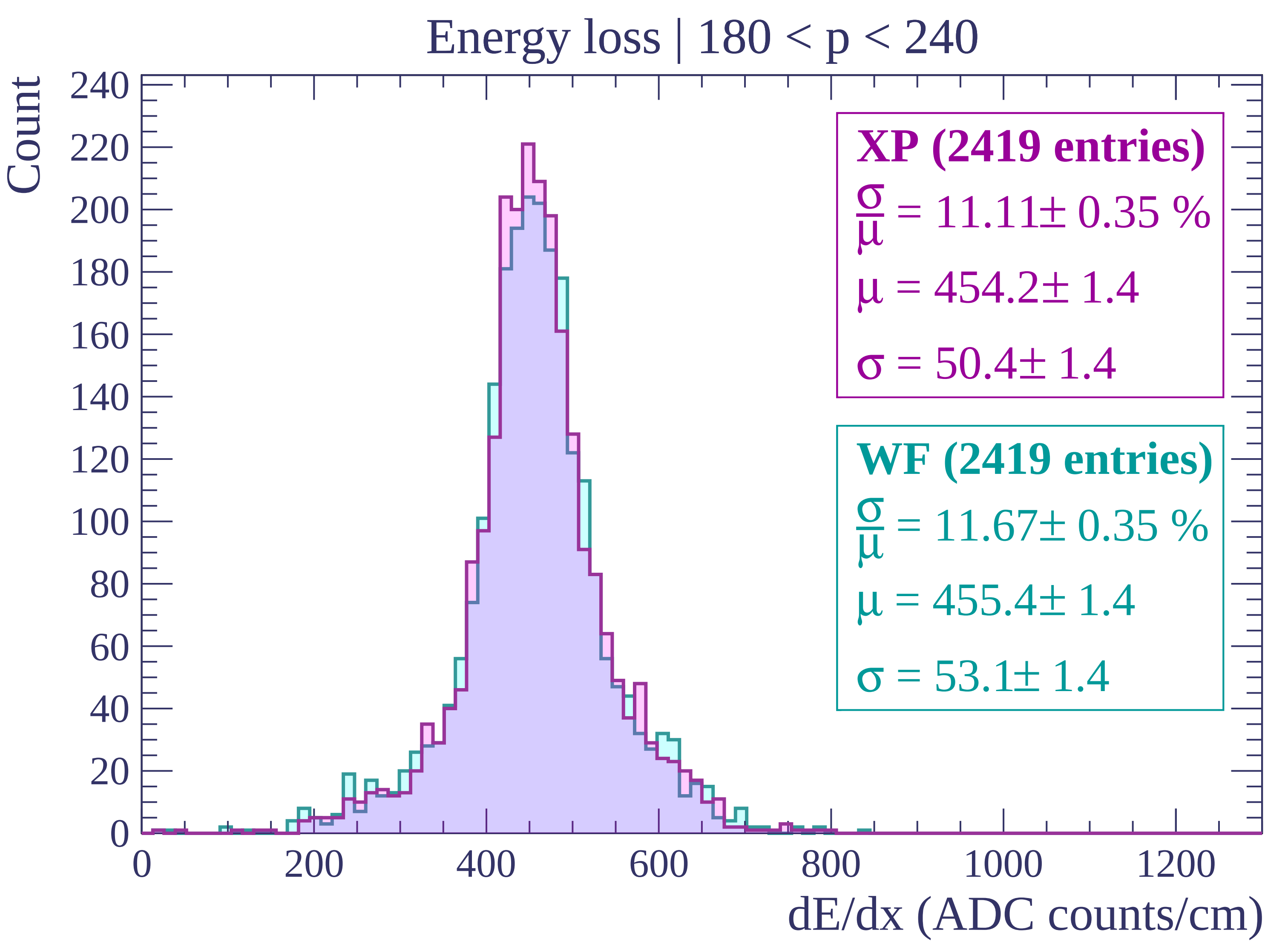


Energy loss | -60 < p < 0
Count
XP (3755 entries)
$\frac{\sigma}{\mu}$ = 9.57± 0.21 %
μ = 557.7± 1.1
σ = 53.3± 1.1
WF (3755 entries)
$\frac{\sigma}{\mu}$ = 10.04± 0.22 %
μ = 553.7± 1.1
σ = 55.6± 1.1
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200

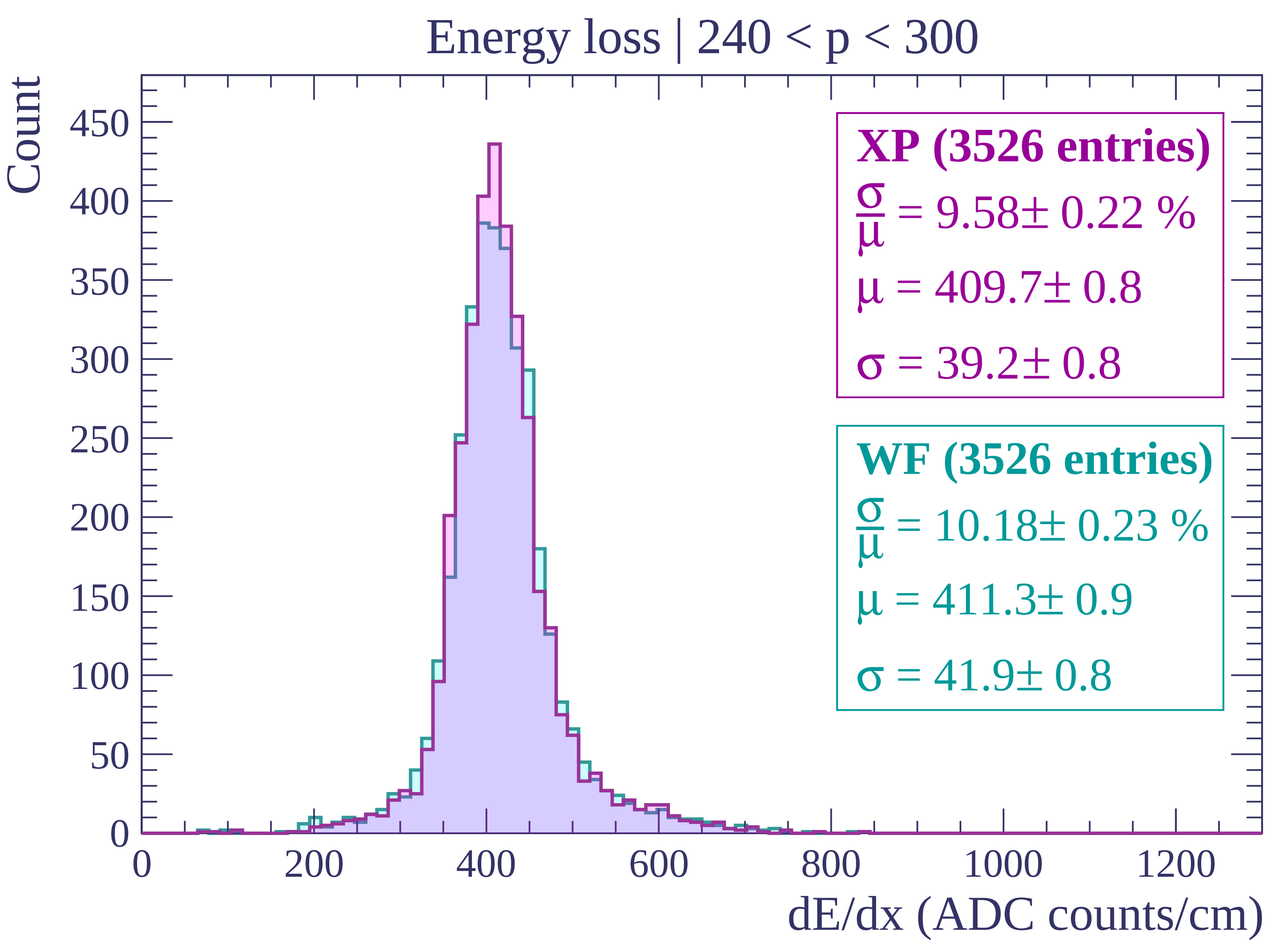

Energy loss | 0 < p < 60
Count
450
400
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (4417 entries)
σ/μ = 11.35± 0.24 %
μ = 511.4± 1.1
σ = 58.0± 1.1
WF (4417 entries)
σ/μ = 11.62± 0.23 %
μ = 506.5± 1.1
σ = 58.9± 1.0


Energy loss | 60 < p < 120
Count
dE/dx (ADC counts/cm)
XP (2948 entries)
$\frac{\sigma}{\mu}$ = 9.34± 0.22 %
μ = 549.2± 1.2
σ = 51.3± 1.1
WF (2948 entries)
$\frac{\sigma}{\mu}$ = 10.01± 0.24 %
μ = 544.4± 1.2
σ = 54.5± 1.2
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200

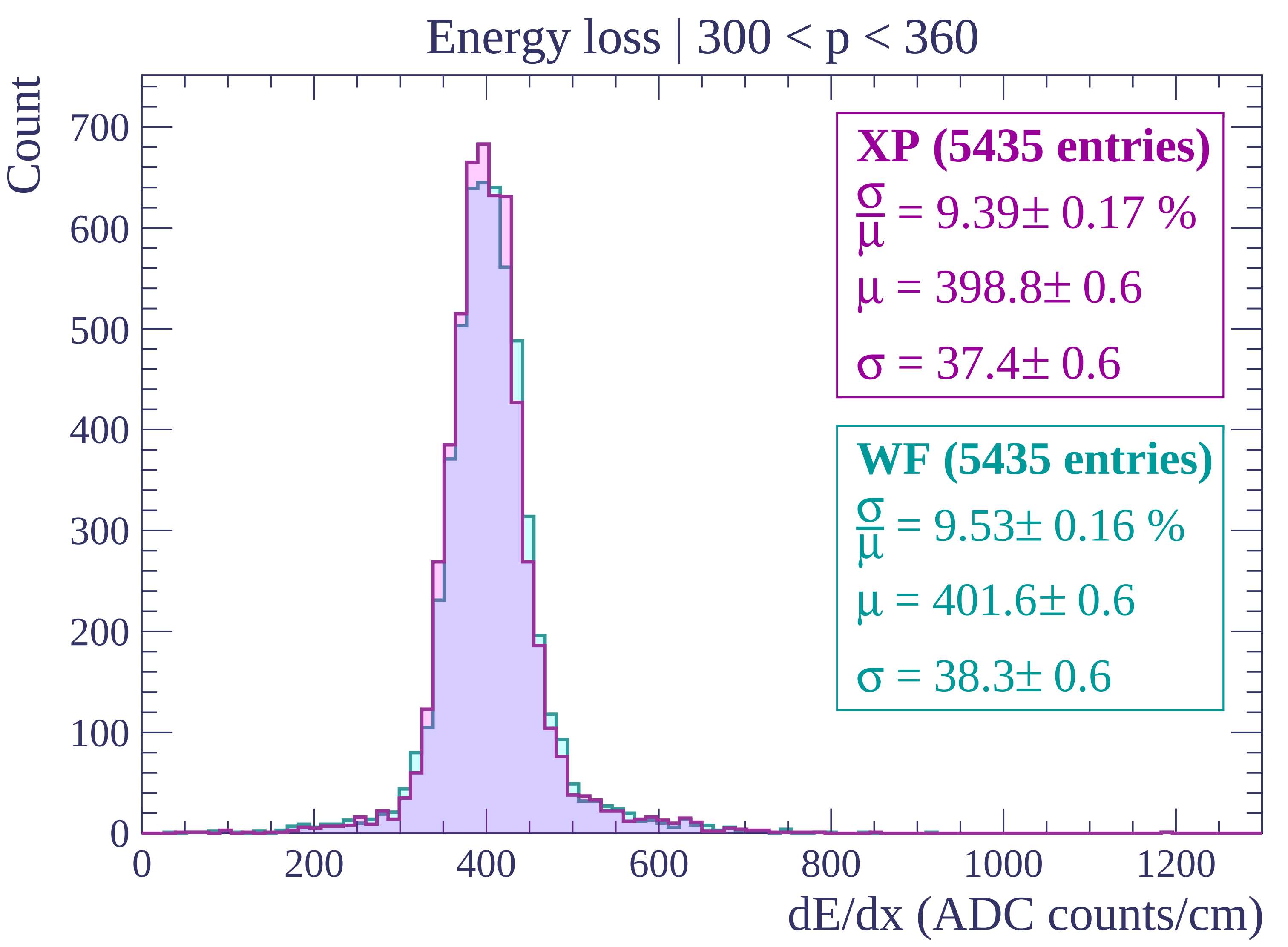

Energy loss | 120 < p < 180
Count
dE/dx (ADC counts/cm)
XP (1613 entries)
σ/μ = 12.45± 0.46 %
μ = 562.7± 2.2
σ = 70.0± 2.3
WF (1613 entries)
σ/μ = 12.50± 0.45 %
μ = 562.9± 2.3
σ = 70.4± 2.2
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200



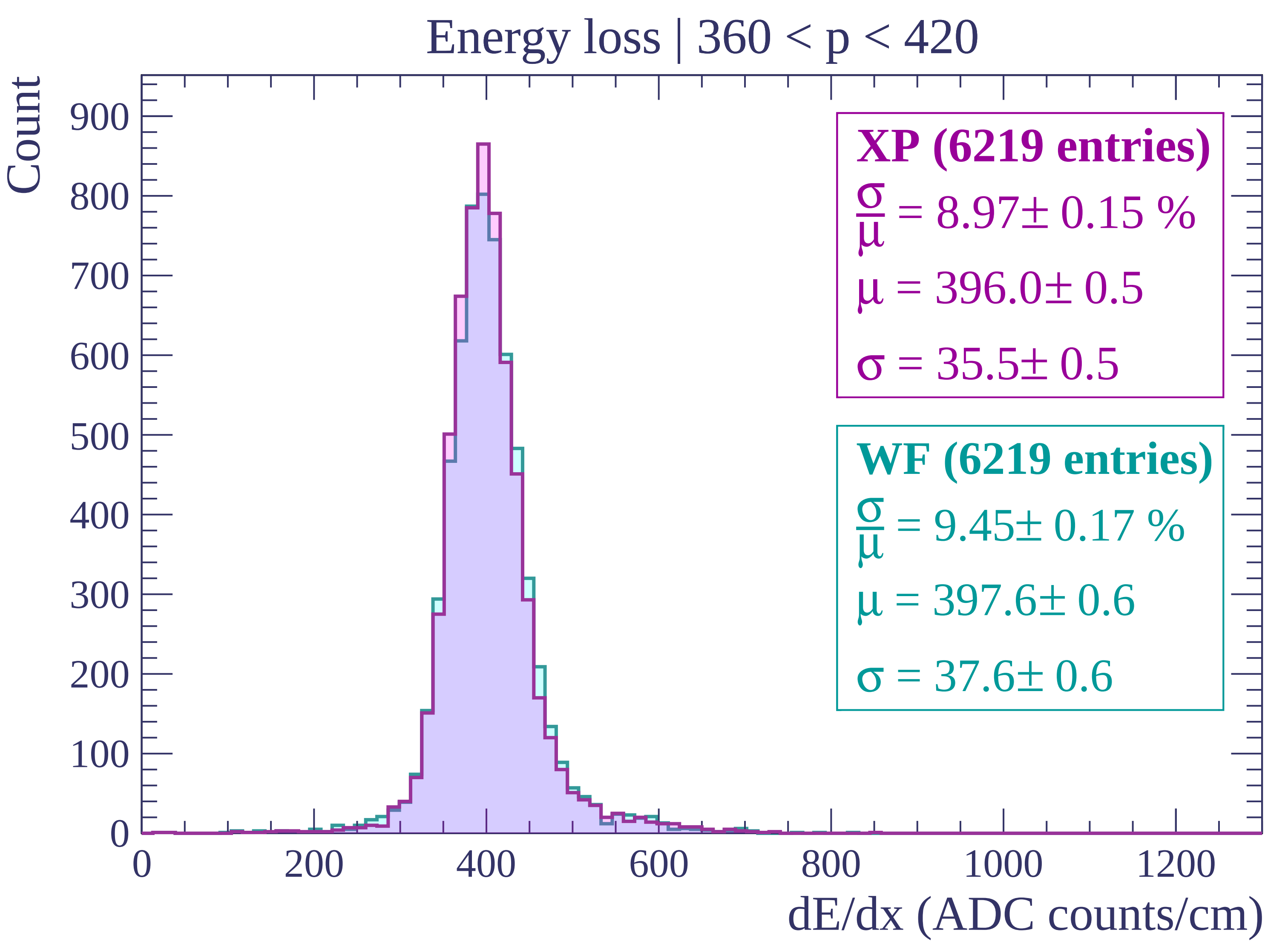
Energy loss | 180 < p < 240
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
240
0
200
400
600
800
1000
1200
XP (2419 entries)
$\frac{\sigma}{\mu}$ = 11.11± 0.35 %
μ = 454.2± 1.4
σ = 50.4± 1.4
WF (2419 entries)
$\frac{\sigma}{\mu}$ = 11.67± 0.35 %
μ = 455.4± 1.4
σ = 53.1± 1.4

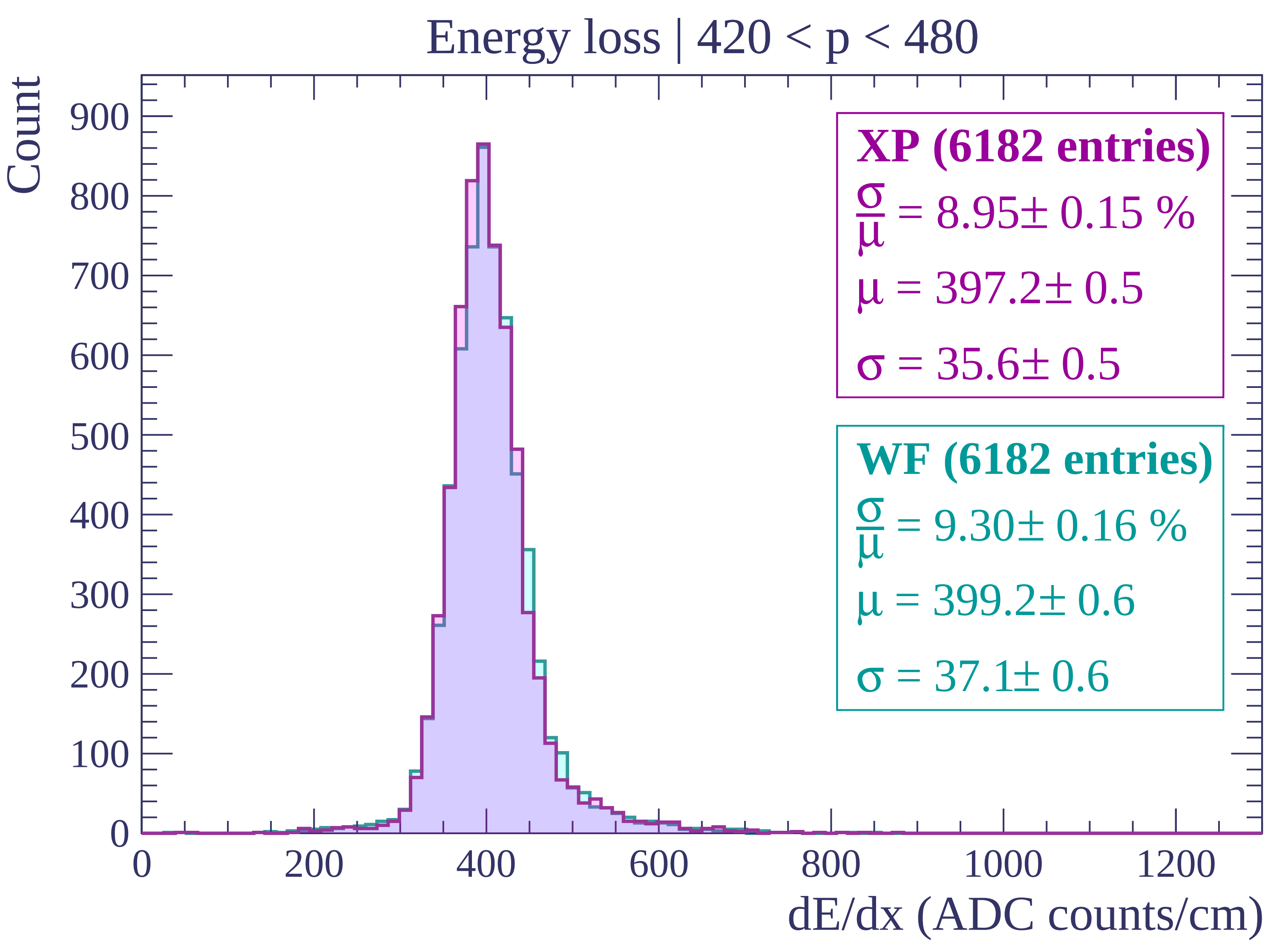


Energy loss | 240 < p < 300
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP (3526 entries)
$\frac{\sigma}{\mu}$ = 9.58± 0.22 %
μ = 409.7± 0.8
σ = 39.2± 0.8
WF (3526 entries)
$\frac{\sigma}{\mu}$ = 10.18± 0.23 %
μ = 411.3± 0.9
σ = 41.9± 0.8

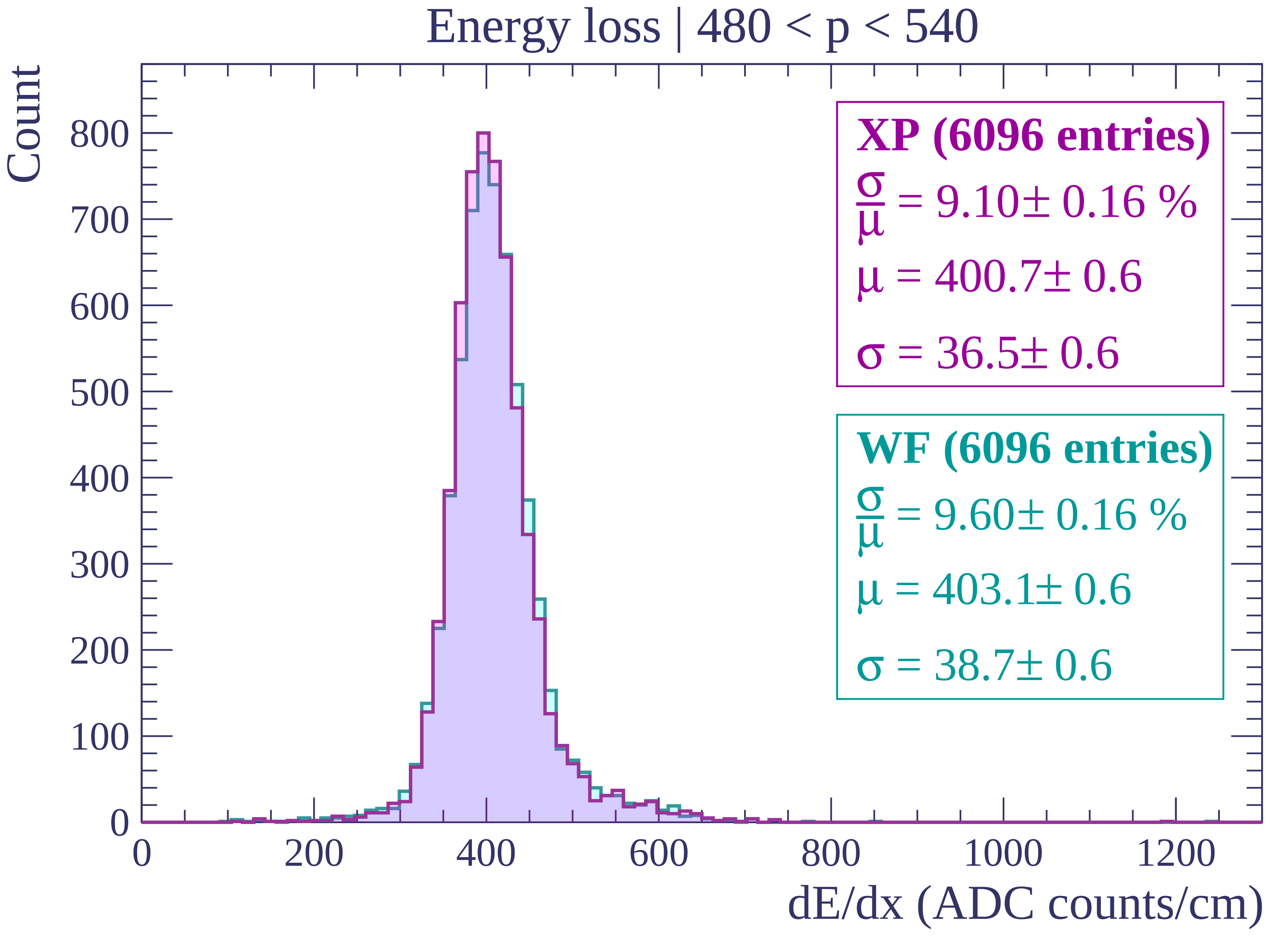


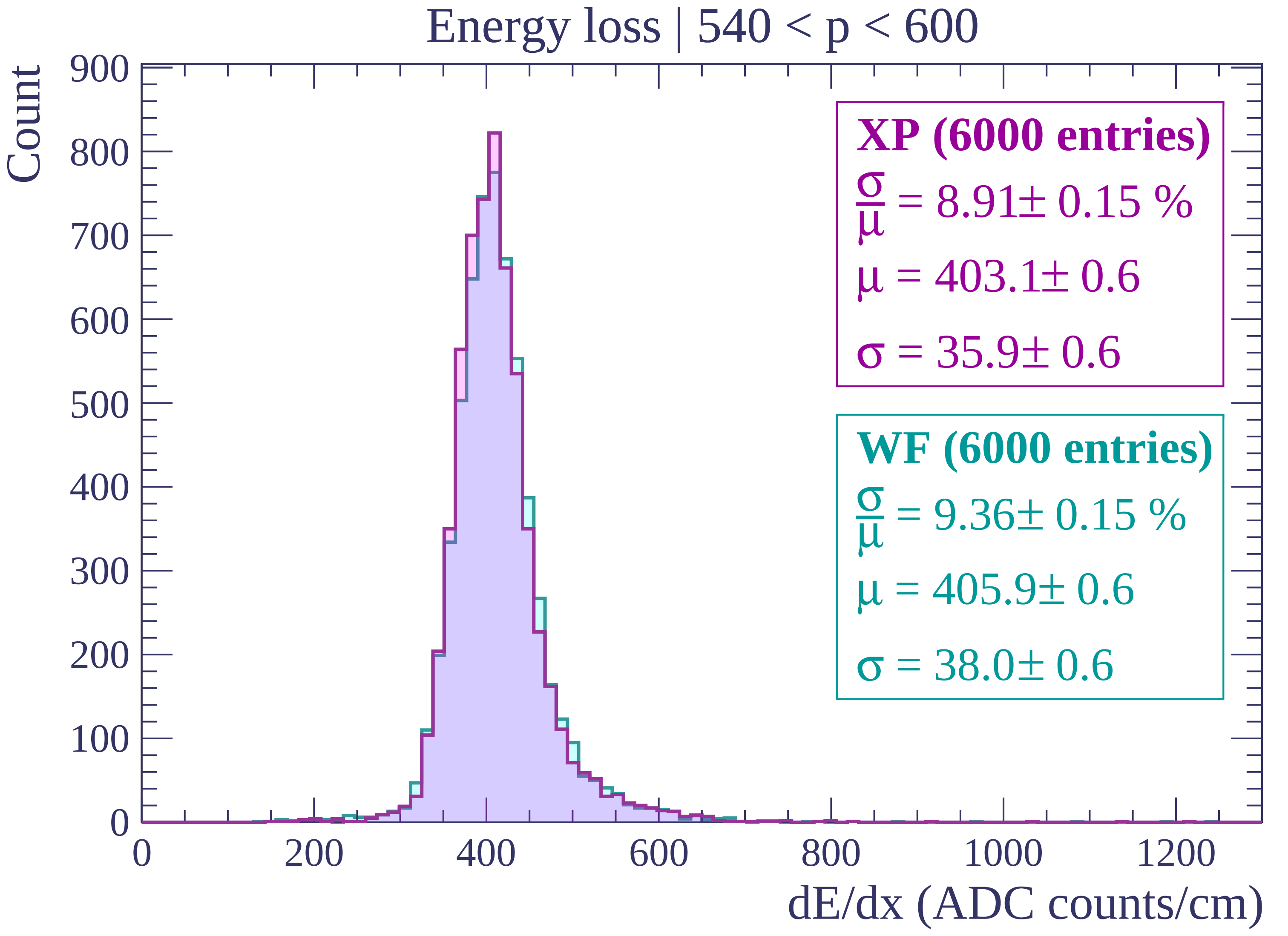
Energy loss | 300 < p < 360
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP (5435 entries)
σ/μ = 9.39± 0.17 %
μ = 398.8± 0.6
σ = 37.4± 0.6
WF (5435 entries)
σ/μ = 9.53± 0.16 %
μ = 401.6± 0.6
σ = 38.3± 0.6


Energy loss | 360 < p < 420
Count
dE/dx (ADC counts/cm)
XP (6219 entries)
σ/μ = 8.97± 0.15 %
μ = 396.0± 0.5
σ = 35.5± 0.5
WF (6219 entries)
σ/μ = 9.45± 0.17 %
μ = 397.6± 0.6
σ = 37.6± 0.6
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200

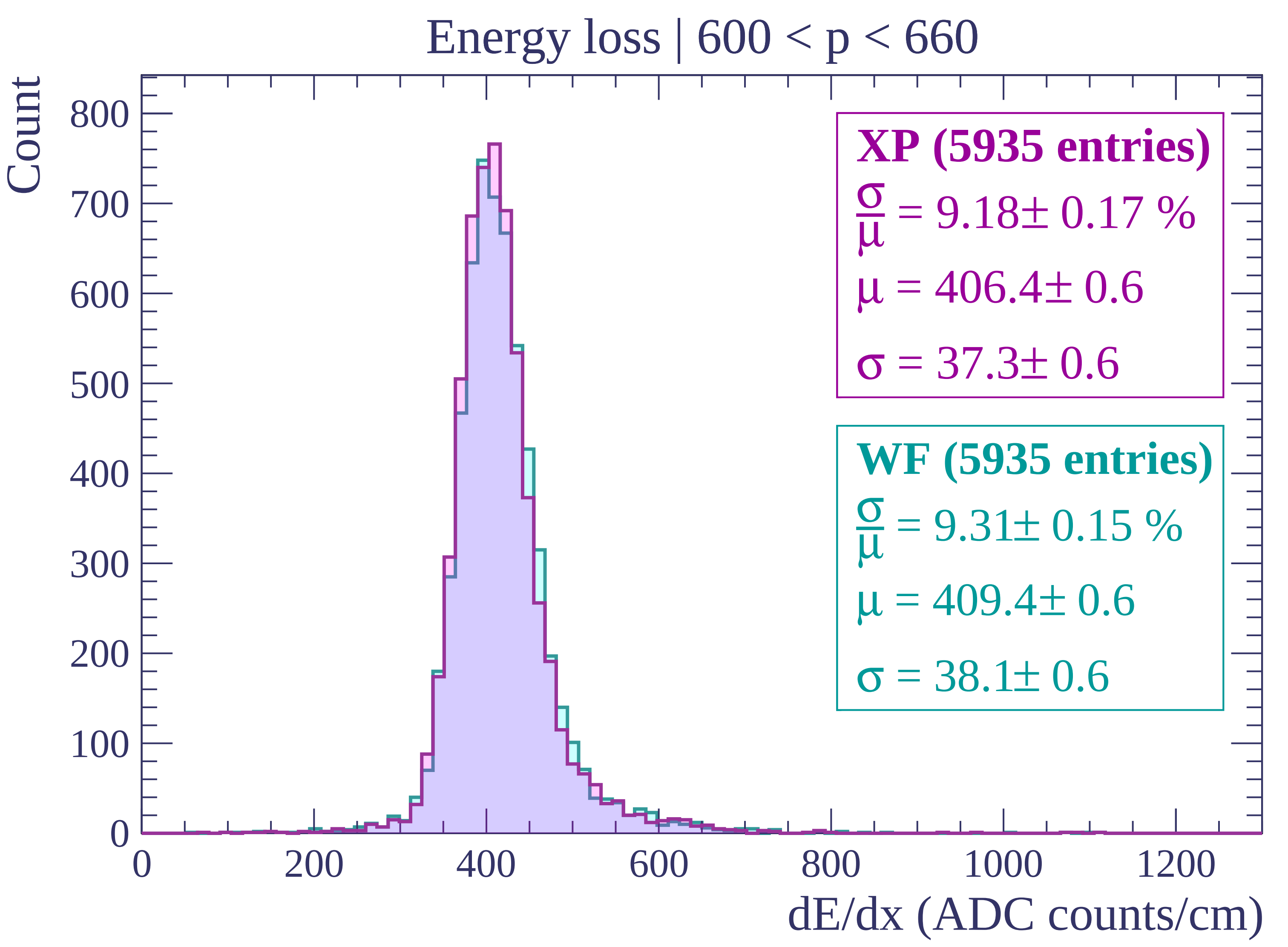


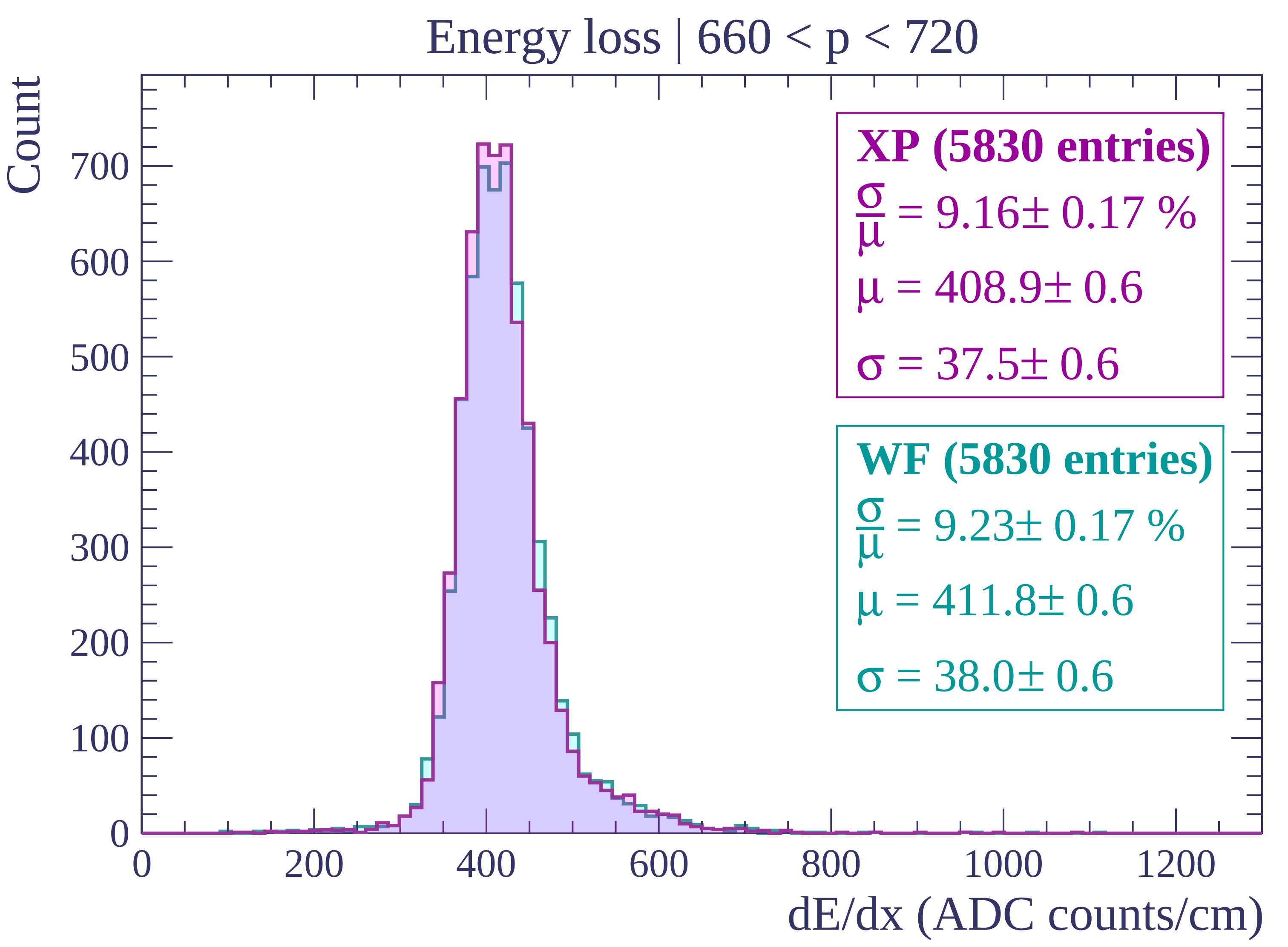
Energy loss | 420 < p < 480
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP (6182 entries)
$\frac{\sigma}{\mu}$ = 8.95± 0.15 %
μ = 397.2± 0.5
σ = 35.6± 0.5
WF (6182 entries)
$\frac{\sigma}{\mu}$ = 9.30± 0.16 %
μ = 399.2± 0.6
σ = 37.1± 0.6


Energy loss | 480 < p < 540
Count
dE/dx (ADC counts/cm)
XP (6096 entries)
σ/μ = 9.10± 0.16 %
μ = 400.7± 0.6
σ = 36.5± 0.6
WF (6096 entries)
σ/μ = 9.60± 0.16 %
μ = 403.1± 0.6
σ = 38.7± 0.6
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200

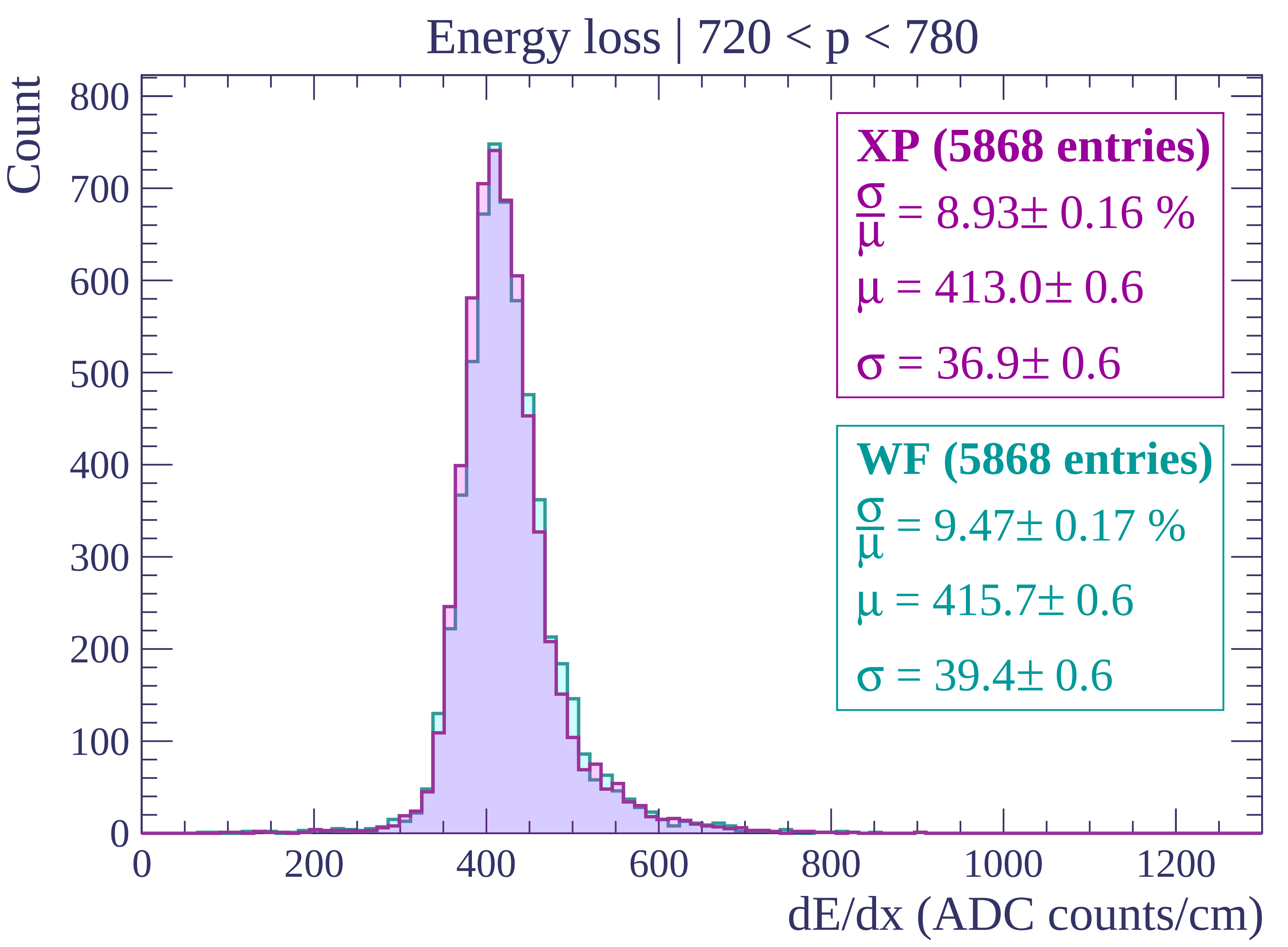


Energy loss | 540 < p < 600
Count
900
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6000 entries)
σ/μ = 8.91± 0.15 %
μ = 403.1± 0.6
σ = 35.9± 0.6
WF (6000 entries)
σ/μ = 9.36± 0.15 %
μ = 405.9± 0.6
σ = 38.0± 0.6

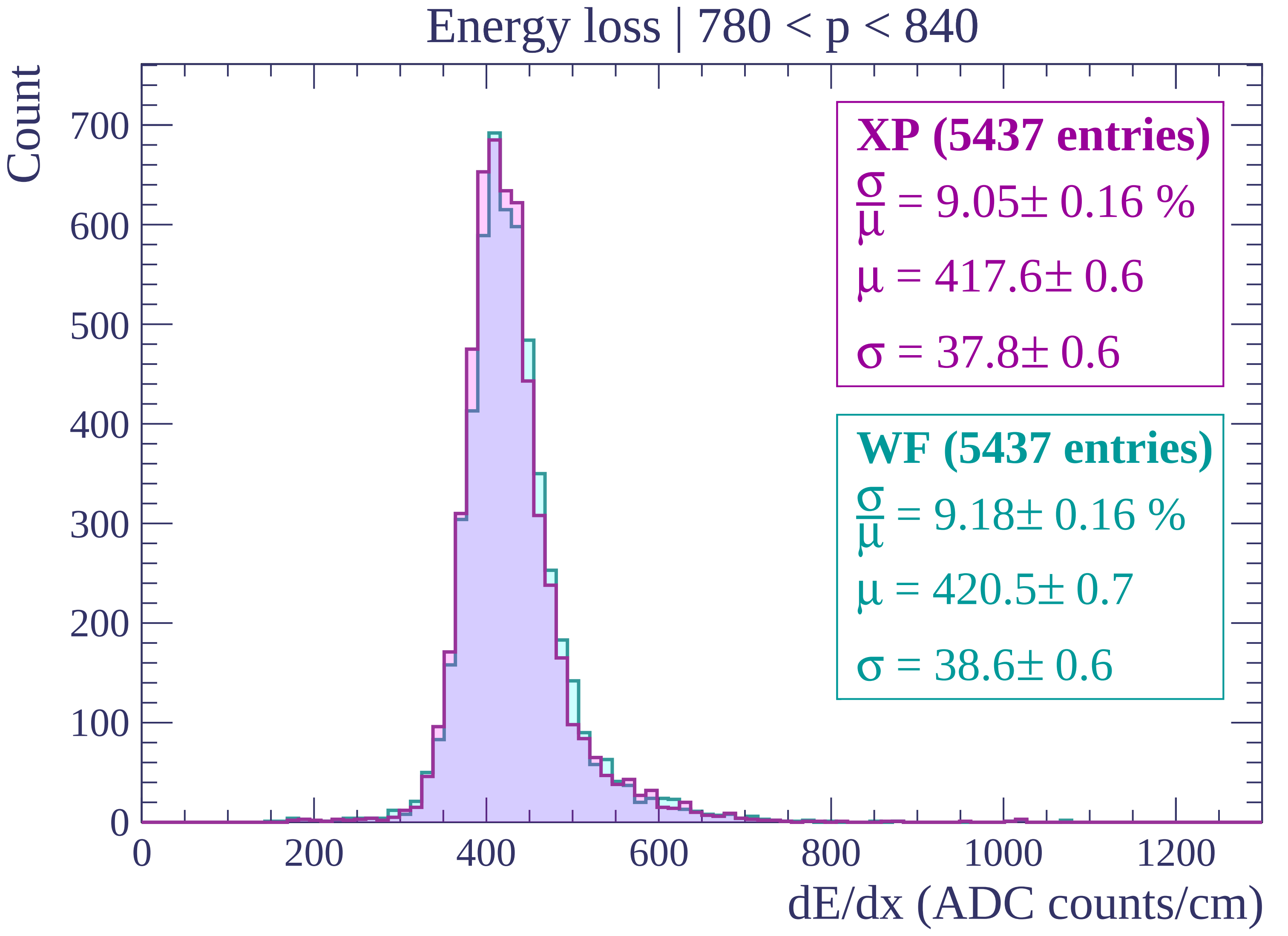

Energy loss | 600 < p < 660
Count
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (5935 entries)
$\frac{\sigma}{\mu}$ = 9.18± 0.17 %
μ = 406.4± 0.6
σ = 37.3± 0.6
WF (5935 entries)
$\frac{\sigma}{\mu}$ = 9.31± 0.15 %
μ = 409.4± 0.6
σ = 38.1± 0.6

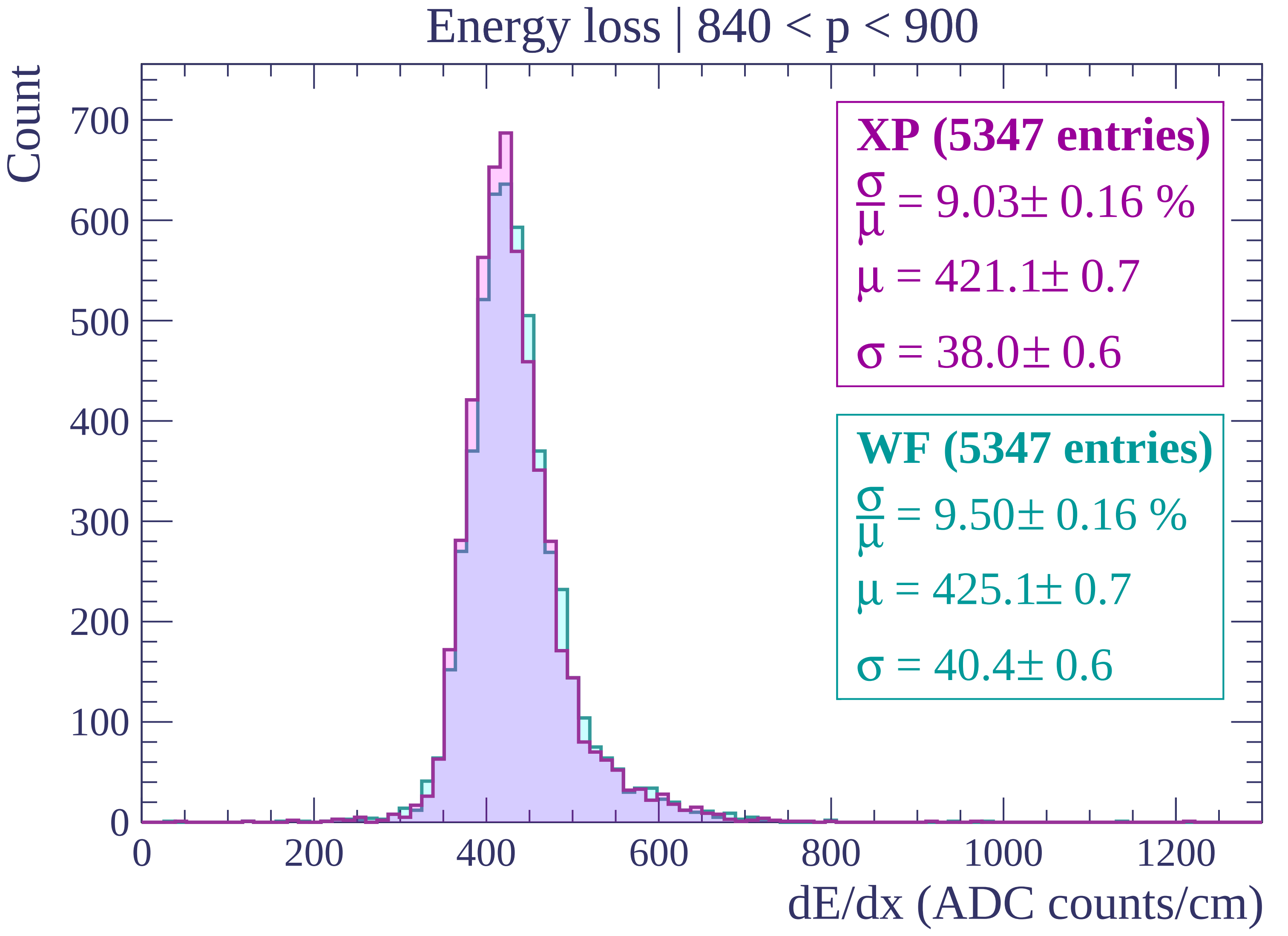


Energy loss | 660 < p < 720
Count
dE/dx (ADC counts/cm)
XP (5830 entries)
σ/μ = 9.16± 0.17 %
μ = 408.9± 0.6
σ = 37.5± 0.6
WF (5830 entries)
σ/μ = 9.23± 0.17 %
μ = 411.8± 0.6
σ = 38.0± 0.6
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200

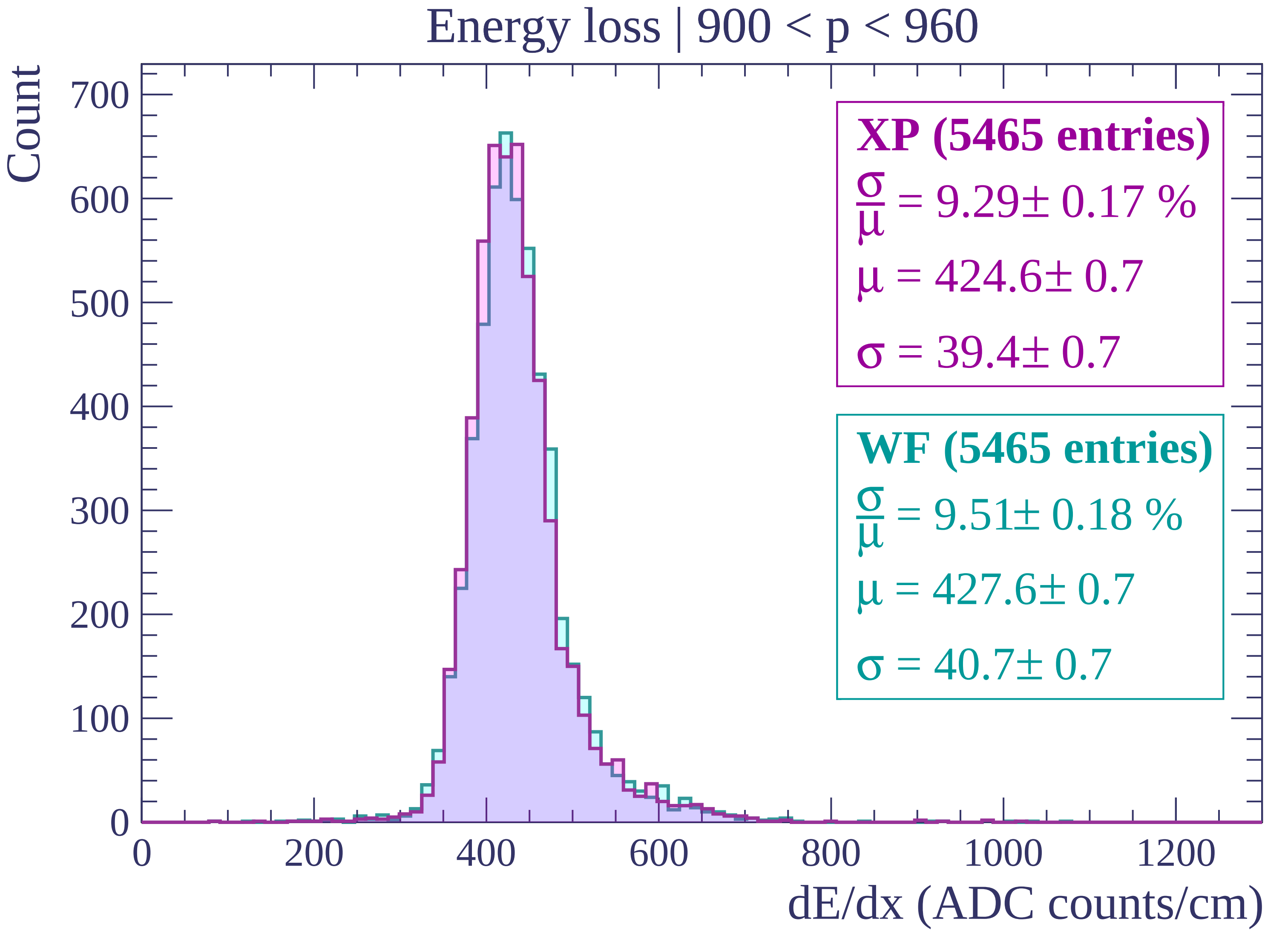


Energy loss | 720 < p < 780
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP (5868 entries)
σ/μ = 8.93± 0.16 %
μ = 413.0± 0.6
σ = 36.9± 0.6
WF (5868 entries)
σ/μ = 9.47± 0.17 %
μ = 415.7± 0.6
σ = 39.4± 0.6

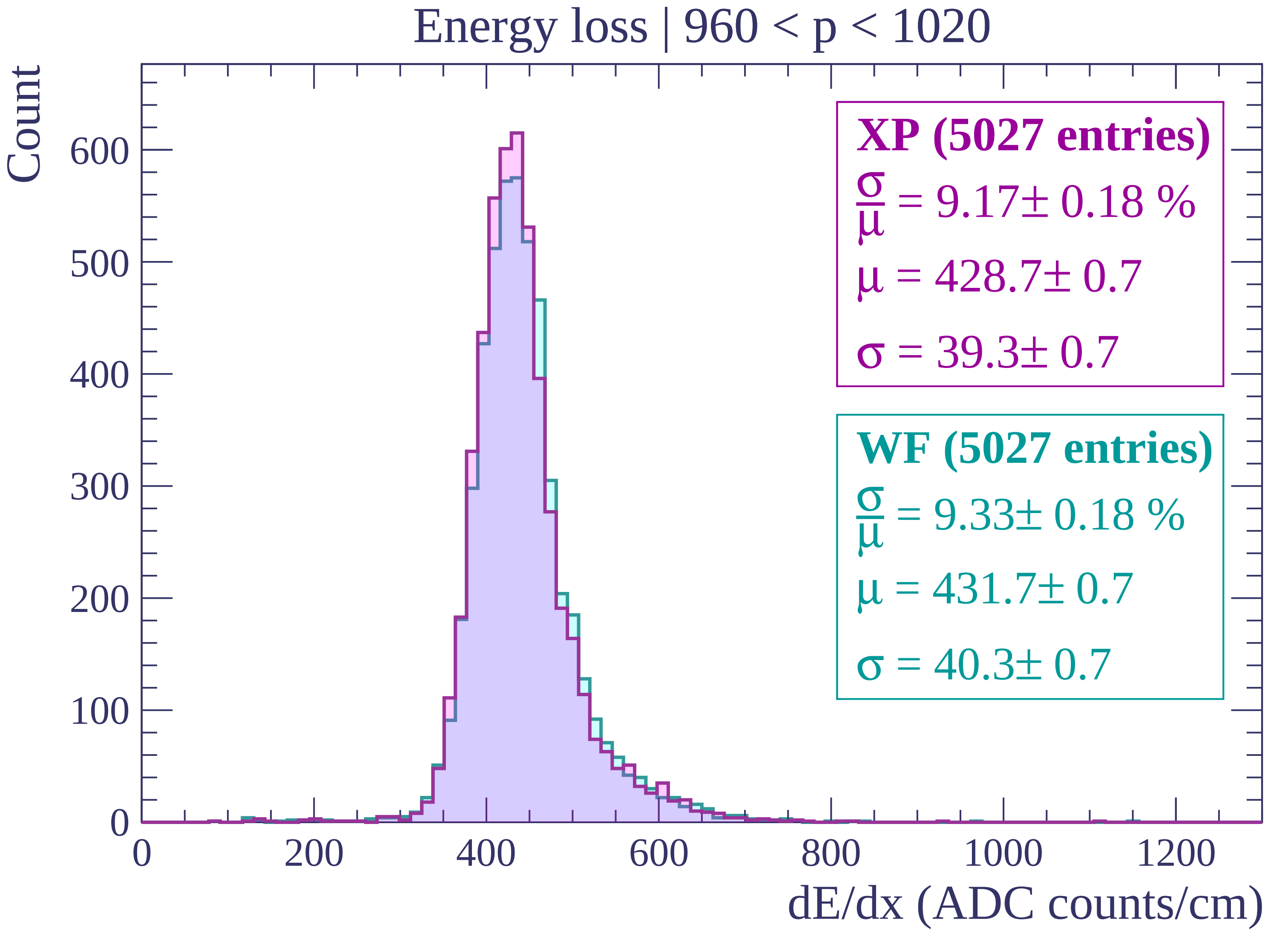

Energy loss | 780 < p < 840
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP (5437 entries)
σ/μ = 9.05± 0.16 %
μ = 417.6± 0.6
σ = 37.8± 0.6
WF (5437 entries)
σ/μ = 9.18± 0.16 %
μ = 420.5± 0.7
σ = 38.6± 0.6

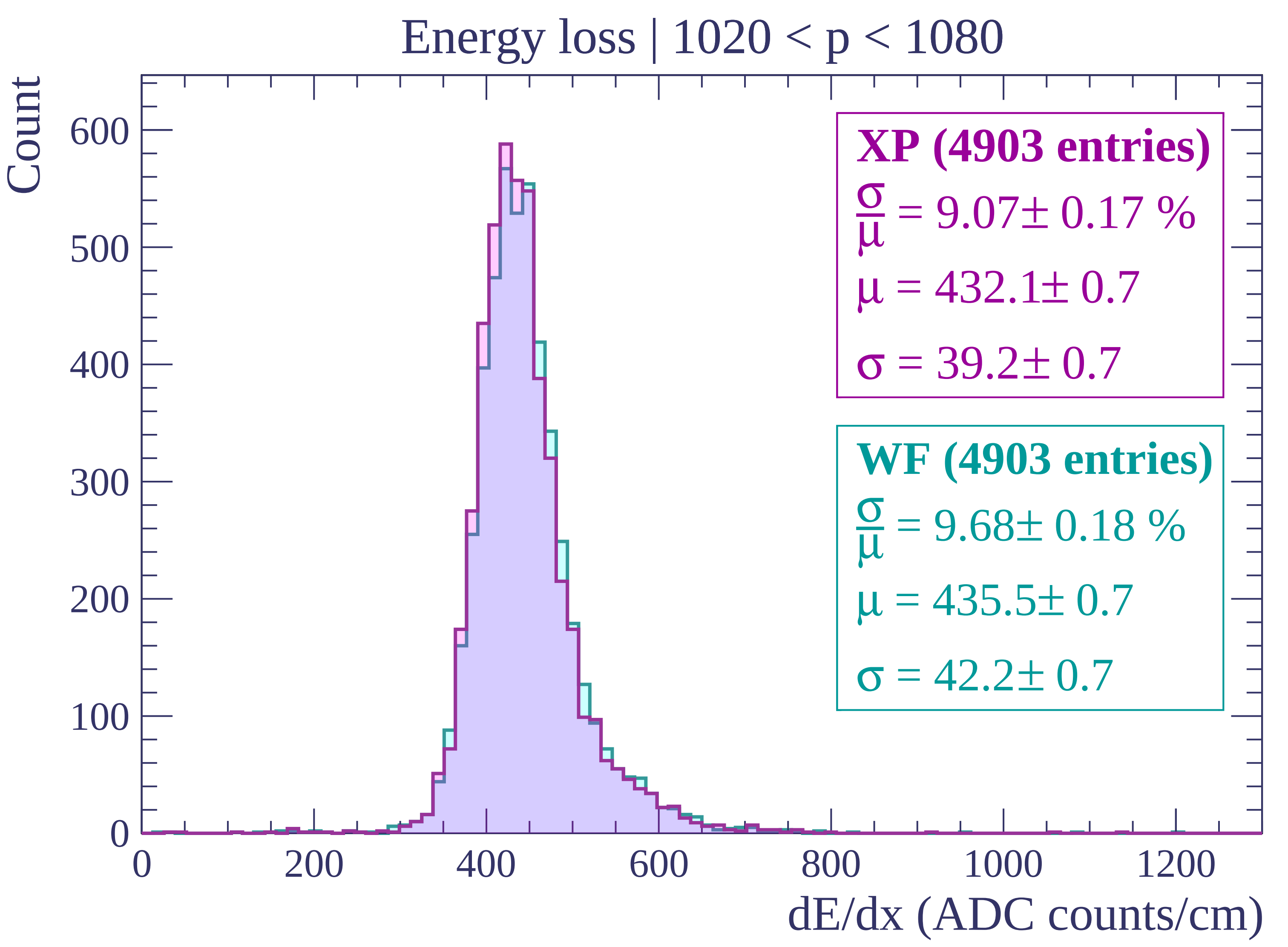


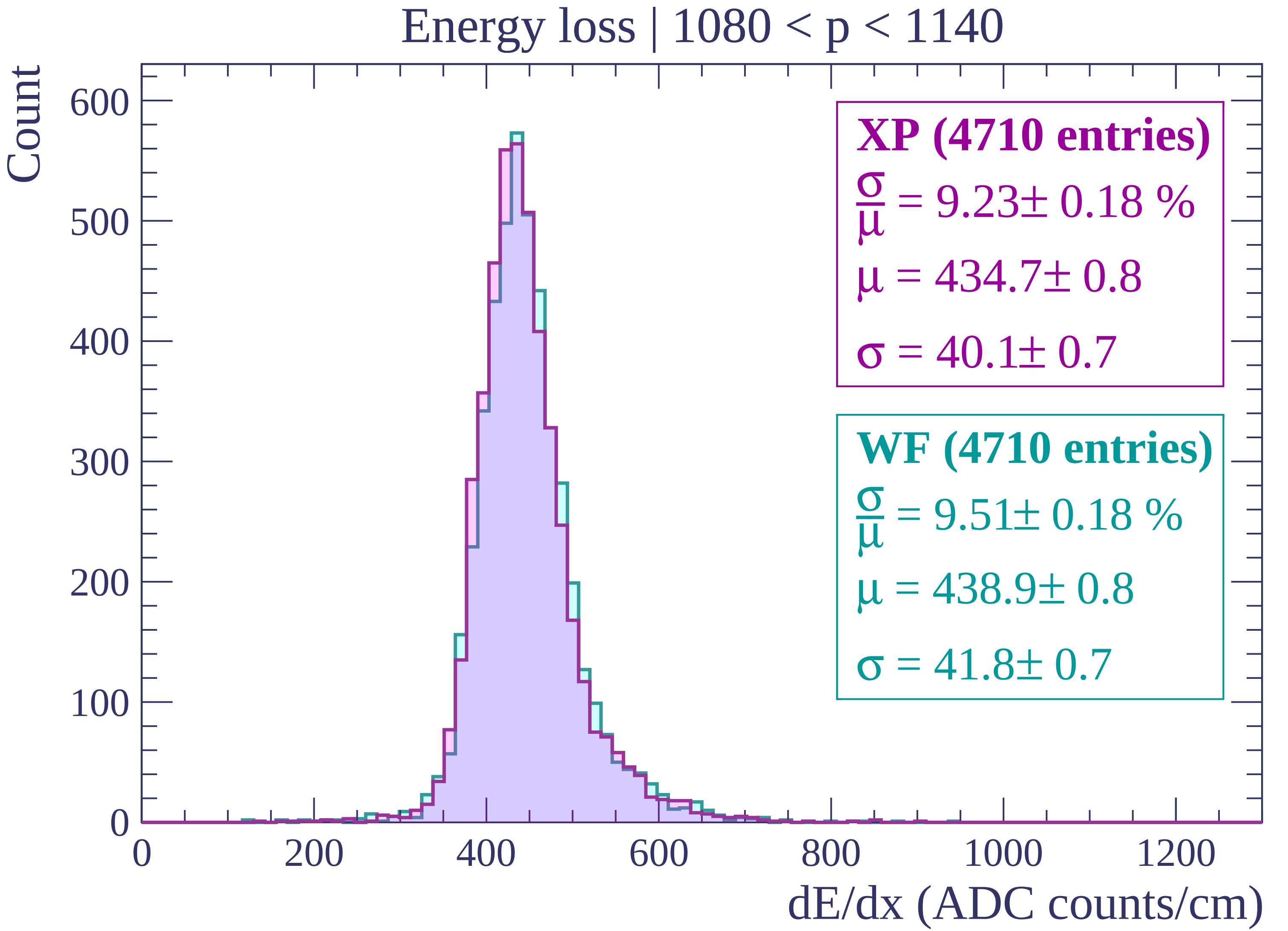
Energy loss | 840 < p < 900
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP (5347 entries)
σ/μ = 9.03± 0.16 %
μ = 421.1± 0.7
σ = 38.0± 0.6
WF (5347 entries)
σ/μ = 9.50± 0.16 %
μ = 425.1± 0.7
σ = 40.4± 0.6



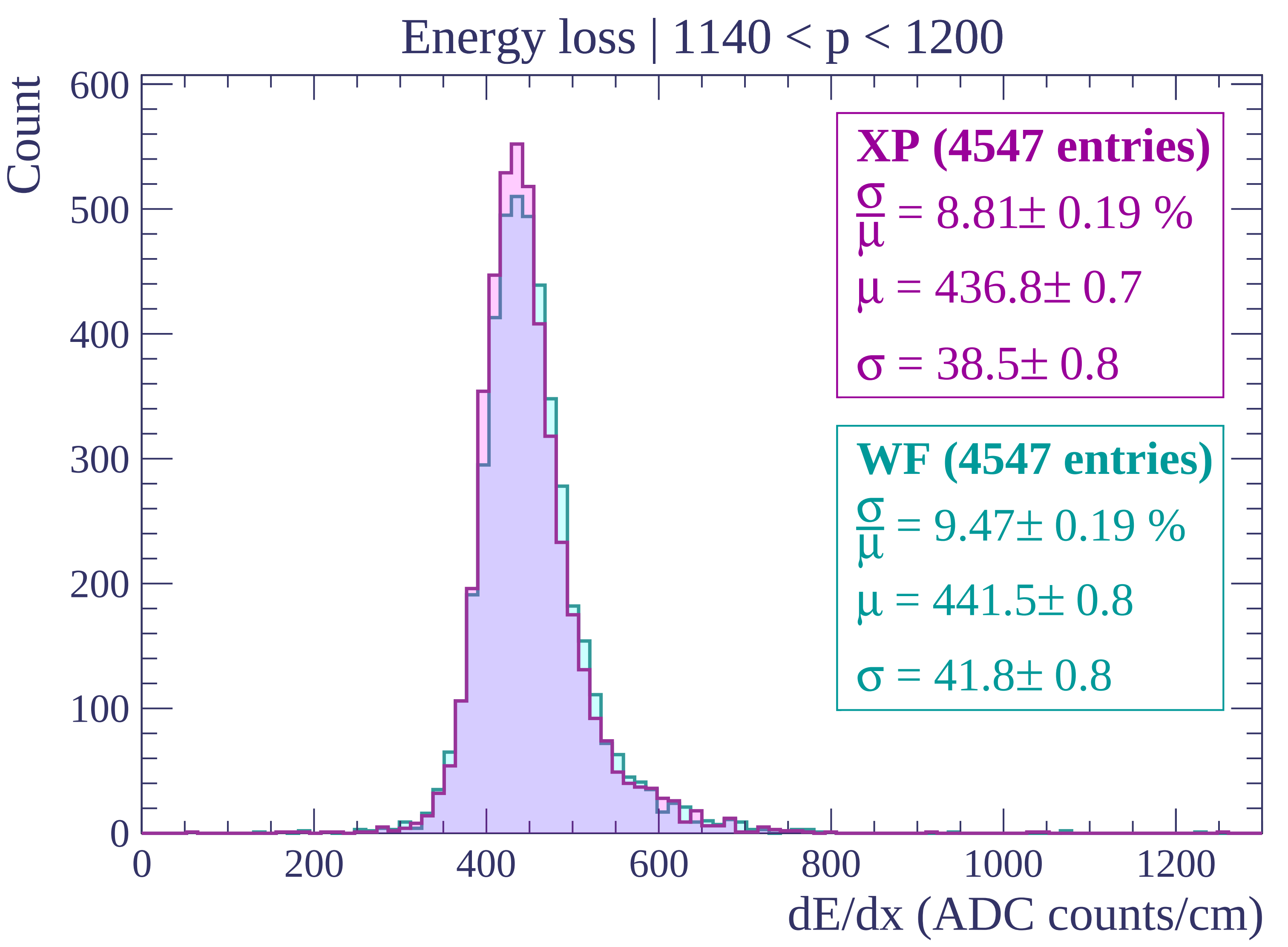
Energy loss | 900 < p < 960
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP (5465 entries)
σ/μ = 9.29± 0.17 %
μ = 424.6± 0.7
σ = 39.4± 0.7
WF (5465 entries)
σ/μ = 9.51± 0.18 %
μ = 427.6± 0.7
σ = 40.7± 0.7


Energy loss | 960 < p < 1020
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP (5027 entries)
$\frac{\sigma}{\mu}$ = 9.17± 0.18 %
μ = 428.7± 0.7
σ = 39.3± 0.7
WF (5027 entries)
$\frac{\sigma}{\mu}$ = 9.33± 0.18 %
μ = 431.7± 0.7
σ = 40.3± 0.7

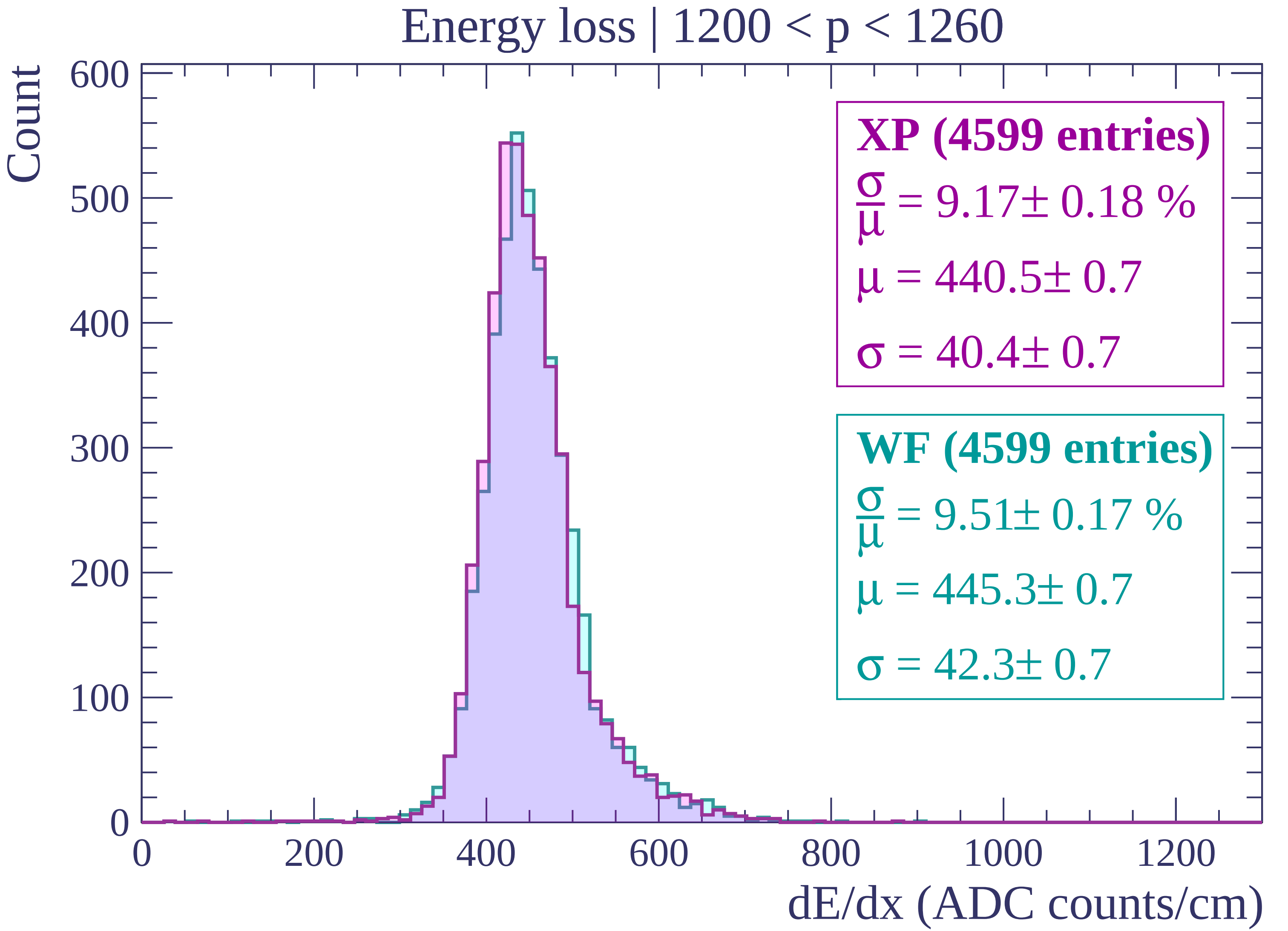

Energy loss | 1020 < p < 1080
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP (4903 entries)
σ/μ = 9.07± 0.17 %
μ = 432.1± 0.7
σ = 39.2± 0.7
WF (4903 entries)
σ/μ = 9.68± 0.18 %
μ = 435.5± 0.7
σ = 42.2± 0.7

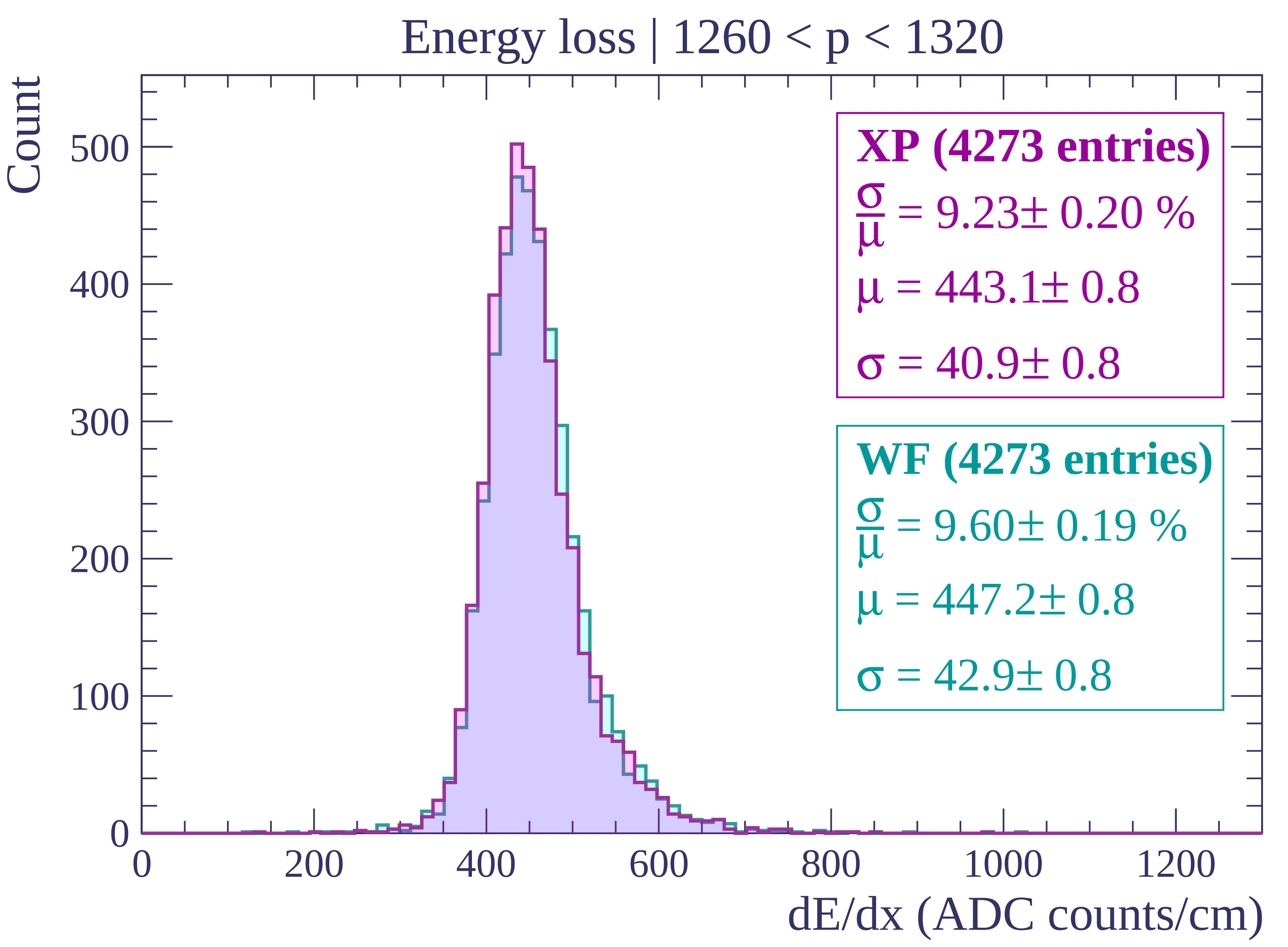


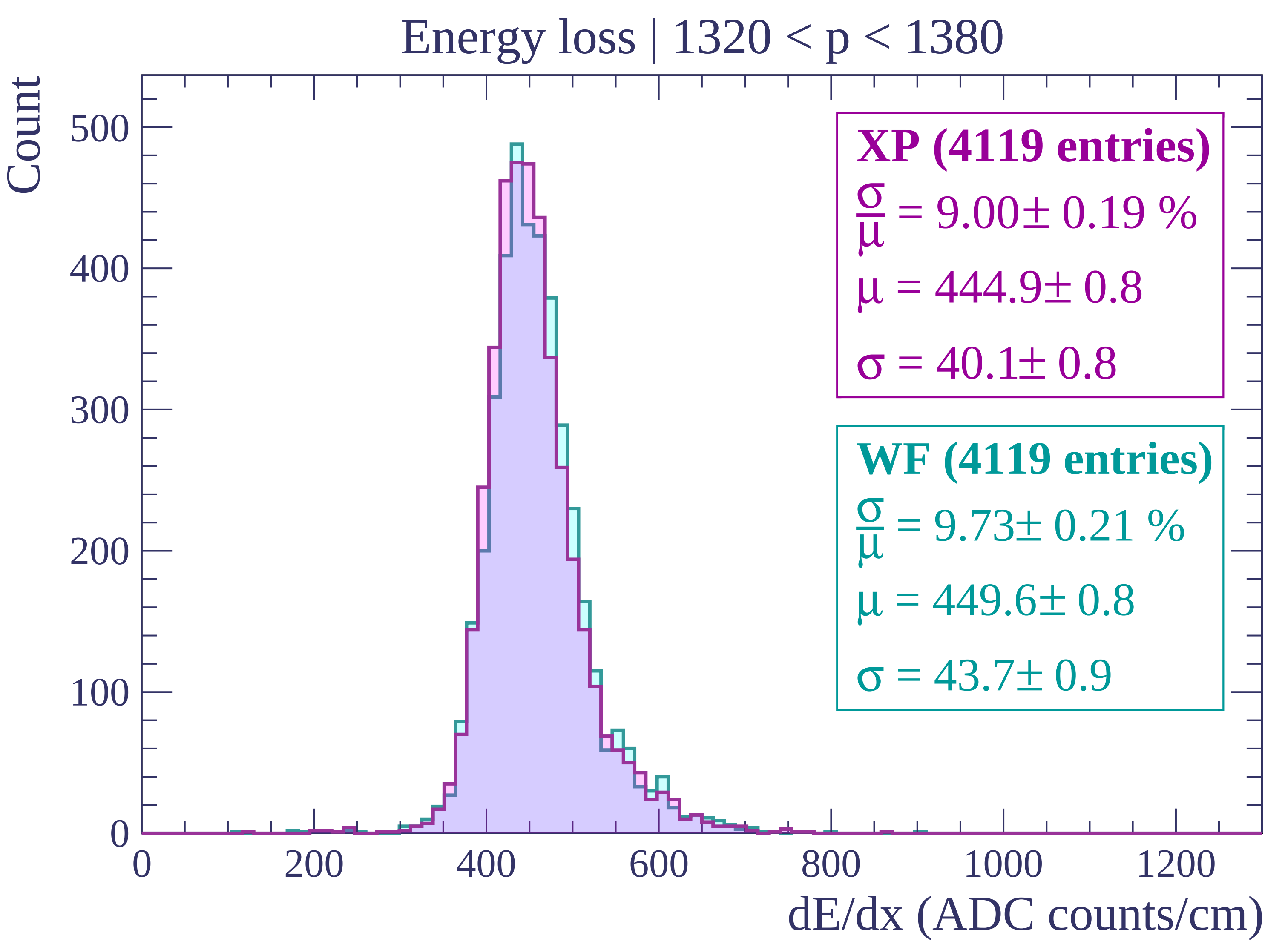
Energy loss | 1080 < p < 1140
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP (4710 entries)
$\frac{\sigma}{\mu}$ = 9.23± 0.18 %
μ = 434.7± 0.8
σ = 40.1± 0.7
WF (4710 entries)
$\frac{\sigma}{\mu}$ = 9.51± 0.18 %
μ = 438.9± 0.8
σ = 41.8± 0.7



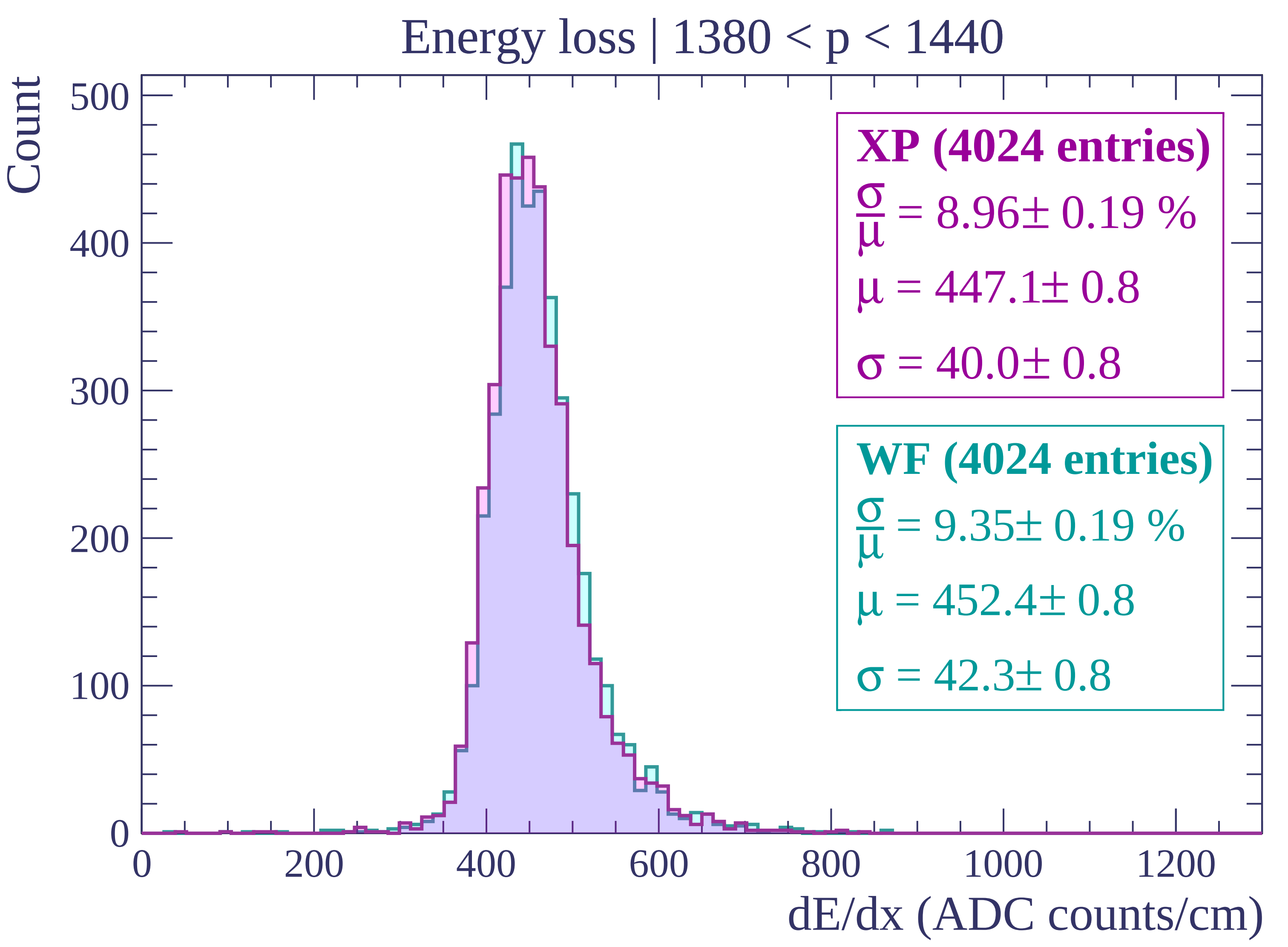
Energy loss | 1140 < p < 1200
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
100
300
500
XP (4547 entries)
$\frac{\sigma}{\mu}$ = 8.81± 0.19 %
μ = 436.8± 0.7
σ = 38.5± 0.8
WF (4547 entries)
$\frac{\sigma}{\mu}$ = 9.47± 0.19 %
μ = 441.5± 0.8
σ = 41.8± 0.8

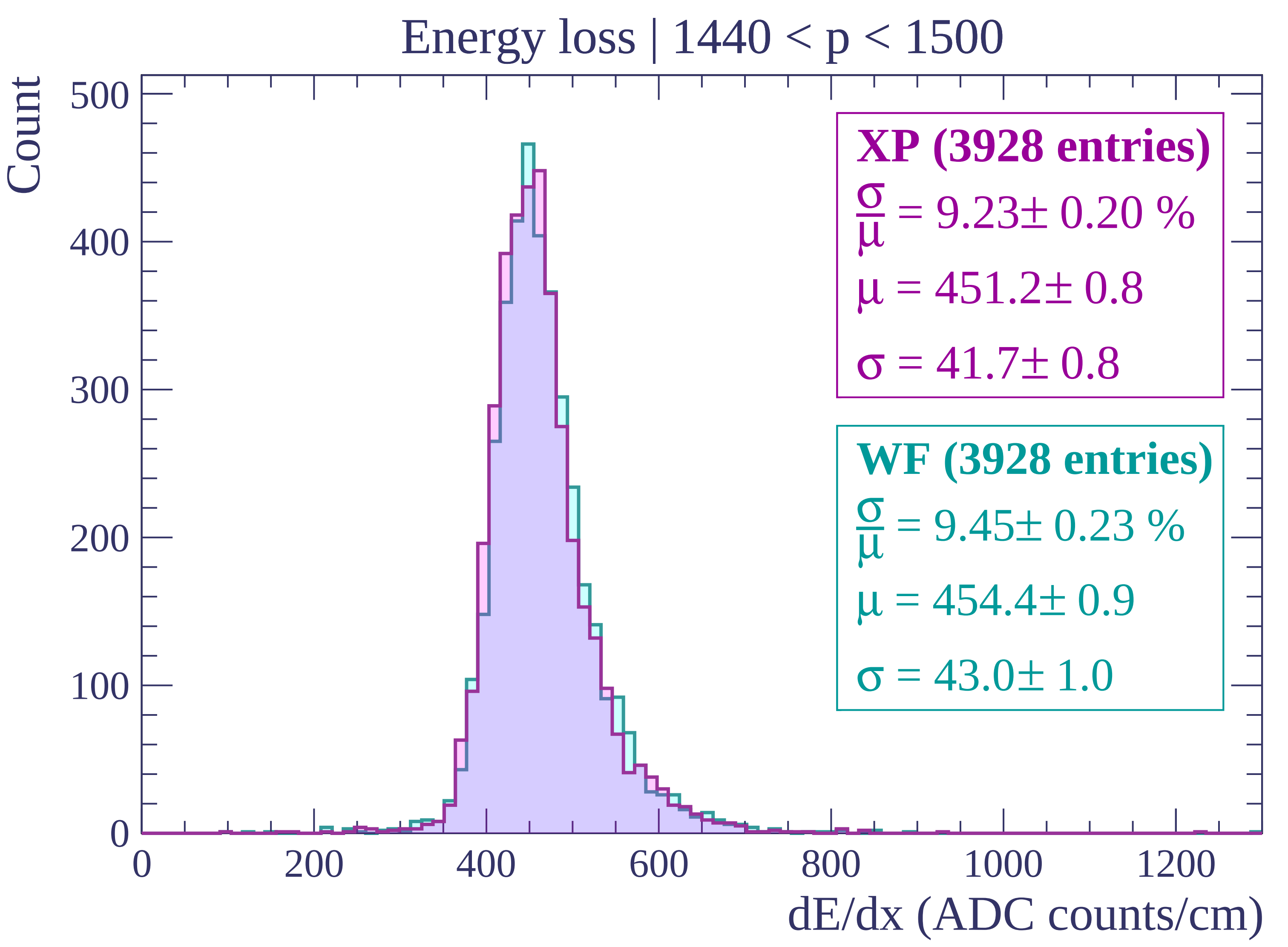

Energy loss | 1200 < p < 1260
Count
dE/dx (ADC counts/cm)
XP (4599 entries)
$\frac{\sigma}{\mu}$ = 9.17± 0.18 %
μ = 440.5± 0.7
σ = 40.4± 0.7
WF (4599 entries)
$\frac{\sigma}{\mu}$ = 9.51± 0.17 %
μ = 445.3± 0.7
σ = 42.3± 0.7
0
200
400
600
800
1000
1200
100
300
500

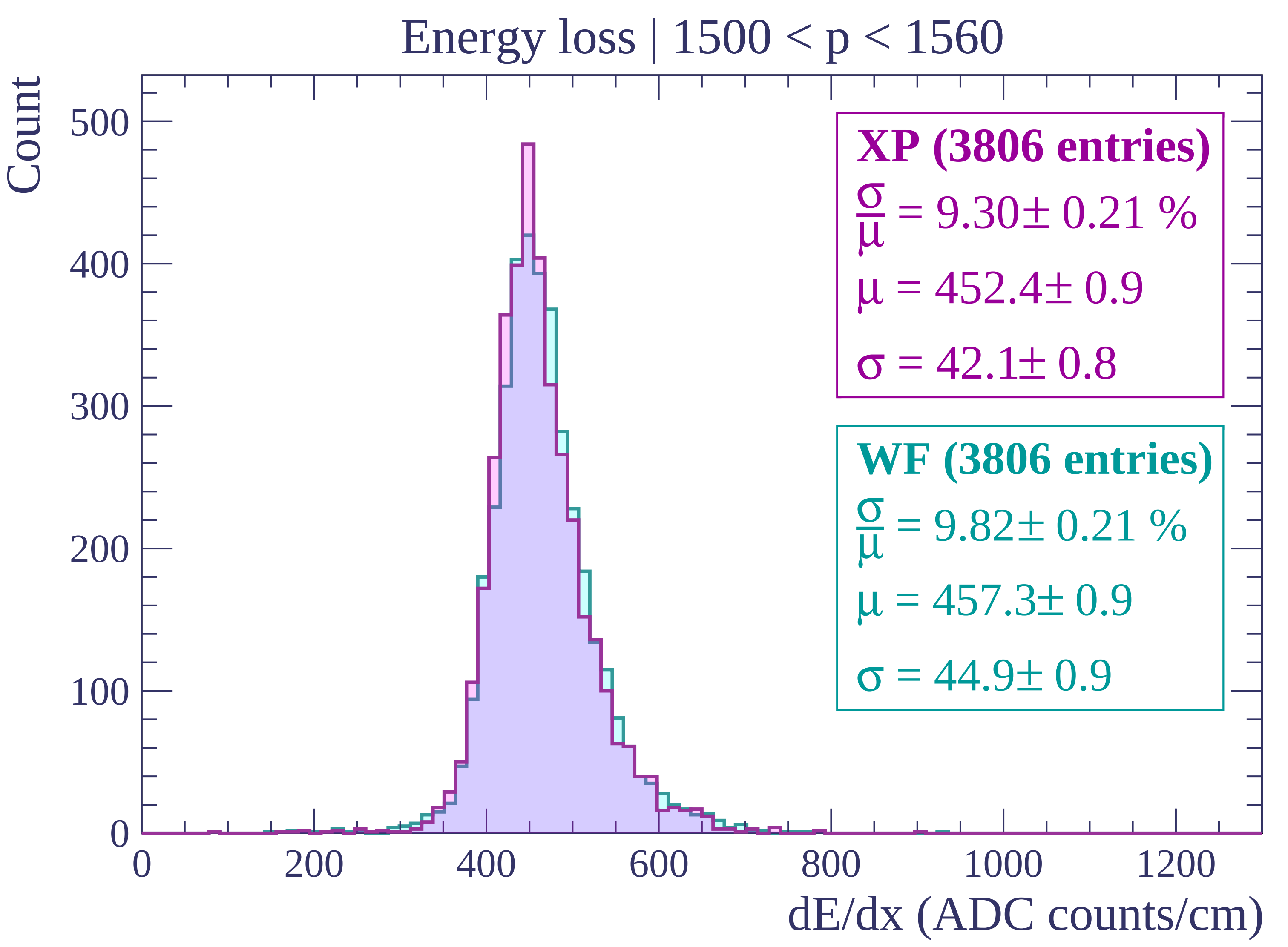

Energy loss | 1260 < p < 1320
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
100
300
500
XP (4273 entries)
σ/μ = 9.23± 0.20 %
μ = 443.1± 0.8
σ = 40.9± 0.8
WF (4273 entries)
σ/μ = 9.60± 0.19 %
μ = 447.2± 0.8
σ = 42.9± 0.8

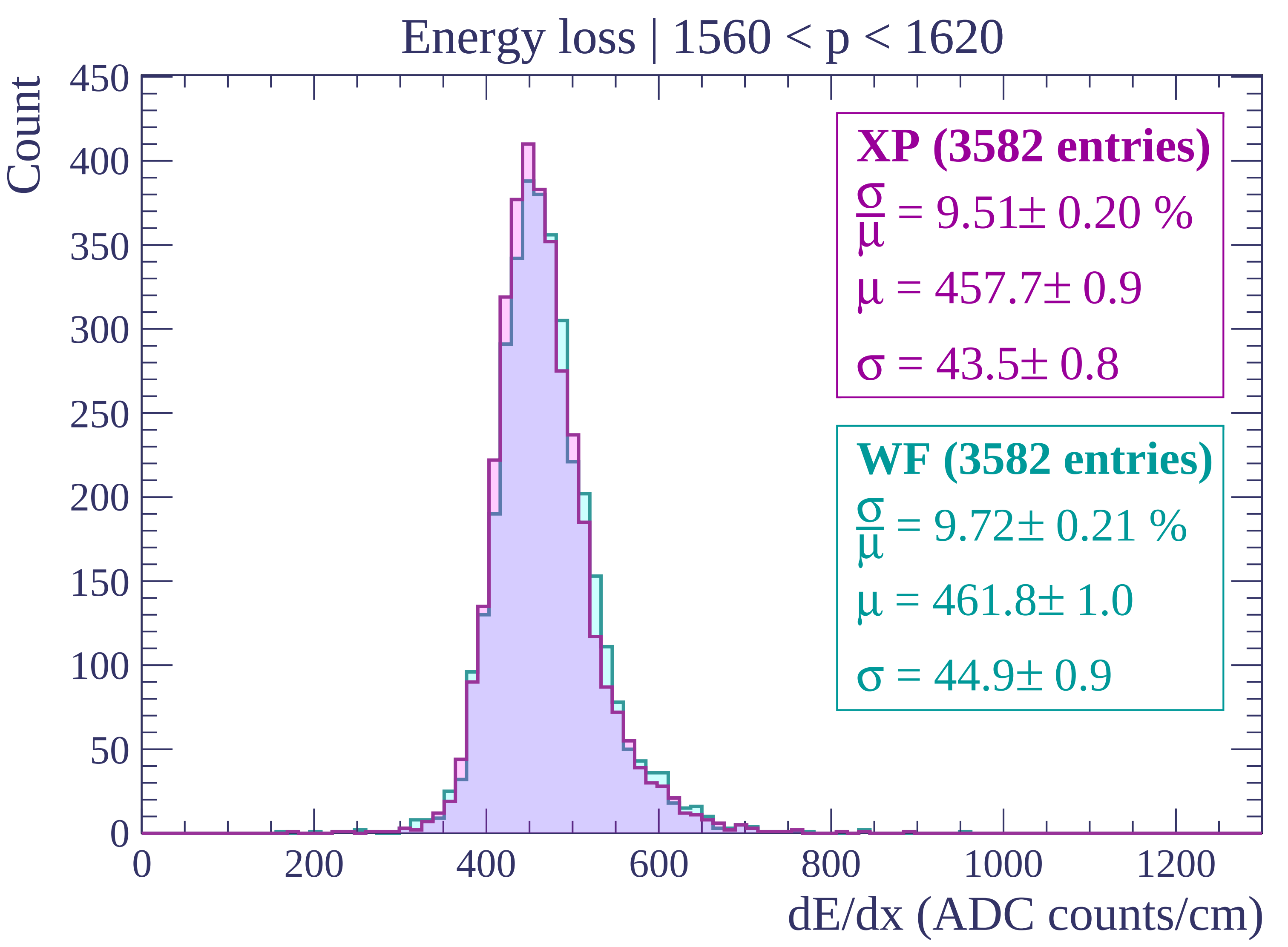


Energy loss | 1320 < p < 1380
Count
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (4119 entries)
σ/μ = 9.00± 0.19 %
μ = 444.9± 0.8
σ = 40.1± 0.8
WF (4119 entries)
σ/μ = 9.73± 0.21 %
μ = 449.6± 0.8
σ = 43.7± 0.9

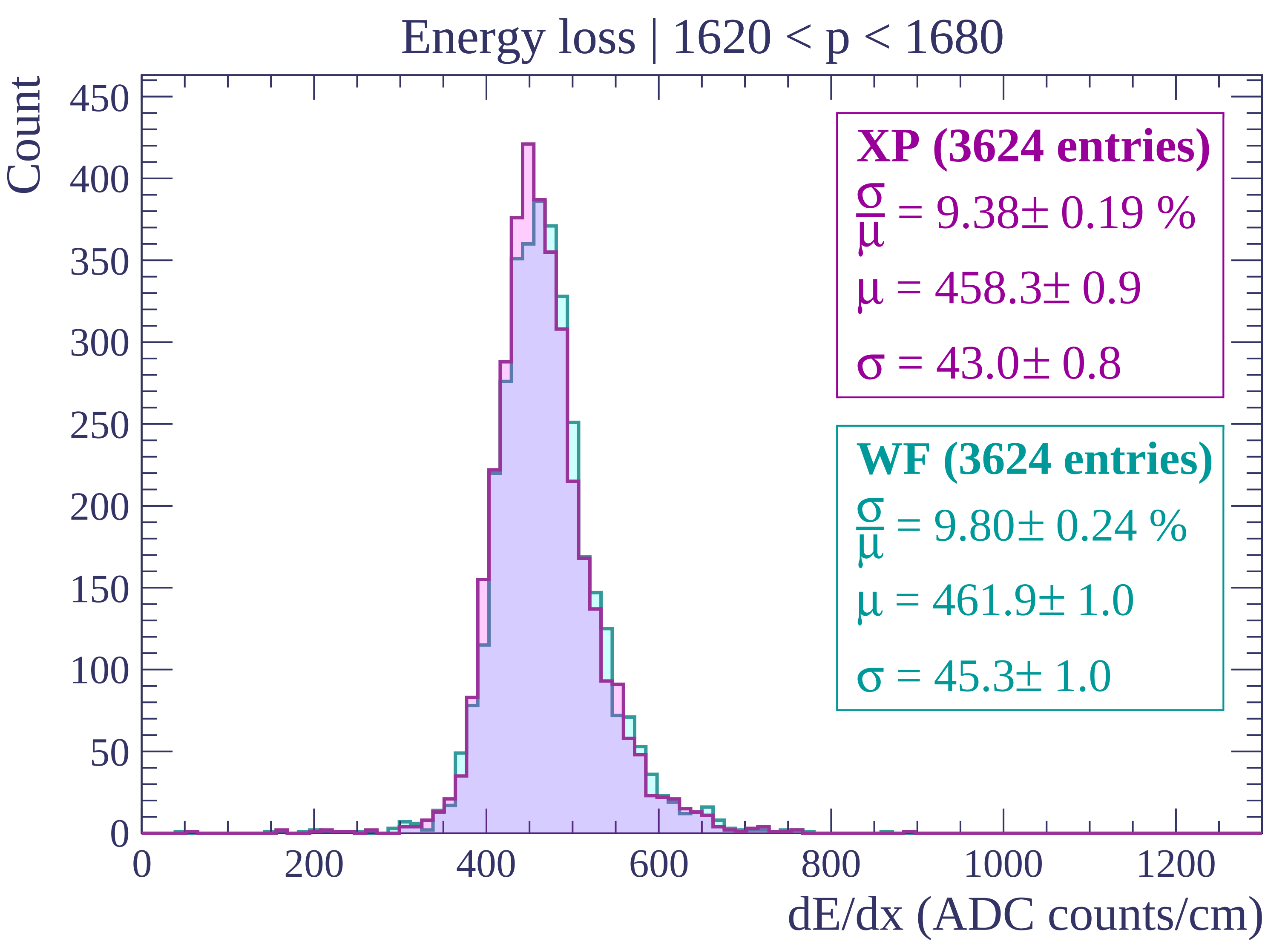

Energy loss | 1380 < p < 1440
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (4024 entries)
σ/μ = 8.96± 0.19 %
μ = 447.1± 0.8
σ = 40.0± 0.8
WF (4024 entries)
σ/μ = 9.35± 0.19 %
μ = 452.4± 0.8
σ = 42.3± 0.8

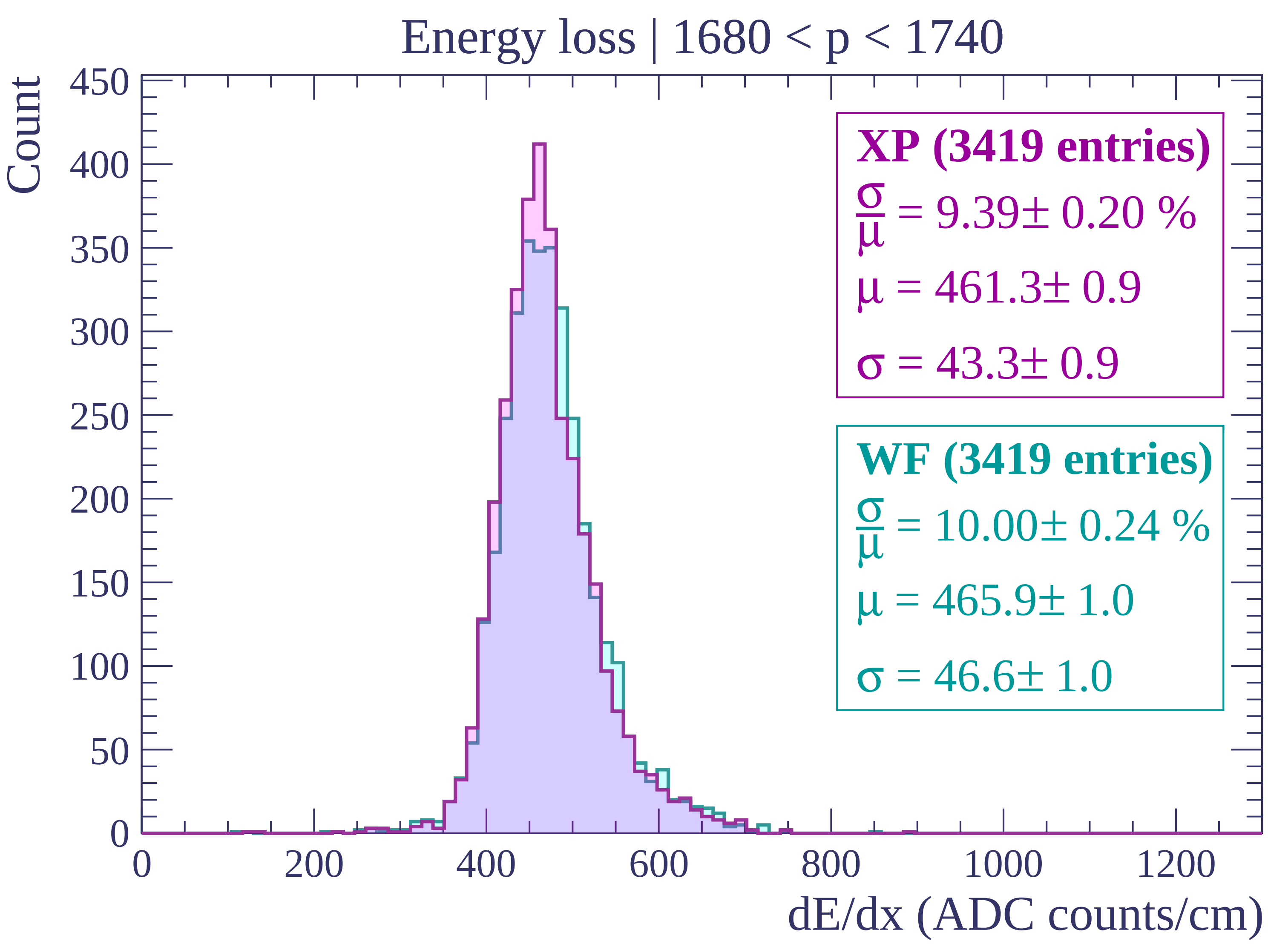

Energy loss | 1440 < p < 1500
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (3928 entries)
σ/μ = 9.23± 0.20 %
μ = 451.2± 0.8
σ = 41.7± 0.8
WF (3928 entries)
σ/μ = 9.45± 0.23 %
μ = 454.4± 0.9
σ = 43.0± 1.0


Energy loss | 1500 < p < 1560
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (3806 entries)
σ/μ = 9.30± 0.21 %
μ = 452.4± 0.9
σ = 42.1± 0.8
WF (3806 entries)
σ/μ = 9.82± 0.21 %
μ = 457.3± 0.9
σ = 44.9± 0.9

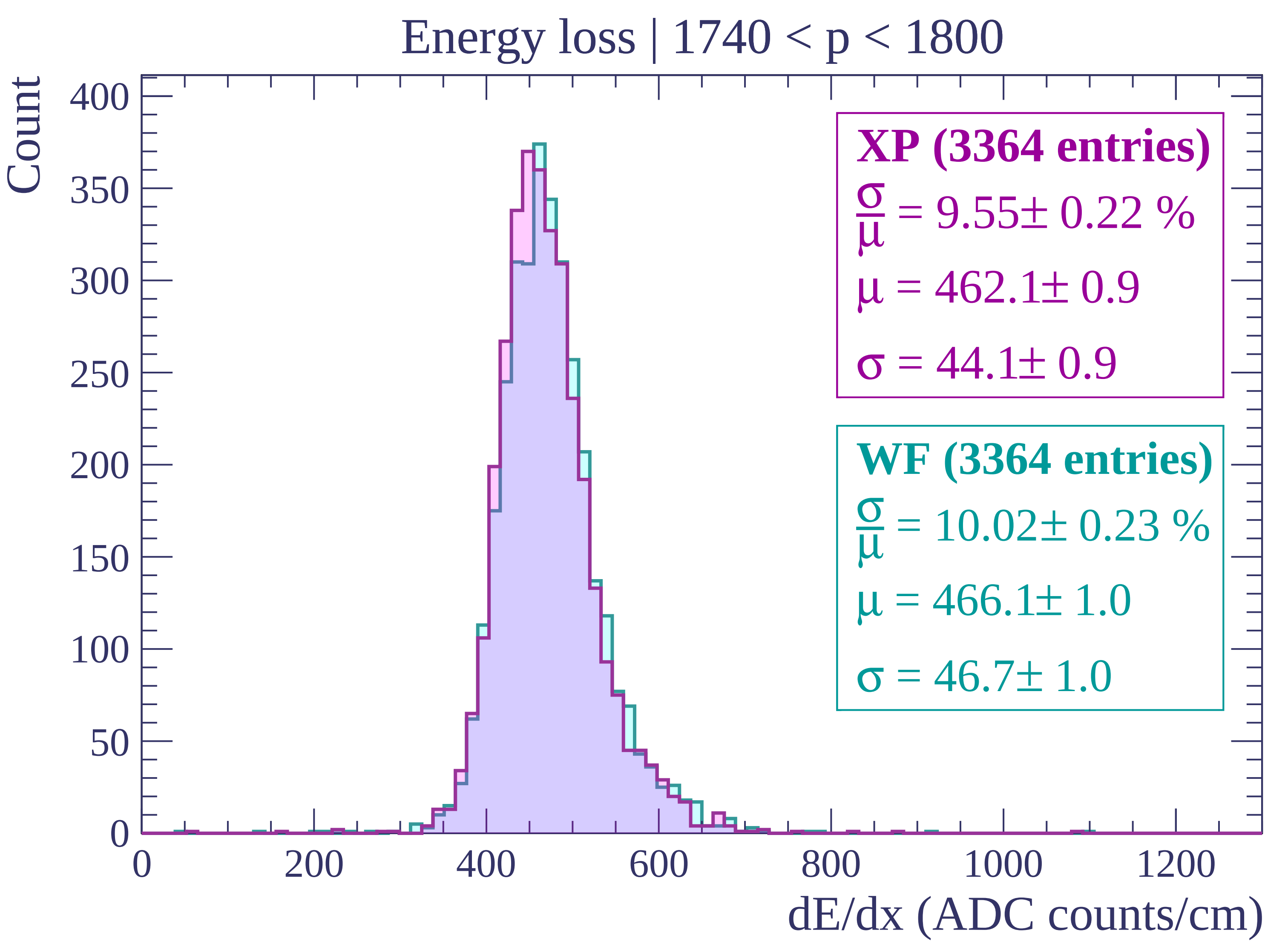


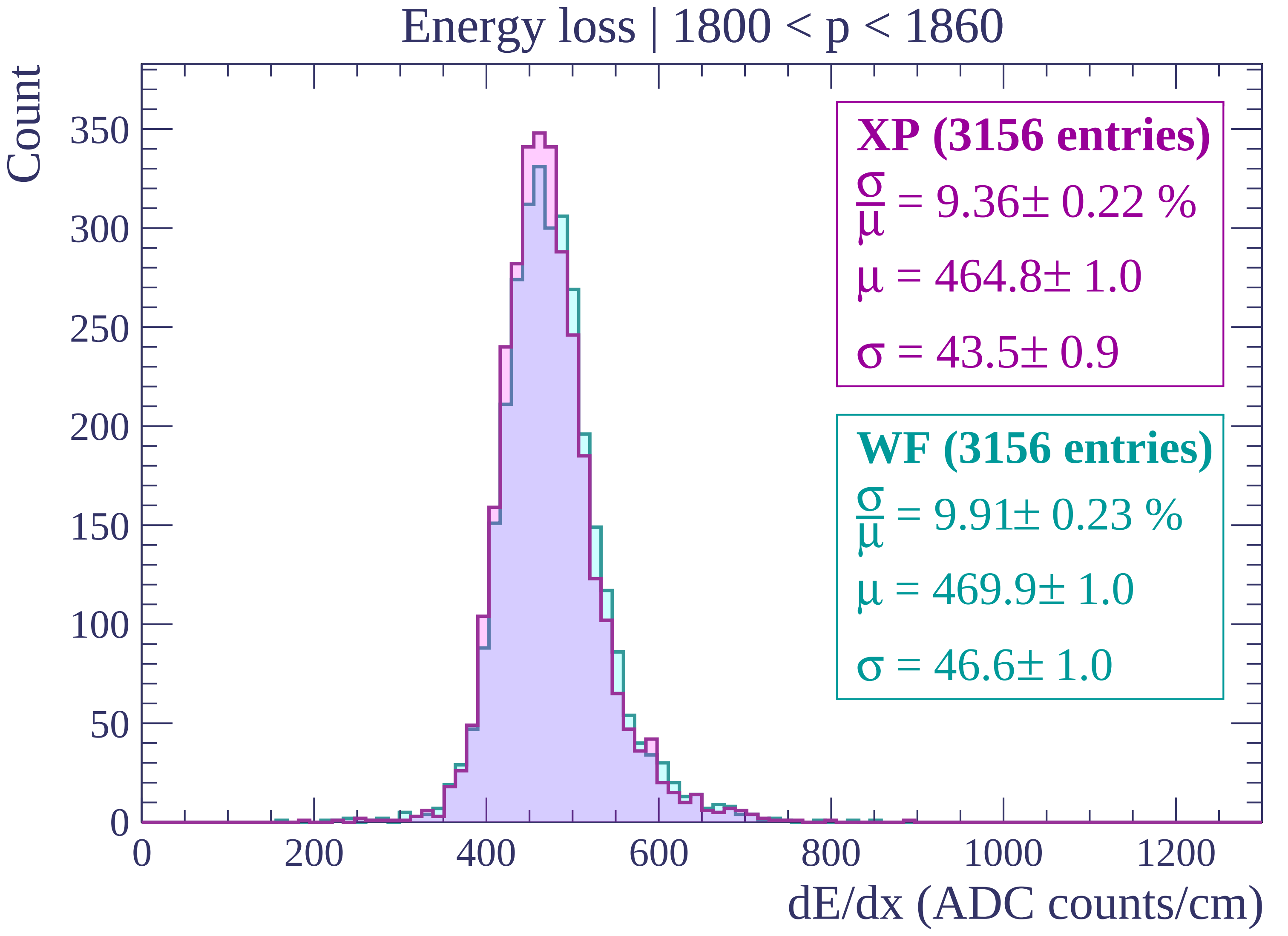
Energy loss | 1560 < p < 1620
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP (3582 entries)
σ/μ = 9.51± 0.20 %
μ = 457.7± 0.9
σ = 43.5± 0.8
WF (3582 entries)
σ/μ = 9.72± 0.21 %
μ = 461.8± 1.0
σ = 44.9± 0.9


Energy loss | 1620 < p < 1680
Count
450
400
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (3624 entries)
σ/μ = 9.38± 0.19 %
μ = 458.3± 0.9
σ = 43.0± 0.8
WF (3624 entries)
σ/μ = 9.80± 0.24 %
μ = 461.9± 1.0
σ = 45.3± 1.0

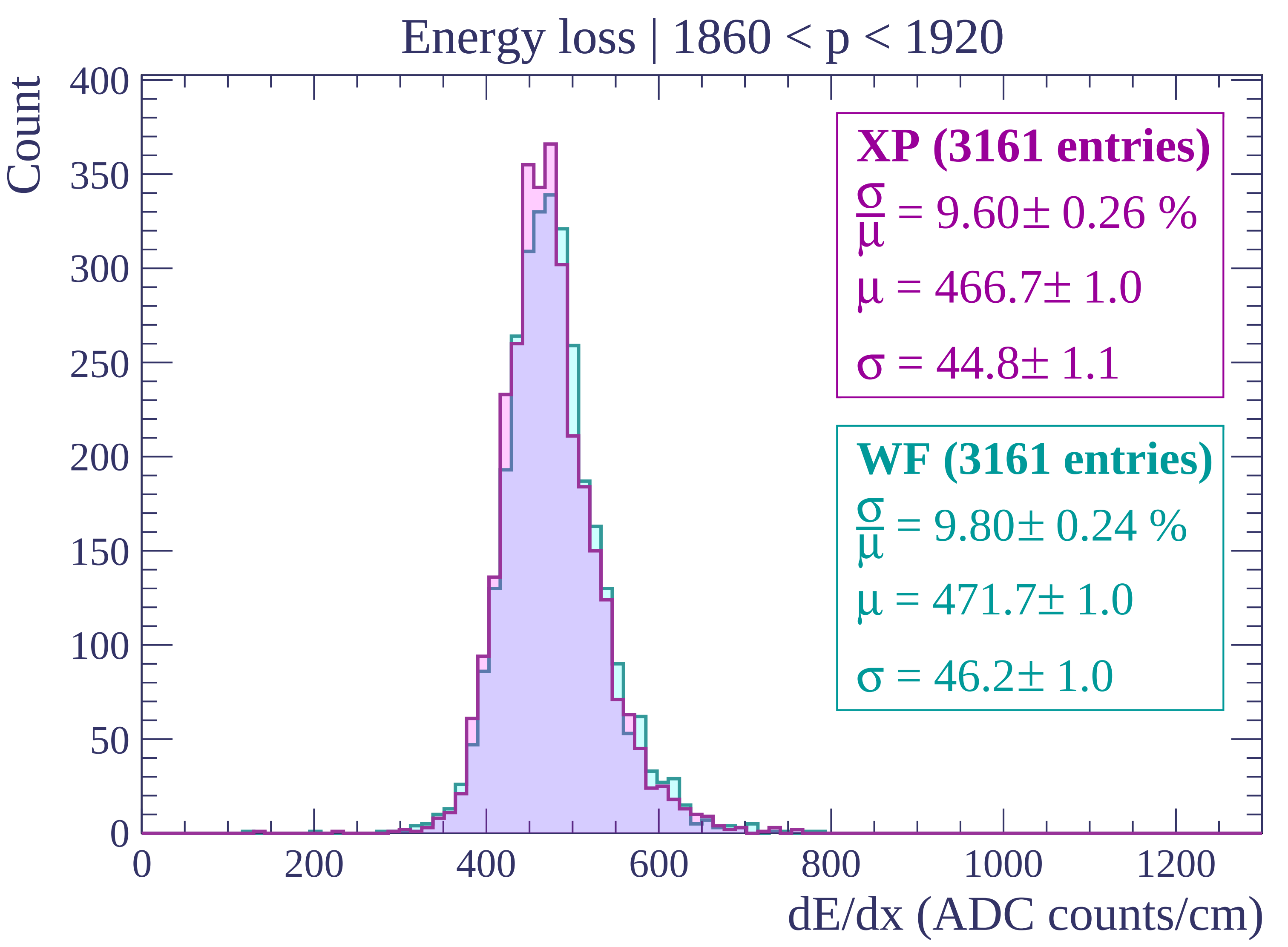


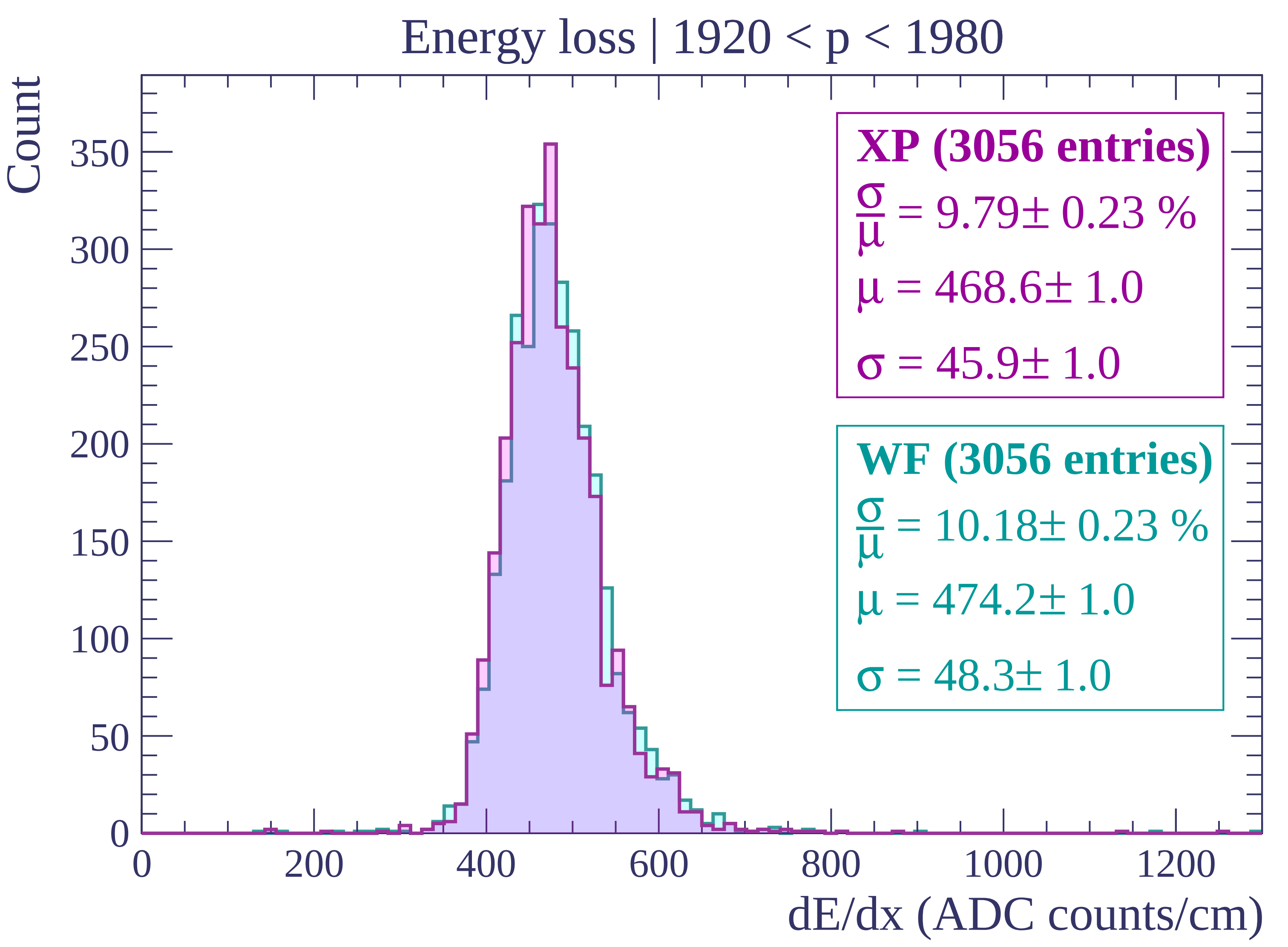
Energy loss | 1680 < p < 1740
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP (3419 entries)
σ/μ = 9.39± 0.20 %
μ = 461.3± 0.9
σ = 43.3± 0.9
WF (3419 entries)
σ/μ = 10.00± 0.24 %
μ = 465.9± 1.0
σ = 46.6± 1.0


Energy loss | 1740 < p < 1800
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
350
400
XP (3364 entries)
σ/μ = 9.55± 0.22 %
μ = 462.1± 0.9
σ = 44.1± 0.9
WF (3364 entries)
σ/μ = 10.02± 0.23 %
μ = 466.1± 1.0
σ = 46.7± 1.0

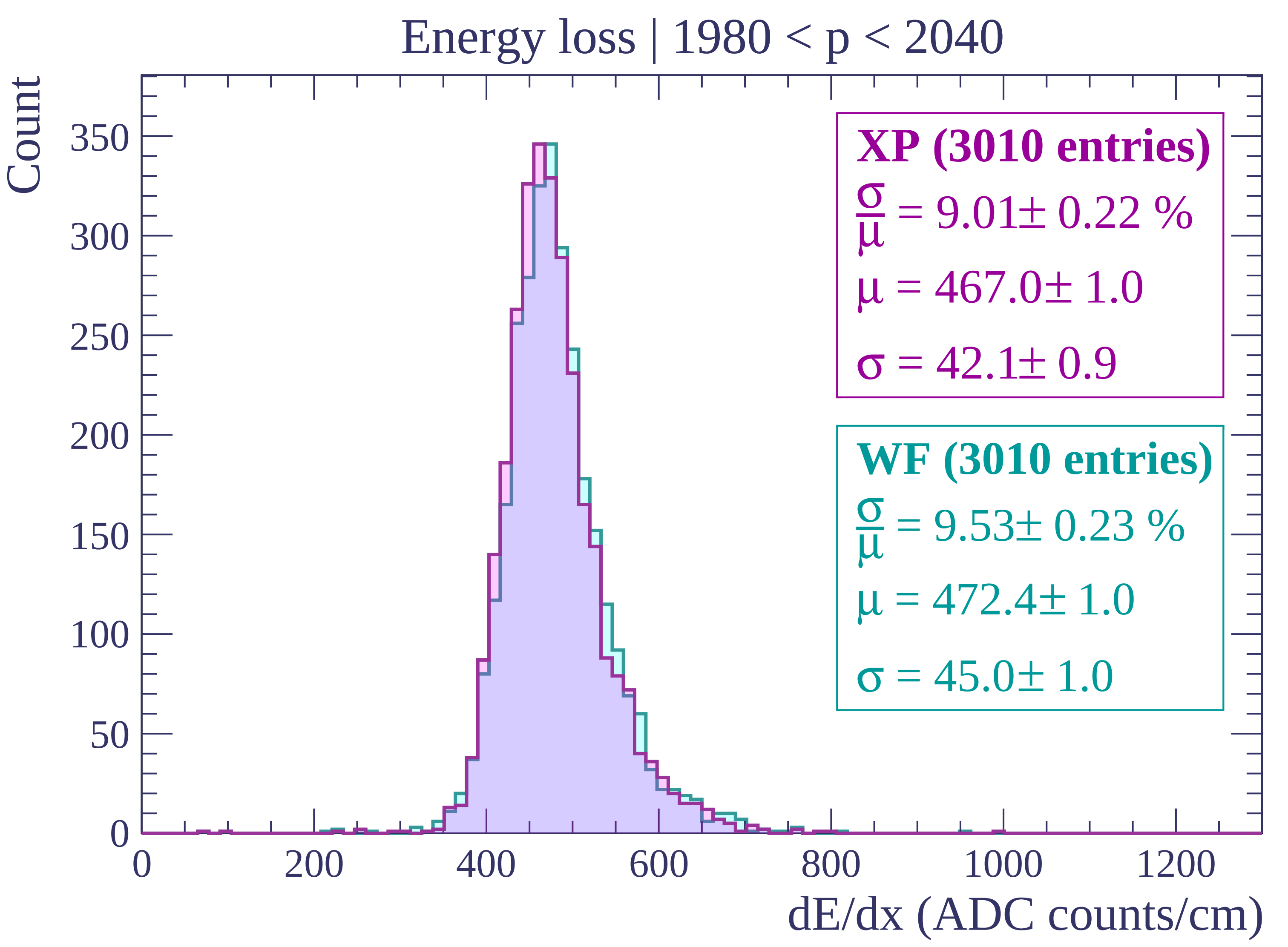


Energy loss | 1800 < p < 1860
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (3156 entries)
σ/μ = 9.36± 0.22 %
μ = 464.8± 1.0
σ = 43.5± 0.9
WF (3156 entries)
σ/μ = 9.91± 0.23 %
μ = 469.9± 1.0
σ = 46.6± 1.0

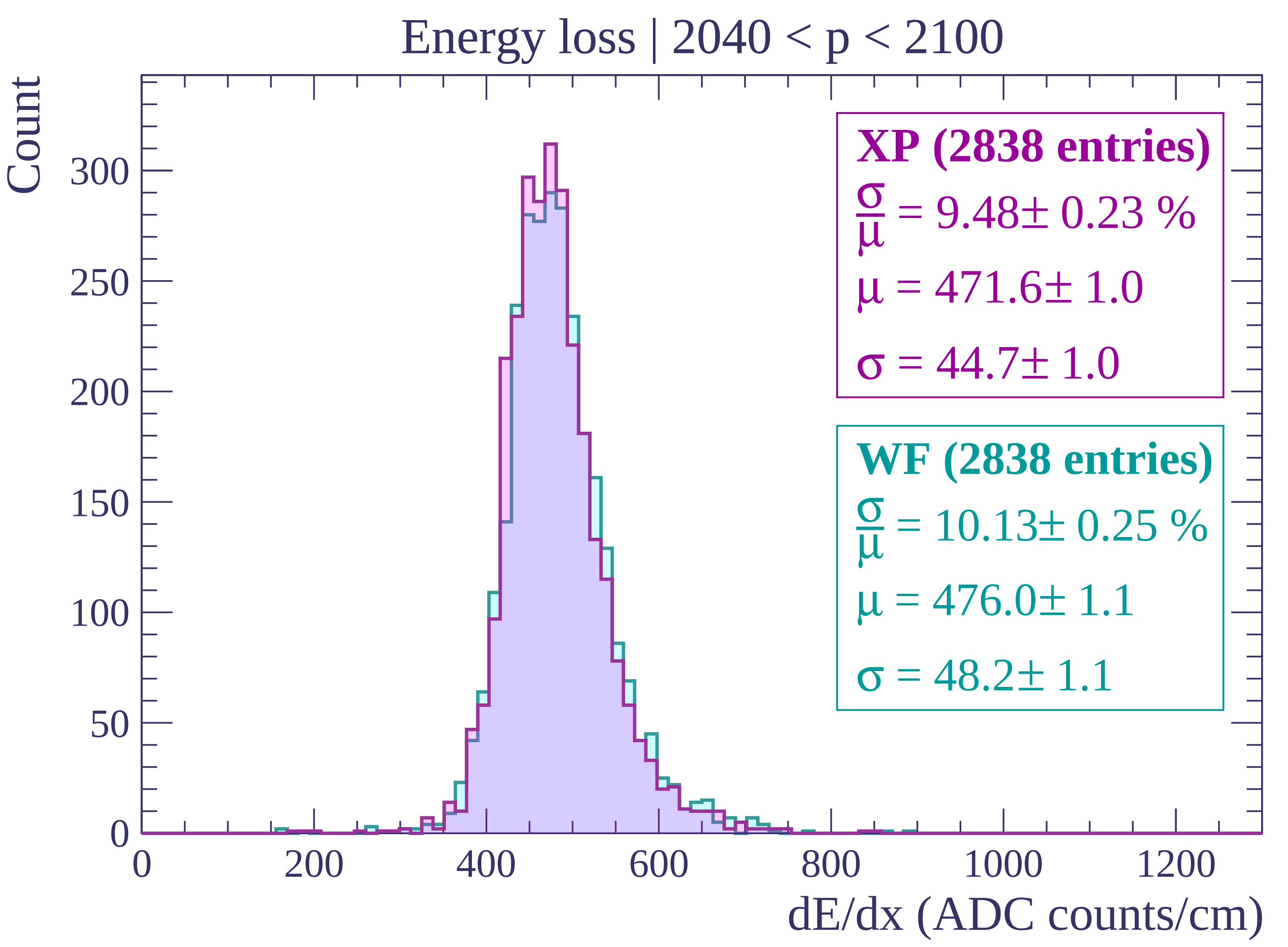

Energy loss | 1860 < p < 1920
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP (3161 entries)
$\frac{\sigma}{\mu}$ = 9.60± 0.26 %
μ = 466.7± 1.0
σ = 44.8± 1.1
WF (3161 entries)
$\frac{\sigma}{\mu}$ = 9.80± 0.24 %
μ = 471.7± 1.0
σ = 46.2± 1.0



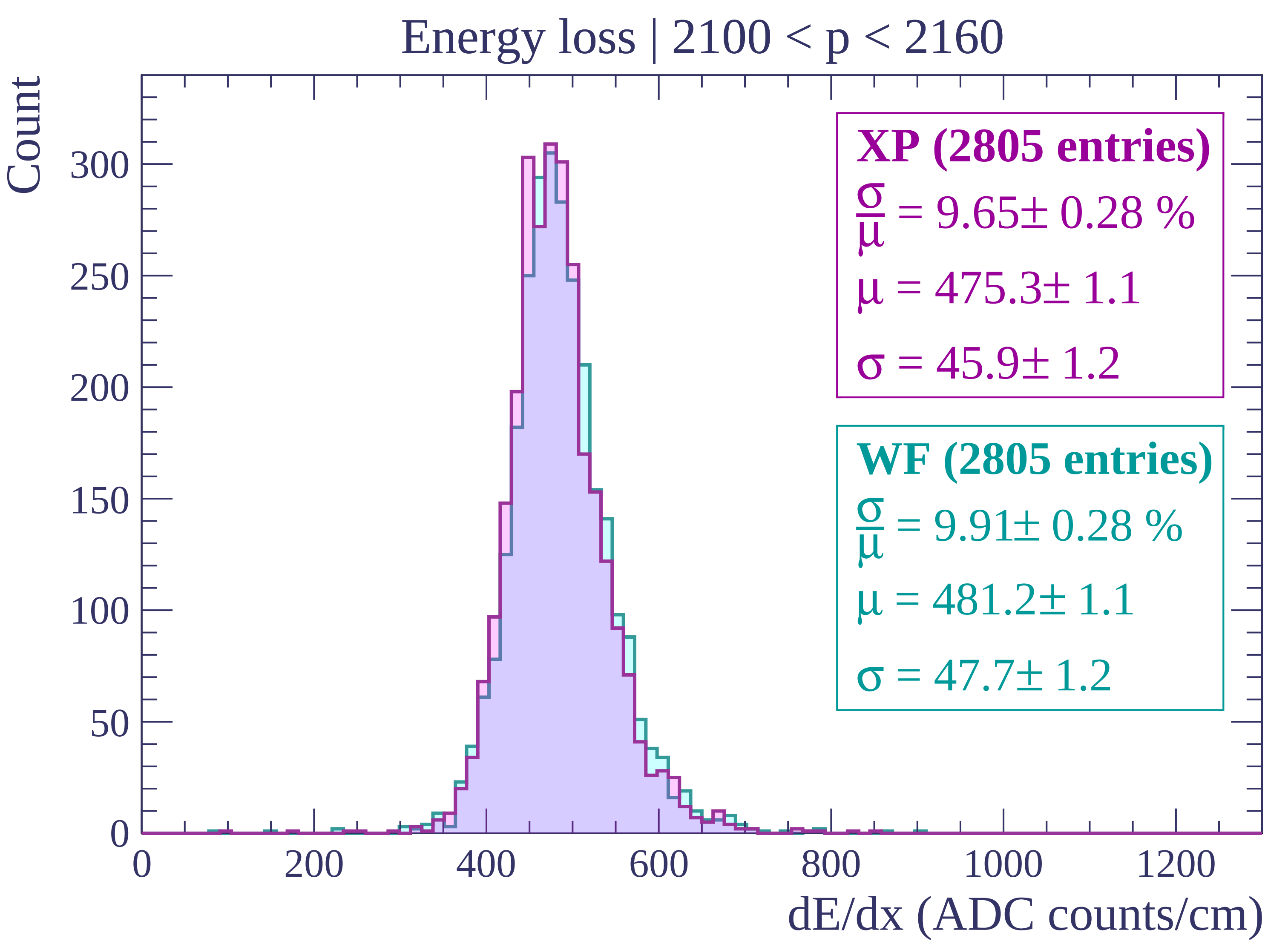
Energy loss | 1920 < p < 1980
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (3056 entries)
σ/μ = 9.79± 0.23 %
μ = 468.6± 1.0
σ = 45.9± 1.0
WF (3056 entries)
σ/μ = 10.18± 0.23 %
μ = 474.2± 1.0
σ = 48.3± 1.0

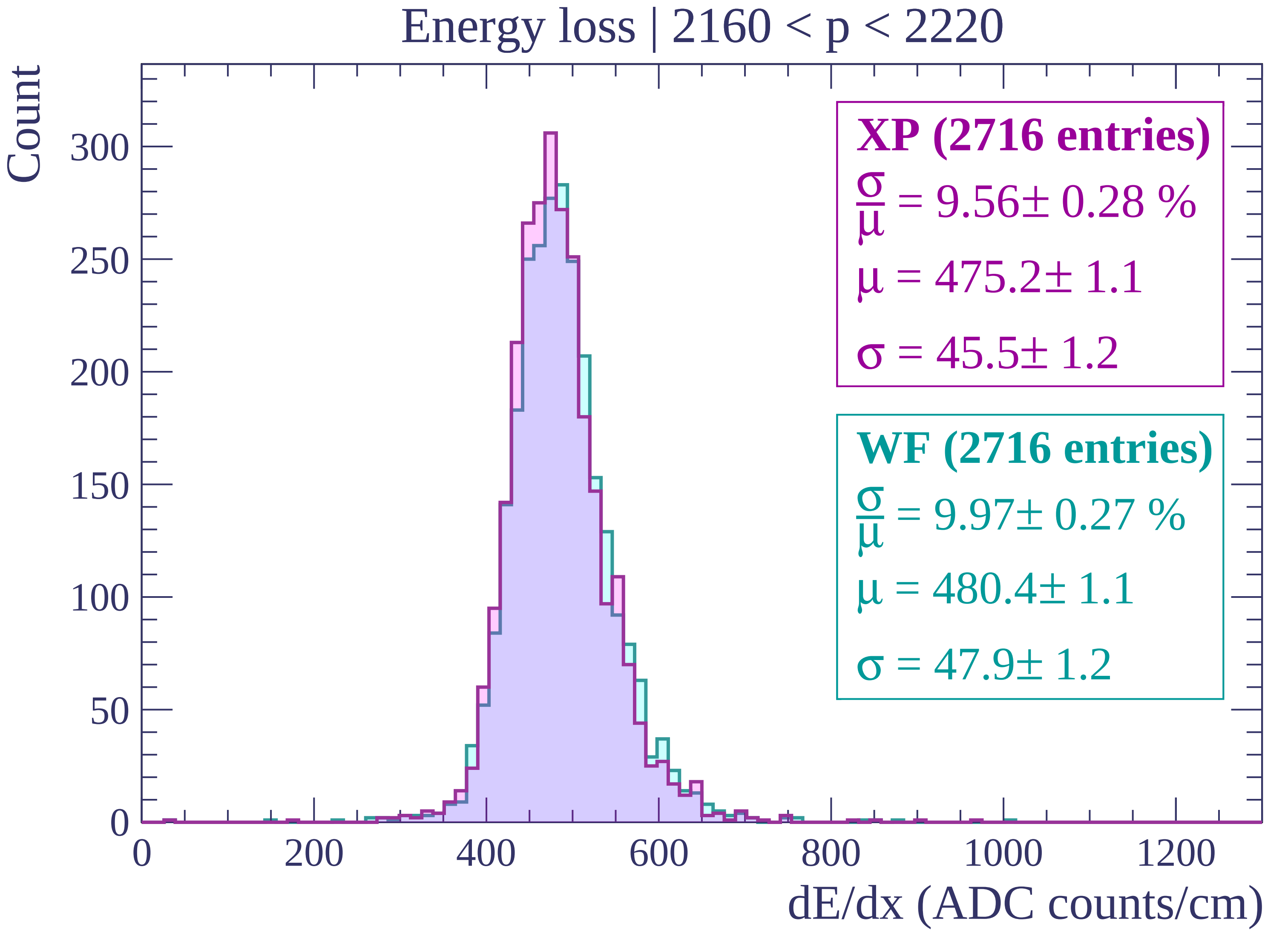


Energy loss | 1980 < p < 2040
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (3010 entries)
σ/μ = 9.01± 0.22 %
μ = 467.0± 1.0
σ = 42.1± 0.9
WF (3010 entries)
σ/μ = 9.53± 0.23 %
μ = 472.4± 1.0
σ = 45.0± 1.0

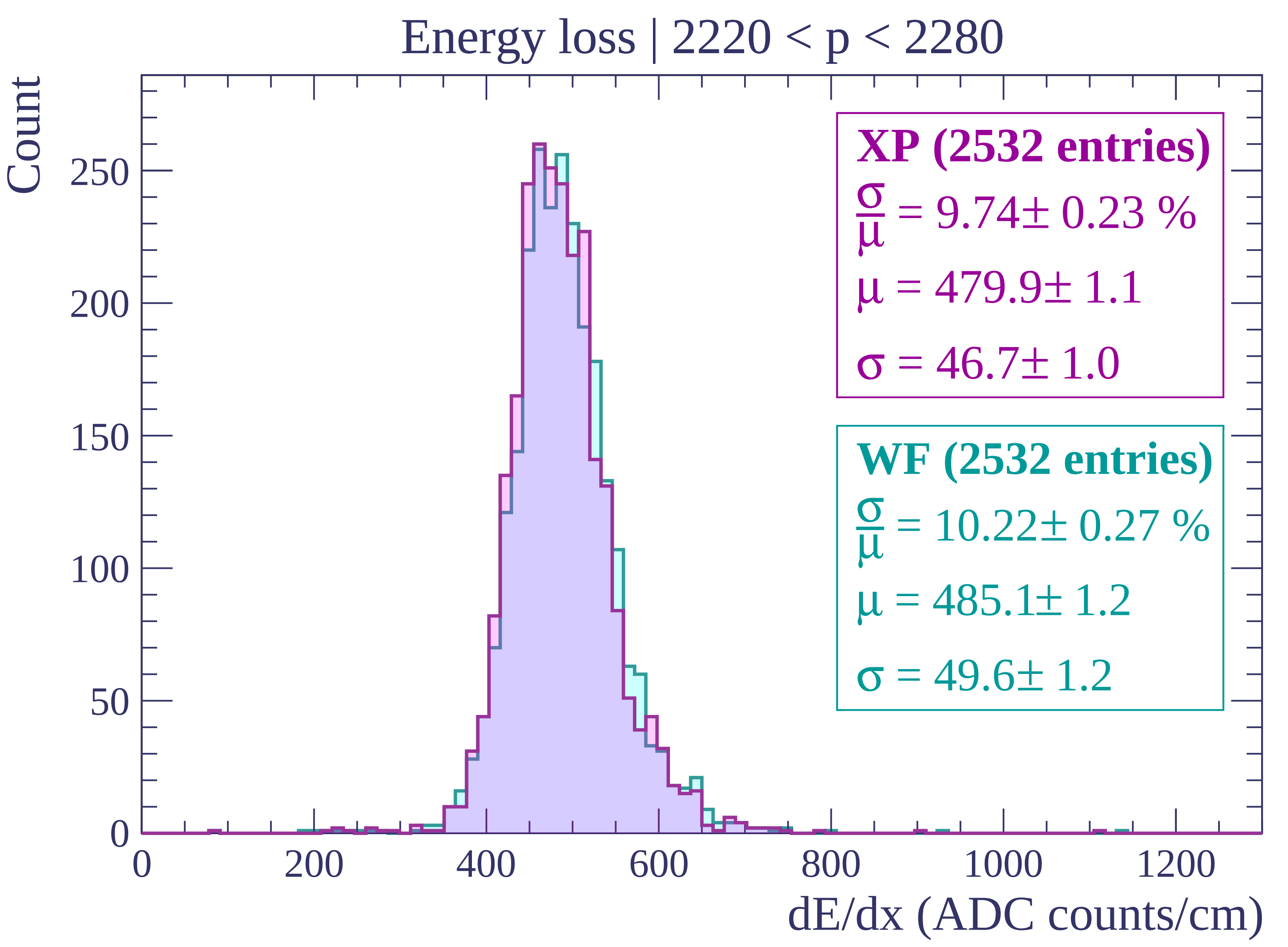


Energy loss | 2040 < p < 2100
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (2838 entries)
$\frac{\sigma}{\mu}$ = 9.48± 0.23 %
μ = 471.6± 1.0
σ = 44.7± 1.0
WF (2838 entries)
$\frac{\sigma}{\mu}$ = 10.13± 0.25 %
μ = 476.0± 1.1
σ = 48.2± 1.1

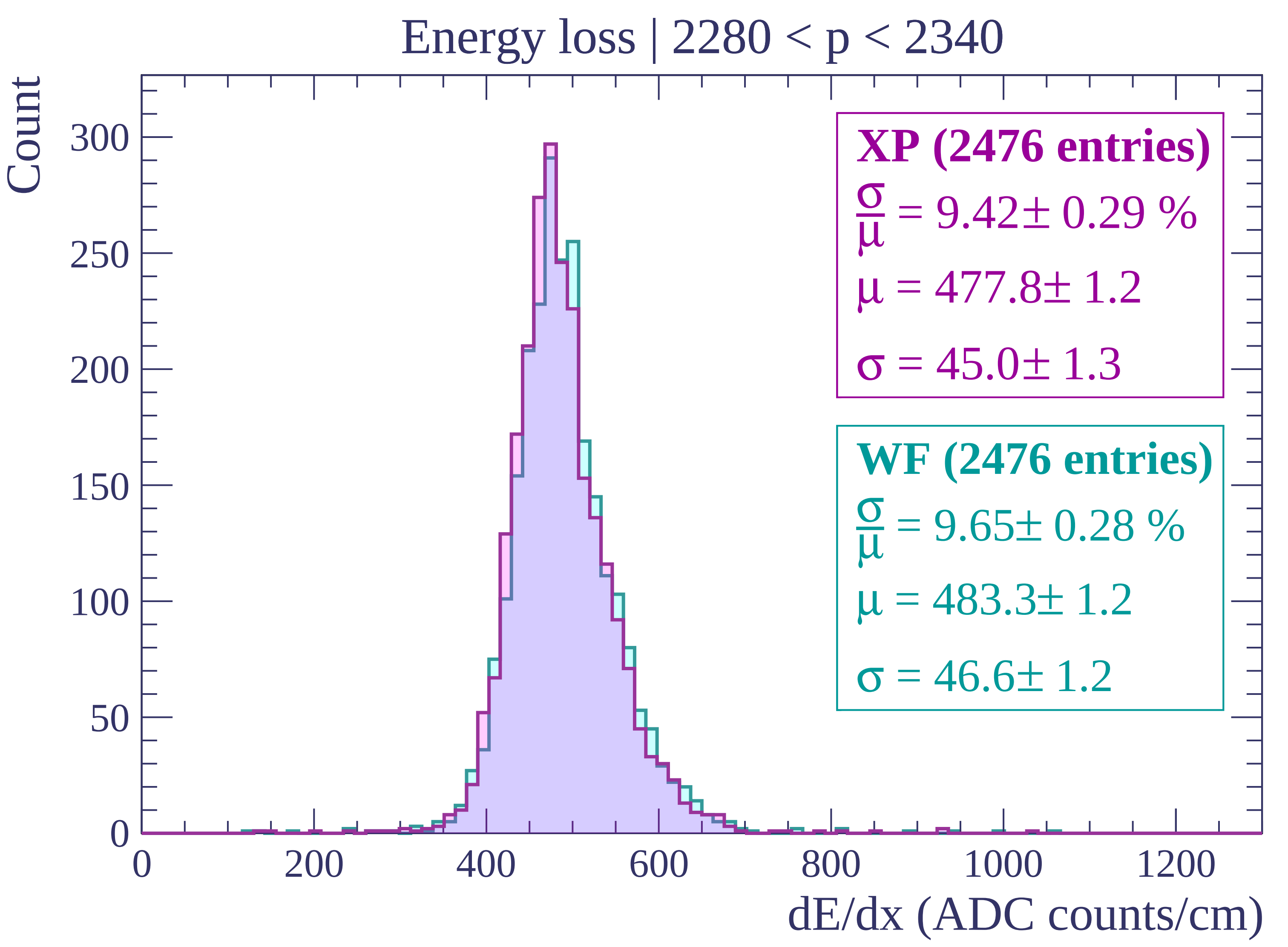

Energy loss | 2100 < p < 2160
Count
dE/dx (ADC counts/cm)
XP (2805 entries)
$\frac{\sigma}{\mu}$ = 9.65± 0.28 %
μ = 475.3± 1.1
σ = 45.9± 1.2
WF (2805 entries)
$\frac{\sigma}{\mu}$ = 9.91± 0.28 %
μ = 481.2± 1.1
σ = 47.7± 1.2



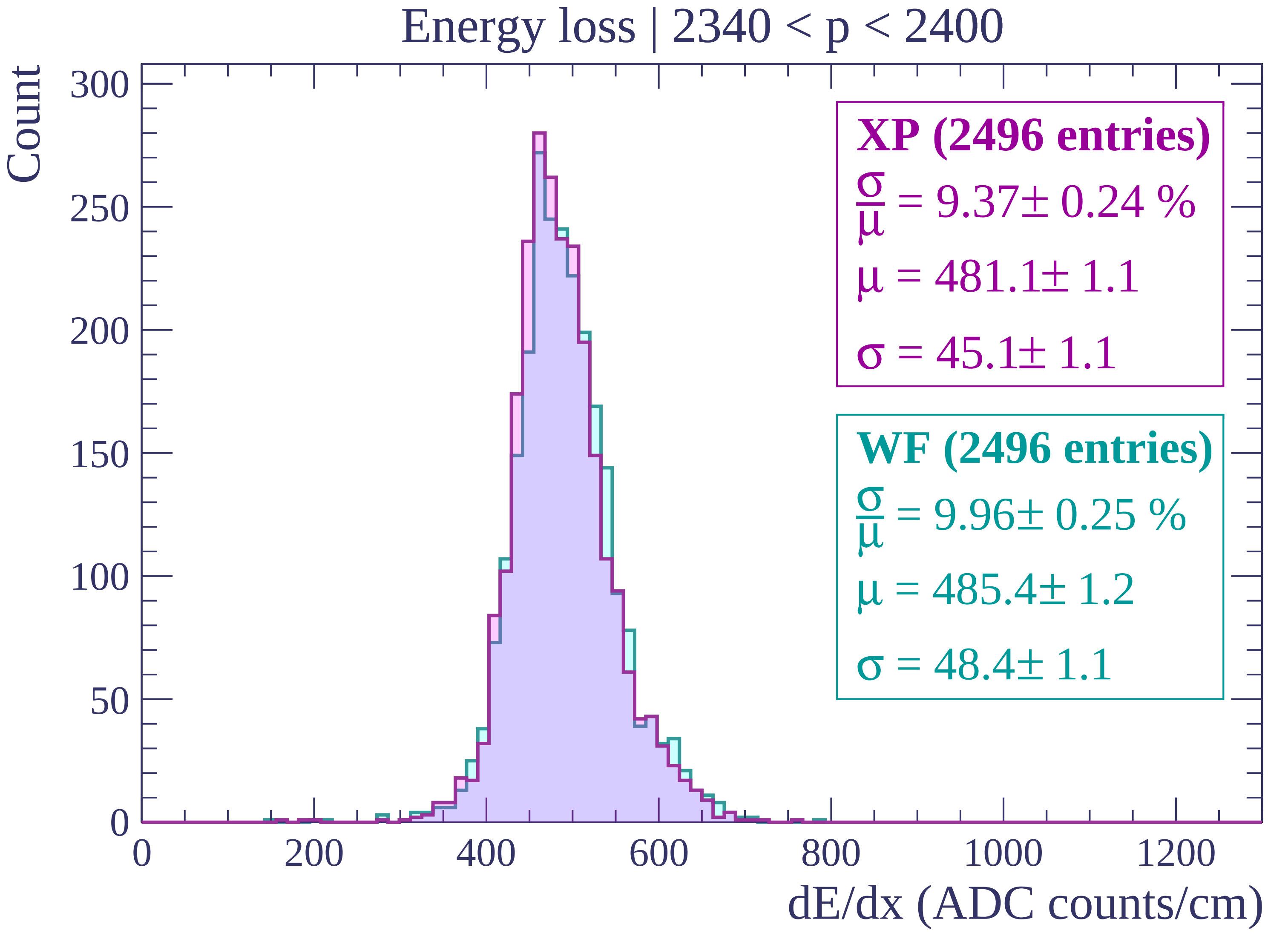
Energy loss | 2160 < p < 2220
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (2716 entries)
σ/μ = 9.56± 0.28 %
μ = 475.2± 1.1
σ = 45.5± 1.2
WF (2716 entries)
σ/μ = 9.97± 0.27 %
μ = 480.4± 1.1
σ = 47.9± 1.2



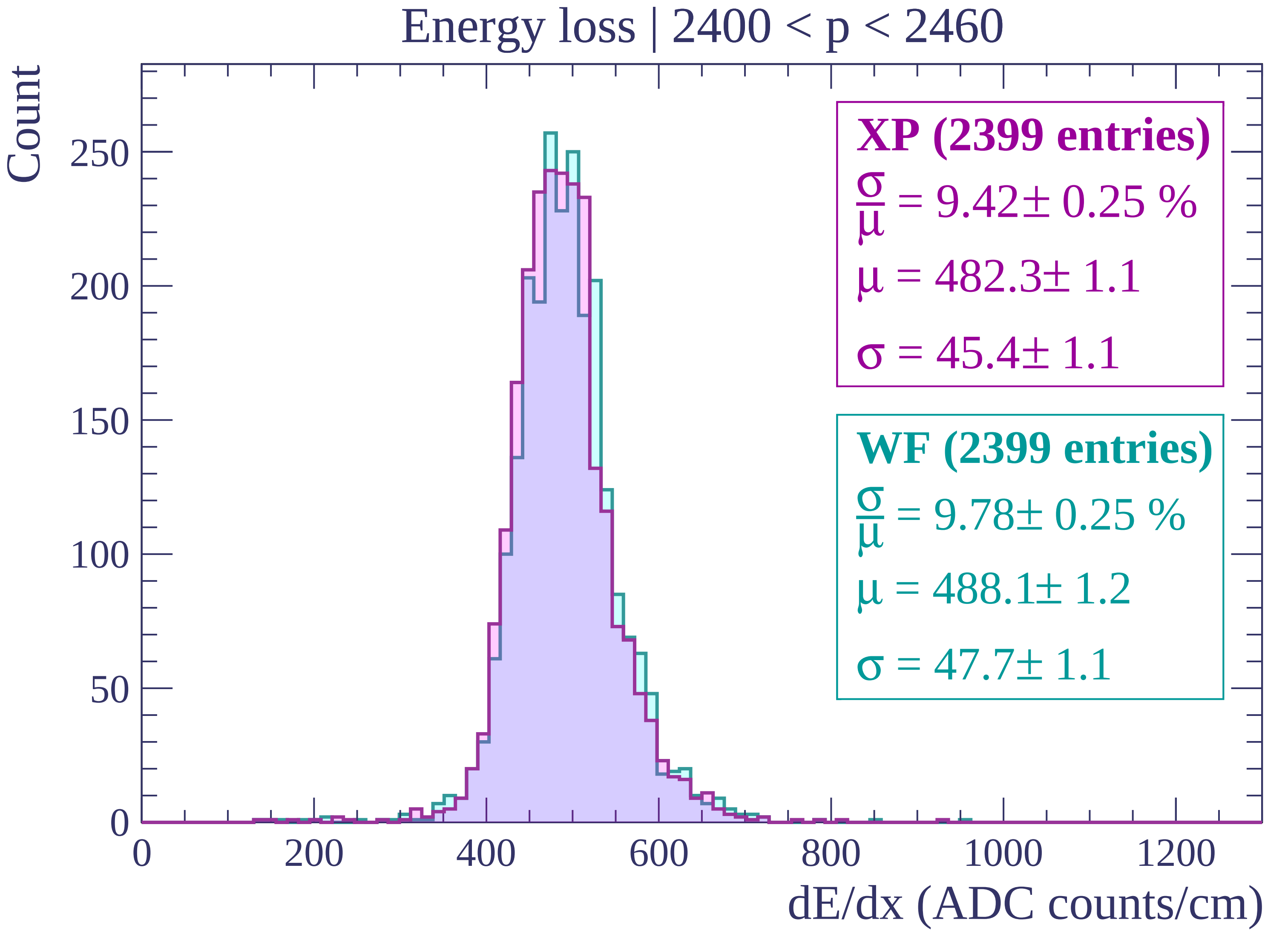
Energy loss | 2220 < p < 2280
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2532 entries)
$\frac{\sigma}{\mu}$ = 9.74± 0.23 %
μ = 479.9± 1.1
σ = 46.7± 1.0
WF (2532 entries)
$\frac{\sigma}{\mu}$ = 10.22± 0.27 %
μ = 485.1± 1.2
σ = 49.6± 1.2



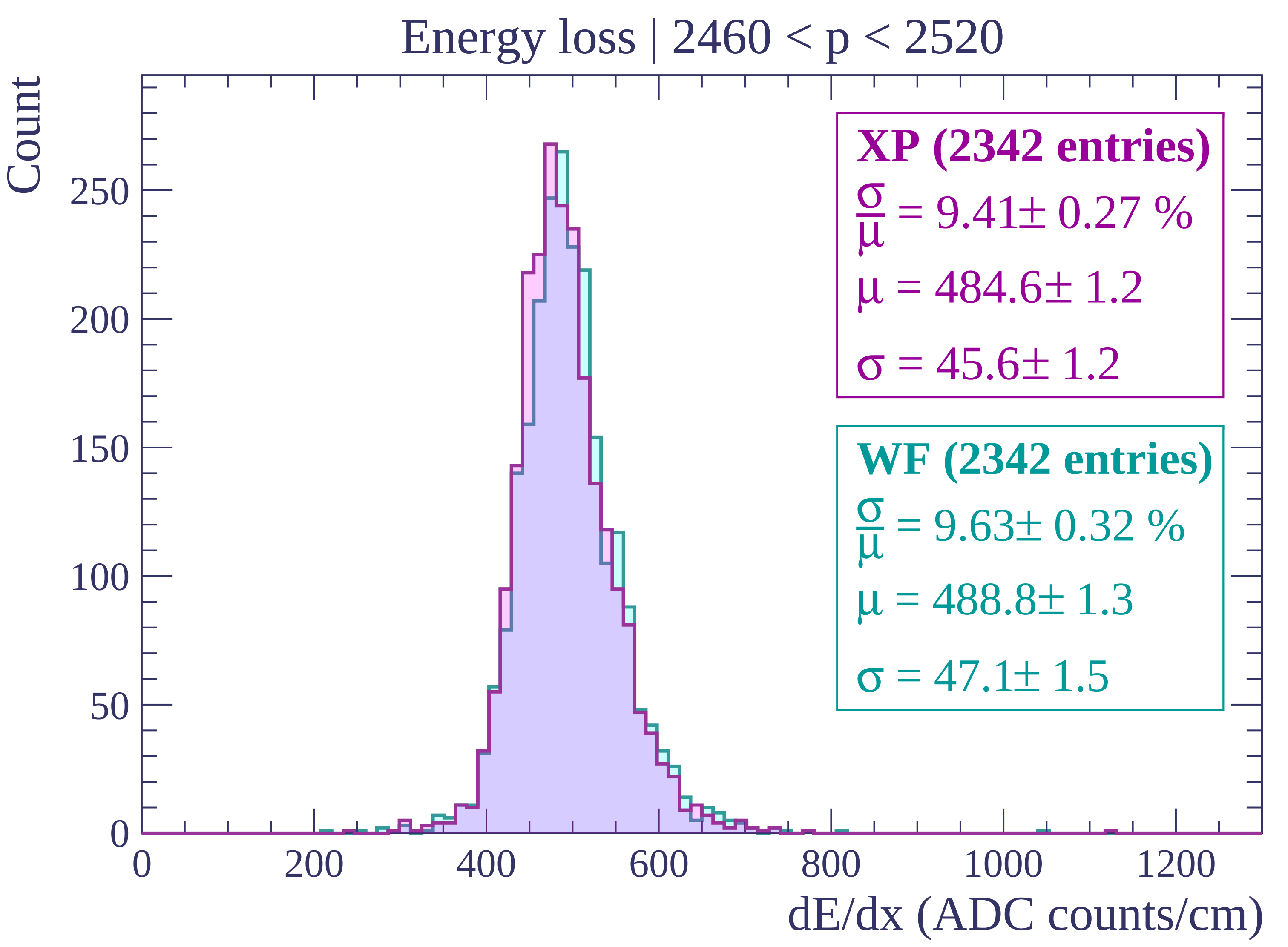
Energy loss | 2280 < p < 2340
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (2476 entries)
σ/μ = 9.42± 0.29 %
μ = 477.8± 1.2
σ = 45.0± 1.3
WF (2476 entries)
σ/μ = 9.65± 0.28 %
μ = 483.3± 1.2
σ = 46.6± 1.2



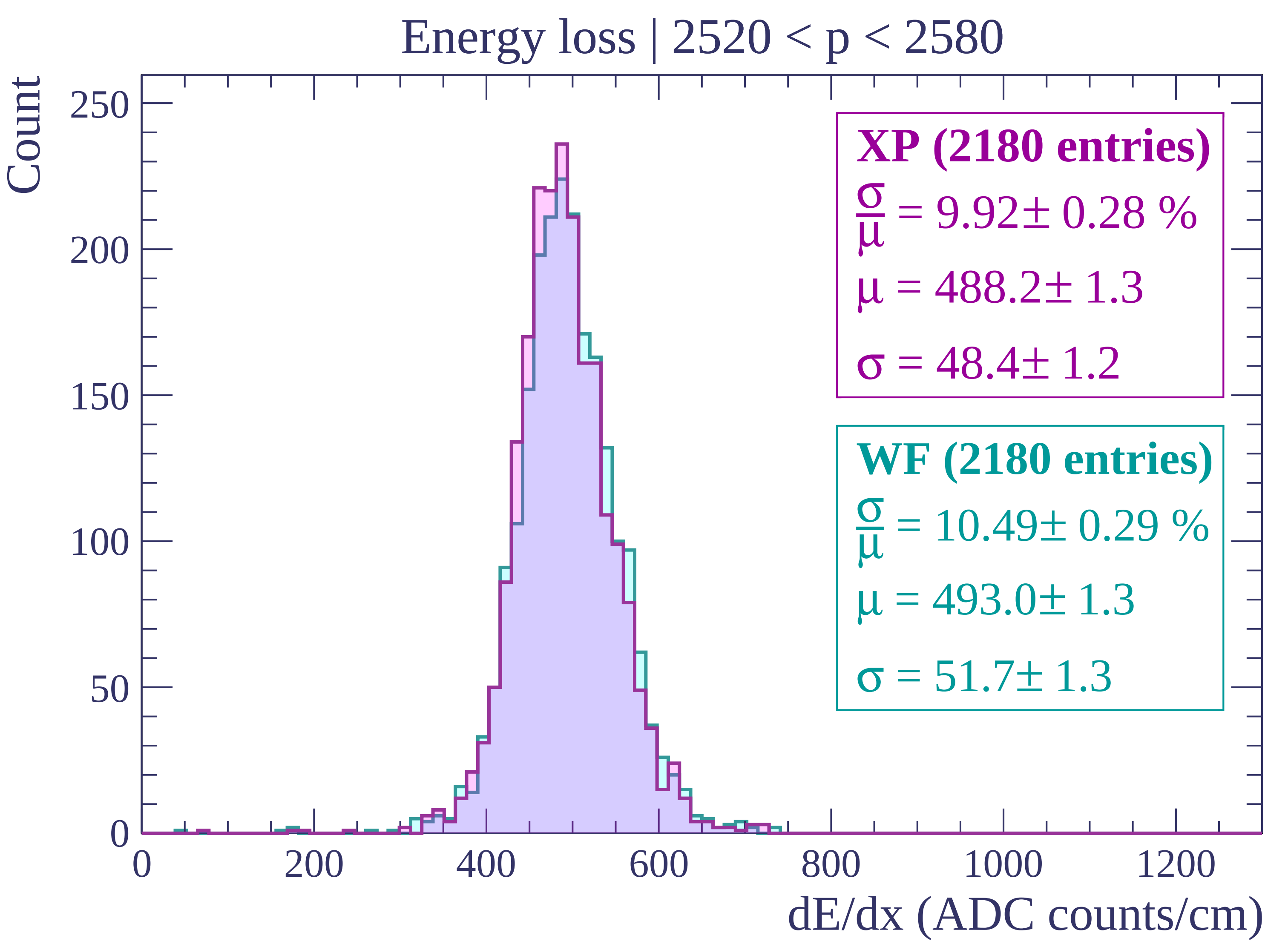
Energy loss | 2340 < p < 2400
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (2496 entries)
σ/μ = 9.37± 0.24 %
μ = 481.1± 1.1
σ = 45.1± 1.1
WF (2496 entries)
σ/μ = 9.96± 0.25 %
μ = 485.4± 1.2
σ = 48.4± 1.1



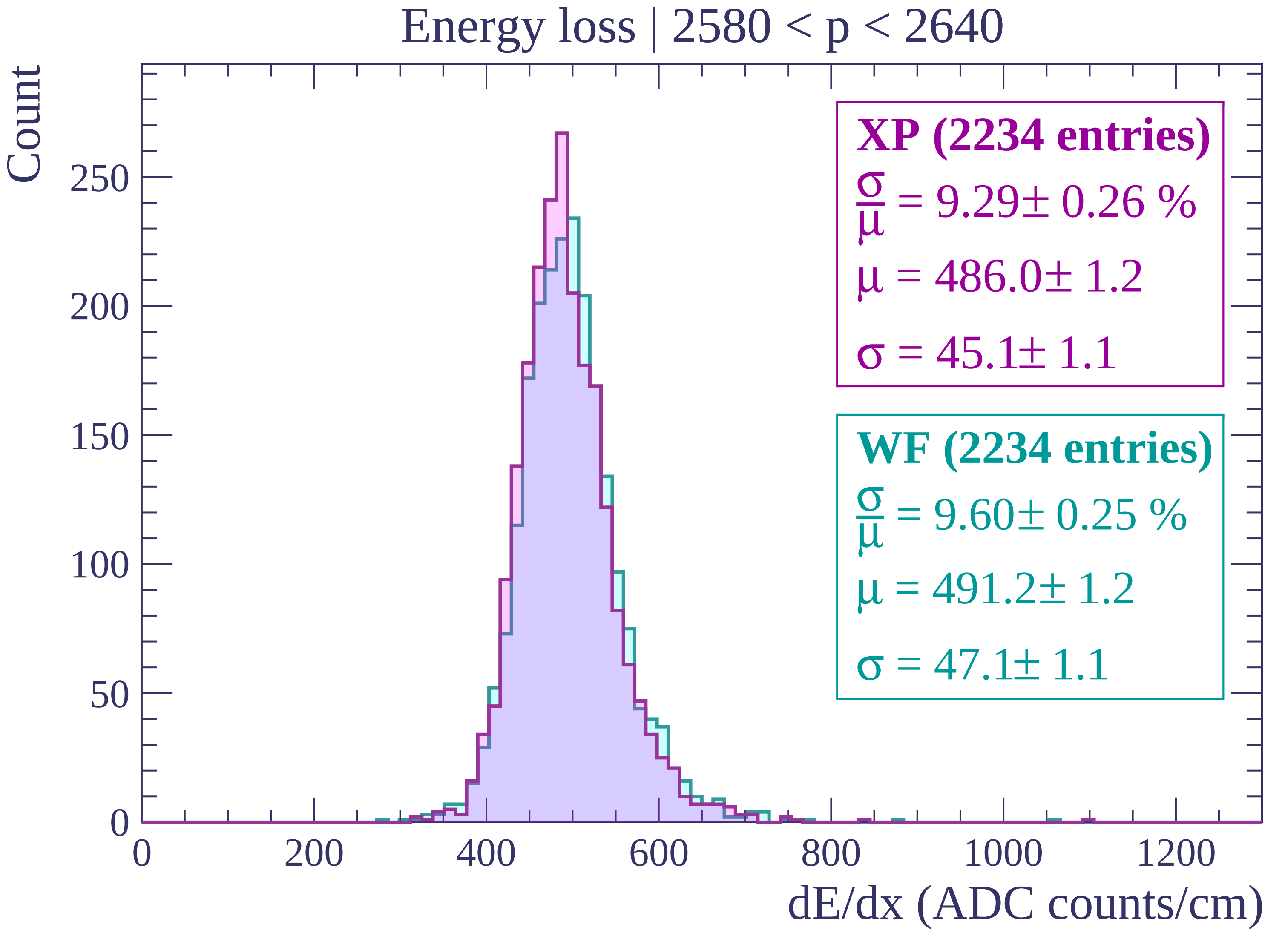
Energy loss | 2400 < p < 2460
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP (2399 entries)
σ/μ = 9.42± 0.25 %
μ = 482.3± 1.1
σ = 45.4± 1.1
WF (2399 entries)
σ/μ = 9.78± 0.25 %
μ = 488.1± 1.2
σ = 47.7± 1.1

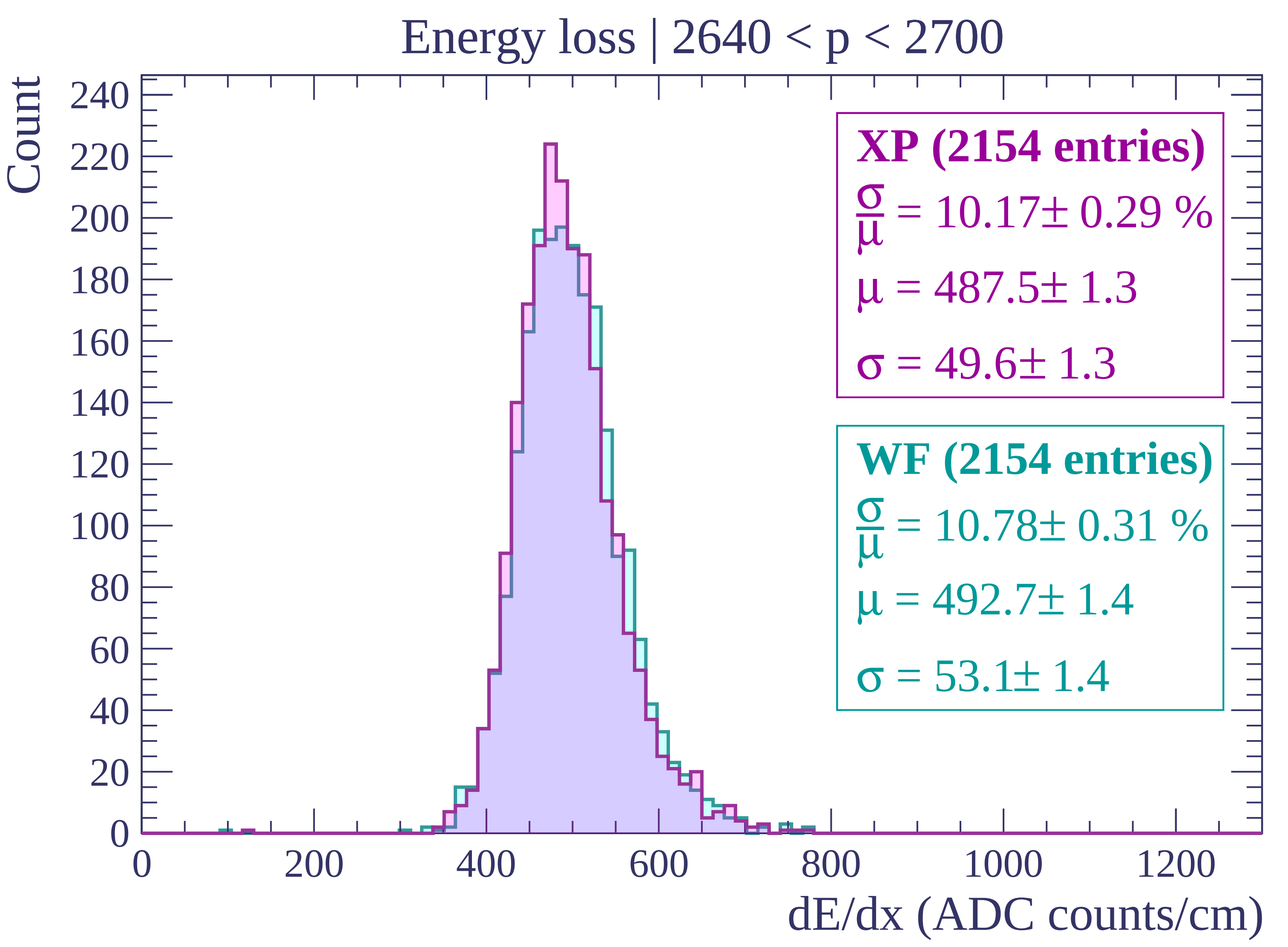


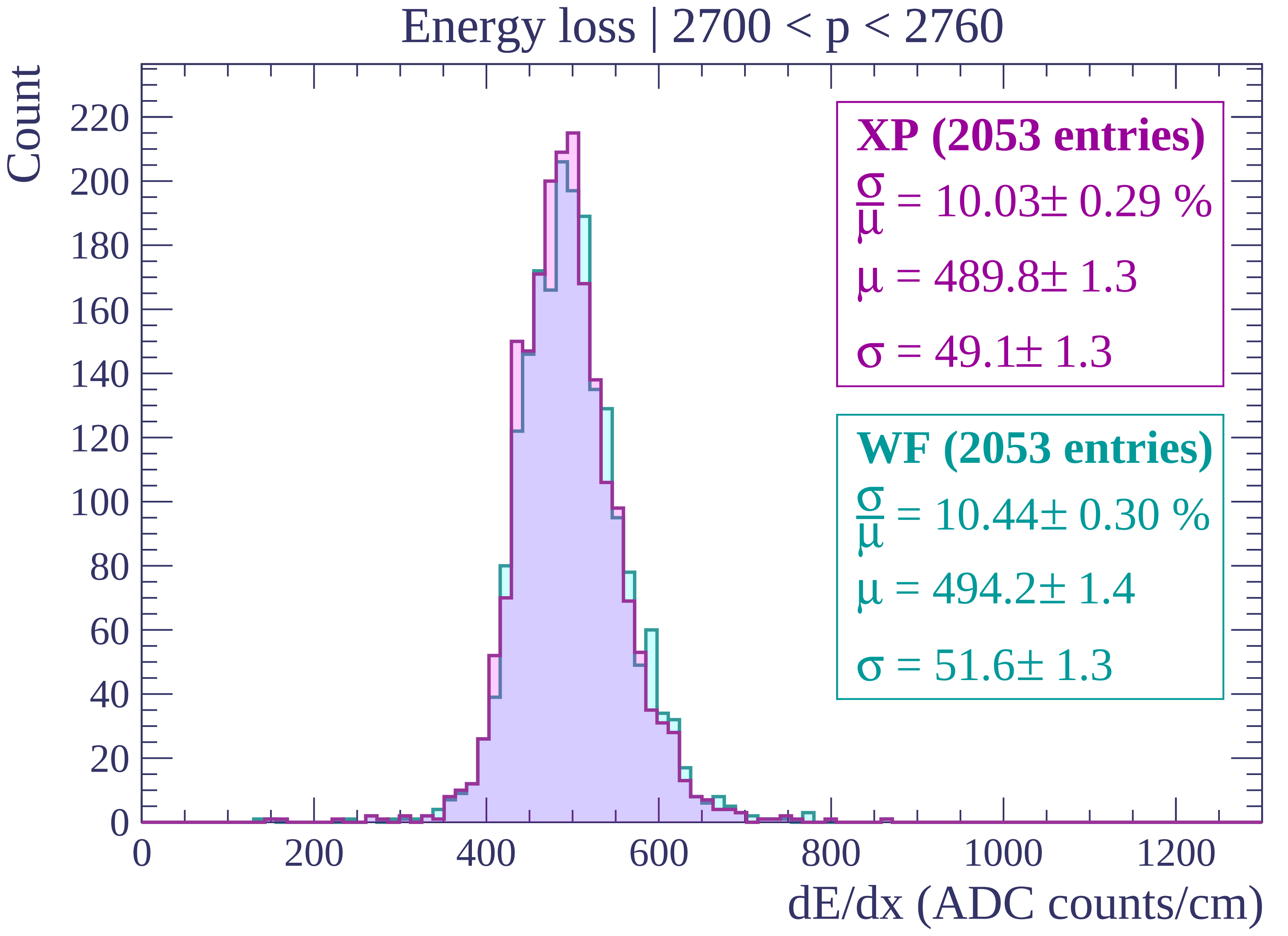
Energy loss | 2460 < p < 2520
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP (2342 entries)
$\frac{\sigma}{\mu}$ = 9.41± 0.27 %
μ = 484.6± 1.2
σ = 45.6± 1.2
WF (2342 entries)
$\frac{\sigma}{\mu}$ = 9.63± 0.32 %
μ = 488.8± 1.3
σ = 47.1± 1.5

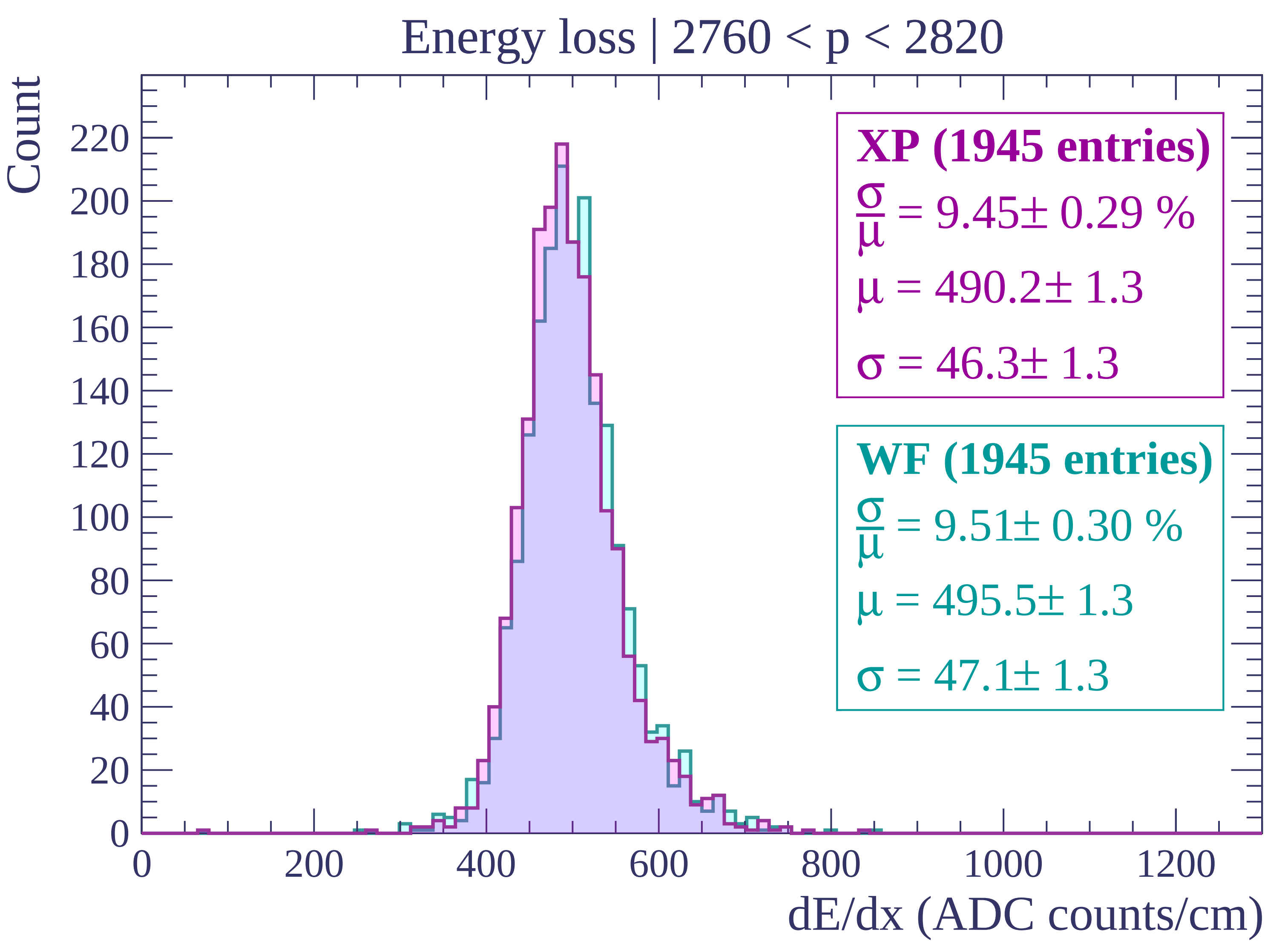


Energy loss | 2520 < p < 2580
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP (2180 entries)
$\frac{\sigma}{\mu}$ = 9.92± 0.28 %
μ = 488.2± 1.3
σ = 48.4± 1.2
WF (2180 entries)
$\frac{\sigma}{\mu}$ = 10.49± 0.29 %
μ = 493.0± 1.3
σ = 51.7± 1.3

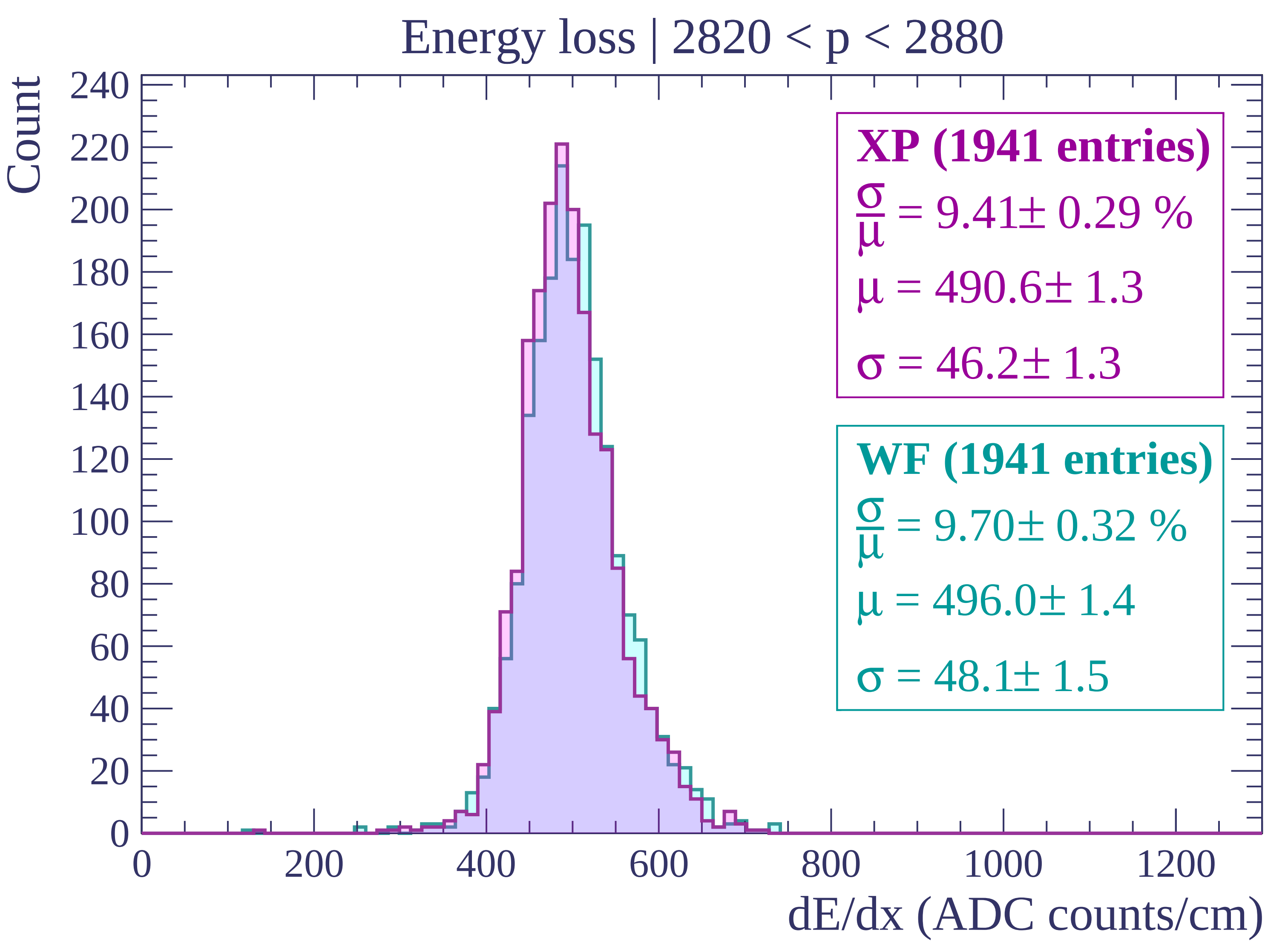

Energy loss | 2580 < p < 2640
Count
dE/dx (ADC counts/cm)
XP (2234 entries)
σ/μ = 9.29± 0.26 %
μ = 486.0± 1.2
σ = 45.1± 1.1
WF (2234 entries)
σ/μ = 9.60± 0.25 %
μ = 491.2± 1.2
σ = 47.1± 1.1
0
50
100
150
200
250
0
200
400
600
800
1000
1200


Energy loss | 2640 < p < 2700
Count
dE/dx (ADC counts/cm)
XP (2154 entries)
σ/μ = 10.17± 0.29 %
μ = 487.5± 1.3
σ = 49.6± 1.3
WF (2154 entries)
σ/μ = 10.78± 0.31 %
μ = 492.7± 1.4
σ = 53.1± 1.4

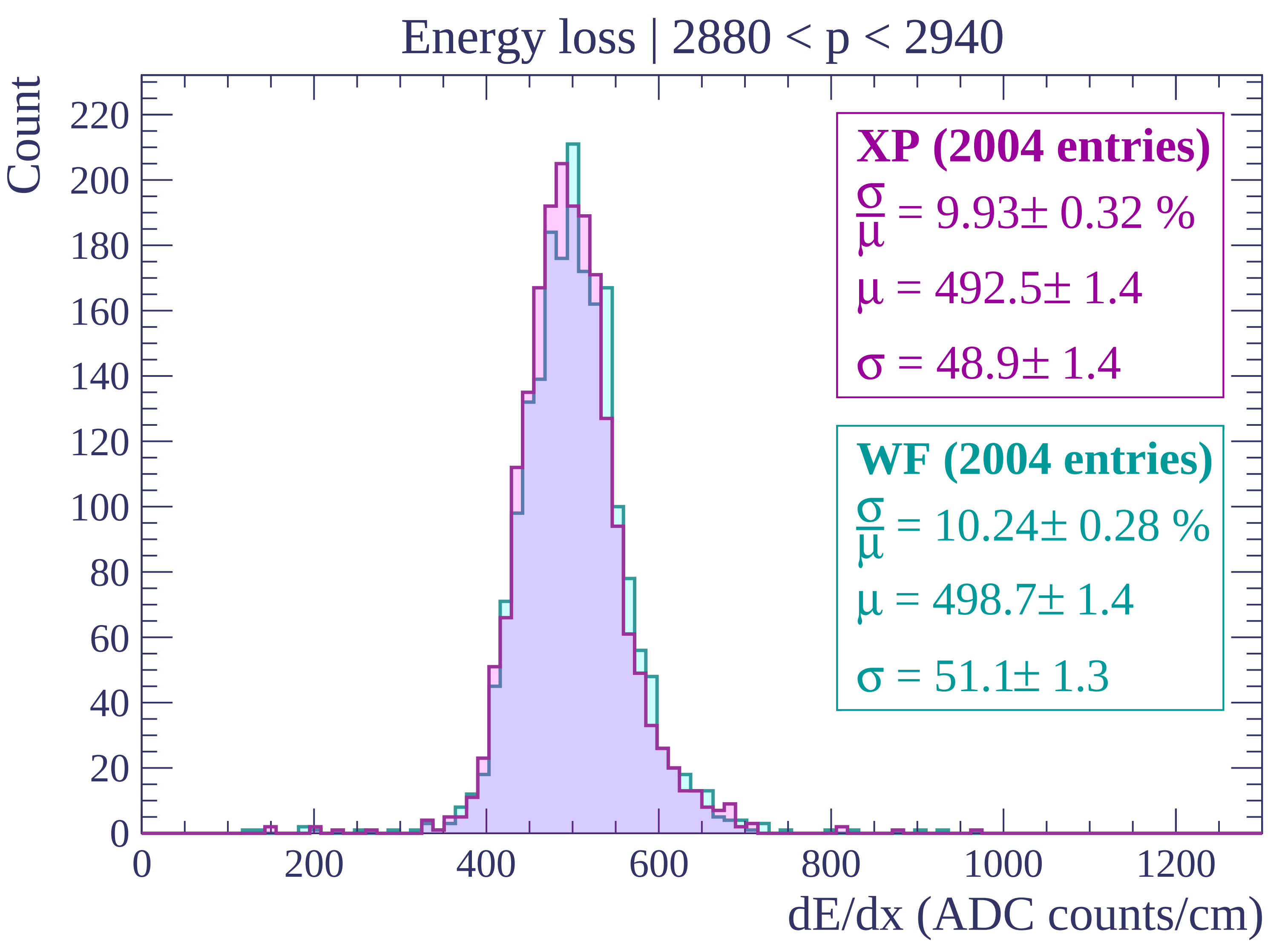

Energy loss | 2700 < p < 2760
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
XP (2053 entries)
σ/μ = 10.03± 0.29 %
μ = 489.8± 1.3
σ = 49.1± 1.3
WF (2053 entries)
σ/μ = 10.44± 0.30 %
μ = 494.2± 1.4
σ = 51.6± 1.3



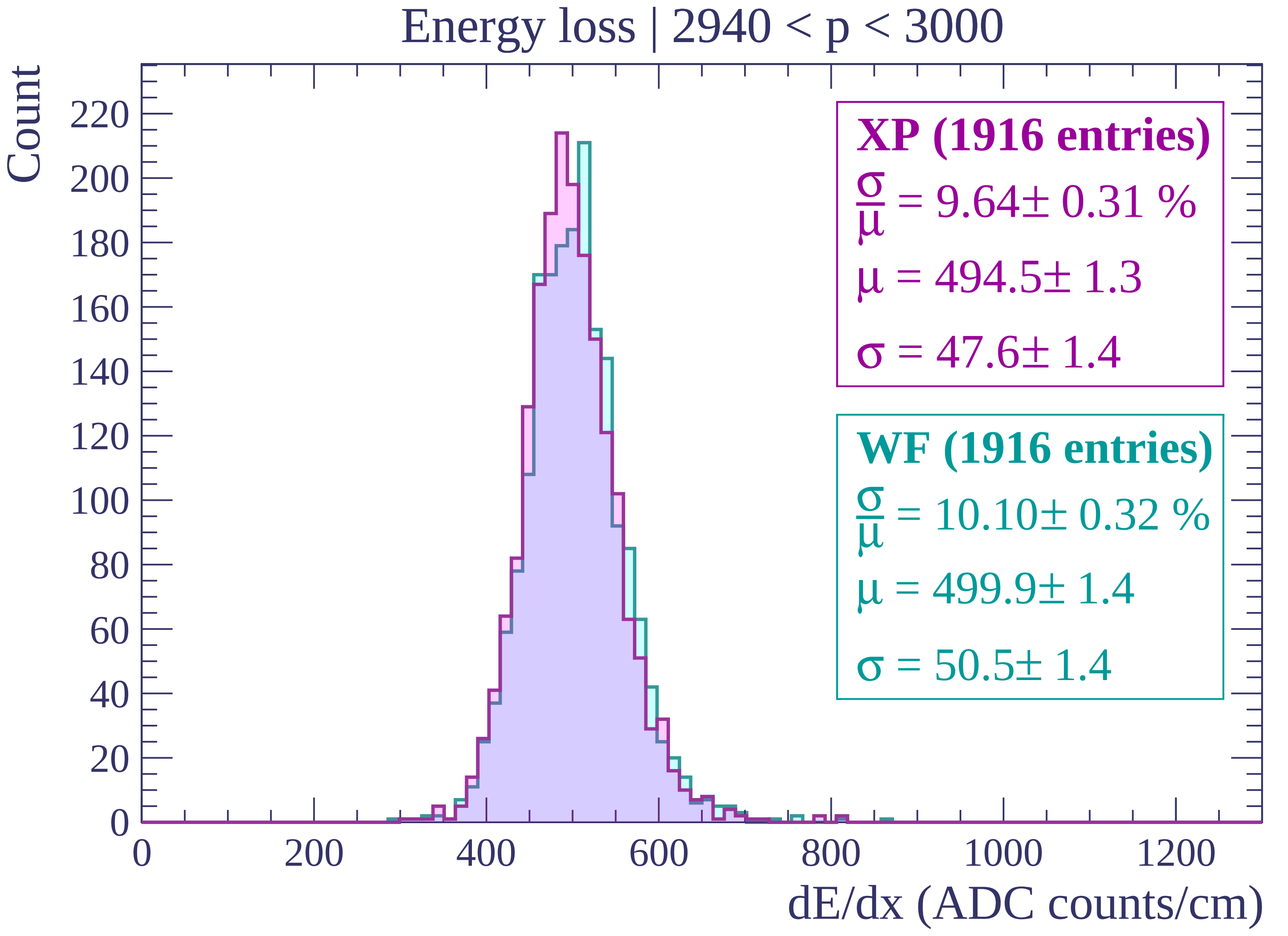
Energy loss | 2760 < p < 2820
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
XP (1945 entries)
$\frac{\sigma}{\mu}$ = 9.45± 0.29 %
μ = 490.2± 1.3
σ = 46.3± 1.3
WF (1945 entries)
$\frac{\sigma}{\mu}$ = 9.51± 0.30 %
μ = 495.5± 1.3
σ = 47.1± 1.3

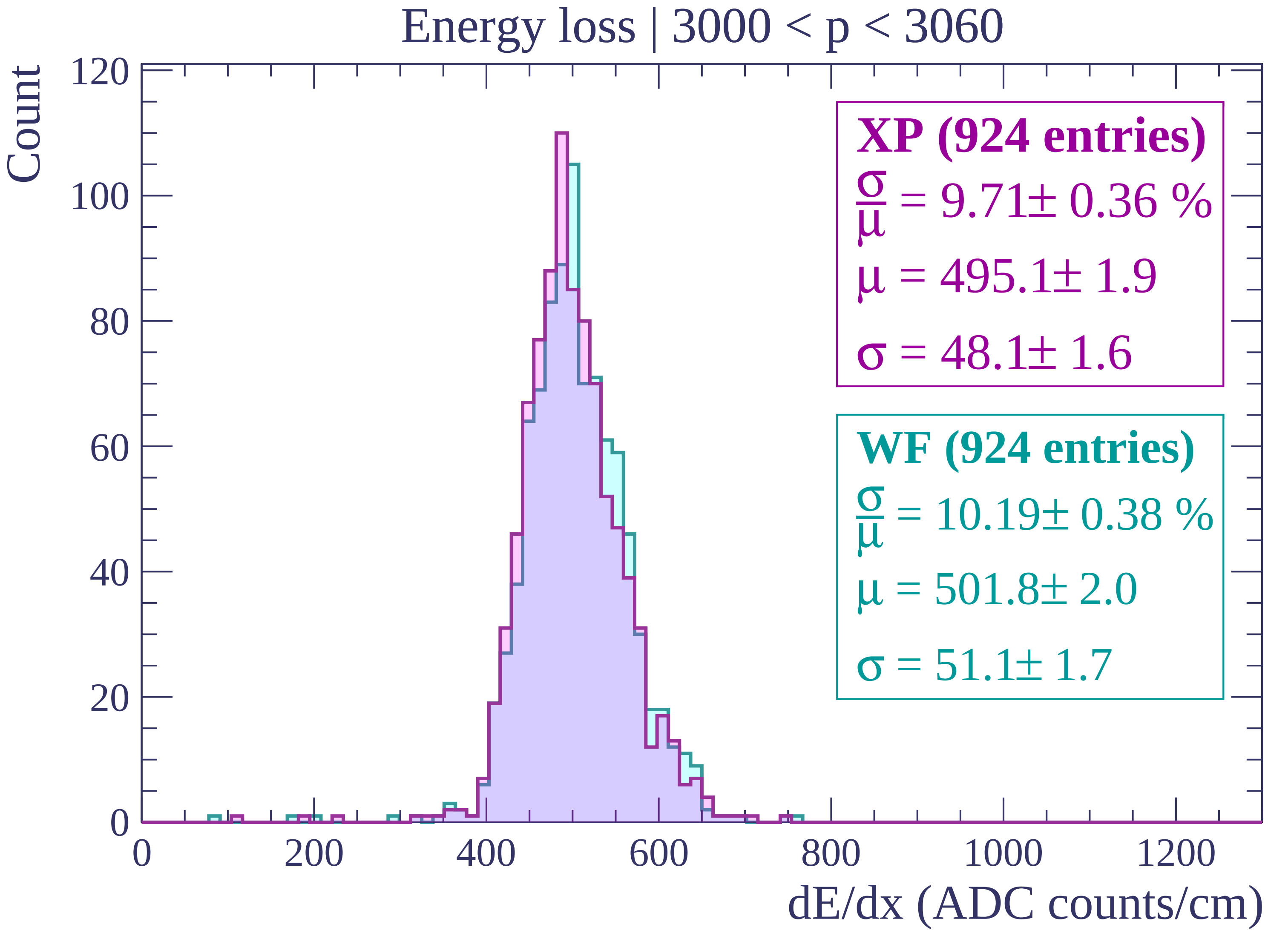


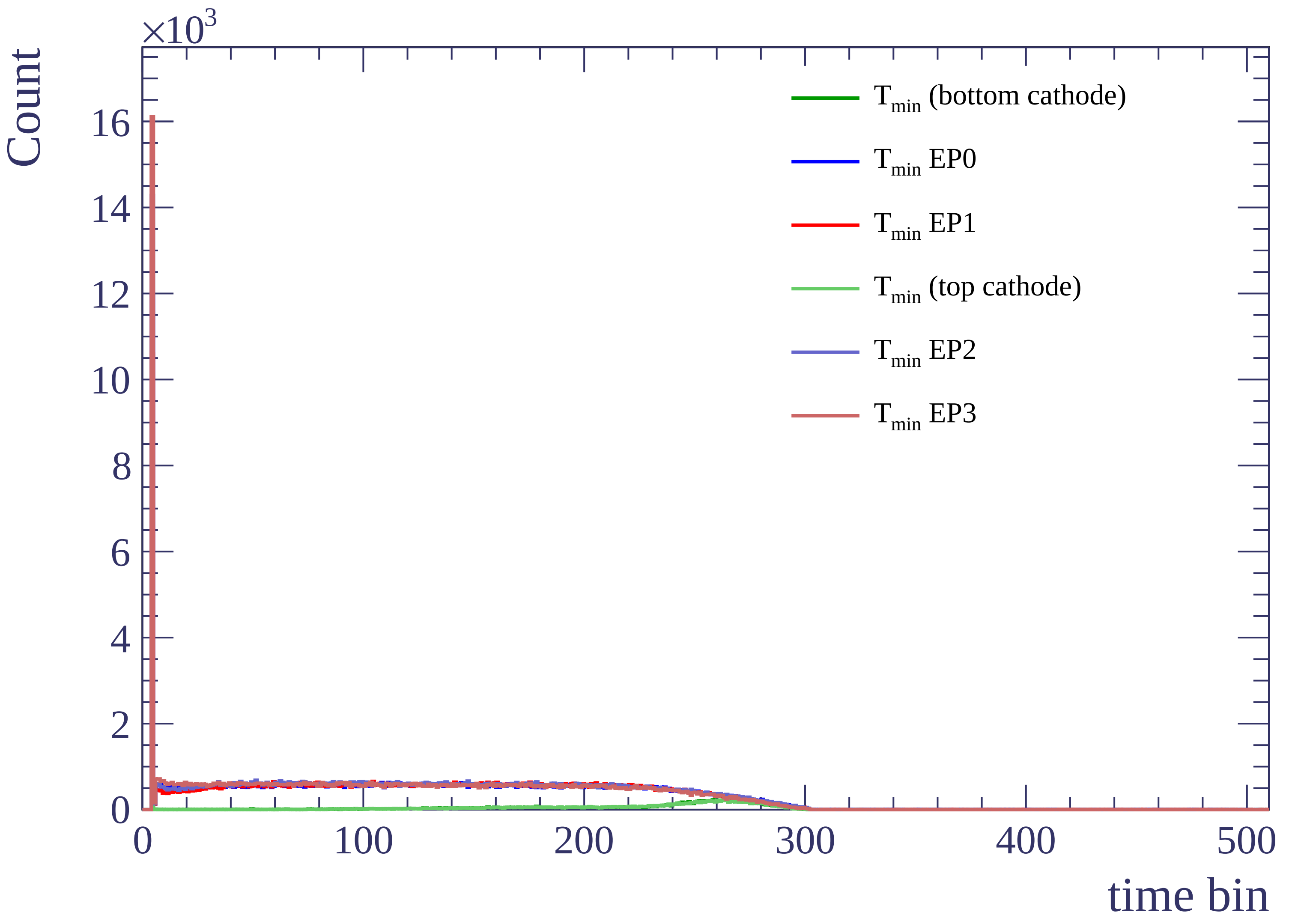
Energy loss | 2820 < p < 2880
Count
dE/dx (ADC counts/cm)
XP (1941 entries)
σ/μ = 9.41± 0.29 %
μ = 490.6± 1.3
σ = 46.2± 1.3
WF (1941 entries)
σ/μ = 9.70± 0.32 %
μ = 496.0± 1.4
σ = 48.1± 1.5



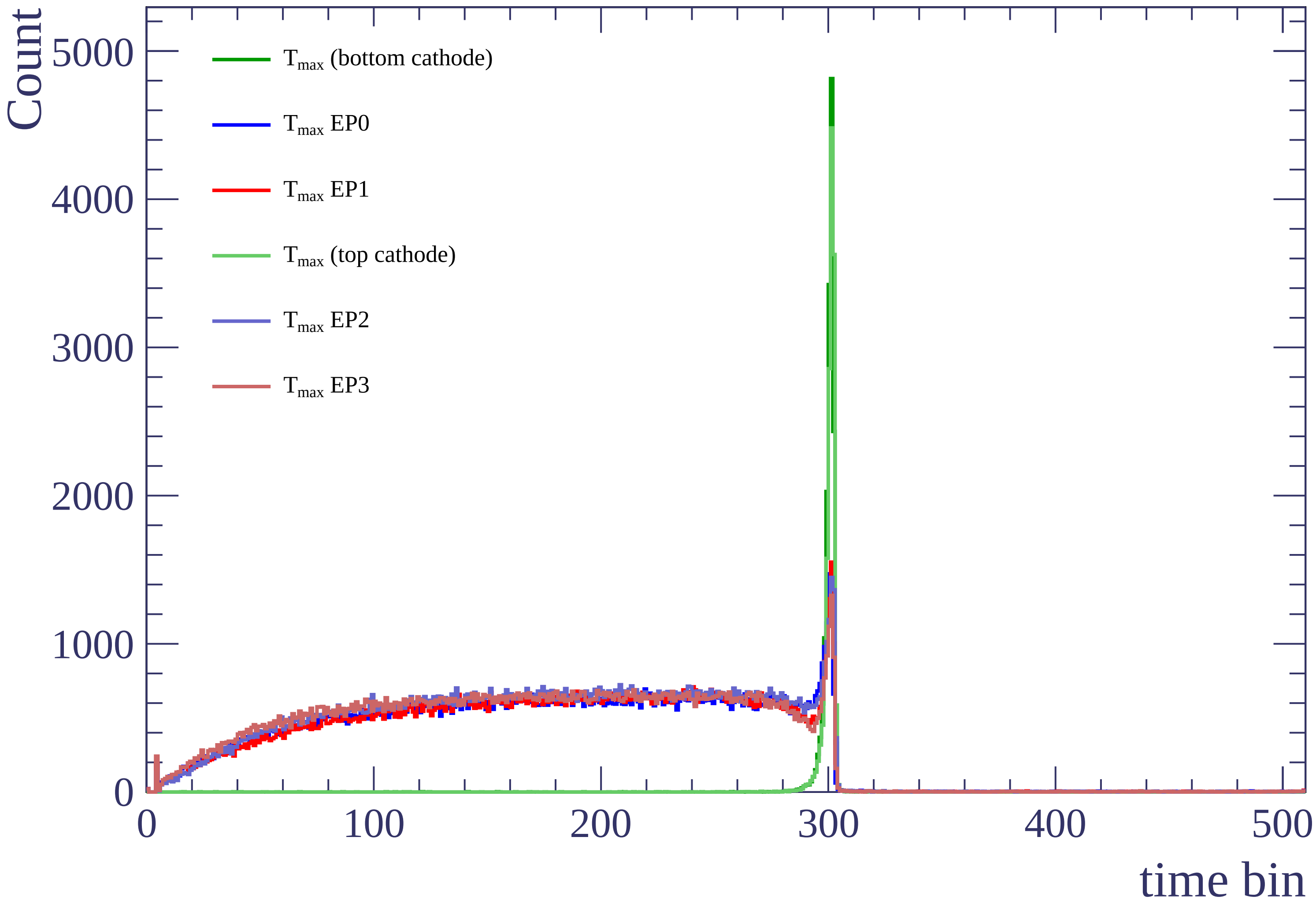
Energy loss | 2880 < p < 2940
Count
dE/dx (ADC counts/cm)
XP (2004 entries)
$\frac{\sigma}{\mu}$ = 9.93± 0.32 %
μ = 492.5± 1.4
σ = 48.9± 1.4
WF (2004 entries)
$\frac{\sigma}{\mu}$ = 10.24± 0.28 %
μ = 498.7± 1.4
σ = 51.1± 1.3


Energy loss | 2940 < p < 3000
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
XP (1916 entries)
σ/μ = 9.64± 0.31 %
μ = 494.5± 1.3
σ = 47.6± 1.4
WF (1916 entries)
σ/μ = 10.10± 0.32 %
μ = 499.9± 1.4
σ = 50.5± 1.4


Energy loss | 3000 < p < 3060
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP (924 entries)
σ/μ = 9.71± 0.36 %
μ = 495.1± 1.9
σ = 48.1± 1.6
WF (924 entries)
σ/μ = 10.19± 0.38 %
μ = 501.8± 2.0
σ = 51.1± 1.7



$\times 10^3$
Count
16
14
12
10
8
6
4
2
0
0
100
200
300
400
500
time bin
T_{min} (bottom cathode)
T_{min} EP0
T_{min} EP1
T_{min} (top cathode)
T_{min} EP2
T_{min} EP3


Count
time bin
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3
0
1000
2000
3000
4000
5000
0
100
200
300
400
500